Texts and Readings in Mathematics 74

Kalyan Sinha
Sachi Srivastava

Theory of Semigroups and Applications

HINDUSTAN BOOK AGENCY

Springer

Texts and Readings in Mathematics

Advisory Editor

C.S. Seshadri, Chennai Mathematical Institute, Chennai

Managing Editor

Rajendra Bhatia, Indian Statistical Institute, New Delhi

Editors

Manindra Agrawal, Indian Institute of Technology, Kanpur
V. Balaji, Chennai Mathematical Institute, Chennai
R.B. Bapat, Indian Statistical Institute, New Delhi
V.S. Borkar, Indian Institute of Technology, Mumbai
T.R. Ramadas, Chennai Mathematical Institute, Chennai
V. Srinivas, Tata Institute of Fundamental Research, Mumbai

Technical Editor

P. Vanchinathan, Vellore Institute of Technology, Chennai

The *Texts and Readings in Mathematics* series publishes high-quality textbooks, research-level monographs, lecture notes and contributed volumes. Undergraduate and graduate students of mathematics, research scholars, and teachers would find this book series useful. The volumes are carefully written as teaching aids and highlight characteristic features of the theory. The books in this series are co-published with Hindustan Book Agency, New Delhi, India.

More information about this series at http://www.springer.com/series/15141

Kalyan B. Sinha · Sachi Srivastava

Theory of Semigroups and Applications

HINDUSTAN BOOK AGENCY

Springer

Kalyan B. Sinha
Jawaharlal Nehru Centre for Advanced Scientific Research
Bangalore
India

Sachi Srivastava
Department of Mathematics
University of Delhi
New Delhi
India

ISSN 2366-8725 (electronic)
Texts and Readings in Mathematics
ISBN 978-981-10-4864-7 (eBook)
DOI 10.1007/978-981-10-4864-7

Library of Congress Control Number: 2017940820

This work is a co-publication with Hindustan Book Agency, New Delhi, licensed for sale in all countries in electronic form only. Sold and distributed in print across the world by Hindustan Book Agency, P-19 Green Park Extension, New Delhi 110016, India. ISBN: 978-93-86279-63-7 © Hindustan Book Agency 2017.
© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017
This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, express or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by Springer Nature
The registered company is Springer Nature Singapore Pte Ltd.
The registered company address is: 152 Beach Road, #21-01/04 Gateway East, Singapore 189721, Singapore

About the Authors

Kalyan B. Sinha is professor and the SERB-fellow at the Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), and at the Indian Institute of Science (IISc), Bengaluru. Professor Sinha is an Indian mathematician who specialised in mathematical theory of scattering, spectral theory of Schrödinger operators, and quantum stochastic processes. He was awarded in 1988 the Shanti Swarup Bhatnagar Prize for Science and Technology, the highest science award in India, in the mathematical sciences category. A Fellow of the Indian Academy of Science (IASc), Bengaluru, Indian National Science Academy (INSA), New Delhi, and The World Academy of Sciences (TWAS), Italy, he completed his PhD from the University of Rochester, New York, U.S.A.

Sachi Srivastava is associate professor at the Department of Mathematics, University of Delhi, India. She obtained her DPhil degree from Oxford University, UK and the MTech degree from the University of Delhi, India. Her areas of interest are functional analysis, operator theory, abstract differential equations, operator algebras. She is also a life member of the American Mathematical Society and Ramanujan Mathematical Society.

Preface

Semigroups (or groups, in many situations) of maps or operators in a linear space have played important roles, mathematically encapsulating the idea of *homogeneous evolution* of many observed systems, physical or otherwise. As an abstract mathematical discipline, the theory of semigroups is fairly old, with the classical text, Functional Analysis and Semigroups by Hille and Phillips [12] being probably the first one of its kind. Indeed, there have been a good number of books and monographs on this topic written over the years, many of which have been referred to in the present text. Perhaps one of the reasons for having so many texts in this one area of advanced mathematical analysis is the fact that the basic theory of semigroups finds many applications in numerous areas of enquiry: partial and ordinary differential equations, the theory of probability and quantum and classical mechanics to name just a few. In the present endeavour, along with the systematic development of the subject, there is an emphasis on the explorations of the contact areas and interfaces, supported by the presentations of explicit computations, wherever feasible.

This book is aimed at the students in the masters level as well as those in a doctoral programme in universities and research institutions and envisages the pre-requisites as: (i) a good understanding

of real analysis with elements of the theory of measures and integration, (for example as in [23]), (ii) a first course in functional analysis and in the theory of operators, say as in [5]. Many examples have been given in each chapter, partly to initiate and motivate the theory developed and partly to underscore the applications. As mentioned earlier, several of these involve detailed analytical computations, many of which have been undertaken in the text and some others left as exercises. Instead of making a separate section on exercises, they appear in line, in bold and in the relevant places as the subject develops and the readers are encouraged to solve as many of them as possible. It is suggested that a beginner may read chapters 1 through 4 (except for sections 3.3 and 3.4) and leave the rest for a second reading. In the Appendix we have collected some standard results from the theory of unbounded operators, Fourier transforms and Sobolev spaces which are required in our treatment of the subject. It is worthwhile to bring to the attention of the reader the fact that we have used the notation $\langle \cdot, \cdot \rangle$ to denote the inner product in Hilbert spaces as well as to represent dual pairing, and $\langle \cdot, \cdot \rangle$ will be taken to be linear in the left and conjugate linear in the right entry. The present text arose out of the notes of the lectures given by the first author (K. B. S.) – twice at the Delhi Centre of the Indian Statistical Institute and once at the Indian Institute of Science, Bangalore and the interaction with the students of those courses has helped shape the final product. Of course, many existing texts on the subject have influenced the authors and a particular mention needs to be made of the classical treatise [12] and the books [11], [15] [19] and [27]. The monographs [2] and [8] have also been referred to frequently. The authors regret that the bibliography is far from exhaustive, instead they were guided only by the need of the topics treated.

The choice of topics in this vastly developed subject is a difficult one and the authors have made an effort to stay closer to applications instead of bringing in too many abstract concepts. While the chapters 2 and 3 make up the fundamentals of any discourse on semigroup theory, the first chapter contains background material, some of which are also of independent interest. Chapter 4 deals with the issue of the stability of classes of semigroups under *small* perturbations as well as the *generalized strong continuity* of semigroups with respect to a parametric dependence. The chapters 5 and 6 deal with special material, opening avenues for many applications: the remarkable theorem of Chernoff leading to the Trotter-Kato product formula which in turn motivates the Feynman-Kac formula for a Schrödinger semigroup, and the Central Limit Theorem. Chapter 6 deals with positivity-preserving (or semi-Markov) semigroups, having its origin in the theory of probability and considers perturbations, *not small* in the sense of Chapter 4. The motivation for some of the material in Chapter 5 and Chapter 6 comes from the theory of probability and for an introduction to elements of that subject, the reader may consult [18]. The last chapter gives a glimpse of how the tools of the semigroup theory can be used to understand partial differential operators in particular the wave and Schrödinger operators.

The first author (K. B. S.) thanks the Indian Statistical Institute, the Indian Institute of Science and most importantly the Jawaharlal Nehru Centre for Advanced Scientific Research, Bangalore, for ready assistance, both direct and indirect, in making this project a reality. He has special words of gratitude for the Department of Science and Technology, Government of India, for the SERB-Distinguished Fellowship, and for his wife Akhila for infinite patience. The second author (S. S.) would like to acknowledge the

support of the Department of Mathematics, University of Delhi in this endeavour and of her husband, Manik. It is also a pleasure to thank Tarachand Prajapati of the Department of Mathematics at the University of Delhi for help, particularly with regards to the drawing of the figure in the book. Last but not the least, the authors are grateful to the anonymous reviewer for many helpful comments for the improvement in the presentation.

Kalyan B. Sinha
Jawaharlal Nehru Centre for
Advanced Scientific Research
Bangalore

Sachi Srivastava
Department of Mathematics
University of Delhi
Delhi

October 2016

Contents

Preface **vii**

1 Vector-valued functions **1**
1.1 Vector-valued functions 1
1.2 The Bochner integral 5
1.3 Measurability implies continuity 10
1.4 Operator valued functions 12
1.5 Semigroups 13

2 C_0-semigroups **21**
2.1 Introduction 21
2.2 The generator 22
2.3 The Hille-Yosida Theorem 32
2.4 Adjoint semigroups 36
2.5 Examples of C_0-semigroups and their generators 39

3 Dissipative operators and holomorphic semigroups **53**
3.1 Dissipative operators 53
3.2 Stone's Theorem 64
3.3 Holomorphic semigroups 66
3.4 Some examples of holomorphic semigroups 77

4 Perturbation and convergence of semigroups **81**
4.1 Perturbation of the generator of a C_0-semigroup 81
4.2 Relative boundedness and some consequences 86
4.3 Convergence of semigroups 90

5 Chernoff's Theorem and its applications **97**
5.1 Chernoff's Theorem 97
5.2 Applications of Trotter-Kato and Chernoff Theorem 100

6 Markov semigroups **115**
6.1 Probability and Markov semigroups 115
6.2 Construction of Markov semigroups on a discrete state space 118

7 Applications to partial differential equations **137**
7.1 Parabolic equations 138
7.2 The wave equation 141
7.3 Schrödinger equation 144

Appendix **147**
A.1 Unbounded operators 147
A.2 Fourier transforms 153
A.3 Sobolev spaces 158

References **161**

Index **165**

Chapter 1

Vector-valued functions

This chapter is mostly of a preliminary nature. In the first section we collect results on measurability and integrability of vector-valued functions that will be useful throughout. For a more comprehensive treatment of these concepts in a general setting, we refer the reader to [2] and [12]. The second section introduces the Bochner integral. The connection between measurability and continuity of subadditive functions is dealt with in Section 3 while operator valued functions and general one-parameter semigroups on Banach spaces are introduced in sections 4 and 5 respectively.

1.1 Vector-valued functions

This section introduces various notions of measurability of vector-valued functions and the connections between them. We assume that the reader is familiar with the basics of the theory of measure and integration for scalar valued functions. In the sequel, X shall denote a Banach space and $\mathcal{B}(X)$ the space of bounded, linear operators on $X, (\Omega, \digamma, \mu)$ will be a σ-finite measure space while $\chi_{\triangle}$ shall denote the indicator function of the set $\triangle$.

Definition 1.1.1. Let $(\Omega, \digamma, \mu)$ be a σ-finite measure space and consider $f_n, f : \Omega \to X,$ where $n \in \mathbb{N}$. The sequence $\{f_n\}$ is said to converge to f

1. **almost everywhere** if there exists a μ-null set $E_0 \in \digamma$ such that given $\epsilon > 0$, for each $t \notin E_0$, there is an $n_{\epsilon, E_0} \in \mathbb{N}$ such that

$$\|f_n(t) - f(t)\| < \epsilon \quad \forall n \geq n_{\epsilon, E_0};$$

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7_1

2. **uniformly almost everywhere** in Ω if for every $\epsilon > 0$ there exists a set $E_\epsilon \in \digamma$ with $\mu(E_\epsilon) < \epsilon$ and for every $\delta > 0$, there exists an integer $n_{\delta,\epsilon}$ such that

$$\|f_n(t) - f(t)\| < \delta \text{ for all } t \notin E_\epsilon \text{ and for all } n > n_{\delta,\epsilon};$$

3. **in measure** if for every $\epsilon > 0$,

$$\mu\big\{t \in \Omega\colon \|f_n(t) - f(t)\| > \epsilon\big\} \to 0 \text{ as } n \to \infty.$$

For the case $X = \mathbb{C}$, it is clear that (2) $\Rightarrow$ (1) and (3). If $\mu(\Omega) < \infty$, then (1) $\Rightarrow$ (2) and (3). Further, if (3) holds, then there exists a subsequence $\{f_{n_k}\}$ converging almost everywhere to f.

Definition 1.1.2. A function $f\colon \Omega \to X$ is said to be

1. **finitely valued** if there exists a sequence $\{\triangle_k : 1 \le k \le n\}$ for some $n \in \mathbb{N}$ of mutually disjoint measurable subsets of Ω and vectors $x_1, ..., x_n$ in X such that
$$f(t) = \begin{cases} \sum_{k=1}^{n} x_k \chi_{\triangle_k}(t), & \text{if } t \in \triangle = \cup \triangle_k, \\ 0 & \text{if } t \notin \triangle; \end{cases}$$

2. **simple** if it is finitely valued and $\mu(\triangle_k) < \infty$;

3. **countably valued** if there exists a sequence $\{\triangle_k : k \in \mathbb{N}\}$ of mutually disjoint measurable subsets of Ω and vectors $x_1, x_2, ...$ in X such that
$$f(t) = \begin{cases} \sum_{n=1}^{\infty} x_n \chi_{\triangle_n}(t), & \text{if } t \in \triangle = \cup_n \triangle_n, \\ 0 & \text{if } t \notin \triangle; \end{cases}$$

4. **separably valued** if $f(\Omega)$ is separable in X;

5. **almost separably valued** if there exists a μ-null set $E \in \digamma$ such that $f(\Omega \setminus E)$ is separable;

6. **weakly measurable** if $t \mapsto F(f(t))$ is measurable for all $F \in X^*$, the dual of X; and

7. **strongly measurable** if there exists a sequence of countably valued functions converging almost everywhere to f.

It is not difficult to see that if $f, g\colon \Omega \to X$ are strongly measurable, then so is the function $f+g$. Further, if $h : X \to Y$ is continuous, Y being any Banach space, then $h \circ f$ is strongly measurable if f is. In particular, this implies that the function $t \to \|f(t)\|$ from Ω to $\mathbb{R}$ is measurable.

A subset Λ of X^* is **determining** for X if

$$\|x\| = \sup\{|x^*(x)| : x^* \in \Lambda\} \text{ for all } x \in X.$$

Clearly, $\|x^*\| \leq 1$ for all $x^* \in \Lambda$.

Lemma 1.1.3. If X is separable, then both X and X^* have countable determining sets.

Proof. Let $\{y_n\} \in X$ be a countable dense set in X. By the Hahn Banach theorem, there exists a countable subset $\Lambda = \{x_n^* : n \in \mathbb{N}\}$ of X^* such that $x_n^*(y_n) = \|y_n\|$ with $\|x_n^*\| = 1$. For any $x \in X$ and $\epsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that

$$\epsilon \geq \|x - y_{n_0}\| > |\|x\| - \|y_{n_0}\||$$

and

$$\|x\| \geq |x_{n_0}^*(x)| = \left|x_{n_0}^*(y_{n_0}) + x_{n_0}^*(x - y_{n_0})\right| \geq \|y_{n_0}\| - \epsilon \geq \|x\| - 2\epsilon.$$

This implies that Λ is determining for X.

Next let Λ_0 be a countable dense set in the unit sphere $S_1(X)$ of X. Then, for $x^* \in X^*$, we have that

$$\sup\left\{|x^*(x_n)| : x_n \in \Lambda_0\right\} = \sup\left\{|x^*(x)|, \|x\| \leq 1\right\} = \|x^*\|.$$

Thus Λ_0 considered as a subset of X^{**} via the canonical embedding of X in X^{**} gives a countable determining set for X^*.

□

Theorem 1.1.4. If $f\colon \Omega \to X$ is weakly measurable and if there exists a countable determining set Λ for X, then $t \mapsto \|f(t)\|$ is measurable.

Proof. Suppose $\Lambda = \{x_n^* : n \in \mathbb{N}\}$. Since

$$\|f(t)\| = \sup_n \left|x_n^*(f(t))\right|$$

and $t \mapsto x_n^*(f(t))$ is measurable for each $n \in \mathbb{N}$, the conclusion follows.

□

The next theorem gives a very useful criterion for strong measurability of a vector-valued function, involving measurability of scalar functions, which is easier to check.

Theorem 1.1.5. An X-valued function is strongly measurable if and only if it is weakly measurable and almost separably valued.

Proof. Suppose first that $f\colon \Omega \to X$ is strongly measurable. Then there exists a null set $E \in \Sigma$ and a sequence $\{f_n\}$ of countably valued functions such that

$$\|f_n(t) - f(t)\| \to 0 \text{ for all } t \notin E.$$

Therefore, for any $x^* \in X^*, x^*(f(t) - f_n(t)) \to 0$ for all $t \notin E$. Now $t \mapsto x^*(f_n(t))$ is a countably $\mathbb{C}$-valued function and hence measurable. This implies that $t \mapsto x^*(f(t))$ is also measurable. Thus f is weakly measurable. Further, the closed linear span $\mathcal{M}$ of the countable subset $\{f_n(\Omega) : n \in \mathbb{N}\}$ of X is a separable subspace of X. Clearly, $f(\Omega \setminus E) \subset \mathcal{M}$. Therefore, f is almost separably valued.

Conversely, assume that f is almost separably valued and weakly measurable. Let $E \in \Sigma$ be a μ-null set such that $f(\Omega \setminus E)$ is separable. Replacing X by the closed linear span of $\{f(\Omega \setminus E)\}$ if required, we can assume that X itself is separable. By Lemma 1.1.3 there exists a countable determining set Λ for X and it follows from Theorem 1.1.4 that $t \mapsto \|f(t)\|$ is measurable. Let $\Omega_0 = \{t \in \Omega \setminus E : \|f(t)\| > 0\}$. Then, $\Omega_0 \in \Sigma$ and the function $t \mapsto f(t) - x_0$ is weakly measurable on Ω_0 for each $x_0 \in X$. Therefore, by Theorem 1.1.4, $t \mapsto \|f(t) - x_0\|$ is measurable on Ω_0. Since $f(\Omega \setminus E)$ is separable, there exists a countable dense subset, say $\{f_n : n \in \mathbb{N}\}$ in $f(\Omega \setminus E)$. Let $\epsilon > 0$. Set

$$E_n = \{t \in \Omega_0 : \|f(t) - f_n\| < \epsilon\}.$$

Then E_n is measurable and since $\{f_n\}$ is dense in $f(\Omega \setminus E), \cup E_n = \Omega_0$. Set $F_n = E_n \setminus \cup_{k<n} E_k$. Then, $\{F_n\}$ is a sequence of pairwise disjoint, measurable sets such that $\cup F_n = \Omega_0$. Define, for each $\epsilon > 0$,

$$g_\epsilon(t) = \begin{cases} f_n & \text{for all} \quad t \in F_n, \\ 0 & \text{for all} \quad t \notin \Omega_0. \end{cases}$$

Clearly, each g_ϵ is countably valued and

$$\|f(t) - g_\epsilon(t)\| = \begin{cases} \|f(t) - f_n\| & \text{if} \quad t \in \Omega_0 \\ \|f(t)\| = 0 & \text{if} \quad t \notin \Omega_0. \end{cases}$$

Therefore, f is the limit of countably valued functions g_ϵ, uniformly with respect to $t \in \Omega \setminus E$, and hence is strongly measurable. □

Corollary 1.1.6. If X is separable, then weak and strong measurability are equivalent.

1.2 The Bochner integral

Once we have a notion of measurable vector-valued functions, it is natural to ask when would such a function be *integrable* or rather, what is meant by an integral of a function in this context. For our purpose it is enough, while answering this question, to restrict attention to functions defined on the real line or its subintervals. Thus, from now on, we shall assume Ω to be $(0, \infty)$ or $\mathbb{R}_+ = [0, \infty)$ or $\mathbb{R}$ or a finite subinterval I of $\mathbb{R}$ and that μ is the Lebesgue measure on Ω, denoted by m. While several extensions of the Riemann and Lebesgue integrals to vector-valued functions exist, here we discuss in detail only the Bochner Integral – a generalisation or extension of the Lebesgue integral to the vector-valued case. It is to be noted that subsequently, while writing the integral of a function with respect to a measure we shall not, in general, mention the underlying measure explicitly and for integrals with Lebesgue measure, the traditional "dt" shall be used. This section ends with Lemma 1.2.5, in which the definition of the Riemann integral of vector-valued functions and a few of its useful properties are given.

For a simple function $f\colon \Omega \to X$ of the form $f(t) = \sum_{k=1}^{n} x_k \chi_{\triangle_k}(t)$, where $x_1, \ldots, x_n \in X$, $\chi_\triangle$ is the real valued indicator (or characteristic) function of the measurable set $\triangle$, and $m(\triangle_k) < \infty,\ \forall k = 1, \ldots, n, n \in \mathbb{N}$, we define its integral by

$$\int_\Omega f(t)\, dt := \sum_{k=1}^{n} x_k m(\triangle_k). \tag{1.1}$$

Definition 1.2.1. A (strongly) measurable function f is called **Bochner integrable** if there exists a sequence $\{f_n\}_{n\in\mathbb{N}}$ of simple functions on Ω such that $f_n \to f$ pointwise almost everywhere and

$$\lim_{n\to\infty} \int_\Omega \|f_n(t) - f(t)\|\, dt = 0. \tag{1.2}$$

Further, if f is Bochner integrable, then its **Bochner integral** is defined to be

$$\int_\Omega f(t)\,dt := \lim_n \int_\Omega f_n(t)\,dt. \tag{1.3}$$

That the limit in (1.3) above exists is a consequence of the fact that (1.2) forces the sequence $\left\{\int_\Omega f_n(t)\,dt\right\}$ to be Cauchy.

This limit is independent of the choice of the sequence f_n. Indeed, suppose $\{f_n\}$ and $\{g_n\}$ are two sequences satisfying (1.2) for the given f, and set

$$a := \lim_n \int_\Omega f_n(t)\,dt \text{ and } b := \lim_n \int_\Omega g_n(t)\,dt.$$

Let $r_{2n}(t) = f_n(t), r_{2n-1}(t) = g_n(t)$. Then we see that the sequence $\{r_n\}$ also satisfies the conditions of Definition (1.2.1), so that $\lim_n \int_\Omega r_n(t)\,dt$ exists. Since a, b are limit points of the convergent sequence $\int_\Omega r_n(t)\,dt$, it follows that $a = b$.

In fact, in the above definition, we can choose f_n to be step functions, that is, simple functions for which the sets $\triangle_k$ are disjoint intervals of $\mathbb{R}$. Note that if $X = \mathbb{C}$, then the Bochner integral of f is nothing but the Lebesgue integral.

Lemma 1.2.2. If $f\colon \Omega \to X$ is strongly measurable, then

(a) f is the limit of a sequence of countably valued functions in ess sup norm, that is, ess $\sup_{t\in\Omega} \|f(t) - f_n(t)\| \to 0$, and

(b) f is the pointwise limit almost everywhere of a sequence of simple functions.

Proof. Let f be strongly measurable. Then, by Theorem 1.1.5 it is weakly measurable and almost separably valued. Let $\epsilon > 0$. Then following the construction as in the proof of the second part of Theorem 1.1.5, define $g_\epsilon := \sum_{n=1}^\infty f_n \chi_{F_{n,\epsilon}}$, where $F_{n,\epsilon} = F_n$ and we have added ϵ to stress the dependence on ϵ. Let $t \in \Omega$. If $t \notin \Omega_0 \cup E$ then $f(t) = 0 = g_\epsilon(t)$. If $t \in \Omega_0$, then there exists $n \in \mathbb{N}$, such that $t \in F_{n,\epsilon}$. Therefore, $\|f(t) - g_\epsilon(t)\| < \epsilon$ for all $t \in \Omega \setminus E$. Since f is the limit of countably valued functions g_ϵ, uniformly with respect to $t \in \Omega \setminus E$, the first part is proved.

For (b), let $\Omega = \cup_n I_n$, where each I_n is an increasing sequence of bounded subintervals of Ω. For each n, let $J_n := I_n \cap \left(\cup_{n=1}^{c_n} F_{n,3^{-n}}\right)$, where c_n is chosen

so that $m(I_n \setminus J_n) < 3^{-n}$. Set $h_n := g_{3^{-n}}\chi_{J_n}, n \in \mathbb{N}$. If $t \in \cap_{n=k}^{\infty} J_n$ for some $k \geq 1$, then

$$\|f(t) - h_n(t)\| = \|f(t) - g_{3^{-n}}(t)\| < 3^{-n}$$

for all $n \geq k$. Thus $\lim_{n\to\infty} h_n(t) = f(t)$ for all $t \in J = \cup_{k=1}^{\infty} \cap_{n=k}^{\infty} J_n$. Moreover, for $k \leq j$,

$$m\big(I_j \setminus \cap_{n=k}^{\infty} J_n\big) \leq \sum_{n=k}^{\infty} m(I_n \setminus J_n) < 3^{-k}.$$

Thus $I_j \setminus J$ is null for each j. Therefore, $\lim_{n\to\infty} h_n(t) = f(t)$, for almost all $t \in \Omega$. □

The class of Bochner integrable functions has a very nice characterisation, making them relatively easy to use.

Theorem 1.2.3. A function $f\colon \Omega \to X$ is Bochner integrable if and only if it is strongly measurable and the function $t \mapsto \|f(t)\|$ is Lebesgue integrable. If f is Bochner integrable, then

$$\Big\| \int_\Omega f(t)\, dt \Big\| \leq \int_\Omega \|f(t)\|\, dt. \tag{1.4}$$

Proof. Suppose first that f is Bochner integrable. Then, there exists a sequence $\{f_n\}$ of simple functions which approximate f in the sense of Definition 1.2.1. Thus, f is strongly measurable and the function $t \mapsto \|f(t)\|$ is measurable. Since

$$\int_\Omega \|f(t)\|\, dt \leq \int_\Omega \|f(t) - f_n(t)\|\, dt + \int_\Omega \|f_n(t)\|\, dt$$

and since

$$\lim_{n\to\infty} \int_\Omega \|f_n(t) - f(t)\|\, dt = 0,$$

it follows that $t \mapsto \|f(t)\|$ is integrable. Further,

$$\begin{aligned} \Big\| \int_\Omega f(t)\, dt \Big\| &= \lim_n \Big\| \int_\Omega f_n(t)\, dt \Big\| \\ &\leq \lim_{n\to\infty} \int_\Omega \|f_n(t)\|\, dt = \int_\Omega \|f(t)\|\, dt. \end{aligned}$$

Conversely, suppose that f is strongly measurable. Then by Lemma 1.2.2 (b), there is a sequence $\{g_n\}$ of simple (or finitely valued) functions converging pointwise to f on a subset $\Omega \setminus \Omega_0$ of Ω, and $m(\Omega_0) = 0$. Set for $t \in \Omega \setminus \Omega_0$,

$$f_n(t) := \begin{cases} g_n(t) & \text{if } \|g_n(t)\| \leq 2\|f(t)\|(1 + n^{-1}), \\ 0, & \text{otherwise.} \end{cases}$$

Note that if for $t \in \Omega \setminus \Omega_0, f(t) = 0$, then $f_n(t) = 0$ for all n. On the other hand, if $f(t) \neq 0$, then the set $\{n \in \mathbb{N} : \|g_n(t)\| > 2\|f(t)\|(1 + n^{-1})\}$ must be finite. Indeed if this set is infinite, then we can find a subsequence $\{n_k\} \subset \mathbb{N}$ such that

$$\|g_{n_k}\| > 2\|f(t)\|(1 + n_k^{-1}). \tag{1.5}$$

Letting $k \to \infty$ in (1.5) gives $\|f(t)\| \geq 2\|f(t)\|$, which is a contradiction. Thus for sufficiently large n, $f_n(t) = g_n(t)$ if $f(t) \neq 0$. Therefore, f_n converges pointwise to f on $\Omega \setminus \Omega_0$. Writing $h_n(t) = f_n(t) - f(t)$ we see that $\|h_n(t)\| \leq 5\|f(t)\|$ and almost everywhere on $\Omega, \lim_{n\to\infty} h_n(t) = 0$. Since $\|f\|$ is integrable, by the scalar Dominated Convergence Theorem, we have that $\lim_{n\to\infty} \int \|h_n(t)\| = 0$. This shows that f is Bochner integrable.

□

Lemma 1.2.4. Let f be a bounded, X-valued strongly measurable function on $\mathbb{R}_+$. Then

$$\int_a^b \|f(t + \delta) - f(t)\|\, dt \to 0 \quad \text{as } \delta \to 0, \text{ for } 0 < a < b < \infty.$$

Proof. Since f is strongly measurable, we may assume without loss of generality that it is separably valued. Also, it follows from Lemma 1.2.2(a) that there exist a sequence $\{f_n\}$ of countably valued functions such that

$$\text{ess sup}_{t\in\Omega} \|f(t) - f_n(t)\| \to 0 \text{ as } n \to \infty.$$

Let $\epsilon > 0$ be given. Thus there exists an $n_0 \in \mathbb{N}$, and $\Omega_0 \subset \Omega$, such that $m(\Omega_0) = 0$ and for all $n \geq n_0$,

$$\sup_{t\in\Omega\setminus\Omega_0} \|f(t) - f_n(t)\| < \epsilon. \tag{1.6}$$

Fix $n = n_0$. Then for any $\delta > 0$,

$$\begin{aligned}\int_a^b \|f(t + \delta) - f(t)\|\, dt &\leq \int_a^b \|f(t + \delta) - f_n(t + \delta)\|\, dt + \int_a^b \|f(t) - f_n(t)\|\, dt \\ &\quad + \int_a^b \|f_n(t + \delta) - f_n(t)\|\, dt \\ &\leq 2(b - a)\epsilon + \int_a^b \|f_n(t + \delta) - f_n(t)\|\, dt. \end{aligned} \tag{1.7}$$

Set

$$f_n(t) = \sum_{k=1}^{\infty} x_k \chi_{\triangle_k}(t). \tag{1.8}$$

The fact that f is bounded together with (1.6) implies that

$$\text{ess sup}_{t\in\Omega\setminus\Omega_0} \|f_n(t)\| \le M, \tag{1.9}$$

for some constant M. Using (1.8) and (1.9) we get that $\sup_{k=1}^{\infty} \|x_k\| \le M_1$. Writing $\triangle_k - \delta = \{s - \delta : s \in \triangle_k\}$, we therefore have

$$\begin{aligned}
&\int_a^b \|f_n(t+\delta) - f_n(t)\|\, dt \qquad (1.10)\\
&= \int_a^b \|\sum_{k=1}^{\infty} x_k[\chi_{\triangle_k-\delta}(t) - \chi_{\triangle_k}(t)]\|\, dt\\
&\le \int_a^b \sum_{k=1}^{\infty} \|x_k\| \big|\chi_{\triangle_k-\delta}(t) - \chi_{\triangle_k}(t)\big|\, dt\\
&\le \sup_k \|x_k\| \sum_{k=1}^{\infty} m([a,b] \cap [(\triangle_k \setminus (\triangle_k - \delta)) \cup ((\triangle_k - \delta) \setminus \triangle_k)]\\
&\le M_1 \sum_{k=1}^{\infty} m([a,b] \cap [(\triangle_k \setminus (\triangle_k - \delta)) \cup ((\triangle_k - \delta) \setminus \triangle_k)]\\
&\to 0 \text{ as } \delta \to 0, \qquad (1.11)
\end{aligned}$$

by the Dominated Convergence Theorem. Now (1.7) together with (1.11) establishes the claim.

□

The next lemma defines the vector-valued Riemann integral and collects a few simple but useful results.

Lemma 1.2.5. (i) Let $a, b \in \mathbb{R}$, and let $f\colon [a,b] \to X$ be a continuous function. For a partition $P = \{a = s_0 < s_1 < s_2 < \ldots < s_n = b\}$ of I let $\Psi(f; P, a, b) := \sum_{j=1}^{n} f(s_j)(s_j - s_{j-1})$ denote the Riemann sum. Then $\Psi(f; P, a, b)$ converges, as the partition width $|P| = \max_{j=1}^{n}(s_j - s_{j-1})$ approaches 0, to an element in X, which shall be called the Riemann integral of f over the interval $[a,b]$ and written as $\int_a^b f(s)\, ds$. This f is also Bochner integrable and the two integrals coincide.

(ii) Let A be a closed operator in X and let f be as in (i) above. Assume that $f(s) \in \mathcal{D}(A)$ for every $s \in [a,b]$ such that the map $s \mapsto Af(s)$ from $[a,b]$ to X is continuous. Then

$$A\int_a^b f(s)\,ds = \int_a^b Af(s)\,ds. \tag{1.12}$$

Proof. The proof of (i) is identical to that in the scalar case, using the fact that the continuity of f implies uniform continuity over I.

Now suppose A and f are as in (ii). Then by (i) both the Riemann integrals $\int_a^b f(s)\,ds$ and the integral $\int_a^b Af(s)\,ds$ exist. Thus the sequence $\{\Psi(f, P_n, a, b)\} \subset \mathcal{D}(A)$ where

$$P_n = \{a < a + (b-a)/n < a + 2(b-a)/n < \ldots, s_n = b\},$$

and converges to $\int_a^b f(s)\,ds$ while the sequence $A\Psi(f, P_n, a, b) = \Psi(Af, P_n, a, b)$ converges to $\int_a^b Af(s)\,ds$ as $n \to \infty$. Since A is closed, this implies that $\int_a^b f(s)\,ds$ is in $\mathcal{D}(A)$ and (1.12) holds.

□

1.3 Measurability implies continuity

We now explore the relation between measurability and continuity for vector-valued functions defined on $(0,\infty)$. If $f\colon (0,\infty) \to X$ is continuous, then clearly f is weakly measurable and the countable set $\{f(t) : t \in \mathbb{Q} \cap (0,\infty)\}$ is dense in the range of f. Therefore, by Theorem 1.1.5, f is strongly measurable. Thus, as in the scalar case, continuity implies measurability. The following result shows that the converse is also true for some special functions. For the purpose of the next theorem, we define a Banach algebra: An algebra X which is also a Banach space with respect to a norm $\|\cdot\|$ such that $\|xy\| \le \|x\|\|y\|$, $\forall x, y \in X$, is called a **Banach algebra**. Recall that a function $g\colon (0,\infty) \to \mathbb{R}$ is said to be **subadditive** if $g(t+s) \le g(t) + g(s)$ $\forall\ t, s \in (0,\infty)$.

Lemma 1.3.1. Let X be a real or complex Banach algebra, possibly without a unit, and let $f\colon (0,\infty) \to X$ be a strongly measurable function satisfying

$$f(t_1)f(t_2) = f(t_1 + t_2), \text{ for all } t_1, t_2 \in (0,\infty).$$

Then f is bounded in every bounded interval in $(0,\infty)$ and is continuous on $(0,\infty)$.

Proof. Since f is strongly measurable, it follows from Theorem 1.1.5 and Theorem 1.1.4 that $t \mapsto \|f(t)\|$ is measurable.

Suppose first that $f(t) \neq 0$ for all $t \in (0,\infty)$. Since for $t_1, t_2 \in (0,\infty)$, $\|f(t_1+t_2)\| \leq \|f(t_1)\|\|f(t_2)\|$, we have that

$$\log \|f(t_1+t_2)\| \leq \log \|f(t_1)\| + \log \|f(t_2)\|.$$

Therefore, the function $\alpha\colon (0,\infty) \to \mathbb{R}$ defined by $\alpha(t) = \log \|f(t)\|$ is subadditive on $(0,\infty)$, that is, $\alpha(t_1+t_2) \leq \alpha(t_1)+\alpha(t_2)$ for all $t_1, t_2 \in (0,\infty)$. We claim that α is bounded above on any subinterval (c,d) of $(0,\infty)$ where $0 < c < d < \infty$. Let $a > 0$ and $\alpha(a) = A$. For $t+s = a$, and $t, s > 0$, $A = \alpha(a) \leq \alpha(t)+\alpha(s)$. If we set

$$E = \Big\{t \in (0,a) : \alpha(t) \geq \frac{A}{2}\Big\},$$

then

$$(0,a) = E \cup (a-E). \tag{1.13}$$

Indeed, for $r \in (0,a)$, if $\alpha(r) \geq \frac{A}{2}$, then $r \in E$. Otherwise, $\alpha(r) < 2^{-1}\alpha(a)$, so that $\alpha(a-r) \geq \alpha(a) - \alpha(r) > \frac{A}{2}$. Thus $a - r \in E$ whence $r \in a - E$. Now (1.13) implies that

$$a \leq m(E) + m(a-E) = 2m(E) \text{ so that } m(E) \geq \frac{a}{2}.$$

Suppose if possible that α is unbounded in some interval (c,d) where $0 < c < d < \infty$. Then there exists a sequence $\{t_n\} \subset (c,d)$ such that $t_n \to t_0 \geq c$ and

$$\alpha(t_n) \geq 2n, \quad \text{for each } n \in \mathbb{N}.$$

Therefore, by an argument similar to one used above, we have that for every $n \in \mathbb{N}$, the set

$$E_n = \{t \in (0,d) : \alpha(t) \geq n\}$$

has measure $m(E_n) > \frac{c}{2}$. This implies that the function α takes the value ∞ on a set of measure at least $c/2$. This is a contradiction. Thus α is bounded above on $(\epsilon, \epsilon^{-1})$ for all $1 > \epsilon > 0$. Let

$$\alpha(t) \leq M_\epsilon \text{ for all } t \in (\epsilon, \epsilon^{-1}),$$

and it follows that $t \mapsto \|f(t)\|$ is a bounded measurable function in $(\epsilon, \epsilon^{-1})$. Choose a, b, c such that $0 < a < b < c < \infty$. Then the integral $\displaystyle\int_a^b f(c-t)f(t)\,dt$

exists as a strong Bochner integral and is equal to $f(c)(b-a)$. Therefore, if $\epsilon > 0$,

$$(b-a)[f(c+\epsilon)-f(c)] = \int_a^b [f(c+\epsilon-t)-f(c-t)]f(t)\,dt,$$

and hence

$$\begin{aligned}(b-a)\|f(c+\epsilon)-f(c)\| &\leq \Big\| \int_{c-a}^{c-b} [f(\tau+\epsilon)-f(\tau)]f(c-\tau)\,d\tau \Big\| \\ &\leq M \int_{c-a}^{c-b} \|f(\tau+\epsilon)-f(\tau)\|\,d\tau \\ &\to 0, \text{ as } \epsilon \to 0,\end{aligned}$$

where $M = \sup_{a\leq t\leq b} \|f(t)\|$, and the convergence to zero is a consequence of Lemma 1.2.4.

Next we consider the case when there exists a $t_0 \in (0,\infty)$ such that $f(t_0) = 0$. Then $f(t) = 0$ for all $t \geq t_0$. The conclusion of the theorem in this case is arrived at by following the same proof as before with the open interval $(0,\infty)$ replaced by $(0,t_0)$.

□

Remark 1.3.2. Caution: Note that the conclusion of the above Lemma is *only* for the *open* right half line $(0,\infty)$ and *not* for $[0,\infty)$.

1.4 Operator valued functions

We now consider the special vector-valued functions which assume values in $\mathcal{B}(X)$, the space of bounded linear operators on some Banach space X. Since these functions take values in $\mathcal{B}(X)$ they are referred to as operator valued functions. Such functions are of particular relevance to us since a semigroup of operators on a Banach space X is an operator valued function $T\colon [0,\infty) \to \mathcal{B}(X)$, satisfying the semigroup property (see (1.5.1) below).

The following definition makes precise the notion of uniform, strong and weak measurability for operator valued functions.

Definition 1.4.1. An operator valued function $T\colon (\Omega, \mathcal{F}, \mu) \to \mathcal{B}(X)$ is

1. **uniformly measurable** if there exists a sequence $\{T_n\}$ of countably (operator) valued functions on Ω converging almost everywhere to T in the operator norm;

2. **strongly measurable** if the vector-valued function $t \mapsto T(t)x$ is strongly measurable for every $x \in X$;

3. **Weakly measurable** if $t \mapsto y^*(T(t)x)$ is measurable for every $x \in X$ and $y^* \in X^*$.

Similarly, we may consider continuity for operator valued functions in various topologies on $\mathcal{B}(X)$. However, the uniform, strong and weak forms of continuity are the ones we will work with most of the time. Therefore, we make precise the definitions here:

Definition 1.4.2. An operator valued function T on Ω where Ω is either $\mathbb{R}_+$ or I, a finite interval in $\mathbb{R}$, is

1. **uniformly continuous** if the function $t \mapsto T(t)$ from Ω to $\mathcal{B}(X)$ is continuous with respect to the operator norm;

2. **strongly continuous** if the vector-valued function $t \mapsto T(t)x$ from Ω to X is continuous for every $x \in X$;

3. **weakly continuous** if the function $t \mapsto y^*(T(t)x)$ from Ω to $\mathbb{C}$ is continuous for every $x \in X$ and $y^* \in X^*$.

1.5 Semigroups

In this section we look at semigroups of bounded operators on a Banach space. As mentioned earlier they may be thought of as operator valued functions with a particular property.

Definition 1.5.1. A **semigroup of operators** on the Banach space X is an operator valued function $T : [0, \infty) \to \mathcal{B}(X)$ satisfying

$$T(t)T(s) = T(t + s) \text{ for all } s, t \geq 0. \tag{1.14}$$

For semigroups, the three types of measurability and continuity we have defined above are closely connected. Since $\mathcal{B}(X)$ is a Banach algebra, a direct

consequence of Lemma 1.3.1 is that a uniformly measurable semigroup is uniformly continuous, that is, the map $t \mapsto T(t)$ from $(0,\infty)$ to $\mathcal{B}(X)$ is continuous in the norm topology of $\mathcal{B}(X)$.

Theorem 1.5.2. Let T be a uniformly measurable semigroup on a Banach space X. Then T is uniformly continuous in $(0,\infty)$.

In fact, the above result remains true if we replace *uniform* by *strong*. However, the proof of this requires some further work. We first establish the following lemma.

Lemma 1.5.3. Let T be a semigroup on X which is strongly measurable on $(0,\infty)$. Then the function $t \mapsto \|T(t)\|$ is bounded on $[\alpha,\beta]$ for all α,β such that $0 < \alpha < \beta < \infty$.

Proof. By the Uniform Boundedness Principle, it suffices to show that for every $x \in X$, the set $\{T(t)x : \alpha \le t \le \beta\}$ is bounded. Suppose that this is not true. Then for some $x \in X$ there exists a $c \in [\alpha,\beta]$ and a sequence $\big(t_n\big) \subset [\alpha,\beta]$ such that $t_n \to c$ as $n \to \infty$ and $\|T(t_n)x\| \ge n$ for all n. Strong measurability of T together with Theorem 1.1.4 and Theorem 1.1.5 implies that $t \mapsto \|T(t)x\|$ is measurable. An application of Lusin's Theorem [23, Lusin's Theorem, page 66] yields that there exists an $M > 0$ and a measurable set $E \subset [0,c]$ with measure $m(E) > \dfrac{c}{2}$ such that

$$\sup_{t\in E} \|T(t)x\| \le M.$$

Now set $E_n = \{t_n - \eta : \eta \in E \cap [0,t_n]\}$. This is a measurable set and for large enough n, $m(E_n) \ge \dfrac{c}{2}$. Therefore,

$$n \le \|T(t_n)x\| \le \|T(t_n-\eta)\|\|T(\eta)x\| \le M\|T(t_n-\eta)\| \tag{1.15}$$

Thus, $\|T(t)\| \ge \frac{n}{M}$ for all $t \in E_n$. Denoting $\limsup_n E_n$ by F, it follows that $\|T(t)\| = \infty \ \forall\ t \in F$. But $m(F) \ge \frac{c}{2} > 0$, implying that $T(t)$ is not defined for t in a set of strictly positive measure, leading to a contradiction. □

Theorem 1.5.4. Let T be a semigroup on X which is strongly measurable on $(0,\infty)$. Then T is strongly continuous on $(0,\infty)$.

Proof. Let $x \in X$ and $0 < a < t < b < s$. Suppose $\epsilon > 0$ is so small that $\epsilon < s - t$. Using the identity $T(s)x = T(t)T(s-t)x$, we have

$$(b-a)[T(s \pm \epsilon) - T(s)]x = \int_a^b T(t)[T(s \pm \epsilon - t) - T(s-t)]x\, dt.$$

From Lemma 1.5.3 it follows that there exists an $M > 0$ such that $\|T(t)\| \leq M$ for all $t \in [a, b]$. Therefore,

$$\begin{aligned}\|(b-a)[T(s \pm \epsilon) - T(s)]x\| &= \Big\| \int_a^b T(t)[T(s \pm \epsilon - t) - T(s-t)]x \, dt \Big\| \\ &\leq M \int_a^b \|[T(s \pm \epsilon - t) - T(s-t)]x\| \, dt \\ &= M \int_{s-b}^{s-a} \|[T(u \pm \epsilon) - T(u)]x\| \, dt \\ &\to 0 \text{ as } \epsilon \to 0,\end{aligned}$$

by Lemma 1.2.4. □

Remark 1.5.5. Thus the hypothesis of strong measurability is enough to render a semigroup into a strongly continuous family on the open interval $(0, \infty)$. However, in general, it may not be possible to extend this continuity to $[0, \infty)$. Those semigroups for which this is valid form the most useful class, viz, C_0-semigroups.

Corollary 1.5.6. Weak one-sided continuity on $(0, \infty)$ of a semigroup T on X implies strong continuity of T on $(0, \infty)$.

Proof. Recall that weak one-sided continuity implies weak measurability. For any a, b with $0 < a < b < \infty$, and $x \in X$ fixed,

$$\begin{aligned}&\text{the closed linear span of } \{T(t)x : t \in [a, b]\} \\ &\qquad \equiv \text{the closed linear span of } \{T(t)x : t \in \mathbb{Q} \cap [a, b]\}.\end{aligned}$$

Since every strongly closed linear subspace is weakly closed (as a consequence of the Hahn-Banach Theorem), it follows therefore that the *weakly closed linear span* of $\{T(t)x : t \in \mathbb{Q} \cap [a, b]\}$ is equal to the *strongly closed linear span* of $\{T(t)x : t \in \mathbb{Q} \cap [a, b]\}$.

Therefore $t \mapsto T(t)x$ from $[a, b]$ to X is separably valued. It follows from Theorem 1.1.5 that this map is strongly measurable and then from Theorem 1.5.4 that it is strongly continuous. Since $a, b \in (0, \infty)$ are arbitrary, the result follows. □

The following example shows that strong continuity of a semigroup does not, in general, imply uniform continuity.

Example 1.5.7. Let $X = C_0(\mathbb{R}_+)$, the Banach space of continuous functions on $\mathbb{R}_+$ which vanish at ∞. Let T be the semigroup defined by setting

$$(T(t)f)(s) = e^{-s^2t}f(s), \text{ for all } t, s \geq 0, \text{ and for } f \in X.$$

Then T is strongly continuous but not uniformly continuous. Indeed, for $h, s > 0, t \geq 0$ fixed, $f \in C_0(\mathbb{R}_+)$, and $\|f\| \leq 1$,

$$\begin{aligned} T(t+h)f(s) - T(t)f(s) &= (e^{-s^2(t+h)} - e^{-s^2t})f(s) \\ &= e^{-s^2t}(e^{-s^2h} - 1)f(s). \end{aligned}$$

$$\text{Therefore,} \quad \|T(t+h)f - T(t)f\| = \sup_{s\geq 0} |e^{-s^2t}(1 - e^{-s^2h})f(s)|. \tag{1.16}$$

Let $\epsilon > 0$. Then there is a compact set $K_\epsilon \subset [0, \infty)$, such that

$$|f(s)| < \epsilon/2 \ \forall s \in [0, \infty) \setminus K_\epsilon.$$

Since the map $t \to e^{-s^2t}$ is uniformly continuous for s in any compact set, there exists $\delta \in (0, 1]$ such that

$$|1 - e^{-s^2h}| < \epsilon \quad \forall s \in K_\epsilon, \ 0 < h < \delta.$$

Then, (1.16) gives

$$\|T(t)f - T(t+h)f\| \leq \max\big(\sup_{s\in[0,\infty)\setminus K_\epsilon} |f(s)|, \sup_{s\in K_\epsilon} |(1 - e^{-s^2h})|\big) < \epsilon,$$

for all h such that $0 < h < \delta$. Thus for each $f \in C_0(\mathbb{R}_+)$, $\|T(t)f - T(t+h)f\| \to 0$ as $h \downarrow 0$. On the other hand,

$$\|T(t) - T(t+h)\| = \sup_{s\geq 0} |1 - e^{-s^2h}| = 1 \text{ for all } h \neq 0.$$

Therefore, T is not uniformly continuous.

However, strong continuity of T does imply that $t \mapsto \|T(t)x\|$ is continuous on $(0, \infty)$ for each $x \in X$. Therefore, $t \mapsto \|T(t)\| = \sup_{\|x\|\neq 0} \dfrac{\|T(t)x\|}{\|x\|}$ is lower semi-continuous and hence measurable. Consider first the case when $T(t) \neq 0$ for all $t \in (0, \infty)$. Then the function $t \mapsto \log \|T(t)\| = \alpha(t)$ is a measurable function on $(0, \infty)$. Since $\|T(t+s)\| \leq \|T(t)\|\|T(s)\|, t, s \geq 0$, the above function α is subadditive and by proof of Lemma 1.3.1, is different from $+\infty$ on $(0, \infty)$. In the case that there exists a $t_0 \in (0, \infty)$ such that $T(t_0) = 0, T(t) = 0$ for all $t > t_0$, and the same conclusion as before may be arrived at by considering α as a function defined on $(0, t_0)$ instead of $(0, \infty)$.

Lemma 1.5.8. Let $f\colon (a,\infty)\to\mathbb{R}$, where $a\geq 0$, be a subadditive measurable function. Then

$$\lim_{t\to\infty}\frac{f(t)}{t}=\inf_{t>a}\frac{f(t)}{t}<\infty.$$

Proof. Let $\beta=\inf_{t>a}\dfrac{f(t)}{t}$. Then β is either finite or $-\infty$. Suppose first that β is finite. Let $\epsilon>0$. Choose $b>a$ such that $f(b)<(\beta+\epsilon)b$ and $n\in\mathbb{N}$ such that $(n+2)b\leq t\leq(n+3)b$. Then, for $t>a$,

$$\begin{aligned}\beta\leq\frac{f(t)}{t}&\leq\frac{f(t-nb)+f(nb)}{t}\\&\leq\frac{nb}{t}\frac{f(b)}{b}+\frac{f(t-nb)}{t}\\&<\frac{nb}{t}(\beta+\epsilon)+\frac{f(t-nb)}{t}.\end{aligned}\tag{1.17}$$

Since $t-nb\in[2b,3b]$, it follows from Lemma 1.3.1 that $|f(t-nb)|$ is bounded. Therefore,

$$\lim_{t\to\infty}\frac{f(t)}{t}\leq\lim_{t\to\infty}\big(\frac{nb}{t}(\beta+\epsilon)+\frac{f(t-nb)}{t}\big)=\beta+\epsilon.$$

Thus,

$$\beta\leq\lim_{t\to\infty}\frac{f(t)}{t}\leq\beta+\epsilon$$

and since $\epsilon>0$ is arbitrary, it follows that

$$\lim_{t\to\infty}\frac{f(t)}{t}=\beta.$$

If $\beta=-\infty$, then for any $m\in\mathbb{N}$ we find $b\geq a$ such that $\dfrac{f(b)}{b}<-m$ and inequality (1.17) shows that for t sufficiently large, $\dfrac{f(t)}{t}<-m$. This implies that $\lim_{t\to\infty}\dfrac{f(t)}{t}=-\infty$.

□

Lemma 1.5.8, when applied to the function $f(t)=\log\|T(t)\|$, where $t\in(0,\infty)$, gives

$$w_0(T):=\inf_{t>0}\frac{\log\|T(t)\|}{t}=\lim_{t\to\infty}\frac{\log\|T(t)\|}{t}<\infty.\tag{1.18}$$

This $w_0(T)$ is called the **type** of the semigroup T. It is also referred to as the **exponential growth bound** of the semigroup T. The reason for this is apparent from the next result where some simple properties of the type of a semigroup are listed.

Theorem 1.5.9. Let T be a semigroup on X and $w_0 = w_0(T)$ denote its type. The following hold:

1. If T is strongly continuous on $(0,\infty)$, then $w > w_0$ implies the existence of $M_w > 0$ such that

$$\|T(t)\| \leq M_w e^{wt} \quad \text{for all } t > 0.$$

   In fact,

$$w_0 = \inf\{w \in \mathbb{R} : \text{ there exists } M_w \geq 0 \text{ with } \|T(t)\| \leq M_w e^{wt}\}.$$

2. If $w_0 > -\infty$, then the spectral radius of $T(t)$ is given by $e^{w_0 t}$ for each $t \in (0,\infty)$.

Proof. Let $\alpha = \inf\{w \in \mathbb{R} : \text{ there exists } M_w \geq 0 \text{ with } \|T(t)\| \leq M_w e^{wt}\}$ and let $w > w_0$. Since $w_0 = \lim_{t\to\infty} \dfrac{\log \|T(t)\|}{t}$, for $\epsilon = w - w_0$ there exists $t_0 > 0$ such that

$$\frac{\log \|T(t)\|}{t} < w_0 + \epsilon \quad \text{for all } t > t_0.$$

Thus

$$\|T(t)\| \leq e^{(w_0+\epsilon)t} = e^{wt} \quad \text{for } t > t_0. \tag{1.19}$$

Using Lemma 1.5.3 and (1.19) we can find an M_w such that $\|T(t)\| \leq M_w e^{wt}$ for all $t > 0$. Thus $w \geq \alpha$. Since $w > w_0$ was arbitrary, it follows that $w_0 \geq \alpha$. Conversely, let $w \in \mathbb{R}$ be such that $\|T(t)\| \leq M_w e^{wt}$ for all $t > 0$. Therefore, for $t > 0$,

$$\frac{\log \|T(t)\|}{t} \leq \frac{\log M_w}{t} + w.$$

So $\lim_{t\to\infty} \dfrac{\log \|T(t)\|}{t} \leq w$. This implies that $w_0 \leq w$ so that $w_0 \leq \alpha$. Therefore, $\alpha = w_0$.

By the spectral radius formula (see [5]) and definition of type, for $t \in (0,\infty)$,

$$\begin{aligned} r(T(t)) = \lim_{n\to\infty} \|T(t)^n\|^{1/n} &= \lim_{n\to\infty} \|T(nt)\|^{1/n} \\ &= \lim_{n\to\infty} \exp\left(\frac{t}{nt} \log \|T(nt)\|\right) = e^{tw_0}. \end{aligned}$$

□

A semigroup T for which $\|T(t)\| \leq 1$, for all $t > 0$, that is, the choice $w = 0$ and $M_w = 1$ is permissible, is called a **contraction semigroup** . Note that we have not yet assumed strong continuity at $t = 0$. A semigroup T which is strongly continuous (equivalently, measurable) on $[0, \infty)$ and $T(0) = I$ is called a **C_0-semigroup**, which will be the subject of discussion in the next chapter. The following is an example of a contraction C_0-semigroup.

Example 1.5.10. Let $X = BUC(\mathbb{R}_+)$, the space of all bounded uniformly continuous functions on the half line. Define the semigroup T on X by setting

$$(T(t)f)(s) = f(t + s), \text{ for all } s, t \geq 0, f \in X.$$

Then $T(0) = I$ and $\|T(t)f\| = \|f\|$ for all $f \in X$, so that $\|T(t)\| = 1$ for all $t \geq 0$. Thus T is a contraction semigroup with $w_0(T) = 0$. Further, as a consequence of uniform continuity, for any $f \in X$,

$$\|T(t)f - f\| = \sup_{s \geq 0} \|f(t + s) - f(s)\| \to 0 \text{ as } t \to 0.$$

Therefore, for any fixed $s > 0$,

$$\begin{aligned} \|T(s + h)f - T(s)f\| = \|T(s)(T(h)f - f)\| &\leq \|T(s)\|\|T(h)f - f\| \qquad (1.20) \\ &\to 0 \text{ as } h \to 0. \end{aligned}$$

Thus T is strongly continuous.

Note that the inequality (1.20) actually holds for any semigroup. This shows that if a semigroup $\{T(t)\}_{t \geq 0}$ is exponentially bounded on $[0, \infty)$, then strong continuity at 0 is sufficient for the semigroup to be strongly continuous on all of $[0, \infty)$. We postpone giving further examples of C_0-semigroups and illustrations of the concept of type until the next chapter, where C_0-semigroups are studied in detail.

Chapter 2

C_0-semigroups

In this chapter we concentrate on strongly continuous or more specifically C_0-semigroups of bounded operators on a Banach space. The notion of the generator of a C_0-semigroup is introduced and their properties are dealt with in detail.

2.1 Introduction

Consider a function $T\colon [0,\infty) \to M_n(\mathbb{C})$, satisfying the following properties: (i) $T(0) = I, (ii)$ $T(t+s) = T(t)T(s)$ $\forall$ $t, s \geq 0$ and (iii) $T(t) \to I$ as $t \to 0$. Then it is not difficult to see that the map $t \to T(t)$ is differentiable and $T(t) = e^{At}$ for some $A \in M_n(\mathbb{C})$ (**Exercise** 2.1.1). Here continuity, together with the semigroup property implies differentiability. This implication carries over to the infinite-dimensional case also, but with a qualification – A may not be defined everywhere. (Recall that $A \in M_n(\mathbb{C})$ may be considered as a linear operator on $\mathbb{C}^n$, defined everywhere). We prove the above assertion in this section.

The following simple result will be used repeatedly in the text and is given for the sake of completeness.

Lemma 2.1.2. Let X be a Banach space and let $f \in [0,a] \to X$ be a continuous function. Then

$$\lim_{t\to 0^+} t^{-1} \int_0^t f(s)\, ds = f(0).$$

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7_2

Proof. Note that the integral exists as a Riemann integral as well as a Bochner integral (Lemma 1.2.5) and that

$$\Big[t^{-1}\int_0^t f(s)\,ds - f(0)\Big] = t^{-1}\int_0^t [f(s)-f(0)]\,ds.$$

Since continuity implies uniform continuity on a compact interval, given $\epsilon > 0$, one can find a $\delta > 0$ such that $\|f(s)-f(0)\| < \epsilon$ whenever $0 < s < \delta$. Therefore,

$$\big\|t^{-1}\int_0^t f(s)\,ds - f(0)\big\| \le t^{-1}\int_0^t \big\|f(s)-f(0)\big\|\,ds < \epsilon \quad \text{for } 0 < t < \delta.$$

□

2.2 The generator

Assume that $T\colon (0,\infty) \to \mathcal{B}(X)$ is a semigroup of operators and is strongly continuous on $(0,\infty)$. The infinitesimal generator A_0 of T is defined in the following manner. Set

$$A_\eta x = \frac{T(\eta)x - x}{\eta},\ \eta > 0, \text{ and} \tag{2.1}$$

$$A_0 x = \lim_{\eta\to 0^+} A_\eta x, \tag{2.2}$$

whenever the limit exists. From now on we shall refer to the infinitesimal generator as simply the generator.

The domain $\mathcal{D}(A_0)$ of A_0 is the set of all $x \in X$ such that the limit in (2.2) above exists. Then $\mathcal{D}(A_0)$ is a linear subspace and A_0 is a linear operator. In general, the operator A_0 may not be closed, nor densely defined. But, $\mathcal{D}(A_0)$ is always non-empty:

Lemma 2.2.1. $\mathcal{D}(A_0)$ is non-empty.

Proof. For $y \in X$ and $0 < \alpha < \beta < \infty$, set $x_{\alpha,\beta} = \int_\alpha^\beta T(t)y\,dt$, which exists by Lemma 1.2.5. We shall establish that $\lim_{\eta\to 0} A_\eta x_{\alpha,\beta}$ exists, thus proving the result. Indeed for $\eta > 0$, by a change of variable in one of the integrals, we have

that

$$\begin{aligned} A_\eta x_{\alpha,\beta} &= \frac{1}{\eta}\int_\alpha^\beta T(t+\eta)y\,dt - \frac{1}{\eta}\int_\alpha^\beta T(t)y\,dt \\ &= \frac{1}{\eta}\int_\beta^{\beta+\eta} T(t)y\,dt - \frac{1}{\eta}\int_\alpha^{\alpha+\eta} T(t)y\,dt \\ &= \frac{1}{\eta}\int_0^\eta T(t+\beta)y\,dt - \frac{1}{\eta}\int_0^\eta T(t+\alpha)y\,dt \\ &\longrightarrow (T(\beta)y - T(\alpha)y), \end{aligned}$$

as $\eta \to 0^+$, by Lemma 2.1.2. □

Next, we set, for $\alpha > 0$,

$$X_\alpha = T(\alpha)(X) \text{ and } X_0 = \bigcup_{\alpha>0} X_\alpha.$$

The semigroup property clearly implies that $X_\beta \subset X_\alpha$, for $\alpha < \beta$, and X_0 is the smallest linear subspace containing the range spaces of T.

Theorem 2.2.2. If T is a semigroup which is strongly continuous for $t > 0$, then for all $x \in \mathcal{D}(A_0)$, we have $T(t)x \in \mathcal{D}(A_0)$ and

$$T(t)A_0x = A_0T(t)x = \frac{d}{dt}T(t)x. \tag{2.3}$$

Proof. For $x \in \mathcal{D}(A_0)$ and $t, \eta > 0$,

$$\frac{T(t+\eta)x - T(t)x}{\eta} = T(t)A_\eta x = A_\eta T(t)x.$$

Since $\lim_{\eta\to 0^+} A_\eta x$ exists and equals A_0x and $T(t)$ is bounded, $\lim_{\eta\to 0^+} T(t)A_\eta x$ exists and equals $T(t)A_0x$. This implies that both the limits

$$\lim_{\eta\to 0^+} A_\eta T(t)x \quad \text{and} \quad \lim_{\eta\to 0^+} \frac{T(t+\eta)x - T(t)x}{\eta}$$

exist. It follows therefore that $T(t)x \in \mathcal{D}(A_0)$ and

$$\lim_{\eta\to 0^+} \frac{T(t+\eta)x - T(t)x}{\eta} = T(t)A_0x = A_0T(t)x.$$

Thus the right hand derivative of $T(\cdot)x$ exists at t and equals $A_0T(t)x = T(t)A_0x$. Also, for $\eta > 0$ sufficiently small and $t > 0$,

$$\frac{T(t-\eta)x - T(t)x}{-\eta} = T(t-\eta)A_\eta x. \tag{2.4}$$

The term on the right in equation (2.4) approaches $T(t)A_0x$ as $\eta \to 0^+$, due to the strong continuity of T at t and Lemma 1.5.3. Thus, the left hand derivative of $T(t)x$ also exists for $t > 0$ and we have

$$\frac{d}{dt}T(t)x = T(t)A_0x.$$

□

Theorem 2.2.3. If $T = \{T(t)\}_{t>0}$ is a strongly continuous semigroup, then

(a) $\mathcal{D}(A_0)$ is dense in X_0;

(b) $\overline{\mathcal{D}(A_0)} = \overline{X_0}$ and

(c) the range of A_0 is contained in $\overline{X_0}$.

Proof. (a) Let $x \in X_0 = \bigcup_{\alpha>0} T(\alpha)X$. Then there exists $y \in X$ and $\alpha > 0$ such that $x = T(\alpha)y$. For $\beta > \alpha > 0, x_{\alpha,\beta}$ defined in the proof of Lemma 2.2.1 exists and is in $\mathcal{D}(A_0)$. Moreover,

$$x_{\alpha,\beta} = T(\alpha/2)\int_\alpha^\beta T(t-\alpha/2)y\,dt = T(\alpha/2)\int_{\alpha/2}^{\beta-\alpha/2} T(t)y\,dt$$
$$= T(\alpha/2)x_{\alpha/2,\beta-\alpha/2} \in X_0.$$

Since $\lim\limits_{\beta\to\alpha} \dfrac{1}{\beta-\alpha}x_{\alpha,\beta} = T(\alpha)y = x$, it follows that $\mathcal{D}(A_0)$ is dense in X_0, that is,

$$X_0 \subset \overline{\mathcal{D}(A_0)} \cap \overline{X_0}. \tag{2.5}$$

(b) Let $x \in \mathcal{D}(A_0)$. Then by Theorem 2.2.2,

$$T(t)x - x = \int_0^t T(s)A_0x\,dt$$

(see Lemma 1.2.5) and therefore, $\lim_{t\to0^+} T(t)x$ exists and equals x. Thus $x \in \overline{X_0}$ and $\mathcal{D}(A_0) \subset \overline{X_0}$. This along with (2.5) implies that $\overline{\mathcal{D}(A_0)} = \overline{X_0}$.

(c) If $x \in \mathcal{D}(A_0) \subset \overline{X_0}$, then $A_\eta x \in \overline{X_0}$ so that $A_0x \in \overline{X_0}$.

□

Since by definition of $X_0, T(t)$ maps $\overline{X_0}$ into $\overline{X_0}$, the restriction $T(t)|_{\overline{X_0}}$ is itself a strongly continuous semigroup for $t > 0$. Also, if we assume additionally, that $T(0) = I$, then $\overline{X_0} = X$, and then there is continuity for $t \geq 0$. However, in general the following holds.

Theorem 2.2.4. Let $\{T(t)\}_{t>0} \in \mathcal{B}(X)$ be a semigroup. Assume that $t \mapsto T(t)$ is strongly measurable on $(0,\infty)$ and $\lim_{t\to 0^+} T(t)x = Jx$ exists for each $x \in X$. Then J is an idempotent, that is, $J^2 = J$, $RanJ = \overline{X_0}$ and $JT(t) = T(t)J = T(t)$ for $t > 0$. In fact, $Jx = x$ for all $x \in \overline{X_0}$.

Proof. By Theorem 1.5.4, $T(t)$ is strongly continuous on $(0,\infty)$. By the Uniform Boundedness Theorem, J is a bounded operator on X. Moreover, for $t, s > 0$ and $x \in X, T(s)T(t)x = T(s+t)x = T(t)T(s)x$. Letting $t \to 0^+$ in this equation yields,

$$T(s)Jx = T(s)x = JT(s)x, \text{ for all } s > 0, \ x \in X. \tag{2.6}$$

Again, letting $s \to 0^+$ in (2.6) above yields $J^2x = Jx$ for all $x \in X$. Now for $x \in X, T(t)x \in X_0$ for all $t > 0$. Therefore, $Jx \in \overline{X_0}$. Thus, $J(X) \subset \overline{X_0}$. Now let $x \in \mathcal{D}(A_0)$. Then $T(t)x \to x$ as $t \to 0^+$. This implies that $Jx = x$ for all $x \in \mathcal{D}(A_0)$. Since J is bounded in X we have that $Jx = x$ for all $x \in \overline{\mathcal{D}(A_0)} = \overline{X_0}$. Therefore, $\overline{X_0} \subset J(\overline{X_0}) \subset J(X)$. Hence, $J(X) = \overline{X_0}$. □

In view of the above theorem, there are the following alternatives: if $T(t)x \to x$ as $t \to 0^+$ for all $x \in X$, then $J = I = T(0)$ and $\overline{X_0} = X$, or $\{T(t)\}_{t\geq 0}$ is a strongly continuous semigroup for $t \geq 0$, and $J \neq I$. In the second case, we can restrict $T(t)$ to $\overline{X_0}$, and work with this space.

From now on, we shall work with strongly continuous semigroups $\{T(t)\}_{t\geq 0}$ with $T(0) = I$. As mentioned earlier, these are called C_0-semigroups. We formalise the definitions in the following.

Definition 2.2.5. A family $\{T(t)\}_{t\geq 0} \in \mathcal{B}(X)$ is a **C_0-semigroup** if

1. $T(t+s) = T(t)T(s) = T(s)T(t)$ for all $s, t \geq 0$.
2. $t \mapsto T(t)$ is strongly continuous on $[0,\infty)$, that is, $\lim_{t\to s} T(t)x = T(s)x$, for all $s \geq 0$ and for each $x \in X$.
3. $T(0) = I$.

As has been observed at the end of Chapter 1, property (2) in Definition 2.2.5 is equivalent to strong continuity at $t = 0$, that is, for each $x \in X$, $\lim_{t\to 0} T(t)x = T(0)x$.

Definition 2.2.6. The (infinitesimal) **generator** of a C_0-semigroup $\{T(t)\}_{t\geq 0}$ on a Banach space X is defined as follows :

$$\mathcal{D}(A) = \left\{x \in X : \lim_{t\to 0^+} t^{-1}(T(t)x - x) \text{ exists } \right\}$$
$$Ax = \lim_{t\to 0^+} t^{-1}(T(t)x - x)$$

Note from A.1.4 that for a closed operator A, the **resolvent set** of A is given by $\rho(A) = \{\lambda \in \mathbb{C} : (\lambda - A)^{-1} \in \mathcal{B}(X)\}$ and $R(\lambda, A) = (\lambda - A)^{-1}$ is called the **resolvent** of A, while the **spectrum** of A is given by $\sigma(A) := \mathbb{C} \setminus \rho(A)$.

The next theorem sums up some of the important properties of a C_0-semigroup and its generator.

Theorem 2.2.7. Let $T = \{T(t)\}_{t\geq 0}$ be a C_0-semigroup defined on the Banach space X. Then the following properties hold.

(a) There exists $M > 0$ and $\beta \in \mathbb{R}$ such that $\|T(t)\| \leq Me^{\beta t}$ for all $t \geq 0$, that is, $T(t)$ is exponentially bounded.

(b) $T(t)Ax = AT(t)x = \dfrac{d}{dt}T(t)x$ for all $x \in \mathcal{D}(A)$, where A is the generator of T.

(c) $\displaystyle\int_0^t T(s)x\,ds \in \mathcal{D}(A)$, for all $t \geq 0, x \in X$. Furthermore,

$$T(t)x - x = \begin{cases} A\displaystyle\int_0^t T(s)x\,ds & \text{if } x \in X \\ \displaystyle\int_0^t T(s)Ax\,ds & \text{if } x \in \mathcal{D}(A). \end{cases} \tag{2.7}$$

(d) The generator A of the semigroup is a densely defined, closed linear operator.

(e) The half plane $H_\beta = \{z \in \mathbb{C} : \operatorname{Re} z > \beta\}$ is contained in the resolvent set $\rho(A)$ and the resolvent $R(z, A)$ is given by

$$R(z, A)x = \int_0^\infty e^{-zt}T(t)x\,dt, \text{ for all } z \in H_\beta.$$

(f) $\|R(z, A)^n\| \leq M(\operatorname{Re} z - \beta)^{-n}$ for $\operatorname{Re} z > \beta$ and $n = 1, 2 \ldots$.

Proof. (a) The semigroup property of T combined with the hypothesis of strong continuity at 0 implies (a) on using Theorem 1.5.9, while

(b) follows from Theorem 2.2.2.

(c) Let $x \in X$. Writing $x_{0,t} = \int_0^t T(s)x\,ds$ as before, and following the proof of Lemma 2.2.1, we have that

$$\begin{aligned}\frac{1}{h}\big(T(h)x_{0,t} - x_{0,t}\big) &= \frac{1}{h}\int_0^t T(s+h)x\,ds - \int_0^t T(s)x\,ds \\ &= \frac{1}{h}\int_t^{t+h} T(s)x\,ds - \frac{1}{h}\int_0^h T(s)x\,ds \\ &\longrightarrow (T(t)x - x), \text{ as } h \to 0^+, \end{aligned} \tag{2.8}$$

by Lemma 2.1.2. Therefore, $x_{0,t} \in \mathcal{D}(A)$, and

$$T(t)x - x = A\int_0^t T(s)x\,ds = Ax_{0,t}.$$

Now suppose $x \in \mathcal{D}(A)$. Then by Theorem 2.2.2, $T(t)x \in \mathcal{D}(A)$, and we set,

$$\begin{aligned} g(t) &= AT(t)x = T(t)Ax \text{ and} \\ g_h(t) &= \frac{1}{h}(T(t+h)x - T(t)x). \end{aligned}$$

Then the strong continuity of T implies that g, g_h are continuous on $[0,\infty)$ and using (a), we have, for $t \geq 0$, that

$$\begin{aligned}\|g_h(t) - g(t)\| &= \big\|T(t)\big(\frac{1}{h}\big(T(h)x - x\big) - Ax\big)\big\| \\ &\leq Me^{\beta t}\big\|\frac{1}{h}\big(T(h)x - x\big) - Ax\big\|. \end{aligned}$$

Thus $g_h \to g$ as $h \to 0$, uniformly on compact subintervals of $[0,\infty)$ and by (2.8) one gets that

$$T(t)x - x = \lim_{h\downarrow 0}\int_0^t g_h(s)\,ds = \int_0^t g(s)\,ds = \int_0^t AT(s)x\,ds = \int_0^t T(s)Ax\,ds.$$

(d) In view of Theorem 2.2.3 and the discussion following immediately after Theorem 2.2.4, $\overline{\mathcal{D}(A)} = \overline{X_0} = X$, so that A is densely defined. To see that A is closed, let $\{x_n\}_n$ be a sequence in $\mathcal{D}(A)$, converging to x and suppose that $Ax_n \to y$ as $n \to \infty$ for some $y \in X$. By (c) we have that

$$T(t)x_n - x_n = \int_0^t T(s)Ax_n\,ds \tag{2.9}$$

for all $n = 1, 2, 3...$ Since $T(t)$ is bounded, $T(t)x_n \to T(t)x$ as $n \to \infty$ so that the left hand side of (2.9) converges to $T(t)x - x$ as $n \to \infty$. On the other hand,

$$\Big\| \int_0^t (T(s)Ax_n - T(s)y)\, ds \Big\| \leq M\|Ax_n - y\| \int_0^t e^{\beta s}\, ds \to 0 \text{ as } n \to \infty.$$

Therefore, $T(t)x - x = \displaystyle\int_0^t T(s)\, y\, ds$, so that by Lemma 2.1.2

$$A_\eta x = \eta^{-1} \int_0^\eta T(s) y\, ds \to y \text{ as } \eta \to 0^+.$$

This implies that $x \in \mathcal{D}(A)$ and $Ax = y$. Thus A is closed.

(e) Let $z \in H_\beta$ and $x \in \mathcal{D}(A)$. Then

$$\frac{d}{dt}T(t)x = T(t)Ax = T(t)(A - z)x + zT(t)x.$$

For $x^* \in X^*$, we have that $\dfrac{d}{dt}e^{-zt}\langle x^*, T(t)x\rangle = e^{-zt}\langle x^*, T(t)(A - z)x\rangle$. Since $T(0) = I$, we have on integrating, that for $x \in \mathcal{D}(A)$,

$$e^{-zt}\langle x^*, T(t)x\rangle - \langle x^*, x\rangle = \int_0^t e^{-zs}\langle x^*, T(s)(A - z)x\rangle\, ds$$

$$\text{or, } \langle x^*, x\rangle = e^{-zt}\langle x^*, T(t)x\rangle - \int_0^t e^{-zs}\langle x^*, T(s)(A - z)x\rangle\, ds.$$

Thus, for all $x^* \in X^*, x \in \mathcal{D}(A)$,

$$|\langle x^*, x\rangle| \leq M\|x^*\|\big\{e^{(\beta - \operatorname{Re} z)t}\|x\| + \int_0^t e^{(\beta - \operatorname{Re} z)s}\|(A - z)x\|\big\}.$$

Therefore, by an application of the Hahn-Banach theorem, one has that

$$\|x\| \leq M\big\{e^{(\beta - \operatorname{Re} z)t}\|x\| + \|(A - z)x\|(\operatorname{Re} z - \beta)^{-1}[1 - e^{-(\operatorname{Re} z - \beta)t}]\big\}.$$

Letting $t \to \infty$ in the above equation, we get, since $\operatorname{Re} z > \beta$, that for $x \in \mathcal{D}(A)$,

$$\|x\| \leq M\|(A - z)x\|(\operatorname{Re} z - \beta)^{-1}$$

so that,

$$\|(A - z)x\| \geq M^{-1}(\operatorname{Re} z - \beta)\|x\|, \tag{2.10}$$

for all $z \in \mathbb{C}$ such that Re $z > \beta$. Thus, $(A - z)$ is injective. Next we show that *Ran* $(A - z)$ is a closed subspace of X for Re $z > \beta$. Let $\{x_n\}_{n\in\mathbb{N}}$ be a sequence in *Ran* $(A - z)$ converging to $x \in X$ as $n \to \infty$. Then there exist $y_n \in \mathcal{D}(A)$ such that $x_n = (A - z)y_n, n = 1, 2...$ and (2.10) leads to

$$\|x_n - x_m\| \geq M^{-1}(\text{Re } z - \beta)\|y_n - y_m\|.$$

Therefore, $\{y_n\}$ is Cauchy and hence converges to some $y \in X$, while $\{(A-z)y_n\}$ converges, by assumption to x. Since A is closed, this implies that $y \in \mathcal{D}(A)$ and $(A - z)y = x$. Thus, x lies in *Ran* $(A - z)$, so that *Ran* $(A - z)$ is closed.

Now suppose that $x^* \in X^*$ is such that $\langle x^*, (A - z)y\rangle = 0$, for all $y \in \mathcal{D}(A)$. For such an x^*, and any $y \in \mathcal{D}(A)$,

$$\begin{aligned}\frac{d}{dt}\langle x^*, T(t)y\rangle &= \langle x^*, AT(t)y\rangle \\ &= \langle x^*, (A - z)T(t)y\rangle + z\langle x^*, T(t)y\rangle = z\langle x^*, T(t)y\rangle,\end{aligned}$$

since by (b), $T(t)$ maps $\mathcal{D}(A)$ into itself. This implies that

$$\langle x^*, T(t)y\rangle = e^{zt}\langle x^*, y\rangle.$$

Therefore,

$$\left|\langle x^*, y\rangle\right| \leq M\|x^*\|\|y\|e^{-(\text{Re } z-\beta)t} \to 0 \text{ as } t \to \infty, \text{ for Re } z > \beta.$$

Thus $\langle x^*, y\rangle = 0$, for all $y \in \mathcal{D}(A)$. Since A is densely defined, that is, $\overline{\mathcal{D}(A)} = X$, it follows that $\langle x^*, x\rangle = 0$ for all $x \in X$. This forces x^* to be 0.

If $\overline{Ran\ (A - z)} \neq X$, then there exists $x_0 \neq 0$ such that $x_0 \notin \overline{Ran\ (A - z)}$. The Hahn-Banach theorem then shows that there exists $x^* \in X^*$ such that $\langle x^*, x_0\rangle = 1, \langle x^*, x\rangle = 0$ for all $x \in \overline{Ran\ (A - z)}$. But this implies, from the discussion in the last paragraph that $x^* = 0$, leading to a contradiction. Therefore, $\overline{Ran\ (A - z)} = X$ and the closedness of $Ran(A - z)$ implies that *Ran* $(A - z) = X$. Thus $H_\beta \subset \rho(A)$ and $\|(A - z)^{-1}\| \leq M(\text{Re } z - \beta)^{-1}$.

Suppose again that $\operatorname{Re} z > \beta$ and $x \in \mathcal{D}(A)$, and let $a > 0$. Since A is closed, it follows from Lemma 1.2.5 that $\displaystyle\int_0^a e^{-zt}T(t)x\,dt \in \mathcal{D}(A)$ and

$$\begin{aligned} A\int_0^a e^{-zt}T(t)x\,dt &= \int_0^a e^{-zt}AT(t)x\,dt = \int_0^a e^{-zt}\frac{d}{dt}T(t)x\,dt \\ &= e^{-za}T(a)x - x + z\int_0^a e^{-zt}T(t)x\,dt. \end{aligned}$$

Letting $a \to \infty$ in the above shows that $\displaystyle A\int_0^\infty e^{-zt}T(t)x\,dt$ exists and that

$$\begin{aligned} A\int_0^\infty e^{-zt}T(t)x\,dt &= -x + z\int_0^\infty e^{-zt}T(t)x\,dt \\ \text{or, } (z-A)\int_0^\infty e^{-zt}T(t)x\,dt &= x. \end{aligned}$$

where we have used the facts that A is a closed operator and that T is of type β so that $\|e^{-zt}T(t)x\| \le Me^{-(\operatorname{Re} z-\beta)t}\|x\|$. Since $z \in \rho(A)$, this implies that for $x \in \mathcal{D}(A)$,

$$(z-A)^{-1}x = \int_0^\infty e^{-zt}T(t)x\,dt. \tag{2.11}$$

Since $(z-A)^{-1}$ is bounded and since the integral in (2.11) is well defined for all $x \in X$ it follows that (2.11) holds for all $x \in X$.

(f) The function $z \mapsto R(z,A) \in \mathcal{B}(X)$ is strongly differentiable (in fact, in operator norm) for $\operatorname{Re} z > \beta$. This can be seen from (2.11) by differentiating inside the integral in the right hand side, which is permitted by an application of the Dominated Convergence Theorem using the property (a). Then one has, for $x \in \mathcal{D}(A)$, that

$$\frac{d}{dz}\big[R(z,A)x\big] = -(z-A)^{-2}x = -\int_0^\infty te^{-zt}T(t)x\,dt. \tag{2.12}$$

Differentiating (2.11) $(n-1)$ times with respect to z in the half-plane H_β, gives

$$(n-1)!(z-A)^{-n}x = \int_0^\infty e^{-zt}t^{n-1}T(t)x\,dt,$$

where the interchange of differentiation in z and integration with respect to t is done as before. Therefore, for all $\operatorname{Re} z > \beta$,

$$\begin{aligned}\|(z-A)^{-n}\| &\leq \frac{M}{(n-1)!}\int_0^\infty e^{-(\operatorname{Re} z-\beta)t}t^{n-1}dt \\ &\leq \frac{M}{(n-1)!(\operatorname{Re} z-\beta)^n}\Gamma(n) = \frac{M}{(\operatorname{Re} z-\beta)^n}.\end{aligned}$$

□

Remark 2.2.8. It is not difficult to see that the generator A of a C_0 semigroup $\{T(t)\}_{t\geq 0}$ defined on a Banach space X is a bounded operator on X if and only if the semigroup is norm continuous on $[0,\infty)$, (that is, the map $t \to T(t)$ from $[0,\infty)$ to $\mathcal{B}(X)$ is continuous with respect to the operator norm topology in $\mathcal{B}(X)$). In such a case, the semigroup $\{T(t)\}_{t\geq 0}$ is referred to as a *norm continuous* or *uniformly continuous* semigroup.

So far, we have talked about operator valued functions T defined on $[0,\infty)$ satisfying the semigroup property. It is perfectly possible to talk of operator valued functions having all of $\mathbb{R}$ as their domain and demanding that they satisfy properties similar to those in Definition 2.2.5. This would, of course, result in a much more restricted class of operators; such a family of operators is called a C_0 group. Precisely, we have

Definition 2.2.9. A **C_0-group** on a Banach space X is a family $\{T(t)\}_{t\in\mathbb{R}}$ of operators in $\mathcal{B}(X)$ satisfying the following properties.

1. $T(t+s) = T(t)T(s) = T(s)T(t)$ for all $s,t \in \mathbb{R}$.
2. $t \mapsto T(t)$ is strongly continuous on $\mathbb{R}$, that is, $\lim_{t\to s} T(t)x = T(s)x$, for all $s \in \mathbb{R}$ and every $x \in X$.
3. $T(0) = I$.

The generator A of this group is defined by setting

$$\begin{aligned}\mathcal{D}(A) &= \big\{x \in X : \lim_{t\to 0} t^{-1}(T(t)x - x) \text{ exists } \big\}, \\ Ax &= \lim_{t\to 0} t^{-1}(T(t)x - x).\end{aligned}$$

Remark 2.2.10. Note that the limit in the definition of the generator of a C_0-group is a two-sided one, not just right-sided, which is the case for C_0-semigroups. Also see Remark 2.3.4(2).

2.3 The Hille-Yosida Theorem

Theorem 2.2.7 shows that for a linear operator A to be the generator of a C_0-semigroup it is necessary that A be densely defined and closed. But this is not sufficient. Conditions like (e) and (f) of Theorem 2.2.7 are not only necessary but also sufficient, as is shown by the Hille-Yosida Theorem. This is perhaps the most important result in the theory of operator semigroups and was proven independently by Hille [12] and Yosida [26].

Theorem 2.3.1 (Hille-Yosida)**.** Let A be a densely defined, closed linear operator on X. Let $H_\beta = \{z \in \mathbb{C} : Re\, z > \beta\} \subset \rho(A)$ for some $\beta \in \mathbb{R}$ and suppose that there exists $M > 0$ such that

$$\left\|(z-A)^{-n}\right\| \leq M(Re\, z - \beta)^{-n} \quad \text{for all} \quad z \in H_\beta \text{ and } n = 1, 2, ... \tag{2.13}$$

Then there exists a unique C_0-semigroup $\{T(t)\}_{t\geq 0}$ such that A is its generator and $\|T(t)\| \leq Me^{\beta t}$.

Proof. Fix $\beta \in \mathbb{R}$, and set $B = A - \beta$ and $w = z - \beta$. Then the hypothesis imply that $H_0 \subset \rho(B), w - B = z - A$ and

$$\|(w-B)^{-n}\| \leq M(\text{ Re } w)^{-n} \text{ for all } w \in H_0 \text{ and } n = 1, 2....$$

So we may assume without loss of generality that $\beta = 0$. Set, for $n = 1, 2, ...$

$$A_n = nA(n-A)^{-1}. \tag{2.14}$$

Then, $A_n \in \mathcal{B}(X)$ for each n and for every $x \in \mathcal{D}(A), \lim_{n\to\infty} A_n x = Ax$. Indeed, for $x \in \mathcal{D}(A)$,

$$\|A_n x - Ax\| = \|(n(n-A)^{-1} - 1)Ax\| = \|(I_n - I)Ax\|,$$

where $I_n x = n(n-A)^{-1}x$. But, for any $y \in \mathcal{D}(A)$,

$$\|I_n y - y\| = \|(n-A)^{-1}Ay\| \leq \frac{M}{n}\|Ay\| \to 0 \text{ as } n \to \infty.$$

Since $\mathcal{D}(A)$ is dense in X and $\|I_n\| \leq M$ by the hypothesis, it follows that $I_n \to I$ strongly on X as $n \to \infty$. Hence, for each $x \in \mathcal{D}(A), A_n x \to Ax$ as $n \to \infty$. Let $T_n(t) = e^{tA_n}$ be defined by the convergent power series $\sum_{k=0}^{\infty} \frac{t^k}{k!}(A_n)^k \in \mathcal{B}(X)$. We note that T_n is an entire function of t for every $n \in \mathbb{N}$ and

$$\frac{d}{dt}T_n(t) = A_n T_n(t) \text{ and } T_n(0) = I.$$

Since $A_n = n(I_n - I), n \in \mathbb{N}$,

$$T_n(t) = e^{tn(I_n - I)} = e^{-nt} e^{tnI_n}$$

for each n. By (2.13), we have that

$$\|(nI_n)^m\| = \|n^{2m}(n - A)^{-m}\| \leq Mn^{2m}n^{-m} = Mn^m,$$

for all $m \in \mathbb{N}$. Therefore, for $t \geq 0$,

$$\|e^{tnI_n}\| \leq \sum_{m=0}^{\infty} \frac{1}{m!} t^m \|(nI_n)^m\| \leq \sum_{m=0}^{\infty} \frac{1}{m!} t^m M n^m = Me^{nt},$$

so that $\|T_n(t)\| \leq M$. Now, $T_m(t) = e^{tA_m}$ and A_n, being functions of I_m and I_n respectively, commute. Therefore, for $x \in \mathcal{D}(A)$ and $n, m \in \mathbb{N}$, one gets that

$$\begin{aligned} T_n(t)x - T_m(t)x = T_n(t-s)T_m(s)x\big|_t^0 &= -\int_0^t \frac{d}{ds}\big[T_n(t-s)T_m(s)x\big]\, ds \\ &= \int_0^t T_n(t-s)(A_n - A_m)T_m(s)x\, ds \\ &= \int_0^t T_n(t-s)T_m(s)(A_n x - A_m x)\, ds. \end{aligned}$$

Therefore,

$$\|T_n(t)x - T_m(t)x\| \leq M^2 t \|A_n x - A_m x\| \to 0 \text{ as } n, m \to \infty.$$

Thus $\{T_n(t)x\}_n$ is a Cauchy sequence for each $t \in [0, \infty)$ and $x \in \mathcal{D}(A)$. Moreover, the sequence is uniformly Cauchy for all t in compact subsets of $[0, \infty)$. Since $\|T_n(t)\|$ is uniformly bounded and $\mathcal{D}(A)$ is dense in X it follows that $\{T_n(t)x\}_n$ is a Cauchy sequence for each $t \in [0, \infty)$ and $x \in X$. Set

$$T(t)x = \lim_{n \to \infty} T_n(t)x,$$

for all $x \in X$ and $t \in [0, \infty)$. As we have noted earlier, the convergence $T_n(t)x \to T(t)x$ as $n \to \infty$ is uniform for all t in compact subsets of $[0, \infty)$, for each $x \in \mathcal{D}(A)$. This uniform convergence extends to all of $x \in X$. Indeed, for $\epsilon > 0, x \in X$, and a compact subset K of $[0, \infty)$, one can choose a $y \in \mathcal{D}(A)$, such that $\|x - y\| < \dfrac{\epsilon}{4M}$ and an n_0 depending on ϵ and y such that

$$\|T_n(t)y - T(t)y\| < \frac{\epsilon}{2} \quad \text{for all } n > n_0$$

and $t \in K$. Then, for all $t \in K$,

$$\begin{aligned}\|T_n(t)x - T(t)x\| &\le \|(T_n(t) - T(t))(x-y)\| + \|(T_n(t) - T(t))y\| \\ &\le 2M\|x-y\| + \|(T_n(t) - T(t))y\| < \epsilon\end{aligned}$$

for all $n > n_0$ and for all $t \in K$, leading to the strong convergence, uniformly with respect to t, in compact subsets of $[0,\infty)$. Since $T_n(t)T_n(s)x = T_n(t+s)x$ for all $n \in \mathbb{N}, x \in X$, taking strong limit as $n \to \infty$ in the above, it follows that $T(t)T(s) = T(t+s)$ for $t, s \ge 0$. Furthermore, for any $x \in X$,

$$\lim_{t\to 0^+} T(t)x = \lim_{t\to 0^+}\lim_{n\to\infty} T_n(t)x = \lim_{n\to\infty}\lim_{t\to 0^+} T_n(t)x = x$$

where the interchange in the order of limits is permissible due to uniform convergence, and $T(0)x = \lim_{n\to\infty} T_n(0)x = x$. Thus, $\{T(t)\}_{t\ge 0}$ is a C_0-semigroup. We show next that A is the generator of this semigroup. For $x \in \mathcal{D}(A)$ and $n \in \mathbb{N}$ we have from Theorem 2.2.7 (c) that

$$T_n(t)x - x = \int_0^t T_n(s)A_n x\, ds. \tag{2.15}$$

The left hand side of (2.15) converges to $T(t)x - x$. On the right hand side of (2.15), $A_n x \to Ax$ while $T_n(s) \to T(s)$, strongly and uniformly for $s \in [0,t]$, as $n \to \infty$. Therefore, taking limit as $n \to \infty$ in (2.15), we obtain,

$$T(t)x - x = \int_0^t T(s)Ax\, ds \quad \text{for all } x \in \mathcal{D}(A),$$

and hence, by Lemma 2.1.2,

$$\lim_{t\to 0^+} t^{-1}(T(t)x - x) = \lim_{t\to 0^+} t^{-1}\Big(\int_0^t T(s)Ax\, ds\Big) = Ax \text{ for all } x \in \mathcal{D}(A).$$

Thus $\tilde{A}x = Ax$ for all $x \in \mathcal{D}(A)$, where $\tilde{A}$ denotes the generator of the semigroup $T(t)$ constructed above. Now, let $x \in \mathcal{D}(\tilde{A})$ and set $v = (\tilde{A} - 1)x$. By hypothesis, $1 \in \rho(A)$. Therefore, there exists $w \in \mathcal{D}(A)$ such that $v = (A-1)w$. This implies that,

$$(\tilde{A} - 1)x = (A-1)w = (\tilde{A} - 1)w. \tag{2.16}$$

By Theorem 2.2.7 (e), and since $\|T(t)\| \le M$ it follows that $1 \in \rho(\tilde{A})$, that is, $(\tilde{A} - 1)^{-1} \in \mathcal{B}(X)$. Therefore, (2.16) implies that $x = w \in \mathcal{D}(A)$. Thus $A = \tilde{A}$.

Finally, for uniqueness, let $\{\tilde{T}(t)\}_{t\geq 0}$ be another semigroup generated by A. For $0 < s < t$, and $x \in \mathcal{D}(A)$,

$$\frac{d}{ds}(\tilde{T}(t-s)T(s)x) = -\tilde{T}(t-s)AT(s)x + \tilde{T}(t-s)AT(s)x = 0.$$

Thus, $\tilde{T}(t-s)T(s)x$ is independent of s for $0 < s < t$. Therefore, for $x \in \mathcal{D}(A)$,

$$\tilde{T}(t)x = \lim_{s\to 0^+} \tilde{T}(t-s)T(s)x = \lim_{s\to t-} \tilde{T}(t-s)T(s)x = T(t)x.$$

The boundedness of T and $\tilde{T}$ allows us to extend this equality to all $x \in X$.

□

The following version of the Hille-Yosida Theorem is often useful and its proof is left as an exercise (**Exercise** 2.3.2).

Theorem 2.3.3. A linear operator A is the infinitesimal generator of a

(i) contraction C_0-semigroup $\{T(t)\}_{t\geq 0}$ if and only if

  (a) A is closed and densely defined in X;

  (b) the resolvent set $\rho(A)$ of A contains the interval $(0, \infty)$ and

$$\big\|R(\lambda, A)\big\| \leq \lambda^{-1} \quad \text{for all } \lambda > 0.$$

(ii) C_0-semigroup $\{T(t)\}_{t\geq 0}$ satisfying $\|T(t)\| \leq Me^{\beta t}$ if and only if

  (a) A is closed and densely defined in X;

  (b) the resolvent set $\rho(A)$ of A contains the interval (β, ∞) and

$$\big\|R(\lambda, A)^n\big\| \leq M(\lambda - \beta)^{-n} \quad \text{for all } \lambda > \beta, \text{ and } n = 1, 2, \ldots.$$

Remark 2.3.4. 1. In Theorem 2.3.1, instead of approximating $T(t)$ by e^{tA_n}, where $A_n = nA(n - A)^{-1}$, we can try a different approximation:

$$V_n(t) = \left(1 - \frac{tA}{n}\right)^{-n}.$$

This is not a semigroup for any n. But, by the hypotheses of Theorem 2.3.1,

(i) $\|V_n(t)\| \leq M, \ \forall n$ and $t \geq 0$;

(ii) $V_n(t)$ is differentiable with respect to t for all $t > 0$, with

$$V_n^{'}(t) = A\left(1 - \frac{tA}{n}\right)^{-n-1} \in \mathcal{B}(X);$$

(iii) $V_n(t)$ is not, in general, differentiable in $\mathcal{B}(X)$ at $t = 0$, but is strongly continuous:

$$V_n(t) \to V_n(0) = I \text{ as } t \to 0^+;$$

(iv) $\{V_n(t)x\}_n$ is strongly Cauchy for each $t \geq 0$ and $x \in X$, uniformly for t in compact subsets of $[0, \infty)$.

The verification of the above claims and using them, the writing of an alternative proof for the Hille-Yosida Theorem are left as an exercise (**Exercise** 2.3.5).

2. The approximation in (1) above may be used to show that A is the generator of a C_0-group $\{T(t)\}_{t\in\mathbb{R}}$ if and only if $\pm A$ each generate respectively the C_0-semigroup $\{T_\pm(t)\}_{t\geq 0}$ where $T_+(t) := T(t)$, for all $t \geq 0$, and $T_-(t) := T(-t)$, for all $t \geq 0$. The details of the proof are left as an exercise (**Exercise** 2.3.6).

2.4 Adjoint semigroups

A C_0-semigroup $\{T(t)\}_{t\geq 0}$ defined on a Banach space X induces in a natural way a family of operators $\{T^*(t)\}_{t\geq 0}$ on the dual space X^*, where $T^*(t)$ is the adjoint of $T(t)$ for each $t \geq 0$. This family clearly satisfies the semigroup law:

$$\begin{aligned}\langle T^*(t)T^*(s)x^*, x\rangle &= \langle T^*(s)x^*, T(t)x\rangle = \langle x^*, T(s)T(t)x\rangle \\ &= \langle x^*, T(t+s)x\rangle = \langle T^*(s+t)x^*, x\rangle\end{aligned}$$

for all $t, s \geq 0, x^* \in X^*$ and $x \in X$. Also, $\langle T^*(0)x^*, x\rangle = \langle x^*, T(0)x\rangle = \langle x^*, x\rangle$ so that $T^*(0) = I$. This semigroup may not be strongly continuous in general. However, if X is reflexive, then $\{T^*(t)\}_{t\geq 0}$ is a C_0-semigroup with generator A^*.

Theorem 2.4.1. Let X be a reflexive Banach space. If $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup on X, with generator A, then $\{T^*(t)\}_{t\geq 0}$ is a C_0-semigroup with generator A^*. Conversely, if A is a closed, densely defined operator on X and

$\beta \in \mathbb{R}$ is such that A satisfies the resolvent estimates as in (2.13) for $z \in H_\beta = \{z \in \mathbb{C} : \text{Re } z > \beta\} \subset \rho(A)$, then so does A^*. Also, if $\{T(t)\}_{t\geq 0}$ is the semigroup generated by A, then $\{T^*(t)\}_{t\geq 0}$ is the semigroup generated by A^*.

Proof. Since X is reflexive, we shall identify $x \in X$ with its embedded image in X^{**} without any new notation for the same. We have already seen that $T^*(t)T^*(s) = T^*(t+s)$ and $T^*(0) = I$. Since $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup, it is exponentially bounded. Suppose that $\|T(t)\| \leq Me^{\beta t}$ for some $\beta \in \mathbb{R}$ and $M > 0$. Then, for $x \in X$ and $x^* \in X^*$,

$$\big|\langle T^*(t)x^*, x\rangle\big| = \big|\langle x^*, T(t)x\rangle\big| \leq \|x^*\|\|x\|Me^{\beta t}.$$

This implies that $\big\|T^*(t)\big\| \leq Me^{\beta t}$. Also, for $t \geq s \geq 0$,

$$\big\langle T^*(t)x^* - T^*(s)x^*, x\big\rangle = \big\langle x^*, (T(t) - T(s))x\big\rangle \to 0$$

as $t \to s$. Therefore by the reflexivity of X, $\{T^*(t)\}_{t\geq 0}$ is a weakly continuous semigroup. It follows, on using Corollary 1.5.6, that $\{T^*(t)\}$ is strongly continuous. Thus, $\{T^*(t)\}_{t\geq 0}$ is a C_0-semigroup and

$$\Big\langle \frac{(T^*(t) - I)}{t}x^*, x\Big\rangle = \Big\langle x^*, \frac{(T(t) - I)}{t}x\Big\rangle \to \langle x^*, Ax\rangle$$

as $t \to 0^+$ for all $x \in \mathcal{D}(A)$. Let $\tilde{A}$ be the generator of $\{T^*(t)\}_{t\geq 0}$. Then, for $x^* \in \mathcal{D}(\tilde{A})$,

$$\big\langle \tilde{A}x^*, x\big\rangle = \big\langle x^*, Ax\big\rangle,$$

which implies $\tilde{A} \subset A^*$ since $\mathcal{D}(A)$ is dense. For all $x^* \in \mathcal{D}(A^*)$ and $x \in \mathcal{D}(A)$,

$$\begin{aligned}\big\langle T^*(t)x^* - x^*, x\big\rangle = \big\langle x^*, T(t)x - x\big\rangle &= \int_0^t \big\langle x^*, AT(s)x\big\rangle ds \\ &= \int_0^t \big\langle T^*(s)A^*x^*, x\big\rangle\, ds,\end{aligned}$$

leading to the identity $T^*(t)x^* - x^* = \displaystyle\int_0^t T^*(s)A^*x^*ds$, since $\mathcal{D}(A)$ is dense in X. But this implies, by Lemma 2.1.2, that $x^* \in \mathcal{D}(\tilde{A})$ and $\tilde{A}x^* = A^*x^*$. Therefore, $A^* \subset \tilde{A}$ and hence $A^* = \tilde{A}$.

□

The following is an example of a C_0-semigroup whose adjoint is not a C_0-semigroup.

Example 2.4.2. Let X be the Banach space $BUC(\mathbb{R}_+)$ of bounded, uniformly continuous functions on $[0,\infty)$ equipped with the supremum norm. Define the family $\{T(t)\}_{t\geq 0}$ on X as follows:

$$(T(t)f)(s) = f(s+t) \text{ for all } f \in X \text{ and } s,t \geq 0.$$

We have seen in Example 1.5.10 that $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup. Let A denote its generator. Set

$$BUC^1(\mathbb{R}_+) = \{f \in BUC(\mathbb{R}_+), f \text{ is differentiable}, f' \in BUC(\mathbb{R}_+)\}.$$

We will show that $\mathcal{D}(A) = BUC^1(\mathbb{R}_+), Af = f'$ and that T^* is not a C_0-semigroup.

Now $f \in \mathcal{D}(A)$, implies that $\lim_{t\to 0^+} \dfrac{T(t)f - f}{t}$ exists in $BUC(\mathbb{R}_+)$, that is, there exists a $g \in BUC(\mathbb{R}_+)$ such that $\lim_{t\to 0^+} \dfrac{T(t)f - f}{t} = g$ in the supremum norm. It follows therefore that for each $s \geq 0$,

$$\frac{(T(t)f)(s) - f(s)}{t} - g(s) = \frac{f(t+s) - f(s)}{t} - g(s) \to 0$$

as $t \to 0$. Thus we may conclude that f is differentiable on $(0,\infty)$ and for all $s > 0, f'(s) = g(s) = Af(s)$. In other words, $f \in BUC^1[0,\infty)$. Therefore, $\mathcal{D}(A) \subset BUC^1[0,\infty)$. On the other hand, if $f \in BUC^1[0,\infty)$, then

$$f(t+s) - f(s) = \int_0^t f'(u+s)d\,u.$$

Using this, we obtain that

$$\left|\frac{f(t+s) - f(s)}{t} - f'(s)\right| \leq \frac{1}{t}\int_0^t \|(T(u) - I)f'\|du \to 0 \text{ as } t \to 0$$

due to the strong continuity of the semigroup and Lemma 2.1.2. Thus, $f \in \mathcal{D}(A)$ and $f' = Af$.

Let $\mu = \delta_a$, the delta measure concentrated at $a \in \mathbb{R}$. It is easy to see that μ is in the dual space of $BUC(\mathbb{R}_+)$. Then, for $f \in X$,

$$\langle T^*(t)\mu, f\rangle = \langle \mu, T(t)f\rangle = (T(t)f)(a) = f(a+t) = \langle \delta_{a+t}, f\rangle.$$

Therefore, $T^*(t)\mu = \delta_{a+t}$, so that

$$\big\|(T^*(t) - I)\mu\big\| = \big\|\delta_{a+t} - \delta_a\big\| = 2$$

for all $t > 0$. Thus T^* is not strongly continuous, and it follows that $BUC(\mathbb{R}_+)$ is not a reflexive Banach space.

2.5 Examples of C_0-semigroups and their generators

Example 2.5.1. Let $X = L^p(\mathbb{R}_+)$ for some p be such that $1 \leq p < \infty$ and set

$$(T(t)f)(s) = f(s+t) \text{ for all } f \in X \text{ and } s, t \geq 0.$$

Then $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup with generator A given by

$$\begin{aligned} \mathcal{D}(A) &= \{f \in L^p(\mathbb{R}_+) : f \text{ absolutely continuous and } f' \in L^p(\mathbb{R}_+)\} \\ &= W^{1,p}(\mathbb{R}_+) \text{ (see Appendix A.3)}, \\ Af &= f'. \end{aligned} \tag{2.17}$$

For $f \in L^p(\mathbb{R}_+)$,

$$\|(T(t) - I)f\|_p^p = \int_0^\infty |f(s+t) - f(s)|^p \, ds.$$

Recall that $C_c^\infty(\mathbb{R}_+)$, the linear space of arbitrarily often differentiable (or smooth) functions on $\mathbb{R}_+$ with compact support, is dense in $L^p(\mathbb{R}_+)$ ([18, Proposition 5.5.9]). Let $f \in C_c^\infty(\mathbb{R}_+)$ and suppose that the support of f, $\operatorname{supp} f \subset [a, b]$ for some $0 < a < b < \infty$. Then, for $|t| < 1$,

$$\int_0^\infty |f(s+t) - f(s)|^p \, ds = \int_{a-1}^{b+1} |f(s+t) - f(s)|^p \, ds.$$

Since $f(s+t) \to f(s)$ as $t \to 0^+$ pointwise, and f is bounded, it follows by the Dominated Convergence Theorem that

$$\lim_{t\to 0^+} \|(T(t) - I)f\|_p^p = \lim_{t\to 0^+} \int_0^\infty |f(s+t) - f(s)|^p \, ds = 0.$$

This is true for all $f \in C_c^\infty(\mathbb{R}_+)$. The density of $C_c^\infty(\mathbb{R}_+)$ now allows us to extend the above convergence to all $f \in L^p(\mathbb{R}_+)$, on observing that $\|T(t)\| \leq 1$ for all $t \geq 0$. Thus T is a strongly continuous contraction semigroup on $L^p(\mathbb{R}_+)$. We shall now determine its generator A. Recall that (see Appendix A.3)

$$W^{1,p}(\mathbb{R}_+) = \big\{f \in L^p[0, \infty) : \int_0^\infty |f'(s)|^p \, ds < \infty\big\},$$

and that the two sets on the right hand side of (2.17) coincide due to Lemma A.3.2.

Let $\phi \in C_c^\infty(\mathbb{R}_+)$ with $\operatorname{supp}\phi \subset [c,d]$ for $0 < c < d < \infty$, and let $f \in \mathcal{D}(A)$. Then

$$\begin{aligned}\langle t^{-1}(T(t)f - f), \phi\rangle &= \int_0^\infty t^{-1}[f(s+t) - f(s)]\overline{\phi}(s)\,ds \\ &= \int_0^\infty f(s)\big[\frac{\overline{\phi}(s-t) - \overline{\phi}(s)}{t} + \overline{\phi}'(s)\big]\,ds \\ &\quad - \int_0^\infty f(s)\overline{\phi}'(s)\,ds - \int_0^t \frac{f(s)\overline{\phi}(s-t)}{t}\,ds \qquad (2.18)\\ &= I_1 - \int_0^\infty f(s)\overline{\phi}'(s)\,ds + I_2, \qquad (2.19)\end{aligned}$$

where I_1 and I_2 represent the first and the last integrals respectively appearing on the right hand side of (2.18). Note that $\overline{\phi}(s-t) = 0$ for $0 \le s \le t$ because of the support properties of ϕ, rendering $I_2 = 0$. Since both the sets $\operatorname{supp}\phi$ and $\operatorname{supp}\phi' \subset [c,d]$ and since $t^{-1}(\overline{\phi}(s-t) - \overline{\phi}(s))$ converges to $-\overline{\phi}'(s)$ uniformly in $s \in [c,d]$, as $t \to 0^+$, I_1 converges to 0 as $t \to 0^+$. Hence, taking the limit as $t \to 0^+$ in (2.19) we obtain that

$$\langle Af, \phi\rangle = -\int_0^\infty f(s)\overline{\phi}'(s)\,ds.$$

Next, we set $g(s) = \displaystyle\int_0^s (Af)(\tau)\,d\tau$ which makes g a well defined absolutely continuous function and an integration by parts leads to the equality

$$-\int_0^\infty f(s)\overline{\phi}'(s)\,ds = \langle Af, \phi\rangle = -\int_0^\infty g(s)\overline{\phi}'(s)\,ds$$

for all $\phi \in C_c^\infty(\mathbb{R}_+)$. This implies that $g = f$ or equivalently, $Af = f'$. Therefore, $\mathcal{D}(A) \subset W^{1,p}(\mathbb{R}_+)$.

Conversely, if $f \in W^{1,p}(\mathbb{R}_+)$, then f' exists in $L^p(\mathbb{R}_+)$, and we have from (2.19) that for $\phi \in C_c^\infty(\mathbb{R}_+)$,

$$\langle f', \phi\rangle = -\int_0^\infty f(s)\overline{\phi}'(s)\,ds = -I_1 + \langle t^{-1}(T(t)f - f), \phi\rangle - I_2.$$

Since $I_1, I_2 \to 0$ as $t \to 0^+$, it follows that

$$\lim_{t\to 0^+} \langle t^{-1}(T(t)f - f), \phi\rangle = -\langle f, \phi'\rangle = \langle f', \phi\rangle$$

for all $\phi \in C_c^\infty(\mathbb{R}_+)$. Hence $f \in \mathcal{D}(A)$, and $Af = f'$.

Remark 2.5.2. (1) The semigroups of operators in Example 2.5.1 above and Example 2.4.2 are called **translation semigroups** or **shifts** (left translations or left shifts to be precise). Different translation semigroups with distinct generators can be constructed by changing the underlying Banach space. For example, one could replace $L^p(\mathbb{R}_+)$ or $BUC(R_+)$ in the above examples by $L^p(\mathbb{R})$ or $BUC(\mathbb{R})$ respectively, or with $C_0(\mathbb{R}), C_0(\mathbb{R}_+)$ and other such function spaces. Here we shall restrict our attention to L^p and BUC spaces. In both theses cases, the generator is again differentiation, but with suitably altered domains. The verification of these properties is left as an exercise (**Exercise** 2.5.3).

(a) Let $X = BUC(\mathbb{R})$, and $(T(t)f)(s) = f(s+t)$ for all $f \in X$ and $s, t \in \mathbb{R}$. Then $\{T(t)\}_{t\in\mathbb{R}}$ is a C_0-group of isometries with generator A given by $Af = f',$ for all $f \in \mathcal{D}(A)$, where

$$\begin{aligned}\mathcal{D}(A) &= \{f \in BUC(\mathbb{R}), f \text{ is differentiable, } f' \in BUC(\mathbb{R})\} \\ &= BUC^1(\mathbb{R}).\end{aligned}$$

(b) Let $X = L^p(\mathbb{R})$ where $1 \leq p < \infty$ and let $(T(t)f)(s) = f(s+t)$ for all $f \in X$ and $s, t \in \mathbb{R}$. Then $\{T(t)\}_{t\in\mathbb{R}}$ is a C_0-group of isometries with generator A given by $Af = f', \forall\ f \in \mathcal{D}(A) = W^{1,p}(\mathbb{R})$ (see Appendix A.3).

(2) Let $X = BUC(\mathbb{R})$ or $L^p(\mathbb{R})$. The inverse of $T(t)$, the left shift operator on X, is the right shift operator $S(t)$, given by $(S(t)f)(s) = f(s-t)$ for all $f \in X$ and $t, s \in \mathbb{R}$. Moreover, $\{S(t)\}_{t\in\mathbb{R}}$ forms a C_0-group of isometries with generator $-A$, where A is the generator of the left shift group defined in (1a) or (1b) respectively.

(3) The right translation semigroup on $L^p(\mathbb{R}_+)$ is defined as follows:

$$(S(t)f)(s) = \begin{cases} f(s-t) & \text{if } s-t \geq 0 \\ 0 & \text{if } s-t < 0. \end{cases} \tag{2.20}$$

Then $\{S(t)\}_{t\geq 0}$ defines a C_0-semigroup of isometries on $L^p(\mathbb{R}_+)$ with generator A given by $\mathcal{D}(A) = \{f \in W^{1,p} : f(0) = 0\}$ and $(Af)(s) = -f'(s)$ if $s > 0$, for any $f \in \mathcal{D}(A)$. Every function in $Ran\, S(t)$ vanishes in the interval $[0, t]$ and therefore $Ran\, S(t) \neq L^p(\mathbb{R}_+)$. A similar result holds on $BUC(\mathbb{R}_+)$.

(4) Translation semigroups can also be defined on function spaces on finite intervals using the same ideas as above and making appropriate modifications. For example, on $L^p[a,b]$, where $a,b \in \mathbb{R}$, we define the left translation semigroup $\{T(t)\}_{t\geq 0}$ by

$$(T(t)f)(s) := \begin{cases} f(s+t) & \text{if } a \leq s+t \leq b \\ 0 & \text{if } s+t > b \text{ or } s+t < a. \end{cases} \quad \forall\ f \in L^p[a,b]. \tag{2.21}$$

Then $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup which is *nilpotent*, that is, $T(t) = 0$ for all $t \geq b-a$.

Let us now revisit the semigroup in Example 1.5.7. Here $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup and for each $t > 0, T(t)$ is multiplication by the function $e^{t\mathfrak{q}}$, where $\mathfrak{q}(s) = -s^2$, for all $s \geq 0$. This is an example of a **multiplication semigroup**, which is usually defined on spaces of continuous or measurable functions. In the next example we look at general multiplication semigroups on L^p spaces. Recall that for a measurable function $\mathfrak{q}\colon \Omega \to \mathbb{C}$, where $(\Omega, \mathcal{F}, \mu)$ is a σ-finite measure space, the *essential range* of $\mathfrak{q}$ is defined to be

$$\text{ess ran}\,\mathfrak{q} := \bigcap_{\epsilon>0} \{\lambda : \mu(\{s \in \Omega : |\mathfrak{q}(s) - \lambda| < \epsilon\}) > 0\}.$$

The associated *multiplication operator* $M_\mathfrak{q}$ on $L^p(\Omega,\mu)$ is defined as follows:

$$\begin{aligned} \mathcal{D}(M_\mathfrak{q}) &:= \{f \in L^p(\Omega,\mu) : \mathfrak{q}f \in L^p(\Omega,\mu)\}, \\ M_\mathfrak{q} &= \mathfrak{q}f. \end{aligned} \tag{2.22}$$

Then $M_\mathfrak{q}$ is a closed and densely defined operator. $M_\mathfrak{q}$ is bounded if and only if $\mathfrak{q} \in L^\infty(\Omega,\mu)$ and in this case

$$\|M_\mathfrak{q}\| = \|\mathfrak{q}\|_\infty = \sup\{|\lambda| : \lambda \in \text{ess ran}\,\mathfrak{q}\}.$$

Further, $M_\mathfrak{q}$ has a bounded inverse if and only if $0 \notin \text{ess ran}\,\mathfrak{q}$. In such a case, $\mu\{s \in \Omega : \mathfrak{q}(s) = 0\} = 0$ and then $(M_\mathfrak{q})^{-1} = M_\psi$ where $\psi(s) = 1/\mathfrak{q}(s)$, μ-almost everywhere. It can be seen that

$$\sigma(M_\mathfrak{q}) = \text{ess ran}\,\mathfrak{q}, \text{ and for } \lambda \notin \sigma(M_\mathfrak{q}), R(\lambda, M_\mathfrak{q}) = M_\phi, \tag{2.23}$$

where ϕ is given by

$$\phi(s) = (\mathfrak{q}(s) - \lambda)^{-1},\ \mu\text{-almost everywhere.} \tag{2.24}$$

We refer the reader to [1] and [21] for a detailed discussion of such operators. See also Theorem A.1.13.

Example 2.5.4. Let $X = L^p(\Omega, \mu)$, where $(\Omega, \mathcal{F}, \mu)$ is a σ-finite measure space and $p \in [1, \infty)$ is fixed. Further, let $\mathfrak{q}\colon \Omega \to \mathbb{C}$ be a measurable function satisfying $\sup_{\lambda \in \text{ess ran}\, \mathfrak{q}} \operatorname{Re} \lambda < \infty$. Define $T(t) \in \mathcal{B}(X)$ by

$$(T(t)f)(s) = e^{t\mathfrak{q}(s)} f(s), \text{ for all } s \in \Omega, t \geq 0 \text{ and } f \in X.$$

Then $\{T(t)\}_{t \geq 0}$ is a C_0-semigroup with generator $A = M_{\mathfrak{q}}$.

We first check strong continuity of the semigroup $\{T(t)\}_{t \geq 0}$. Let $k = \sup_{\lambda \in \text{ess ran}\, \mathfrak{q}} \operatorname{Re} \lambda$. Then we have, for $f \in X$ and $t > 0$, that

$$\|T(t)f - f\|^p = \int_\Omega \left|(e^{tq(s)} - 1) f(s)\right|^p \mu(ds),$$

and since

$$\left|e^{t\mathfrak{q}(s)} - 1\right| \leq 1 + e^{tk} \text{ for } 0 \leq t \leq 1,$$

it follows, by using the Dominated Convergence Theorem that $\|T(t)f - f\| \to 0$ as $t \to 0^+$. Moreover, it is easy to check that

$$\|T(t)\| \leq e^{tk}, \ \forall\ t \geq 0. \tag{2.25}$$

Let A be the generator of this semigroup. To show that $A = M_{\mathfrak{q}}$ let us first take $f \in \mathcal{D}(A)$. For $s \in \Omega$, we have

$$\lim_{t \to 0^+} t^{-1}\big((T(t)f)(s) - f(s)\big) = \lim_{t \to 0^+} t^{-1}(e^{t\mathfrak{q}(s)} - 1) f(s) = \mathfrak{q}(s) f(s). \tag{2.26}$$

Since $f \in \mathcal{D}(A)$ implies that $\lim_{t \to 0^+} t^{-1}\big((T(t)f) - f\big)$ exists in $X = L^p(\Omega, \mu)$, which implies pointwise convergence μ-almost everywhere, possibly for a subsequence, it follows from (2.26) that the limit equals $M_{\mathfrak{q}} f$ and $M_{\mathfrak{q}} f \in L^p(\Omega, \mu)$. Thus $f \in \mathcal{D}(M_{\mathfrak{q}})$ and $Af = M_{\mathfrak{q}} f$. Hence, $A \subset M_{\mathfrak{q}}$. We now show that $M_{\mathfrak{q}} \subset A$.

Let $f \in \mathcal{D}(M_{\mathfrak{q}})$. Since

$$\begin{aligned}\left|e^{t\mathfrak{q}(s)} - 1 - t\mathfrak{q}(s)\right| &= \left|\mathfrak{q}(s) \int_0^t \left(e^{r\mathfrak{q}(s)} - 1\right) dr\right| \leq |\mathfrak{q}(s)|\Big(t + \int_0^t e^{r \operatorname{Re}\ \mathfrak{q}(s)}\, dr\Big) \\ &\leq |\mathfrak{q}(s)|\big|t + te^{tk}\big|,\end{aligned}$$

we have, for μ-almost all s and for $0 < t < 1$,

$$\left|t^{-1}\big(e^{t\mathfrak{q}(s)} - 1 - t\mathfrak{q}(s)\big) f(s)\right| \leq (1 + e^k)|\mathfrak{q}(s) f(s)|.$$

Moreover,

$$t^{-1}\big(e^{t\mathfrak{q}(s)} - 1 - t\mathfrak{q}(s)\big) \to 0 \text{ for } \mu\text{-almost all } s \text{ as } t \to 0^+.$$

Therefore, by the Dominated Convergence Theorem,

$$\int_\Omega \left|t^{-1}(e^{t\mathfrak{q}(s)} - 1)f(s) - \mathfrak{q}(s)f(s)\right|^p \mu(ds) \to 0 \text{ as } t \to 0^+.$$

This implies that $\lim_{t\to 0^+} t^{-1}\big(T(t)f - f\big) = \mathfrak{q}f$ in X. Hence $f \in \mathcal{D}(A)$ and $Af = \mathfrak{q}f = M_\mathfrak{q} f$. Thus $M_\mathfrak{q} \subset A$.

The example we discuss next – Gaussian semigroups – are representatives of a very important class of semigroups, called **convolution semigroups.** Gaussian semigroups, also called heat semigroups or diffusion semigroups, by themselves are an important class of C_0-semigroups, occurring very frequently in applications.

Example 2.5.5 (Convolution Semigroups)**.** Convolution semigroups occur in many areas of applications, particularly in probability theory and are also of interest by themselves. Special examples of some of them, particularly the heat semigroup, play important roles in the theory of Brownian motion, of diffusion processes and in geometry.

Definition 2.5.6. A *convolution semigroup* on $\mathbb{R}^d$ is a family of probability measures $\{\mu_t\}_{t\in\mathbb{R}}$ satisfying, $(i)\,\mu_t * \mu_s = \mu_{t+s}, (ii)\,\mu_0 = \lim_{t\to 0^+}\mu_t = \delta_0$, where δ_0 is the Dirac delta measure concentrated at $0 \in \mathbb{R}^d$ and the limit exists in the sense that for every $f \in C_c(\mathbb{R}^d)$, $\lim\limits_{t\to 0^+}\int_{\mathbb{R}^d} \mu_t(dx)f(x) = f(0)$.

In the above, the convolution $\mu*\nu$ of two finite measures μ and ν is defined as the unique finite measure such that

$$\int f(x)(\mu * \nu)(dx) = \int\int f(x+y)\mu(dx)\nu(dy), \text{ for all } f \in C_c^\infty(\mathbb{R}^d),$$

or equivalently, for a Borel set $\triangle \subset \mathbb{R}$, $(\mu*\nu)(\triangle) = \int \mu(\triangle - y)\nu(dy)$. If μ and ν are probability measures, then so is their convolution.

The next theorem gives the basic structure of such objects, justifying *semigroup* in its name.

Theorem 2.5.7. Let $\{\mu_t\}_{t\geq 0}$ be a convolution semigroup on $\mathbb{R}^d$ and let $1 \leq p < \infty$. Define a map $T(t)$ on $L^p(\mathbb{R}^d)$ by setting

$$(T(t)f)(x) = \int f(x-y)\,\mu_t(dy) \text{ and } T(0)f = f \text{ for all } f \in L^p(\mathbb{R}^d) \text{ and } x \in \mathbb{R}^d.$$

Then $\{T(t)\}_{t\in\mathbb{R}_+}$ is a C_0 contraction semigroup on $L^p(\mathbb{R}^d)$, and is positive, that is, $(T(t)f)(x) \geq 0$ if $f(x) \geq 0$, almost everywhere for all $t \in \mathbb{R}_+$. Furthermore, $T(t)$ commutes with the group of translations on $L^p(\mathbb{R}^d)$ and if $f \in L^1_+(\mathbb{R}^d)$ (the positive elements of $L^1(\mathbb{R}^d)$), then $\|T(t)f\|_1 = \|f\|_1$.

Proof. Since

$$|(T(t)f)(x)|^p \leq \Big(\int_{\mathbb{R}^d} |f(x-y)|\mu_t(dy)\Big)^p$$

and since $[0,\infty) \ni \lambda \mapsto \lambda^p$ is a convex function for each $p \in [1,\infty)$, we have by Jensen's inequality [23, Jensen's Inequality, page 133] that

$$\big|(T(t)f)(x)\big|^p \leq \int |f(x-y)|^p \, \mu_t(dy)$$

and therefore

$$\|T(t)f\|_p^p \leq \int dx \int |f(x-y)|^p \, \mu_t(dy).$$

An application of Fubini's theorem and the observation that the Lebesgue measure is translation invariant leads to the result that

$$\big\|T(t)f\big\|_p^p \leq \int \Big(\int |f(x-y)|^p \, dx\Big)\mu_t(dy) = \|f\|_p^p.$$

This means that $T(t)$ is a contraction on $L^p(\mathbb{R}^d)$. Next, for $t, s > 0$, $f \in L^p(\mathbb{R}^d)$,

$$\begin{aligned}(T(t)T(s)f)(x) &= \int (T(s)f)(x-y)\mu_t(dy) = \int \Big(\int f(x-y-z)\mu_s(dz)\Big)\mu_t(dy) \\ &= \int f(x-y') \int \mu_s(d(y'-y))\mu_t(dy) = \int f(x-y)(\mu_s * \mu_t)(dy),\end{aligned}$$

where we have made use of Fubini's theorem to interchange the order of integration, made a change of variable and used the definition of the convolution of two measures μ_s and μ_t. Thus we get that

$$(T(t)T(s)f)(x) = \int f(x-y)\mu_{t+s}(dy) = (T(t+s)f)(x),$$

proving the semigroup property. Next we show strong continuity. Noting that

$$(T(t)f - f)(x) = \int [f(x-y) - f(x)]\mu_t(dy)$$

and applying Jensen's inequality [23, Jensen's Inequality, page 133], the above leads to

$$\big\|T(t)f - f\big\|_p^p \leq \int dx \int |f(x-y) - f(x)|^p \mu_t(dy). \tag{2.27}$$

We now restrict f to be in $C_c^\infty(\mathbb{R}^d)$. Recall that $\mu_0 = \lim_{t\to 0^+} \mu_t = \delta_0$ so that $\lim_{t\to 0^+} \mu_t\{x \in \mathbb{R}^d : |x| < \alpha\} = 1$ for all $\alpha > 0$, and therefore observe that for small positive t, both the y- and x-integrals are effectively over bounded subsets of $\mathbb{R}^d$ over which f is bounded. Given $\epsilon > 0$, choose $\beta > 0$ such that $|f(x-y) - f(x)| < \epsilon$ for $|y| < \beta$, for x in such bounded sets, where $|y|$ is the Euclidean norm of $y \in \mathbb{R}^d$. Then choose $t_0 > 0$ such that $\mu_t(|y| > \beta) < \epsilon$ for $0 < t < t_0$. Thus,

$$\begin{aligned}\int dx \int |f(x-y) - f(x)|^p \, \mu_t(dy) &= \int dx \int_{|y|<\beta} |f(x-y) - f(x)|^p \, \mu_t(dy) \\ &\quad + \int dx \int_{|y|>\beta} |f(x-y) - f(x)|^p \, \mu_t(dy) \\ &= I_1 + I_2.\end{aligned}$$

Now

$$\begin{aligned}I_1 &= \int_{x\in(\operatorname{supp} f+\beta)} dx \int_{|y|<\beta} |f(x-y) - f(x)|^p \mu_t(dy) \\ &< \epsilon^p m(\operatorname{supp} f + \beta),\end{aligned}$$

where we have overestimated by using $\int_{|y|<\beta} \mu_t(dy) \le 1$. For the second integral I_2, we have

$$I_2 \le 2^p \|f\|_p^p \mu_t(|y| > \beta) \le \epsilon 2^p \|f\|_p^p.$$

Combining these two estimates and (2.27), we get that

$$\|T(t)f - f\|_p \to 0, \text{ as } t \to 0^+ \text{ for any } f \in C_c^\infty(\mathbb{R}^d).$$

Since $C_c^\infty(\mathbb{R}^d)$ is dense in $L^p(\mathbb{R}^d)$ for $1 \le p < \infty$, and $\|T(t)\|$ is uniformly bounded, we are able to extend the said convergence to the whole of $L^p(\mathbb{R}^d)$, showing that $\{T(t)\}_{t\ge 0}$ is a C_0-semigroup. That $\{T(t)\}_{t\ge 0}$ is positive is clear from the definition and the fact that the μ_t's are probability measures. Let $(S(a)f)(x) = f(x-a)$ for all $a \in \mathbb{R}^d, x \in \mathbb{R}^d$ and $f \in L^p(\mathbb{R}^d)$, so that $\{S(a)\}_{a\in\mathbb{R}^d}$ is the group of translations. In fact, $\{S(a)\}_{a\in\mathbb{R}}$ is a C_0-group of isometries on $L^p(\mathbb{R}^d)$ (compare with Remark 2.5.2 (2)). One notes that

$$\begin{aligned}(S(a)T(t)f)(x) = (T(t)f)(x-a) &= \int f(x-a-y)\mu_t(dy) \\ &= \int (S(a)f)(x-y)\mu_t(dy) = (T(t)S(a)f)(x). \qquad (2.28)\end{aligned}$$

For $f \in L^1_+(\mathbb{R}^d)$, using Fubini's theorem, we get that

$$\begin{aligned}\|T(t)f\|_1 &= \int (T(t)f)(x)\,dx = \int dx \int f(x-y)\mu_t(dy)\\ &= \int \mu_t(dy) \int f(x-y)\,dx = \|f\|_1.\end{aligned}$$

□

The most well-known example of a convolution semigroup is the **heat semigroup** given by the Normal $(0,t)$ probability measure on $\mathbb{R}^d$:

$$\mu_t(dx) = (2\pi t)^{-d/2} e^{-\frac{|x|^2}{2t}}\,dx, \tag{2.29}$$

where $x \in \mathbb{R}^d$, $|x|$ is the Euclidean norm of x and dx is the Lebesgue measure on $\mathbb{R}^d$. Similarly, the Cauchy measure also is of importance in probability theory: on $\mathbb{R}$,

$$\gamma_t(dx) = \pi^{-1} t(t^2 + x^2)^{-1}\,dx.$$

It is left as an exercise (**Exercise** 2.5.8) to show that

(i) the heat and Cauchy measures satisfy the convolution semigroup properties of measures, as given in Definition 2.5.6, and

(ii) that the proof of Theorem 2.5.7 holds with $L^p(\mathbb{R})$ replaced by $C_0(\mathbb{R})$, the Banach space with supremum norm of continuous functions on $\mathbb{R}$ which vanish at infinity.

Theorem 2.5.9. (Computation of the generator for the heat semigroup in $L^p(\mathbb{R})$.) The generator of the heat semigroup, associated with the convolution semigroup given by μ_t as in (2.29) is as follows:

$$\mathcal{D}(A) = \Big\{ f \in L^p(\mathbb{R}^d) : \sum_{i=1}^{d} \frac{\partial^2 f}{\partial {x_j}^2} \in L^p(\mathbb{R}^d) \Big\} = W^{2,p}(\mathbb{R}^d),$$

$$Af = \frac{1}{2}\Delta f = \frac{1}{2}\sum_{j=1}^{d} \frac{\partial^2 f}{\partial {x_j}^2}, \text{ for all } f \in \mathcal{D}(A),$$

that is, the generator of the heat semigroup is the Laplacian with maximal domain, multiplied by $1/2$ (see Section A.3).

Proof. For simplicity we will work with $d = 1$. Fix $p \in [1, \infty)$. From Theorem 2.5.7 we know already that the heat semigroup associated with the convolution

semigroup given by (2.29) is a C_0 contraction semigroup on $L^p(\mathbb{R})$ and hence, by Theorem 2.2.7, its infinitesimal generator A is a densely defined closed operator in $L^p(\mathbb{R})$. For any $f \in L^p(\mathbb{R})$, almost all $x \in \mathbb{R}$ and any $t > 0$,

$$\begin{aligned}(T(t)f - f)(x) &= \int_{\mathbb{R}} (f(x-y) - f(x))(2\pi t)^{-1/2} e^{-y^2/2t}\, dy \\ &= \int_{\mathbb{R}} (f(x-\sqrt{t}u) - f(x))(2\pi)^{-1/2} e^{-u^2/2}\, du \\ &= \int_{\mathbb{R}} (f(x-\sqrt{t}u) - f(x)) N(u)\, du, \end{aligned} \tag{2.30}$$

where we have made a change of variable $y = \sqrt{t}u$, and have set $N(u) = (2\pi)^{-1/2} e^{-u^2/2}$ as the standard normal distribution function. Formally, we can expand the expression in the parenthesis in (2.30) to get, for a sufficiently smooth function f, that

$$f(x-\sqrt{t}u) - f(x) = -(\sqrt{t}u) f'(x) + \frac{1}{2}(tu^2) f''(x) + O(t^{3/2}), \tag{2.31}$$

as $t \to 0^+$. Therefore,

$$(T(t)f - f)(x) = t/2 f''(x) + O(t^{3/2}),$$

since $\int uN(u)\, du = 0$ and $\int u^2 N(u)\, du = 1$. Now we make this formal argument rigorous. Let $f \in \mathcal{D}(\frac{d^2}{dx^2}) \subset L^p(\mathbb{R})$ (see Appendix A.3). Then proceeding as above, we have

$$\begin{aligned} & t^{-1}(T(t)f - f)(x) - \frac{1}{2} f''(x) \\ &\quad = \int_{\mathbb{R}} N(u)\, du\, t^{-1} \big[f(x-\sqrt{t}u) - f(x) - \frac{t}{2} f''(x) \big] \\ &\quad = \int_{\mathbb{R}} N(u)\, du\, t^{-1} \big[f(x-\sqrt{t}u) - f(x) + \sqrt{t}u f'(x) - \frac{t}{2} u^2 f''(x) \big] \\ &\quad = \int_{\mathbb{R}} N(u)\, du\, t^{-1} \int_0^{-u\sqrt{t}} d\alpha \int_0^{\alpha} d\beta \big[f''(x+\beta) - f''(x) \big], \end{aligned}$$

where we used the fact that $f'' \in L^1_{\mathrm{loc}}(\mathbb{R})$, the space of locally integrable functions defined on $\mathbb{R}$, and hence the fundamental theorem of integral calculus is valid for it. Next, using the fact that $\mathbb{R}_+ \ni \lambda \mapsto \lambda^p$ is a convex function for

$p \geq 1$ and also using Jensen's inequality twice in succession, we get that

$$\begin{aligned}
&\big\|t^{-1}(T(t)f - f) - \frac{1}{2}f''\big\|_p^p \\
&= \int_{\mathbb{R}} dx \Big| \int_{\mathbb{R}} N(u)\, du\, t^{-1} \int_0^{-u\sqrt{t}} d\alpha \int_0^{\alpha} d\beta \big[f''(x+\beta) - f''(x)\big] \Big|^p \\
&\leq \int_{\mathbb{R}} N(u)\, du \int dx \Big| t^{-1} \int_0^{-\sqrt{t}u} d\alpha \int_0^{\alpha} d\beta \big[f''(x+\beta) - f''(x)\big] \Big|^p \\
&\leq \int_{\mathbb{R}} N(u)\, du\, t^{-p} \big(\frac{1}{2}tu^2\big)^{(p-1)} \int_0^{|u|\sqrt{t}} d\alpha \int_0^{\alpha} d\beta \int_{\mathbb{R}} \big|f''(x+\beta) - f''(x)\big|^p\, dx \\
&= \int_{|u|>\delta t^{-1/2}} N(u)\, du\, t^{-1} \big(\frac{1}{2}u^2\big)^{(p-1)} \int_0^{|u|\sqrt{t}} d\alpha \int_0^{\alpha} d\beta \big\|f''(\cdot+\beta) - f''(\cdot)\big\|_p^p \\
&\quad + \int_{|u|<\delta t^{-1/2}} N(u)\, du\, t^{-1} \big(\frac{1}{2}u^2\big)^{(p-1)} \int_0^{|u|\sqrt{t}} d\alpha \int_0^{\alpha} d\beta \big\|(S(\beta) - I)f''\big\|_p^p \\
&= I_1 + I_2,
\end{aligned}$$

where $\delta > 0$. We have used the fact that the simplex $0 \leq \beta \leq \alpha \leq u\sqrt{t}$ has the total area $= (u^2 t/2)$ and set $\{S(\beta)\}_{\beta\in\mathbb{R}}$ as the C_0-group of isometries of translation in $L^p(\mathbb{R})$. Next, for any $\epsilon > 0$,

$$I_1 \leq 2^p \|f''\|_p^p \int_{|u|>\delta t^{-1/2}} \big(\frac{u^2}{2}\big)^p N(u)\, du$$

which we can make less than ϵ, by choosing $t_0 > 0$ such that for $0 < t < t_0$, the integral in I_1 is less than $\dfrac{\epsilon}{2^p \|f''\|_p^p}$ which is possible since $\displaystyle\int_{\mathbb{R}} \big(\frac{u^2}{2}\big)^p N(u)\, du < \infty$. On the other hand, for I_2 we note that $S(\beta)$, the translation group on $L^p(\mathbb{R})$, is strongly continuous and hence uniformly strongly continuous for $0 \leq \beta \leq 1$, that is,

$$\sup_{0\leq\beta\leq\alpha\leq|u|\sqrt{t}\leq\delta} \|(S(\beta) - I)f''\|_p^p < \epsilon \quad \text{for } 0 < t < t_0.$$

Therefore,

$$I_2 \leq \epsilon \int_{|u|<\delta t^{-1/2}} N(u) \Big(\frac{u^2}{2}\Big)^{(p-1)} t^{-1} \Big(\frac{tu^2}{2}\Big)\, du < \epsilon \int N(u) \Big(\frac{u^2}{2}\Big)^p\, du$$

showing that $\frac{1}{2}\frac{d^2}{dx^2} \subset A$. For the other inclusion, that is, $A \subset \frac{1}{2}\frac{d^2}{dx^2}$, we start with the resolvent formula

$$(z - A)^{-1} f = \int_0^{\infty} e^{-zt} T(t) f\, dt, \text{ for } z \in \mathbb{C} \text{ such that } Re z > \omega, \tag{2.32}$$

where ω is the exponential growth bound of $T(t)$. Since $T(t)$ is a contraction, $\omega = 0$, and we may choose $z = 1$. Furthermore, since the heat semigroup is given by the integral kernel

$$K(t; x-y) = (2\pi t)^{-1/2} e^{-\frac{(x-y)^2}{2t}}$$

so that

$$(T(t)f)(x) = \int K(t; x-y) f(y)\, dy,$$

it is easily verified that the resolvent $(1-A)^{-1}$ is also an integral operator given as

$$((1-A)^{-1}f)(x) = \int dy \Big[\int_0^\infty e^{-t} K(t; x-y)\, dt \Big] f(y). \tag{2.33}$$

Using the fact that $\displaystyle\int_0^\infty (2\pi t)^{-1/2} e^{-(t+\frac{x^2}{2t})}\, dt = 2^{-1/2} e^{-\sqrt{2}|x|}$ (see Lemma 2.5.10), and the above form of the resolvent, we have that

$$((1-A)^{-1}f)(x) = 2^{-1/2} \int e^{-\sqrt{2}|x-y|} f(y)\, dy \tag{2.34}$$

$$= 2^{-1/2} \int_{-\infty}^x e^{-\sqrt{2}(x-y)} f(y)\, dy + 2^{-1/2} \int_x^\infty e^{\sqrt{2}(x-y)} f(y)\, dy \tag{2.35}$$

where both the integrals converge absolutely for each $x \in \mathbb{R}$, by an application of the Hölder inequality along with the observation that $f \in L^p(\mathbb{R}) \subset L^1_{\text{loc}}(\mathbb{R})$. Since $(1-A)^{-1}$ maps $L^p(\mathbb{R})$ onto $\mathcal{D}(A)$ we conclude from (2.35) that every element $g \in \mathcal{D}(A)$ is absolutely continuous and hence differentiable almost everywhere. Furthermore, by differentiating on both sides of (2.35), we get that for every $f \in L^p(\mathbb{R})$ and for almost all $x \in \mathbb{R}$,

$$((1-A)^{-1}f)'(x) = -\int_{-\infty}^x e^{-\sqrt{2}(x-y)} f(y)\, dy + \int_x^\infty e^{\sqrt{2}(x-y)} f(y)\, dy$$

which is again absolutely continuous and is differentiable almost everywhere and

$$((1-A)^{-1}f)''(x) = 2[((1-A)^{-1}f)(x) - f(x)]. \tag{2.36}$$

The equation (2.36) establishes the facts that (i) every vector $g \in \mathcal{D}(A) \subset L^p(\mathbb{R})$ is twice differentiable almost everywhere and (ii) $g'' \in L^p(\mathbb{R})$. Thus, $\mathcal{D}(A) \subset \mathcal{D}(\frac{d^2}{dx^2})$, leading to the equality of the two domains and that $A = \frac{1}{2}\frac{d^2}{dx^2}$ on that domain.

□

Next we prove the lemma that was used in the proof in the preceding paragraph.

Lemma 2.5.10. For $x \in \mathbb{R}$,

$$\int_0^\infty (2\pi t)^{-1/2} e^{-(t+\frac{x^2}{2t})}\, dt = 2^{-1/2} e^{-\sqrt{2}|x|}.$$

Proof. We begin with the well-known Gamma integral, viz,

$$\int_0^\infty e^{-y^2}\, dy = \frac{\sqrt{\pi}}{2}$$

and substitute $y = \alpha - c/\alpha$ with $c > 0$ and $\sqrt{c} \le \alpha < \infty$. Thus,

$$\begin{aligned}\frac{\sqrt{\pi}}{2} &= \int_{\sqrt{c}}^\infty e^{-(\alpha - c/\alpha)^2} \Big(1 + \frac{c}{\alpha^2}\Big)\, d\alpha \\ &= e^{2c} \int_{\sqrt{c}}^\infty e^{-\left(\alpha^2 + \frac{c^2}{\alpha^2}\right)}\, d\alpha + e^{2c} \int_{\sqrt{c}}^\infty e^{-(\alpha^2 + \frac{c^2}{\alpha^2})} \Big(\frac{c}{\alpha^2}\Big)\, d\alpha.\end{aligned}$$

In the second integral on the right hand side another change of variable $\sigma = c\alpha^{-1}$ converts the said integral into

$$-\int_{\sqrt{c}}^0 e^{-\left(\sigma^2 + \frac{c^2}{\sigma^2}\right)}\, d\sigma$$

and combining all these together we get that

$$\int_0^\infty e^{-\left(\sigma^2 + \frac{c^2}{\sigma^2}\right)}\, d\sigma = e^{-2c} \frac{\sqrt{\pi}}{2}.$$

Finally, in the given integral in the statement, we set $t = \sigma^2$ to get that

$$\int_0^\infty (2\pi t)^{-1/2} e^{-\left(\frac{x^2}{2t} + t\right)}\, dt = \frac{2}{\sqrt{2\pi}} \int_0^\infty e^{-\left(\sigma^2 + \frac{x^2}{2\sigma^2}\right)}\, d\sigma = 2^{-1/2} e^{-\sqrt{2}|x|}.$$

□

Chapter 3

Dissipative operators and holomorphic semigroups

In this chapter we continue the study of C_0-semigroups concentrating on contractive and holomorphic semigroups. A brief summary of the frequently used concepts and properties of densely defined closed or closable linear operators in a Banach or Hilbert space can be found in Appendix A.1.

3.1 Dissipative operators

The Hille Yosida Theorem 2.3.1 and another version of it in Theorem 2.3.3 gives a characterisation for the generator of a contraction semigroup in terms of certain resolvent estimates. Another, very different characterisation for such generators is available via the notion of dissipative operators. We first define dissipative operators on Hilbert spaces and then generalise to Banach spaces before presenting the Lumer-Phillips characterisation, the primary goal of this section.

Definition 3.1.1. Let X be a Hilbert space. A linear operator A on X is said to be **dissipative** if $\operatorname{Re}\langle u, Au\rangle \leq 0$ for all $u \in \mathcal{D}(A)$. If $\operatorname{Re}\langle u, Au\rangle \geq 0$, then A is called **accretive**.

Definition 3.1.2. A dissipative operator A on a Hilbert space X is called **maximal dissipative** if A does not admit any proper, dissipative extension.

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7_3

Example 3.1.3. Let $X = L^2(0,1)$ and A be the operator given by

$$\mathcal{D}(A) = \{f \in W^{1,2}(0,1) : f(0) = 0\}$$
$$Af = f'.$$

(For definition of $W^{1,2}(0,1)$ see Appendix A.3.) Then, for $f \in \mathcal{D}(A), |ff'| \in L^1[0,1]$ and f is absolutely continuous. Therefore,

$$\operatorname{Re}\langle f, Af\rangle = \operatorname{Re}\int_0^1 f(t)\overline{f'(t)}\,dt = \frac{1}{2}\int_0^1 (f(t)\overline{f'(t)} + \overline{f(t)}f'(t))\,dt$$
$$= \frac{1}{2}\int_0^1 \frac{d}{dt}(f(t)\overline{f(t)})\,dt = \frac{1}{2}|f(1)|^2 \geq 0.$$

Thus, A is accretive and $-A$ is a dissipative operator.

The proof of the following useful lemma is very simple and we leave the details to the reader (**Exercise** 3.1.4).

Lemma 3.1.5. An operator A on a Hilbert space is dissipative if and only if

$$\|(A+I)f\| \leq \|(A-I)f\| \text{ for all } f \in \mathcal{D}(A).$$

Theorem 3.1.6. For an operator A on a Hilbert space, the following are equivalent:

(a) A is dissipative;

(b) $\|(A-\lambda)f\| \geq \operatorname{Re}\lambda\|f\|$ for all $f \in \mathcal{D}(A)$ and $\lambda \in \mathbb{C}$ with $\operatorname{Re}\lambda > 0$;

(c) $\|(A-\lambda)f\| \geq \lambda\|f\|$ for all $f \in \mathcal{D}(A)$ and $\lambda > 0$.

Proof. (a) $\Rightarrow$ (b): For $f \in \mathcal{D}(A)$, we have, for $\operatorname{Re}\lambda > 0$,

$$\operatorname{Re}\langle f, (A-\lambda)f\rangle = \operatorname{Re}\langle f, Af\rangle - \operatorname{Re}\lambda\|f\|^2 \leq -\operatorname{Re}\lambda\|f\|^2 < 0.$$

Therefore,

$$\big\|(A-\lambda)f\big\|\|f\| \geq \big|\operatorname{Re}\big\langle f, (A-\lambda)f\big\rangle\big| = -\operatorname{Re}\big\langle f, (A-\lambda)f\big\rangle \geq \operatorname{Re}\lambda\|f\|^2.$$

(b) $\Rightarrow$ (c): This is obvious.

(c) $\Rightarrow$ (a): Let $\lambda > 0$ and $f \in \mathcal{D}(A)$. Since

$$\|Af\|^2 - 2\lambda\operatorname{Re}\langle f, Af\rangle = \|Af - \lambda f\|^2 - \lambda^2\|f\|^2 \geq 0,$$

we have $2\lambda \operatorname{Re}\langle f, Af\rangle \leq \|Af\|^2$. But $\lambda > 0$ is arbitrary, so it follows that $\operatorname{Re}\langle f, Af\rangle \leq 0$.

□

Remark 3.1.7. As a direct consequence of Theorem 3.1.6, it follows that for every λ with $\operatorname{Re}\lambda > 0, Ran(A-\lambda)$ is a closed subspace of H whenever A is a closed, dissipative operator on H. Furthermore, for a dissipative operator $A, (A-\lambda)$ is injective for $\operatorname{Re}\lambda > 0$, but may not be surjective. Suppose that A is a closed, densely defined dissipative operator on a Hilbert space. Let λ be such that $\operatorname{Re}\lambda > 0$ and $f_n \in \mathcal{D}(A)$ be such that $(A-\lambda)f_n$ is a Cauchy sequence in H. From (b) of Theorem 3.1.6 it follows that (f_n) is Cauchy and so there exists $f \in H$ with $f_n \to f$ as $n \to \infty$. But closedness of A implies that $f \in \mathcal{D}(A)$ and $(A-\lambda)f_n \to (A-\lambda)f$ as $n \to \infty$. This implies that for $\operatorname{Re}\lambda > 0$, the $Ran(A-\lambda)$ is a closed subspace of H. Furthermore,

$$\left\|(A-\lambda)^{-1}g\right\| \leq (\operatorname{Re}\lambda)^{-1}\|g\|, \quad \forall g \in Ran(A-\lambda). \tag{3.1}$$

But $Ran(A-\lambda)$ is not necessarily all of H. As an example one can look at the operator A_0 in $L^2[0,1]$ appearing in Example A.1.10. The domain of A_0^* is easily seen to be

$$\mathcal{D}(A_0^*) = \{\, f \in L^2[0,1] : f \text{ absolutely continuous}, f' \in L^2[0,1] \,\},$$

and therefore any vector orthogonal to $Ran(A_0-\lambda)$ for some $\lambda > 0$ will be the solution in $\mathcal{D}(A_0^*)$ of the equation $A_0^* f = \lambda f$. This equation can be seen to be a classical differential equation and has a solution $f(t) = ce^{i\lambda t}$, where c is a constant. This $f \in L^2[0,1]$ and therefore, $Ran(A_0-\lambda) \neq L^2[0,1]$. The verification of these statements is left as an exercise (**Exercise** 3.1.8).

The property of being dissipative is stable under closure. This is made precise in the following lemma.

Lemma 3.1.9. Any densely defined dissipative operator on a Hilbert space is closable. The closure of A is again dissipative. Thus, a maximal dissipative operator which is densely defined is closed.

Proof. Let A be a densely defined dissipative operator on the Hilbert space H with domain $\mathcal{D}(A)$. Let $\{u_n\}_n \subset \mathcal{D}(A)$ be a sequence converging to 0 such that $Au_n \to v$ as $n \to \infty$. For any $u \in \mathcal{D}(A)$ and $\alpha \in \mathbb{C}$, we have

$$\operatorname{Re}\langle u + \alpha u_n, A(u + \alpha u_n)\rangle \leq 0.$$

Letting $n \to \infty$ in the above gives

$$\operatorname{Re}\langle u, Au\rangle + \operatorname{Re}\alpha\langle u, v\rangle \leq 0.$$

Since α is arbitrary, it follows that $\langle u, v\rangle = 0$ for any $u \in \mathcal{D}(A)$. The density of $\mathcal{D}(A)$ implies therefore that $v = 0$. Thus A is closable. Now $\overline{A}$, the closure of A is given by setting

$$\begin{aligned}\mathcal{D}(\overline{A}) = \{u \in H : &\text{ if } \{u_n\} \subset \mathcal{D}(A) \text{ with } \lim_n u_n = u, \text{ then}\\ &\text{there exists } v \in H \text{ such that } \lim_n Au_n = v\},\\ \overline{A}u = v.&\end{aligned}$$

Then $\operatorname{Re}\langle u, \overline{A}u\rangle = \lim_n \operatorname{Re}\langle u_n, Au_n\rangle \leq 0$, for all $u \in \mathcal{D}(\overline{A})$. Thus $\overline{A}$ is also dissipative. The second part of the lemma now follows from the property of maximality.

□

Lemma 3.1.10. If A is a densely defined, dissipative operator on a Hilbert space H, and $Ran(A - \lambda) = H$, for some λ with $\operatorname{Re}\lambda > 0$, then A is maximal dissipative.

Proof. Let A be densely defined and dissipative, with $Ran(A - \lambda) = H$ for some $\operatorname{Re}\lambda > 0$. Let $\tilde{A}$ be a dissipative extension of A, let $u \in \mathcal{D}(\tilde{A})$ and let $v = (\tilde{A} - \lambda)u$. Then there exists $w \in \mathcal{D}(A)$ such that

$$v = (\tilde{A} - \lambda)u = (A - \lambda)w = (\tilde{A} - \lambda)w.$$

On the other hand, $\tilde{A}$ being dissipative implies $(\tilde{A} - \lambda)$ is injective, so that $u = w$. Thus $u \in \mathcal{D}(A)$ and $A = \tilde{A}$. □

Lemma 3.1.11. Every densely defined, dissipative operator on a Hilbert space admits a maximal dissipative extension.

Proof. In view of Lemma 3.1.9 we may assume, without loss of generality, that A is a closed, densely defined dissipative operator. Let λ be such that $\operatorname{Re}\lambda > 0$ and set $\mathcal{N} = (Ran(A - \lambda))^{\perp}$. Then for $v \in \mathcal{N}$ and $u \in \mathcal{D}(A)$,

$$\langle v, Au\rangle = \langle v, (A - \lambda)u\rangle + \overline{\lambda}\langle v, u\rangle.$$

Since $\langle v, (A - \lambda)u\rangle = 0$, it follows that $\big|\langle v, Au\rangle\big| \leq |\lambda|\|v\|\|u\|$. Therefore, $v \in \mathcal{D}(A^*)$ and $A^*v = \overline{\lambda}v$. If $v \in \mathcal{D}(A) \cap \mathcal{N}$, then $\langle v, (A - \lambda)v\rangle = 0$ which implies

that $\operatorname{Re}\lambda\|v\|^2 = \operatorname{Re}\langle v, Av\rangle \leq 0$. Since $\operatorname{Re}\lambda > 0$, it follows that $v = 0$. Thus, $\mathcal{D}(A) \cap \mathcal{N} = \{0\}$. Set

$$\tilde{\mathcal{D}} := \big\{u + v : u \in \mathcal{D}(A), v \in \mathcal{N}\big\},$$
$$\tilde{A}(u + v) := Au - \overline{\lambda}v.$$

Then $\tilde{A}$ is a linear operator with $\mathcal{D}(\tilde{A}) = \tilde{\mathcal{D}}$ and $A \subset \tilde{A}$. We will now show that $\tilde{A}$ is a maximal dissipative extension of A. Let $u \in \mathcal{D}(A)$ and $v \in \mathcal{N}$. Then

$$\begin{aligned}
\operatorname{Re}\langle u + v, \tilde{A}(u + v)\rangle &= \operatorname{Re}\langle u + v, Au - \overline{\lambda}v\rangle \\
&= \operatorname{Re}\langle u, Au\rangle + \operatorname{Re}\langle v, Au\rangle - \operatorname{Re}\lambda\langle u, v\rangle - \operatorname{Re}\lambda\|v\|^2 \\
&= \operatorname{Re}\langle u, Au\rangle + \operatorname{Re}\langle (A - \lambda)u, v\rangle - \operatorname{Re}\lambda\|v\|^2 \\
&= \operatorname{Re}\langle u, Au\rangle - \operatorname{Re}\lambda\|v\|^2 \\
&\leq 0.
\end{aligned}$$

Thus $\tilde{A}$ is a dissipative extension of A. Since A is densely defined, so is $\tilde{A}$. In view of Lemma 3.1.10, it is now sufficient to show that $Ran\ (\tilde{A} - \lambda) = H$. As discussed in Remark 3.1.7, the facts that A is closed, densely defined and dissipative ensures that $Ran\ (A - \lambda)$ is a closed subspace of H. Thus any $x \in H$ is expressible uniquely as $x = w + v$, where $w \in Ran\ (A - \lambda)$ and $v \in \mathcal{N}$. This implies that $x = (A - \lambda)u + v$ for some $u \in \mathcal{D}(A)$. Assuming that $v \neq 0$, we set $y = u + v_1$, where $v_1 = -(\lambda + \overline{\lambda})^{-1}v \in \mathcal{N}$. Then $y \in \mathcal{D}(\tilde{A})$ and

$$(\tilde{A} - \lambda)y = (\tilde{A} - \lambda)(u + v_1) = (A - \lambda)u - (\lambda + \overline{\lambda})v_1 = x.$$

Thus $x \in Ran\ (\tilde{A} - \lambda)$.

□

Lemma 3.1.12. The following three statements are equivalent for a densely defined, dissipative operator A on a Hilbert space H.

(i) A is maximal dissipative.

(ii) $Ran(A - \lambda) = H$ for all λ with $\operatorname{Re}\lambda > 0$.

(iii) $Ran(A - \lambda) = H$ for some λ with $\operatorname{Re}\lambda > 0$.

Proof. (i) ⇒ (ii): Let $\lambda \in \mathbb{C}$ be such that $\operatorname{Re}\lambda > 0$. Assume that $Ran\ (A - \lambda)$ is not the whole of H. Then $Ran(A - \lambda)$ is a proper, closed subspace of H. Set

$\mathcal{N} = (Ran\ (A-\lambda))^{\perp} \neq \{0\}$. Define $\tilde{A}$ as in the proof of Lemma 3.1.11. This $\tilde{A}$ is then a dissipative extension of A, contradicting (i).

(ii) $\Rightarrow$ (iii): This is trivial.

(iii) $\Rightarrow$ (i): This follows from Lemma 3.1.10. □

Theorem 3.1.13. A densely defined operator A on a Hilbert space H is maximal dissipative if and only if it is closed, $\{z \in \mathbb{C} : \operatorname{Re} z > 0\} \subset \rho(A)$ and

$$\left\|(A-z)^{-1}\right\| \leq (\operatorname{Re} z)^{-1} \text{ whenever } \operatorname{Re} z > 0.$$

Proof. Let A be a densely defined maximal dissipative operator. By Theorem 3.1.9, A is closed. Now $A - z$ is injective since A is dissipative and by Theorem 3.1.12, $A - z$ is also surjective for $\operatorname{Re} z > 0$. The claim then follows from Theorem 3.1.6 and (3.1) of Remark 3.1.7. Conversely, suppose that A is a densely defined, closed operator satisfying $\{z \in \mathbb{C} : \operatorname{Re} z > 0\} \subset \rho(A)$ and $\left\|(A-z)^{-1}\right\| \leq (\operatorname{Re} z)^{-1}$ whenever $\operatorname{Re} z > 0$. Again Theorem 3.1.6 shows that A is dissipative and maximal dissipativity follows from Lemma 3.1.10.

□

Theorem 3.1.14. Let the densely defined closed operator A be dissipative. If A is maximal dissipative, then A^* is maximal dissipative. On the other hand, if A^* is dissipative, then A is maximal dissipative.

Proof. Suppose that A is a closed, densely defined maximal dissipative operator. Since A is closed, $\mathcal{D}(A^*)$ is dense. By Theorem 3.1.13, if $\operatorname{Re}\lambda > 0$, then

$$\left\|\left(A^* - \lambda\right)^{-1}\right\| = \left\|\left(A - \overline{\lambda}\right)^{-1}\right\| \leq (\operatorname{Re}\lambda)^{-1}.$$

In addition, we note that A^* is a densely defined operator which is closed and $\{\lambda : \operatorname{Re}\lambda > 0\} \subset \rho(A^*)$. Therefore, by Theorem 3.1.13 it follows that A^* is maximal dissipative. Conversely, suppose A^* is dissipative. Then it follows from Theorem 3.1.6 that $\left\|\left(A^* - 1\right)u\right\| \geq \|u\|$ for all $u \in \mathcal{D}(A^*)$. Suppose, if possible that $Ran\ (A-1) \neq H$. But $Ran\ (A-1)$ is closed, since A is dissipative, by Remark 3.1.7. Set $N = Ran\ (A-1)^{\perp}$ and let $v \in N$. Then, for $u \in \mathcal{D}(A)$, $\langle v, (A-1)u\rangle = 0$. This implies that $v \in \mathcal{D}(A^*)$ and $\langle (A^*-1)v, u\rangle = 0$. This is true for all $u \in \mathcal{D}(A)$. The density of $\mathcal{D}(A)$ implies that $\left(A^* - 1\right)v = 0$ so that $v = 0$. Thus $N = \{0\}$. So, $Ran\ (A-1) = H$. The maximal dissipativity of A now follows from Theorem 3.1.12.

□

Till now we have dealt with dissipative operators on Hilbert spaces. It is possible to extend these ideas to Banach spaces as well. Let X be a Banach space and recall that, as a consequence of the Hahn Banach Theorem, for any $x \in X$, there exists an $f^x \in X^*$ such that $f^x(x) = \|x\|$, and $\|f^x\| = 1$. This functional f^x is not necessarily unique. Set

$$f_x = \|x\| f^x.$$

Then $f_x(x) = \|x\|^2 = \|f_x\|^2$. This association $x \mapsto f_x$ from $X \to X^*$ such that

$$f_x(x) = \|x\|^2 = \|f_x\|^2$$

is called a **dual injection** (sometimes also called a normalised tangent functional) of X in X^*. Using this association, we define dissipative operators in this general setting:

Definition 3.1.15. An operator A on a Banach space X is said to be **dissipative** if for every $x \in \mathcal{D}(A)$ there exists a dual injection f_x of x such that $\operatorname{Re} f_x(Ax) \leq 0$. A is maximal dissipative if it does not admit any proper dissipative extensions.

Here we note that if X is a Hilbert space then $X^* = X$ and $f_x = x$ in this identification, so the dual injection is canonical and thus this general definition gives back the one in Definition 3.1.1. The following theorem proves properties in Banach spaces, similar to those in Hilbert spaces as proven earlier in Theorem 3.1.6.

Theorem 3.1.16. The following are equivalent for an operator A on a Banach space X.

(i) A is dissipative;

(ii) $\|(A - \lambda)u\| \geq \operatorname{Re}\lambda \|u\|$ for all $u \in \mathcal{D}(A)$ and $\operatorname{Re}\lambda > 0$;

(iii) $\|(A - \lambda)u\| \geq \lambda \|u\|$ for all $u \in \mathcal{D}(A)$ and $\lambda > 0$.

Proof. For $u, v \in X$, with $\|u\| \leq \|u - \alpha v\|$ for all $\alpha > 0$, we claim that there exists $f_u \in X^*$ such that $\operatorname{Re} f_u(v) \leq 0$. Indeed, for $u = 0$, choose $f_u = 0$. If $u \neq 0$, pick the dual injection $f_{u-\alpha v}$ and denote it as f_α for brevity. Set $g_\alpha = f_\alpha / \|f_\alpha\|$. Then $\|g_\alpha\| = 1$ and by the w^*-compactness of the unit ball in

X^*, the net $\{g_\alpha\}_{\alpha>0}$ will have a convergent subnet, converging to g as $\alpha \to 0^+$ in w^*-topology with $\|g\| \leq 1$. Thus,

$$\begin{aligned}\|u\| \leq \|u - \alpha v\| &= \frac{\|u-\alpha v\|^2}{\|f_\alpha\|} = \frac{f_\alpha(u-\alpha v)}{\|f_\alpha\|} \\ &= g_\alpha(u-\alpha v) = \operatorname{Re} g_\alpha(u) - \alpha \operatorname{Re} g_\alpha(v) \\ &\leq \|u\| - \alpha \operatorname{Re} g_\alpha(v), \end{aligned} \tag{3.2}$$

which implies that $\operatorname{Re} g_\alpha(v) \leq 0, \forall\ \alpha > 0$ and thus $\operatorname{Re} g(v) \leq 0$. On the other hand, taking the limit as $\alpha \to 0^+$ in (3.2) we get that $\operatorname{Re} g(u) \geq \|u\|$ which combined with the fact that $\operatorname{Re} g(u) \leq |g(u)| \leq \|u\|$ leads to the conclusion that $\operatorname{Re} g(u) = \|u\|$. Set $f = \|u\| g \in X^*$. Then $f(u) = \|u\| g(u) = \|u\|^2 = \|f\|^2$. Thus $f = f_u$ and $\operatorname{Re} f_u(v) = \|u\| \operatorname{Re} g(v) \leq 0$, thereby establishing the claim.

(iii) $\Rightarrow$ (i): Suppose $\|(A-\lambda)u\| \geq \lambda \|u\|$ for all $\lambda > 0$ and $u \in \mathcal{D}(A)$. Then

$$\left\|u - \lambda^{-1} Au\right\| = \lambda^{-1} \|(A-\lambda)u\| \geq \|u\|$$

or $\|u\| \leq \|u - \alpha Au\|$ for all $\alpha > 0$. From the first part of the proof, this implies that there exists $f_u \in X^*$ such that $\operatorname{Re} f_u(Au) \leq 0$. Thus, A is dissipative.

(i) $\Rightarrow$ (ii): Let $u \in \mathcal{D}(A)$ and $\operatorname{Re} \lambda > 0$. Since A is dissipative, there is a dual injection f_u such that $\operatorname{Re} f_u(Au) \leq 0$. Since $\operatorname{Re} f_u((A-\lambda)u) = \operatorname{Re} f_u(Au) - \operatorname{Re} \lambda f_u(u) \leq -\operatorname{Re} \lambda \|u\|^2$, we have that

$$\begin{aligned}\|(A-\lambda)u\| \|u\| = \|(A-\lambda)u\| \left\|f_u\right\| &\geq \left|f_u((A-\lambda)u)\right| \\ &\geq \left|\operatorname{Re} f_u(A-\lambda)u)\right| \geq \operatorname{Re} \lambda \|u\|^2. \end{aligned}$$

(ii) $\Rightarrow$ (iii): This is obvious. □

As we have seen in the setting of a Hilbert space (Lemmas 3.1.11, 3.1.12 and Theorems 3.1.13 and 3.1.14) the density of the domain of a dissipative operator is necessary to get any complete description of it. The same is true in a Banach space as the next Theorem 3.1.18 will show. Before we prove the theorem, we give an example of an operator that is dissipative, but not densely defined.

Example 3.1.17. Let $X = C[0,1]$ and consider the operator

$$Af = -f', \quad f \in \mathcal{D}(A) = \left\{g \in C^1[0,1] : g(0) = 0\right\}$$

Then A is a closed, dissipative operator on a Banach space which is not densely defined, and $(\lambda - A)\mathcal{D}(A) = X$, for all $\lambda > 0$. For if $\{f_n\} \subset \mathcal{D}(A)$ is a sequence converging to some $f \in X$ and $\{Af_n\}$ converges to g, then this simply means that $\{f_n\}$ is a sequence of functions converging uniformly to f, with $\{f_n'\}$ converging uniformly to g. This implies that f is also differentiable with $f' = g$ and $f(0) = 0$. Thus $f \in \mathcal{D}(A)$ and $Af = g$, showing that A is closed. Clearly, $\overline{\mathcal{D}(A)} \neq X$, as the constant function $\mathbf{1} \in X$ cannot be approximated by any sequence in $\mathcal{D}(A)$. We now establish dissipativity of the operator A.

Let $f \in \mathcal{D}(A)$. For $\lambda > 0$ and $g \in X$,

$$(\lambda - A)f = \lambda f + f' = g.$$

defines a classical linear ordinary differential equation. Solving the above differential equation yields, for $t \in [0, 1]$,

$$f(t) = \int_0^t e^{-\lambda(t-s)} g(s)\, ds. \tag{3.3}$$

Therefore,

$$|f(t)| \leq \|g\| e^{-\lambda t} \int_0^t e^{s\lambda}\, ds \leq \|g\| e^{-\lambda t} \Big(\frac{e^{t\lambda} - 1}{\lambda} \Big),$$

where $\|g\| = \sup_{t\in[0,1]} |g(t)|$.

Thus, $\|f\| \leq \|g\|/\lambda$, or $\|(\lambda - A)f\| \geq \lambda \|f\|$. Since this is true for all $\lambda > 0$ and $f \in \mathcal{D}(A)$, it follows from Theorem 3.1.16 that A is dissipative. Moreover, for every $g \in C[0, 1]$, the function f given by (3.3) is in $\mathcal{D}(A)$ and satisfies $\lambda f + f' = g$, that is, $(\lambda - A)f = g$. Therefore, $Ran(\lambda - A) = X$.

We are now in a position to prove the Lumer-Phillips Theorem [16] for any Banach space.

Theorem 3.1.18 (Lumer-Phillips)**.** Let A be a densely defined operator on a Banach space X. Then A is the generator of a C_0-semigroup of contractions if and only if A is dissipative and $Ran(A - 1) = X$.

Proof. Suppose A is dissipative and $Ran(A - 1) = X$. Then, by Theorem 3.1.16(ii), $1 \in \rho(A)$, $\big\|(A - 1)^{-1}\big\| \leq 1$ and A is a closed operator (see Remark 3.1.19). If we show that $\lambda \in \rho(A)$ for every $\lambda > 0$, and $\big\|(A - \lambda)^{-1}\big\| \leq \lambda^{-1}$, then by the Hille-Yosida Theorem 2.3.3 it will follow that A is the generator of

a C_0 contraction semigroup. Let $0 < \lambda < 2$ and set

$$R' = (A-1)^{-1}\big[1-(\lambda-1)(A-1)^{-1}\big]^{-1} \in \mathcal{B}(X),$$

where we have noted that the inverse of the term in the square parenthesis exists in $\mathcal{B}(X)$ as the limit of a convergent Neumann series, since $|\lambda - 1| < 1$. Then $Ran\ R' = \mathcal{D}(A)$ and

$$(A-\lambda)R' = (A-1-(\lambda-1))(A-1)^{-1}\big[1-(\lambda-1)(A-1)^{-1}\big]^{-1} = I.$$

On the other hand, for $u \in \mathcal{D}(A)$,

$$\begin{aligned} &R'(A-\lambda)u \\ &= (A-1)^{-1}\big[1-(\lambda-1)(A-1)^{-1}\big]^{-1}\big[1-(\lambda-1)(A-1)^{-1}\big](A-1)u \\ &= u. \end{aligned}$$

This implies that $(A-\lambda)$ is bijective and $(A-\lambda)^{-1} = R'$, so that $\lambda \in \rho(A)$, and by Theorem 3.1.16(iii),

$$\big\|(A-\lambda)^{-1}\big\| \leq \lambda^{-1}. \tag{3.4}$$

Thus, $(0,2) \subset \rho(A)$ and every $\lambda \in (0,2)$ satisfies (3.4). Now fix $\lambda_0 \in (1,2)$ and let $\mu > 0$ with $|\lambda_0 - \mu| < \|(A-\lambda_0)^{-1}\|^{-1}$. Proceeding as before, but with 1 replaced by λ_0 we obtain that $\mu \in \rho(A)$ and $\|(A-\mu)^{-1}\| \leq \mu^{-1}$. Continuing in this way, we can cover all of $(0,\infty)$, that is, we obtain $(0,\infty) \subset \rho(A)$ and that (3.4) holds for all $\lambda \in (0,\infty)$. Note that since $\|(A-\lambda_0)^{-1}\|^{-1} \geq \lambda_0 > 1$ initially, all of $(0,\infty)$ can indeed be covered in the above mentioned manner.

Conversely, suppose $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup of contractions with generator A. Let $x \in X$ and let f_x be a dual injection. Then

$$f_x((T(t)-I)x) = f_x(T(t)x) - f_x(x) = f_x(T(t)x) - \|x\|^2.$$

Therefore, since T is a contraction,

$$\mathrm{Re}\, f_x((T(t)-I)x) = \mathrm{Re}\ \ f_x(T(t)x) - \|x\|^2 \leq \|f_x\|\|T(t)x\| - \|x\|^2 \leq 0.$$

If $x \in \mathcal{D}(A), \lim_{t\to 0^+}(T(t)x - x)/t = Ax$. It follows therefore, that for any $x \in \mathcal{D}(A)$, $\mathrm{Re}\, f_x(Ax) \leq 0$. Finally, by Theorem 2.3.1, every $z \in \mathbb{C}$ with $\mathrm{Re}\, z > 0$ is in $\rho(A)$ so that $1 \in \rho(A)$ and $Ran(A-1) = X$.

□

Remark 3.1.19. Note that the statement of Theorem 3.1.18 does not mention explicitly that A is a closed operator. However, it must be so in order to be the generator of a C_0-semigroup. And indeed the conditions of dissipativity and surjectivity of $(A-1)$ imply closedness of A. Let $\{u_n\} \subset \mathcal{D}(A)$ converge to $u \in X$ and suppose that $Au_n \to v$ as $n \to \infty$. Then $(A-1)u_n \to v-u$ as $n \to \infty$. But since A is dissipative, it follows from Theorem 3.1.16 that $\|(A-I)u\| \geq \|u\|$ for all $u \in \mathcal{D}(A)$. This, along with the hypothesis $Ran\ (A-1) = X$ implies that $1 \in \rho(A)$, and $(A-1)^{-1} \in \mathcal{B}(X)$. Therefore, $u_n \to (A-1)^{-1}(v-u)$ as $n \to \infty$, or, $u = (A-1)^{-1}(v-u)$ leading to the conclusion that $u \in \mathcal{D}(A)$ and $Au = v$.

Corollary 3.1.20. Let A be a densely defined closed linear operator in X, such that both A and A^* are dissipative. Then A is the generator of a C_0 contraction semigroup.

Proof. In view of Theorem 3.1.18 we just need to prove that $Ran(A-1) = X$. Since A is closed and dissipative, it follows that $Ran(A-1)$ is a closed subspace of X. If $Ran(A-1) \neq X$, then there exists $x^* \neq 0$ in X^* such that $\langle x^*, (A-I)x\rangle = 0$, for all $x \in \mathcal{D}(A)$. This implies that $x^* \in \mathcal{D}(A^*)$ and $\langle A^*x^*, x\rangle = \langle x^*, x\rangle$ for all $x \in \mathcal{D}(A)$. Density of $\mathcal{D}(A)$ implies that $A^*x^* = x^*$. Then

$$f_{x^*}(A^*x^*) = \|x^*\|^2 = \operatorname{Re} f_{x^*}(A^*x^*) \geq 0,$$

contradicting the dissipativity of A^*. □

Example 3.1.21. We revisit Example 3.1.3. As we saw earlier, the operator $-A$ is dissipative. We now show that the range condition of Theorem 3.1.18 also holds. For $f \in L^2(0,1)$ and $\lambda > 0$, set

$$u(t) = \int_0^t e^{-\lambda(t-s)} f(s)\, ds.$$

Then $u \in \mathcal{D}(A)$ and $\lambda u + u' = f$. Hence $(\lambda + A)\mathcal{D}(A) = X$ for all $\lambda > 0$. Also, $\mathcal{D}(A)$ is dense in $L^2(0,1)$. Thus, $-A$ generates a contraction C_0-semigroup on X.

Example 3.1.22. Let $X = L^2(\mathbb{R}^d)$, and set

$$\mathcal{D}(A) = \left\{f \in L^2(\mathbb{R}^d) : \Delta f \in L^2(\mathbb{R}^d)\right\} = H^2(\mathbb{R}^d),$$
$$Af = \Delta f,$$

where Δ is the Laplacian, defined in Appendix A.3. We shall show that A is a densely defined, dissipative operator with $(1-A)\mathcal{D}(A) = X$ and hence conclude from Theorem 3.1.18 that A generates a C_0 contraction semigroup. Recall from Appendix A.3 that $\Delta f = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_j^2}$, and that for $f \in H^2(\mathbb{R}^d) \subset H^1(\mathbb{R}^d)$,

$$\langle \Delta f, f \rangle_{L^2(\mathbb{R}^d)} = \int_{\mathbb{R}^d} (\Delta f)\overline{f}\, dx = -\int_{\mathbb{R}^d} |\nabla f|^2\, dx \leq 0.$$

Thus A is dissipative. Since $C_c^\infty(\mathbb{R}^d) \subset H^2(\mathbb{R}^d)$, A is densely defined. We prove next that $Ran(1-A) = X$. Let

$$\phi(k) = \frac{1}{|k|^2+1} \text{ and } \psi(k) = |k|^2 \text{ for all } k \in \mathbb{R}^d.$$

From Appendix A.2, one finds that $(\mathcal{F}(\Delta f))(k) = -\psi(k)(\mathcal{F}f)(k)$ and thus, if for any $g \in L^2(\mathbb{R}^d)$, we set $u = \mathcal{F}^{-1}(M_\phi \mathcal{F} g)$, it follows from Lemma A.3.1 that $u \in H^2(\mathbb{R}^d) \subset L^2(\mathbb{R}^d)$. Furthermore, $M_\psi \mathcal{F} u = M_\psi M_\phi \mathcal{F} f$. Therefore,

$$\begin{aligned}(1-\Delta)u &= \mathcal{F}^{-1}(M_\phi \mathcal{F} g) + \mathcal{F}^{-1}(M_\psi \mathcal{F} u) = \mathcal{F}^{-1}(M_\phi \mathcal{F} g) + \mathcal{F}^{-1}(M_\psi M_\phi \mathcal{F} g) \\ &= \mathcal{F}^{-1}((I + M_\psi) M_\phi \mathcal{F} g) = g.\end{aligned}$$

Thus $g \in Ran\ (1-A)$ and hence $X = Ran\ (1-A)$. Therefore, by Theorem 3.1.18 it follows that A generates a C_0-semigroup of contractions.

3.2 Stone's Theorem

In this section, we obtain a classical theorem characterising the generator of a unitary C_0-group on a Hilbert space, as an application of Theorem 3.1.18. By a unitary C_0-group on a Hilbert space H, we mean a family $\{U(t)\}_{t\in\mathbb{R}}$ of unitary operators on H (that is, bounded operators on H satisfying $U(t)^* = U(t)^{-1}\ \forall\ t \in \mathbb{R}$) which forms a C_0-group (see Definition 2.2.9). The group property implies, in particular, that $U(t)^{-1} = U(-t)$.

Theorem 3.2.1. *A is the generator of a C_0 unitary group $\{U(t)\}_{t\in\mathbb{R}}$ in a Hilbert space H if and only if iA is selfadjoint.*

Proof. Suppose first that A generates a C_0-semigroup of unitaries, $\{U(t)\}_{t\in\mathbb{R}}$. Then, setting

$$U_+(t) := U(t) \text{ and } U_-(t) := U(-t) \text{ for all } t \geq 0,$$

we see that $\{U_+(t)\}_{t\geq 0}$ is a C_0-semigroup of contractions, generated by A, while $\{U_-(t)\}_{t\geq 0}$ is a C_0-semigroup of contractions with generator $-A$. Furthermore, since

$$t^{-1}\big[U(t)^*x - x\big] = t^{-1}\big[U(-t)x - x\big], \tag{3.5}$$

it follows, from Theorem 2.4.1 and the above discussion, on letting $t \to 0$ that $x \in \mathcal{D}(A^*)$ if and only if $x \in \mathcal{D}(A)$, and in that case, $A^*x = -Ax$. Thus, $(iA)^* = iA$, that is, iA is selfadjoint.

Conversely, assume that iA is selfadjoint, so that $\mathcal{D}(A)^* = \mathcal{D}(A)$, and $A^*x = -Ax$ for all $x \in \mathcal{D}(A)$. Then

$$\langle Ax, x\rangle = \langle x, A^*x\rangle = -\langle x, Ax\rangle = -\overline{\langle Ax, x\rangle}, \quad \forall x \in \mathcal{D}(A).$$

Thus $\operatorname{Re}\langle Ax, x\rangle = 0$ for all $x \in \mathcal{D}(A)$, so that both A and $-A$ are dissipative. Note that since A^* is closed, being the generator of the C_0-semigroup U_-, A is densely defined, that is, $\overline{\mathcal{D}(A)} = H$. Furthermore, A is closed since every adjoint operator is necessarily closed (see Appendix A.1). By Theorem 3.1.18, A generates a C_0-semigroup, provided $Ran(A-1) = H$. We establish this fact next. From Remark 3.1.7 we have that $(A-1)$ is injective, and that $Ran(A-1)$ is closed. Now suppose that $g \perp Ran(A-1)$. Then

$$\langle g, (A-1)f\rangle = 0 \quad \forall f \in \mathcal{D}(A).$$

Thus, $g \in \mathcal{D}((A-1)^*) = \mathcal{D}(A^*) = \mathcal{D}(A)$, and

$$-Ag - g = A^*g - g = (A-1)^*g = 0.$$

Therefore,

$$\operatorname{Re}\langle Ag, g\rangle = \langle Ag, g\rangle = -\|g\|^2. \tag{3.6}$$

Moreover,

$$\langle Ag, g\rangle = -\langle A^*g, g\rangle = -\langle g, Ag\rangle = -\overline{\langle Ag, g\rangle}.$$

Thus, $\operatorname{Re}\langle Ag, g\rangle = 0$, so that using (3.6), we have that $g = 0$. Thus, by Theorem 3.1.18 it follows that A generates a C_0-semigroup of contractions, say, $\{U_+(t)\}_{t\geq 0}$. Starting with $-A$ instead of A and using arguments similar to above, we similarly conclude that $-A$ also generates a C_0-semigroup of contractions,$\{U_-(t)\}_{t\geq 0}$. Since iA is selfadjoint, using the functional calculus

associated with selfadjoint operators (see [20, Theorem VIII.5, page 262]), we can write

$$U_+(t) = e^{tA} = e^{-it(iA)}, U_-(t) = e^{-tA} = e^{it(iA)}.$$

Define the family $\{U(t)\}_{t\in\mathbb{R}}$ by

$$U(t) = \begin{cases} U_+(t) & \text{if } t > 0 \\ I & \text{if } t = 0 \\ U_-(t) & \text{if } t < 0. \end{cases} \tag{3.7}$$

Using the above mentioned functional calculus one can check that the family $\{U(t)\}_{t\in\mathbb{R}}$ forms a group. Further, since U_+ and U_- are strongly continuous and

$$\begin{aligned} \lim_{t\to 0^+} U(t)x &= \lim_{t\to 0^+} (U_+(t))x = x = \lim_{s\to 0^+} (U_-(s))x \\ &= \lim_{s\to 0^+} U(-s)x = \lim_{t\to 0^-} U(t)x, \end{aligned}$$

this family is strongly continuous on $\mathbb{R}$. Moreover,

$$\lim_{t\to 0} t^{-1}(U(t)x - x) = Ax, \quad \forall x \in \mathcal{D}(A).$$

Thus $\{U(t)\}_{t\in\mathbb{R}}$ forms a C_0-group with generator A. Since $U(t)U(-t) = I = U(-t)U(t)$ it follows that $U(t)^{-1} = U(-t)$ for all $t \in \mathbb{R}$. This implies that $U(t)$ is a unitary operator for each t. □

3.3 Holomorphic semigroups

Holomorphic semigroups play a very important role in the study of partial differential equations. They are C_0-semigroups with several special properties and occur abundantly in applications. In fact, nearly all the semigroups that we have seen thus far are holomorphic. Before introducing a formal definition, we need to introduce the notion of holomorphy for vector-valued functions.

Definition 3.3.1. Let Ω be an open connected set in $\mathbb{C}$, and let $f\colon \Omega \to X$, where X is a Banach space.

1. f is said to be **weakly holomorphic** in Ω if the map $\Omega \ni z \mapsto \langle x^*, f(z)\rangle$ is holomorphic for every $x^* \in X^*$.

2. f is said to be **strongly holomorphic** if the map $\Omega \ni z \mapsto f(z) \in X$ is holomorphic in the strong (norm) topology of X.

It turns out that these two notions of holomorphy are actually equivalent, by an application of Cauchy's Theorem and of the Uniform Boundedness Principle (**Exercise** 3.3.2). For the theory of vector-valued functions of a complex variable, in particular the above mentioned property, the reader is referred to [2, Appendix A].

We denote, for $0 < \theta \leq \pi$, by S_θ, the sector

$$S_\theta := \big\{z \in \mathbb{C} \setminus \{0\} : |\arg z| < \theta\big\}.$$

Definition 3.3.3. A **holomorphic (or analytic) semigroup** of angle θ, where $0 < \theta \leq \pi/2$, defined on a Banach space X, is a family $\{T(z)\}_{z \in S_\theta \cup \{0\}} \subset \mathcal{B}(X)$ satisfying

(1) $T(0) = I$;

(2) the map $z \mapsto T(z)$ is holomorphic in S_θ;

(3) $T(z_1)T(z_2) = T(z_1 + z_2)$ for all $z_1, z_2 \in S_\theta$;

(4) $\lim_{S_{\theta-\epsilon} \ni z \to 0} T(z)x = x$, for all $x \in X$, whenever $0 < \epsilon < \theta$.
If in addition,

(5) $\|T(z)\| \leq M_\epsilon$ for all $z \in S_{\theta-\epsilon}$ and for all ϵ such that $\theta > \epsilon > 0$,
then $\{T(z)\}_{z \in S_\theta \cup \{0\}}$ is called a **bounded holomorphic semigroup.**

It is clear that if $\{T(z)\}_{z \in S_\theta \cup \{0\}}$ is a holomorphic semigroup on X, then $\{T(t)\}_{t \geq 0}$ is a C_0-semigroup on X. The generator A of this C_0-semigroup is referred to as the **generator of the holomorphic semigroup** $\{T(z)\}_{z \in S_\theta \cup \{0\}}$. On the other hand, a given C_0-semigroup may not be extendable as a holomorphic semigroup, as for example, a unitary group with its selfadjoint generator having all of $\mathbb{R}$ as its spectrum.

The special properties of holomorphic semigroups are reflected in the generator of such a semigroup in several ways. Of these, perhaps the most important ones are the shape of the resolvent set and the norm estimates of the resolvent of the generator.

Theorem 3.3.4. Let $\{T(z)\}_{z \in S_\alpha \cup \{0\}}$ be a bounded holomorphic semigroup of angle α, with generator A. Then $\{w \in \mathbb{C} : |\arg w| < \frac{\pi}{2} + \alpha\} \subset \rho(A)$ and for

each $\epsilon \in (0, \alpha)$, there is a constant M_ϵ, such that

$$\big\|R(w, A)\big\| \leq \frac{M_\epsilon}{|w|}, \quad \forall\ w \in S_{\frac{\pi}{2}+\alpha-\epsilon}.$$

Proof. Choose ϵ such that $0 < \epsilon < \alpha$. Fix $|\theta| \leq \alpha - \epsilon$ and set $S(t) = T(te^{i\theta})$ for all $t \geq 0$. Then, as $\{T(z)\}_{z \in S_\theta \cup \{0\}}$ satisfies the semigroup property,

$$S(t_1)S(t_2) = T(t_1 e^{i\theta})T(t_2 e^{i\theta}) = T((t_1 + t_2)e^{i\theta}) = S(t_1 + t_2).$$

Further, $S(0) = T(0) = I$ and $S(t)x = T(te^{i\theta})x \to x$ as $t \to 0^+$. Therefore $(S(t))_{t\geq 0}$ is a C_0-semigroup satisfying $\|S(t)\| = \|T(te^{i\theta})\| \leq M_\epsilon$, where M_ϵ is as in (5) of Definition 3.3.3. Let B be the generator of $\{S(t)\}_{t\geq 0}$. Then for $x \in \mathcal{D}(B)$ and $s > 0$, we have

$$\begin{aligned} S(s)Bx &= \lim_{t\to 0^+} S(s)t^{-1}(S(t)x - x) = \lim_{t\to 0^+} t^{-1}(T((t+s)e^{i\theta})x - T(se^{i\theta})x \\ &= e^{i\theta} \lim_{t\to 0^+} t^{-1}(T(t + se^{i\theta})x - T(se^{i\theta})x) \\ &= e^{i\theta} \lim_{t\to 0^+} t^{-1}(T(t) - I)S(s)x = e^{i\theta} AS(s)x. \end{aligned}$$

This implies that $S(s)\mathcal{D}(B) \subset \mathcal{D}(A)$ for any $s > 0$. As $s \to 0$, the left hand side of the above equation approaches Bx and, as A is closed, it follows that $x \in D(A)$. In a similar manner, one can show that if $x \in \mathcal{D}(A)$, then $\lim_{s\to 0^+} s^{-1}(S(s)x - x)$ exists and equals $e^{i\theta}Ax$, so that

$$Bx = e^{i\theta}Ax \ \ \forall x \in \mathcal{D}(B) = \mathcal{D}(A). \tag{3.8}$$

Since $\|S(s)\| \leq M_\epsilon$, it follows from Theorem 2.2.7(e) and (f) that

$$\rho(B) \supset \{w : \operatorname{Re} w > 0\} \text{ and for such } w, \big\|R(w, B)\big\| \leq \frac{M_\epsilon}{\operatorname{Re}\ w}.$$

Now, $w \in \rho(A)$ if and only if $w' = we^{i\theta} \in \rho(B)$, and in such a case

$$\big\|(R(w, A)\big\| = \big\|R(w', B)\big\| \leq \frac{M_\epsilon}{\operatorname{Re} w'} = \frac{M_\epsilon}{(\operatorname{Re} we^{i\theta})}.$$

Therefore,

$$\{e^{-i\theta}\lambda : \operatorname{Re}\ \lambda > 0\} \subset \rho(A) \quad \text{and} \quad \|(\lambda e^{i\theta} - A)^{-1}\| \leq \frac{M_\epsilon}{\operatorname{Re}\ \lambda}, \ \text{ or}$$

$$\left\{re^{i(\psi-\theta)} : -\frac{\pi}{2} < \psi < \frac{\pi}{2}\right\} \subset \rho(A) \quad \text{and} \quad \|(re^{i(\psi-\theta)} - A)^{-1}\| \leq \frac{M_\epsilon}{r\cos\psi}, \ \text{ or}$$

$$\left\{w = re^{i\gamma} : -\frac{\pi}{2} < \gamma + \theta < \frac{\pi}{2}\right\} \subset \rho(A) \quad \text{and} \quad \|(w - A)^{-1}\| \leq \frac{M_\epsilon}{|w|\cos(\gamma + \theta)}.$$

On the other hand, since $|\theta| \leq \alpha - \epsilon$, with $0 < \alpha \leq \pi/2$, if $-\frac{\pi}{2} < \gamma + \theta < \frac{\pi}{2}$, then

$$-\frac{\pi}{2} - (\alpha - \epsilon) < \gamma + \theta < \frac{\pi}{2} + (\alpha - \epsilon)$$

so that $\cos(\arg w + \theta) \geq \sin(\alpha - \epsilon)$. Thus, it follows that

$$\left\{w : |\arg w| < \frac{\pi}{2} + (\alpha - \epsilon)\right\} \subset \rho(A)$$

and, for all such w,

$$\|(w - A)^{-1}\| \leq \frac{M_\epsilon}{|w| \sin(\alpha - \epsilon)} = \frac{M'_\epsilon}{|w|}.$$

□

The conditions that the generator A in Theorem 3.3.4 above satisfies makes it a sectorial operator of angle α. Precisely, we define sectorial operators as follows.

Definition 3.3.5. A closed linear operator A with domain $\mathcal{D}(A)$ in a Banach space X is called **sectorial** of angle δ where $0 < \delta \leq \frac{\pi}{2}$, if the following conditions hold.

1. The sector $S_{\frac{\pi}{2}+\delta}$ is contained in the resolvent set $\rho(A)$.
2. For each $\epsilon \in (0, \delta)$ there exists $M_\epsilon \geq 1$ such that

$$\|R(\lambda, A)\| \leq \frac{M_\epsilon}{|\lambda|} \quad \forall \lambda \in \overline{S}_{\frac{\pi}{2}+\delta-\epsilon} \setminus \{0\}.$$

The next theorem shows that every sectorial operator of angle less than $\pi/2$ is the generator of a bounded holomorphic semigroup.

Theorem 3.3.6. Let A be a closed operator on X with $S_{\frac{\pi}{2}+\alpha} \subset \rho(A)$, where $0 < \alpha \leq \pi/2$, and such that for all $\epsilon > 0$, there exists $N_\epsilon > 0$ satisfying

$$\|R(w, A)\| \leq \frac{N_\epsilon}{|w|} \text{ for all } w \in S_{\frac{\pi}{2}+\alpha-\epsilon}.$$

Then the integral

$$T(z) := \frac{1}{2\pi i} \int_\Gamma e^{wz} R(w, A)\, dw \tag{3.9}$$

converges for a suitable smooth curve Γ in $S_{\alpha+\frac{\pi}{2}}$, for every $z \in S_\alpha$. Moreover, the family $\{T(z)\}_{S_\alpha \cup \{0\}}$ with $T(0) = I$, satisfies (2), (3), (4) and (5) of Definition 3.3.3, thus forming a bounded holomorphic semigroup, with generator A.

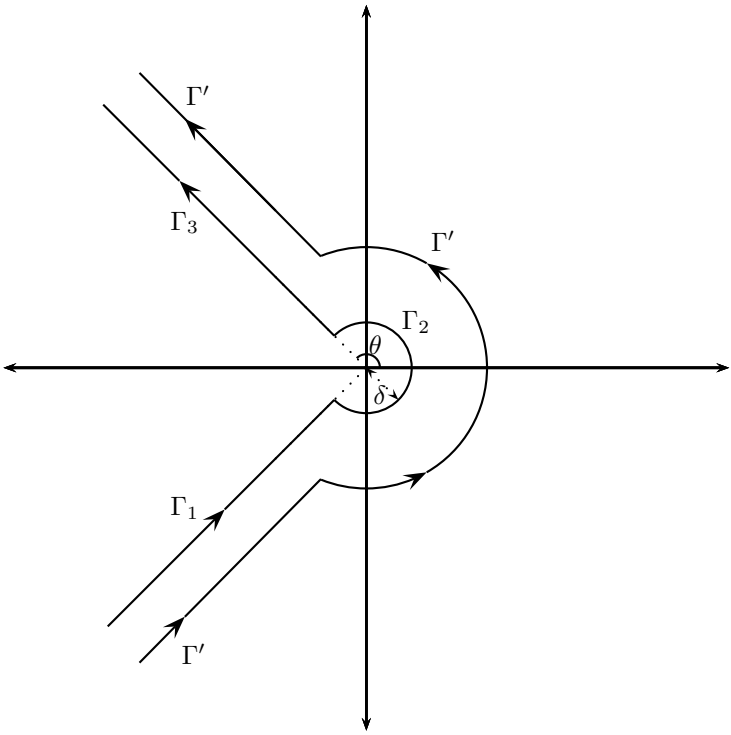


Figure 3.1: Contours Γ and Γ'

Proof. We first note that since the map $w \mapsto e^{wz}R(w, A)$ is holomorphic for $w \in S_{\alpha+\frac{\pi}{2}}$, the integral in (3.9), if it exists, is independent of the choice of Γ due to Cauchy's Integral Theorem. We of course need to verify that the contribution to the difference of the two integrals vanish in the limit as the appropriate radius parameter increases to infinity (**Exercise** 3.3.7). We choose Γ as follows.

Let ϵ be a suitably small positive number. Choose $\theta = \frac{\pi}{2} + \alpha - \epsilon/2$. Let $\Gamma = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3$ oriented anti-clockwise (see Figure 3.1) where

$$\Gamma_1 = \left\{re^{-i\theta} : \delta < r < \infty\right\}$$
$$\Gamma_2 = \left\{\delta e^{i\beta} : -\theta < \beta < \theta\right\}$$
$$\Gamma_3 = \left\{re^{i\theta} : \delta < r < \infty\right\}$$

and δ is a small positive number chosen so that 0 does not lie on Γ.

We first prove that the integral in (3.9) converges uniformly in $B(X)$ for $z \in S_{\alpha-\epsilon}$. Using the hypothesis on the resolvent of A, we have for $z = re^{i\psi}$ where $|\psi| < \alpha - \epsilon$,

$$\left\|e^{wz}R(w,A)\right\| \leq \frac{e^{\operatorname{Re}(wz)}N_\epsilon}{|w|}. \tag{3.10}$$

For $w = re^{i\theta} \in \Gamma_3$ and $z = |z|e^{i\psi}$, $\operatorname{Re}(wz) = r|z|\cos(\theta+\psi)$. Since $|\psi| < \alpha - \epsilon$, it follows that

$$\frac{\pi}{2} + \alpha - \epsilon/2 - (\alpha - \epsilon) < \psi + \theta < \frac{\pi}{2} + \alpha - \epsilon/2 + \alpha - \epsilon;$$

and since $0 < \alpha - \epsilon < \alpha < \frac{\pi}{2}$ we have that $\frac{\pi}{2} + \epsilon/2 < \psi + \theta < 3\pi/2 - \epsilon/2$. Therefore, $\cos(\theta+\psi) \leq -\sin\epsilon/2$ and

$$\left\|e^{wz}R(w,A)\right\| \leq \frac{e^{-|w||z|\sin(\epsilon/2)}N_\epsilon}{|w|} \tag{3.11}$$

for all $z \in S_{\alpha-\epsilon}$ and $w \in \Gamma_3$. Similarly, it can be shown that (3.11) is true for all $z \in S_{\alpha-\epsilon}$ and $w \in \Gamma_1$. For $w \in \Gamma_2$ with $\delta = |z|^{-1}$ and $z \in S_{\alpha-\epsilon}$,

$$\left\|e^{wz}R(w,A)\right\| \leq e\,\frac{N_\epsilon}{|w|} = e\,N_\epsilon|z|. \tag{3.12}$$

Therefore, for all $z \in S_{\alpha-\epsilon}$,

$$\begin{aligned}
&\left\|\frac{1}{2\pi i}\int_\Gamma e^{wz}(w-A)^{-1}\,dw\right\| = \frac{1}{2\pi}\sum_{k=1}^{3}\left\|\int_{\Gamma_k} e^{wz}(w-A)^{-1}\,dw\right\| \\
&\leq \frac{1}{2\pi}\Big(\int_{\Gamma_1}+\int_{\Gamma_3}\Big)\frac{e^{-|w||z|\sin(\epsilon/2)}N_\epsilon}{|w|}\,d|w| + \frac{1}{2\pi}\int_{\Gamma_2} eN_\epsilon|z|\,d|w| \\
&\leq \frac{N_\epsilon}{\pi}\int_{1/|z|}^{\infty}\frac{1}{r}e^{-r|z|\sin(\epsilon/2)}\,dr + \frac{1}{2\pi}e\,N_\epsilon|z|2\pi|z|^{-1} \\
&\leq \frac{N_\epsilon}{\pi}\int_{1}^{\infty}\frac{1}{r}e^{-r\sin\frac{\epsilon}{2}}\,dr + eN_\epsilon
\end{aligned} \tag{3.13}$$

showing that the integral defining $T(z)$ converges in $B(X)$ absolutely and uniformly for all $z \in S_{\alpha-\epsilon}$. Since the above holds for every $\epsilon > 0$, it follows that the map $z \mapsto T(z)$ is holomorphic on S_α, the details of the verification of which is left as (**Exercise** 3.3.8). Moreover, it also follows that there exists a constant $M_\epsilon > 0$ such that $\|T(z)\| \leq M_\epsilon\ \forall z \in S_{\alpha-\epsilon}$. Next we check the semigroup property. Let Γ' be another contour, chosen as shown in Figure 3.1. Then as noted in the beginning of this proof,

$$T(z) = \frac{1}{2\pi i}\int_{\Gamma'} e^{wz}R(w,A)\,dw, \quad z \in S_\alpha. \tag{3.14}$$

Thus for $z_1, z_2 \in S_{\alpha-\epsilon}$, using the resolvent equation (Appendix A.1) we get that

$$\begin{aligned}
T(z_1)T(z_2) &= (2\pi i)^{-2} \int_{w\in\Gamma}\int_{\lambda\in\Gamma'} e^{wz_1}e^{\lambda z_2} R(w,A)R(\lambda,A)\,dw\,d\lambda \\
&= (2\pi i)^{-2} \int_{w\in\Gamma}\int_{\lambda\in\Gamma'} e^{wz_1+\lambda z_2}(\lambda-w)^{-1}\big[R(w,A)-R(\lambda,A)\big]\,dw\,d\lambda \\
&= (2\pi i)^{-2} \int_{w\in\Gamma}\int_{\lambda\in\Gamma'} e^{wz_1+\lambda z_2}(\lambda-w)^{-1}R(w,A)\,dw\,d\lambda \\
&\quad - (2\pi i)^{-2} \int_{w\in\Gamma}\int_{\lambda\in\Gamma'} e^{wz_1+\lambda z_2}(\lambda-w)^{-1}R(\lambda,A)\,dw\,d\lambda \\
&= (2\pi i)^{-2}(I_1+I_2). \qquad (3.15)
\end{aligned}$$

Now by Cauchy's Theorem, for $\lambda \in \Gamma'$,

$$\int_{w\in\Gamma} e^{wz_1}(\lambda-w)^{-1}dw = 0$$

since the completion of Γ does not enclose λ, and

$$\int_{\lambda\in\Gamma'} e^{\lambda z_2}(\lambda-w)^{-1}\,d\lambda = 2\pi i e^{wz_2}.$$

These, together with Fubini's Theorem, leads to:

$$\begin{aligned}
I_1 &= \int_\Gamma e^{wz_1}\Big(\int_{\Gamma'} e^{\lambda z_2}(\lambda-w)^{-1}\,d\lambda\Big)R(w,A)\,dw \\
&= 2\pi i \int_\Gamma e^{wz_1}e^{wz_2}R(w,A)\,dw \\
&= 2\pi i \int_\Gamma e^{wz_1+wz_2}R(w,A)\,dw \\
&= (2\pi i)^2 T(z_1+z_2), \\
\text{while } I_2 &= -\int_{w\in\Gamma}\int_{\lambda\in\Gamma'} e^{wz_1+\lambda z_2}(\lambda-w)^{-1}R(\lambda,A)\,dw\,d\lambda \\
&= \int_{\lambda\in\Gamma'} e^{\lambda z_2}\Big(\int_{w\in\Gamma} e^{wz_1}(\lambda-w)^{-1}\,dw\Big)R(\lambda,A)\,d\lambda \\
&= 0.
\end{aligned}$$

It now follows from (3.15) that $T(z_1)T(z_2) = T(z_1+z_2)$. To complete the proof, we need to show that the map $z \mapsto T(z)$ is strongly continuous in $S_{\alpha-\epsilon}$

for every $\epsilon > 0$. For $x \in \mathcal{D}(A)$ and $z \in S_{\alpha-\epsilon}$, we have that

$$\begin{aligned} T(z)x - x &= (2\pi i)^{-1} \int_\Gamma e^w [(w - zA)^{-1}x - w^{-1}x]\, dw \\ &= (2\pi i)^{-1} \int_\Gamma w^{-1} e^w (w - zA)^{-1} zAx\, dw. \end{aligned}$$

Therefore, proceeding as in the estimation of the integrals in (3.13) with $\delta = |z|^{-1}$ in Figure 3.1, we get that

$$\begin{aligned} \|T(z)x - x\| &\le \frac{1}{2\pi}\|Ax\| \Big[2N_\epsilon \int_{|z|^{-1}}^\infty r^{-2}\, dr + eN_\epsilon |z|^2 |z|^{-1} 2\theta \Big] \\ &= \frac{N_\epsilon}{\pi}\|Ax\|(1 + e\theta)|z| \ \to\ 0 \text{ as } |z| \to 0. \end{aligned}$$

Since A is densely defined and $\|T(z)\|$ is uniformly bounded in $S_{\alpha-\epsilon}$ it follows that $T(z)x - x \to 0$ for all $x \in X$. The continuity for all $z \in S_{\alpha-\epsilon}$ is the consequence of the already established semigroup law. □

Theorem 3.3.6 above shows that a given uniformly bounded C_0-semigroup $\{T(t)\}_{t\ge 0}$ with generator A can be extended as a bounded holomorphic semigroup on a sector containing the positive real axis provided A is sectorial (in the sense of Definition 3.3.5) of angle less than $\pi/2$. Also note that intuitively, the domain of holomorphy (in z) of a bounded holomorphic semigroup $\{T(z)\}$ differs from the domain of holomorphy of the resolvent of its generator A or the angle of sectoriality of A by $\pi/2$. We now look at other, different but related conditions for a C_0-semigroup to admit a holomorphic extension. We shall need the following Lemma for this purpose.

Lemma 3.3.9. Let $\{T(t)\}_{t\ge 0}$ be a C_0-semigroup on a Banach space X with generator A. Suppose that $T(t)X \subset \mathcal{D}(A)$ for every $t > 0$. Then, for each $x \in X, t \mapsto T(t)x$ is arbitrarily often (strongly) differentiable on $\mathbb{R}_+$ and

$$T^{(n)}(t)x := \frac{d^n}{dt^n}T(t)x = \Big(\Big[\frac{d}{ds} T(s) \Big]_{s=\frac{t}{n}} \Big)^n x = A^n T(t)x = [AT(t/n)]^n x, \tag{3.16}$$

and $A^n T(t) \in \mathcal{B}(X)$ for all $t > 0, x \in X$ and $n \in \mathbb{N}$.

Proof. Suppose $t > t_0 > 0$. Then $T(t)x = T(t - t_0)T(t_0)x$. Since $T(t_0)x \in \mathcal{D}(A)$, using (b) of Theorem 2.2.7 we have that $T'(t)x$ exists and $T'(t)x = T(t - t_0)AT(t_0)x = AT(t)x = T(t/2)AT(t/2)x \in \mathcal{D}(A)$. Repeating the same

reasoning again yields that $T''(t)x$ exists and that

$$T''(t)x = [AT(t/2)]^2 x = A^2 T(t)x, \text{ for every } x \in X \text{ and } t > 0.$$

The equality (3.16) follows now by induction (**Exercise** 3.3.10). Next we note that for $t > 0$, the linear operator $AT(t)$ is defined everywhere, that is, $\mathcal{D}(AT(t)) = X$, and since A is closed, it is easy to see that $AT(t)$ is closed. Therefore, by the Closed Graph Theorem, $AT(t) \in \mathcal{B}(X)$, for each $t > 0$. By (3.16), it follows then that for $t > 0$, and $n \in \mathbb{N}$,

$$A^n T(t) = [AT(t/n)]^n = T^{(n)}(t) \in \mathcal{B}(X).$$

□

Theorem 3.3.11. The following are equivalent for a uniformly bounded, C_0-semigroup $\{T(t)\}_{t\geq 0}$ defined on a Banach space X, with generator A.

(i) For all $t > 0$, $T(t)X \subset \mathcal{D}(A)$, and there exists a constant $M > 0$ independent of t such that $\|tAT(t)\| \leq M$.

(ii) $\{T(t)\}_{t\geq 0}$ admits an extension to a bounded holomorphic semigroup $\{T(z)\}_{S_\alpha \cup \{0\}}$, where $\tan\alpha = \frac{1}{Me}$, and for every α', with $0 < \alpha' < \alpha$, there is a constant $C_{\alpha'}$ such that

$$\|T(z)\| \leq C_{\alpha'} \quad \forall z \in S_{\alpha'}.$$

(iii) There exists a positive constant $K > 0$ such that for all $a > 0$ and $b \neq 0$,

$$\|R(a + ib, A)\| \leq \frac{K}{|b|}. \tag{3.17}$$

Proof. (i) ⇒ (ii): Since $T(t)X \subset \mathcal{D}(A)$, it follows from Lemma 3.3.9 that $T(t)$ is infinitely differentiable, and for all $t > 0$ and $n = 1, 2, \ldots$,

$$\Big(\frac{t}{n}\Big)^n T^{(n)}(t) = \Big(\frac{t}{n} T'\big(t/n\big)\Big)^n = \big(t/nAT\big(t/n\big)\big)^n.$$

Hence

$$\Big\|\Big(\frac{t}{n}\Big)^n T^{(n)}(t)\Big\| = \Big\|\Big(\frac{t}{n} AT\big(t/n\big)\Big)^n\Big\| \leq M^n, \tag{3.18}$$

by the hypothesis. Now (3.18) leads to the estimate that

$$\Big\|\frac{(z-t)^n}{n!} T^{(n)}(t)\Big\| = \Big\|\frac{n^n (z-t)^n}{t^n n!}\big((t/n)^n T^{(n)}(t)\big)^n\Big\|$$

$$\leq \frac{(nM)^n}{n!}\Big|\frac{z-t}{t}\Big|^n \leq (Me)^n \Big|\frac{z-t}{t}\Big|^n,$$

where we have used the inequalities

$$(n-1)! = \Gamma(n) = \int_0^\infty e^{-x}x^{n-1}\,dx > \int_0^n e^{-x}x^{n-1}\,dx > e^{-n}\frac{n^n}{n},$$

to get that $\frac{n^n}{n!} < e^n$. Thus, the power series

$$\sum_{k=0}^{\infty} \frac{(z-t)^n T^{(n)}(t)}{n!} \tag{3.19}$$

converges in $\mathcal{B}(X)$ in operator norm, uniformly for all $z \in \mathbb{C}$ such that $|z-t| < t/(Me)$. Setting

$$T(z) := \sum_{n=0}^{\infty} \frac{(z-t)^n T^{(n)}(t)}{n!} \tag{3.20}$$

we see that $T(z) = T(t)$ for $z = t \in \mathbb{R}$ and that $T(t)$ has a strong holomorphic extension $T(z)$ to the sector S_α, where $\tan\alpha = 1/eM$. Since the family $\{T(t)\}_{t\geq 0}$ satisfies the semigroup property, the Identity Theorem for holomorphic functions ([2, Proposition A.2]) ensures that so does $\{T(z)\}_{z\in S_\alpha}$. Indeed, for fixed $t > 0$, the analytic map $S_\alpha \ni z \to T(t)T(z)$ satisfies $T(t)T(z) = T(t+z)$ for all $z \in S_\alpha \cap \mathbb{R}_+$. Therefore, by the Identity Theorem, $T(t)T(z) = T(t+z) \quad \forall\, z \in S_\alpha$. Now, repeating this argument, with t replaced by an arbitrary, but fixed $z_1 \in S_\alpha$ yields finally $T(z_1)T(z_2) = T(z_1+z_2)$ for all $z_1, z_2 \in S_\alpha$.

Next we prove uniform boundedness. Let $0 < \alpha' < \alpha$ and choose $k \in (0,1)$ such that $\tan\alpha' = k/eM$. For $z = a+ib \in S_{\alpha'}$, we have, using the power series for $T(z)$ given in (3.20) and estimates obtained earlier in the proof,

$$\begin{aligned}
\|T(a+ib)\| &= \Big\|T(a) + \sum_{n=1}^{\infty} \frac{1}{n!}(ib)^n T^{(n)}(a)\Big\| \\
&\leq \|T(a)\| + \sum_{n=1}^{\infty} |b|^n \Big(\frac{eM}{a}\Big)^n \\
&\leq \|T(a)\| + \sum_{n=1}^{\infty} \Big(\frac{k}{eM}\Big)^n (eM)^n = \|T(a)\| - 1 + \frac{1}{1-k}, \qquad (3.21)
\end{aligned}$$

proving the uniform boundedness of $\{T(z)\}$ in $S_{\alpha'}$. For $x \in X$, $z \in S_\alpha$ and $t > 0$, we write, using the semigroup law that

$$T(z)x - x = \big[T(z+t)x - T(t)x\big] - (T(z) - I)\big[T(t)x - x\big].$$

This, along with the facts that $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup and that the map $z \to T(z+t)x$ is holomorphic in a neighbourhood of $z = 0$, and the estimate (3.21) leads to the conclusion that $T(z)x \to x$ as $z \to 0$ in S_α for each $x \in X$ (**Exercise** 3.3.12).

(ii) $\Rightarrow$ (iii): By Theorem 3.3.4, A is sectorial. In particular, (3.17) holds.

(iii) $\Rightarrow$ (i): We first show that if (3.17) holds, then A is sectorial in the sense of Definition 3.3.5. Since $\{T(t)\}_{t\geq 0}$ is uniformly bounded, by Theorem 2.2.7, (e) and (f) we have that $\{\lambda \in \mathbb{C} : \operatorname{Re}\lambda > 0\} \subset \rho(A)$ and for $\lambda = a + ib, a > 0$,

$$\big\|R(a+ib, A)\big\| \leq \frac{C}{a}$$

for some constant $C > 0$. This estimate combined with (3.17) implies that

$$\big\|R(\lambda, A)\big\| \leq \frac{K'}{|\lambda|} \text{ for all } \lambda \text{ such that } \operatorname{Re}\lambda > 0. \tag{3.22}$$

Writing now the Taylor expansion for $R(\lambda, A)$ (see (A.2)) around λ_0 where $\lambda_0 = a + ib, a > 0$, we have

$$R(\lambda, A) = \sum_{n=0}^{\infty} R(\lambda_0, A)^{(n+1)}(\lambda_0 - \lambda)^n. \tag{3.23}$$

This series converges in $\mathcal{B}(X)$ for all $\lambda \in \mathbb{C}$ such that

$$\|R(\lambda_0, A)\||\lambda_0 - \lambda| \leq k < 1.$$

Letting $\lambda = \mu + ib$ in above and on using the estimate

$$\|R(\lambda_0, A)\||\lambda_0 - \lambda| = \|R(\lambda_0, A)\||a - \mu| \leq \frac{K}{|b|}|a - \mu|,$$

we have that the series in (3.23) converges uniformly in $\mathcal{B}(X)$ for all $\lambda = \mu + ib$ such that $|a-\mu| \leq \frac{k|b|}{K}$. Thus $\lambda = \mu+ib \in \rho(A)$ for all λ satisfying $|a-\mu| \leq \frac{k|b|}{K}$. Since $a > 0$ and $k < 1$ are arbitrary, it follows that

$$\{\lambda \in \mathbb{C} : \operatorname{Re}\lambda \leq 0, |\operatorname{Re}\lambda|/|\operatorname{Im}\lambda| < 1/K\} \subset \rho(A).$$

In particular, this implies that $S_{\alpha+\pi/2} \subset \rho(A)$, where $\tan\alpha = 1/K$. To obtain the estimate on the resolvent in this region, we use the expansion in (3.23) to get,

$$\begin{aligned}\|R(\lambda, A)\| &\leq \sum_{n=0}^{\infty} \|R(\lambda_0, A)\|^{(n+1)}|\lambda_0 - \lambda|^n \leq \frac{K}{|b|}\sum_{n=0}^{\infty} k^n \\ &= \frac{K}{(1-k)|b|} \leq \frac{\sqrt{K^2+1}}{(1-k)|\lambda|} = \frac{M'}{|\lambda|}.\end{aligned}$$

Thus A is sectorial.

Therefore, by Theorem 3.3.6, A generates a bounded holomorphic semigroup $\{T(z)\}_{z\in S_\alpha}$, which extends the semigroup $\{T(t)\}_{t\geq 0}$. In particular, the map $(0,\infty)\ni t\mapsto T(t)x$ is differentiable for all $x\in X$, implying that the limit

$$\lim_{s\to 0^+} s^{-1}\big(T(t+s)-T(t)\big)x=\lim_{s\to 0^+} s^{-1}\big(T(s)-I\big)T(t)x$$

exists for all $x\in X$ and $t>0$ so that $T(t)X\subset\mathcal{D}(A)$ for all $t>0$. From the proof of Theorem 3.3.6, with the same notations as in Figure 3.1, it follows by using Lemma 1.2.5(ii) that for $t>0$

$$\begin{aligned}AT(t)&=\frac{1}{2\pi i}A\int_\Gamma e^{\lambda t}R(\lambda,A)\,d\lambda=\frac{1}{2\pi i}\int_\Gamma e^{\lambda t}(\lambda R(\lambda,A)-I)\,d\lambda\\&=\frac{1}{2\pi i}\int_\Gamma \lambda e^{\lambda t}R(\lambda,A)\,d\lambda.\end{aligned}$$

As in the proof of Theorem 3.3.6, we choose Γ_2 such that $\delta=t^{-1}$. Thus, using (3.11) and (3.12), we have

$$\begin{aligned}\|AT(t)\|&=\Big\|\frac{1}{2\pi i}\Big(\int_{\Gamma_1}+\int_{\Gamma_2}+\int_{\Gamma_3}\Big)\lambda e^{\lambda t}R(\lambda,A)\,d\lambda\Big\|\\&\leq\frac{1}{2\pi}\Big(2\int_\delta^\infty N_\epsilon e^{-rt\sin(\epsilon/2)}\,dr+\int_{-\theta}^{\theta} eN_\epsilon\delta\,d\alpha\Big)\\&\leq\frac{1}{2\pi}\frac{2N_\epsilon}{t}\Big(\frac{1}{\sin(\epsilon/2)}+\pi e\Big)=\frac{M}{t}.\end{aligned}$$

The proof is now complete. □

3.4 Some examples of holomorphic semigroups

Example 3.4.1. Consider the multiplication semigroup defined in Example 2.5.4 with generator $M_{\mathfrak{q}}$. This semigroup extends to a bounded holomorphic semigroup of angle α if and only if

$$S_{\alpha+\pi/2}\subset\rho(M_{\mathfrak{q}}). \tag{3.24}$$

Recall from (2.23) that $\sigma(M_{\mathfrak{q}})=\operatorname{ess\,ran}\mathfrak{q}$. From Theorem 3.3.4 it follows that if $M_{\mathfrak{q}}$ generates a bounded holomorphic semigroup of angle α then (3.24) holds. To prove the converse, we invoke Theorem 3.3.6 and Theorem A.1.13 (i).

So suppose that $S_{\alpha+\pi/2} \subset \rho(M_{\mathfrak{q}}) = \mathbb{C} \setminus \text{ess ran}\,\mathfrak{q}$. Let $\epsilon > 0$ be arbitrary and suitably small. For $\lambda \in S_{\alpha+\pi/2-\epsilon}$, we have, on using (2.23) and the discussion preceding it, that,

$$\begin{aligned}\|R(\lambda, M_{\mathfrak{q}})\| = \|M_\psi\| = \|\psi\|_\infty &= \sup\left\{\frac{1}{|\mathfrak{q}(s) - \lambda|} : s \in \text{ess ran}\,\mathfrak{q}\right\} \\ &= \frac{1}{\text{dist}(\lambda, \text{ess ran}\,\mathfrak{q})} \leq \frac{1}{\text{dist}(\lambda, \mathbb{C} \setminus S_{\alpha+\pi/2})} \\ &\leq \frac{N_\epsilon}{|\lambda|}\end{aligned}$$

where ψ is as in (2.23) and N_ϵ is a constant depending on α and ϵ. Since $\epsilon > 0$ is arbitrary the claim follows from Theorem 3.3.6.

Example 3.4.2. Suppose A is a selfadjoint operator on a Hilbert space H and there exists $\gamma \in \mathbb{R}$, with $\gamma \leq 0$, such that

$$\langle Ax, x\rangle \leq \gamma\langle x, x\rangle \quad \forall\ x \in \mathcal{D}(A). \tag{3.25}$$

Then A is the generator of a bounded holomorphic semigroup of angle $\pi/2$.

By the Spectral Theorem A.1.15, we can assume that $H = L^2(\Omega, \mu)$ where μ is a σ-finite measure, and that $A = M_{\mathfrak{q}}$, where $\mathfrak{q}\colon \Omega \to \mathbb{C}$ is a measurable function, and $M_{\mathfrak{q}}$ is the multiplication operator with respect to $\mathfrak{q}$. Since A is selfadjoint, so is $M_{\mathfrak{q}}$ and therefore, by Theorem A.1.13, $\mathfrak{q}$ is real valued. Moreover, (3.25) applied to $M_{\mathfrak{q}}$ implies that

$$\text{ess sup}_{s\in\Omega}\ \text{Re}\,\mathfrak{q}(s) \leq \gamma.$$

Thus, by Theorem 2.5.4, $M_{\mathfrak{q}}$ generates a C_0-semigroup. Moreover, (3.25) implies that $\sigma(A) \equiv \sigma(M_{\mathfrak{q}}) \subset (-\infty, 0]$. Therefore condition (3.24) above holds for $M_{\mathfrak{q}}$ with $\alpha = \pi/2$. We conclude from Example 3.4.1 that $M_{\mathfrak{q}}$ and hence A generates a bounded holomorphic semigroup of angle $\pi/2$.

Example 3.4.3. The heat semigroup discussed in Example 2.5.9, on $X = L^2(\mathbb{R}^d)$, extends to a bounded holomorphic semigroup of angle $\pi/2$.

The generator A of this semigroup is $\frac{1}{2}\Delta$ with maximal domain $H^2(\mathbb{R}^d)$ (see Theorem 2.5.9). By Remark A.2.6(d) , $(\mathcal{F}\Delta f)(k) = -|k|^2(\mathcal{F}f)(k)$ for all $f \in \mathcal{D}(\Delta)$, that is, Δ is unitarily equivalent to M_ϕ with $\phi(k) = -|k|^2$ and hence Δ is a non-positive selfadjoint operator in $L^2(\mathbb{R}^d)$ by Theorem A.1.13. Therefore, A is also a non-positive selfadjoint operator. Therefore, it follows

that A satisfies the conditions of Example 3.4.2 with $\gamma = 0$ and the heat semigroup is bounded holomorphic of angle $\pi/2$.

Remark 3.4.4. The probability measure on $\mathbb{R}^d$ associated with the heat semigroup as given in (2.29) makes sense for a complex parameter z replacing t :

$$\mu_z(dx) = (2\pi z)^{-d/2} \exp\Big(- \frac{|x|^2}{2z}\Big)dx, \tag{3.26}$$

with $\operatorname{Re} z > 0$, and the branch of the square root is chosen such that $\operatorname{Re}\sqrt{z} > 0$. We can verify (**Exercise** 3.4.5) that the associated heat semigroup $T(z)$, as defined in Theorem 2.5.7 is well defined in $L^p(\mathbb{R}^d)$ $(1 \leq p < \infty)$ as a holomorphic semigroup of angle $\pi/2$. It is interesting to note that though the line $\operatorname{Re} z = 0$ is not in the domain of holomorphy, $\{T(it)\}_{t\in\mathbb{R}}$ does make sense as a unitary group in $L^2(\mathbb{R}^d)$, and as a family of bounded maps from $L^1(\mathbb{R}^d)$ into $L^\infty(\mathbb{R}^d)$. In fact, we can define

$$\mu_{it}(dx) = \begin{cases} (2\pi|t|)^{-d/2} e^{-id\pi/4} \exp\big(\frac{i|x|^2}{2t}\big)dx & \text{if } t > 0, \\ (2\pi|t|)^{-d/2} e^{id\pi/4} \exp\big(\frac{i|x|^2}{2t}\big)dx & \text{if } t < 0, \end{cases}$$

and it can be shown that the associated semigroup has the above mentioned properties (**Exercise** 3.4.6). The unitary group resulting from this exercise is called the Schrödinger free evolution group and is of interest in Quantum Mechanics. For further reading in this area, the reader is referred to [1] and [21].

Chapter 4

Perturbation and convergence of semigroups

In this chapter, the stability of various classes of semigroups under suitable sets of perturbations will be studied, viz. for general C_0-semigroups and contraction semigroups. The methods involved will be the expansion of either the perturbed semigroup itself in terms of the unperturbed one and the perturbation of the generator or in terms of the resolvents concerned.

4.1 Perturbation of the generator of a C_0-semigroup

Theorem 4.1.1. Let $\{T(t)\}_{t\geq 0}$ be a C_0-semigroup with generator A in a Banach space X satisfying $\|T(t)\| \leq Me^{\beta t}\ \forall\ t\geq 0$ where $M \geq 0$ and $\beta \in \mathbb{R}$, and let $B \in \mathcal{B}(X)$. Then $A + B$ is the generator of a C_0-semigroup $\{S(t)\}_{t\geq 0}$ with bound $\|S(t)\| \leq Me^{(\beta+M\|B\|)t}$.

Proof. For $x \in X, t \geq 0$, set

$$S(t)x = T(t)x + \sum_{n=1}^{\infty} I_n(t)x, \tag{4.1}$$

where

$$I_0(t)x = T(t)x$$
$$I_1(t)x = \int_0^t T(t-t_1)BT(t_1)x\,dt_1$$

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7_4

$$\begin{aligned} I_2(t)x &= \int_0^t T(t-t_2)BI_1(t_2)x\,dt_2 \\ &= \int_0^t T(t-t_2)B\,dt_2 \int_0^{t_2} T(t_2-t_1)BT(t_1)x\,dt_1 \\ &= \int dt_2 \int_{0\le t_1\le t_2\le t} dt_1\, T(t-t_2)BT(t_2-t_1)BT(t_1)x, \end{aligned}$$

and so on. Thus, I_n is recursively defined by setting

$$I_n(t)x = \int_0^t T(t-t_n)BI_{n-1}(t_n)x\,dt_n \text{ for all } n \ge 1.$$

First of all, we note that

$$\begin{aligned} \|I_n(t)x\| &\le (M\|B\|)^n \|x\| M e^{\beta t} \int_{0\le t_1\le t_2 \ldots \le t_n \le t} dt_1\,dt_2 \ldots\, dt_n \\ &\le \|x\| \frac{(M\|B\|t)^n}{n!} M e^{\beta t}, \end{aligned} \tag{4.2}$$

so that the infinite series $\sum_{n=1}^{\infty} I_n(t)$ converges in operator norm (uniformly for t in a compact interval) and defines $S(t)$ as a bounded linear operator for every t, with $\|S(t)\| \le Me^{(\beta+M\|B\|)t}$. Since $I_n(0) = 0\ \forall n \ge 1$, we have that $S(0) = I$, and the strong continuity of $S(t)$ follows from the same property of $T(t)$ and $I_n(t)$, and the fact that $\sum_{n=0}^{\infty} I_n(t)$ converges in operator norm uniformly in compact intervals. Next, to show that $\{S(t)\}_{t\ge 0}$ is a C_0-semigroup, we first need to establish the integral equation,

$$S(t)x = T(t)x + \int_0^t T(t-s)BS(s)x\,ds \text{ for all } x \in X. \tag{4.3}$$

This follows by rewriting the infinite sum on the right hand side of (4.1) as

$$\begin{aligned} S(t)x - T(t)x &= \sum_{n=1}^{\infty} \int_0^t T(t-s)BI_{n-1}(s)x\,ds \\ &= \int_0^t T(t-s)B\Big(\sum_{n=1}^{\infty} I_{n-1}(s)x\Big)\,ds \\ &= \int_0^t T(t-s)B\Big(\sum_{n=2}^{\infty} I_{n-1}(s)x + T(s)x\Big)\,ds \\ &= \int_0^t T(t-s)BS(s)x\,ds, \end{aligned}$$

where the interchange of integration and summation above is justified by uniform convergence as seen before. Therefore,

$$\begin{aligned}
&S(t+s) - S(t)S(s) \\
&\quad = T(t+s) + \int_0^{t+s} T(t+s-v)BS(v)\,dv \\
&\qquad - \Big(T(t) + \int_0^t T(t-u)BS(u)\,du\Big)S(s) \\
&\quad = T(t+s) - T(t)\big(T(s) + \int_0^s T(s-u)BS(u)\,du\big) \\
&\qquad + \Big(\int_0^{t+s} T(t+s-v)BS(v)\,dv - \int_0^t T(t-u)BS(u)S(s)\,du\Big) \\
&\quad = -\int_0^s T(t+s-u)BS(u)\,du + \int_0^{t+s} T(t+s-v)BS(v)\,dv \\
&\qquad - \int_0^t T(t-u)BS(u)S(s)\,du \\
&\quad = \int_0^t T(t-u)B(S(u+s) - S(u)S(s))\,du.
\end{aligned}$$

On iterating the above n times we get the norm estimate

$$\|S(t+s) - S(t)S(s)\| \le \frac{(M\|B\|t)^n}{n!} e^{|\beta|t} \sup_{u\in[0,t]} (\|S(u)\|\|S(s)\| + \|S(u+s)\|).$$

The left hand side is independent of n while the right hand side converges to 0 as $n \to \infty$, showing that S satisfies the semigroup property. Now, using (4.3) for $x \in \mathcal{D}(A)$, we have that

$$t^{-1}(S(t) - I)x = t^{-1}(T(t) - I)x + t^{-1}\int_0^t T(t-s)BS(s)x\,ds, \tag{4.4}$$

and the right hand side of (4.4) converges strongly to $Ax + Bx$ as $t \to 0^+$, where we have applied Lemma 2.1.2 to show that the second term on the right hand side of (4.4) converges to Bx as $t \to 0^+$. It is easy to see that $A + B$, defined on $\mathcal{D}(A)$, is a closed operator since B is bounded. Therefore, it follows that $A + B$ is the generator of the semigroup $\{S(t)\}_{t\ge 0}$.

□

The above result shows that perturbing the generator of a C_0-semigroup by a bounded operator again yields a generator of a C_0-semigroup. The requirement of *bounded* perturbation can be weakened to *relatively bounded* perturbation to obtain the same conclusion. Here we introduce the definition of *relative*

boundedness of operators and use it for proving results on perturbation of contraction semigroups, leaving further discussion about this topic to Section 4.2.

Definition 4.1.2. Let $A\colon \mathcal{D}(A) \subset X \to X$, be a linear operator on a Banach space X. An operator $B\colon \mathcal{D}(B) \subset X \to X$ is said to be **bounded relative to** A if $\mathcal{D}(A) \subset \mathcal{D}(B)$ and if there exist constants $a, b \in \mathbb{R}_+$ such that

$$\|Bx\| \leq a\|Ax\| + b\|x\| \tag{4.5}$$

for all $x \in \mathcal{D}(A)$. The **relative bound** of B with respect to A is defined as

$$a_A(B) := \inf\{a \geq 0 : \text{ there exists } b \in \mathbb{R}_+ \text{ such that (4.5) holds }\}. \tag{4.6}$$

Theorem 4.1.3. Let A and B be generators of C_0 contraction semigroups $\{T(t)\}_{t\geq 0}$ and $\{S(t)\}_{t\geq 0}$ respectively in X and let B be bounded relative to A with relative bound less than $\frac{1}{2}$. Then $A+B$ is the generator of a C_0 contraction semigroup.

Proof. Let a and b be as in (4.5). Set $x(t) = S(t)T(t)e^{-t\lambda}x$ with $x \in \mathcal{D}(A) \subseteq \mathcal{D}(B)$. Then for $t > 0$ and $\lambda > 0$,

$$t^{-1}(x(t) - x) = S(t)t^{-1}(e^{-t\lambda}T(t)x - x) + t^{-1}(S(t) - I)x,$$

which converges strongly to $(A - \lambda)x + Bx$ as $t \longrightarrow 0^+$. On the other hand,

$$\|t^{-1}(x(t) - x)\| \geq t^{-1}(1 - e^{-t\lambda})\|x\| \longrightarrow \lambda\|x\| \text{ as } t \longrightarrow 0^+$$

and therefore

$$\|[\lambda - (A + B)]x\| \geq \lambda\|x\|$$

for all $x \in \mathcal{D}(A)$ and $\lambda > 0$, which implies that $\lambda - (A + B)$ is injective. Recall that $\lambda \in \rho(A)$ for all $\lambda > 0$, hence by Theorem 2.2.7

$$\begin{aligned}\|B(\lambda - A)^{-1}\| &\leq b\|(\lambda - A)^{-1}\| + a\|A(\lambda - A)^{-1}\| \\ &\leq b\lambda^{-1} + a(1 + \lambda\|(\lambda - A)^{-1}\|) \leq b\lambda^{-1} + 2a < 1\end{aligned}$$

for sufficiently large λ, since $a < 1/2$. This along with the equality

$$\lambda - (A + B) = (I - B(\lambda - A)^{-1})(\lambda - A),$$

implies that $Ran(\lambda-(A+B))=X$ for sufficiently large λ, say for $\lambda>\lambda_0$, and hence, for such $\lambda>\lambda_0, \|(\lambda-(A+B))^{-1}\|\leq\lambda^{-1}$. Next, let $0<\mu<\lambda_0<\lambda$ and note that

$$[\mu-(A+B)]=[\lambda-(A+B)][1+(\mu-\lambda)(\lambda-(A+B))^{-1}],$$

yielding that

$$[\mu-(A+B)]^{-1}=[1+(\mu-\lambda)(\lambda-(A+B))^{-1}]^{-1}[\lambda-(A+B)]^{-1},$$

where the Neumann series for $[I+(\mu-\lambda)(\lambda-(A+B))^{-1}]^{-1}$ converges in operator norm since the series $\sum\limits_{n=0}^{\infty}\big(\frac{|\mu-\lambda|}{\lambda}\big)^n$ converges. Thus

$$\big\|(\mu-(A+B))^{-1}\big\|\leq\lambda^{-1}\sum_{n=0}^{\infty}\big(\frac{|\mu-\lambda|}{\lambda}\big)^n=\mu^{-1},$$

proving the necessary estimate for all $\mu>0$. Furthermore, by Theorem 4.2.1(i) $A+B$, defined on $\mathcal{D}(A)$ is closed. An application of The Hille-Yosida Theorem 2.3.3(i) leads to the desired result.

□

Corollary 4.1.4. In the previous theorem, it is sufficient to assume that $a<1$ instead of $<\frac{1}{2}$.

Proof. Suppose for some α with $0\leq\alpha<1$, $A+\alpha B$ is the generator of a C_0 contraction semigroup. For this note that $A+\alpha B$ is defined on $\mathcal{D}(A)$ and is closed by Theorem 4.2.1, since the relative bound a of B is less than 1. Then

$$(1-a\alpha)\|Bx\|\leq a(\|Ax\|-\alpha\|Bx\|)+b\|x\|\leq a\|(A+\alpha B)x\|+b\|x\|$$

or

$$\|Bx\|\leq a(1-a\alpha)^{-1}\|(A+\alpha B)x\|+b(1-a\alpha)^{-1}\|x\|$$

and therefore, if we choose β with $0\leq\beta\leq\frac{1}{2}(1-a)$, then $\beta a(1-a\alpha)^{-1}<\frac{1}{2}$ and we can apply Theorem 4.1.3 with βB as the perturbation to $(A+\alpha B)$ to get a C_0 contraction semigroup with generator $\big(A+(\alpha+\beta)B\big)$.

Iterating this process n times leads to the generator $(A+(\alpha+n\beta)B)$, which is equal to $A+B$ if we choose n such that $\alpha+n\beta=1$. Such a choice can always be made with $\alpha<\frac{1}{2}$ and $\beta<\frac{1}{2}(1-a)$ since $a<1$. □

Observe that in view of Theorem 3.1.18, the condition that the relatively bounded operator B be the generator of a contraction C_0-semigroup in Theorem 4.1.3 may be replaced with the condition that B is dissipative and $Ran(B-1) = X$. However, the range condition may be done away with completely. More precisely, the following holds (see [19, Corollary 3.3.3]):

Theorem 4.1.5. Let A be the generator of a contraction C_0-semigroup $\{T_t\}_{t\geq 0}$ and B be a dissipative operator such that $\mathcal{D}(B) \supset \mathcal{D}(A)$. Suppose B is bounded relative to A with relative bound less than 1. Then $A + B$ is the generator of a contraction semigroup.

Remark 4.1.6. We remark here that perturbation of the generator A of a holomorphic semigroup by a relatively bounded closed operator B again yields a holomorphic semigroup (with generator $A + B$) provided the relative bound is small enough. We refer the interested reader to [19, Section 3.2], for details of this and related results.

4.2 Relative boundedness and some consequences

In Definition 4.1.2, the concept of boundedness of an unbounded operator B relative to another unbounded operator A (in such a case B is said to be A-bounded, for short) and the bound of B relative to A (often called the A-bound of B) are defined. Here we collect some properties as consequences of relative boundedness. For more details, the reader is referred to [1] and [21]. Definition 4.1.2, along with the inequality (4.5), can be rephrased as follows: given two operators A and B with $\mathcal{D}(A) \subset \mathcal{D}(B), B$ is A-bounded if B maps the set $\{x \in \mathcal{D}(A) \colon \|x\| + \|Ax\| \leq 1\}$ into a bounded set in X. Similarly, B is said to be A-compact if $\mathcal{D}(A) \subset \mathcal{D}(B)$ and B maps the set $\{x \in \mathcal{D}(A) \colon \|x\|+\|Ax\| \leq 1\}$ into a pre-compact set or equivalently, for any sequence $\{x_n\} \in \mathcal{D}(A) \subset \mathcal{D}(B)$ such that $\|x_n\|+\|Ax_n\| \leq 1, \{Bx_n\}$ contains a convergent subsequence. Clearly, if B is A-compact, then B is A-bounded. Indeed, let $\{x_n\} \in \mathcal{D}(A)$ be such that $\|x_n\|+\|Ax_n\| \leq 1$ and that $\{\|Bx_n\|\}$ is unbounded, making finding a convergent subsequence in $\{Bx_n\}$ impossible, contradicting the A-compactness of B.

Theorem 4.2.1. Let $A \colon \mathcal{D}(A) \to X$ and $B : \mathcal{D}(B) \to X$ be linear operators with $\mathcal{D}(A) \subset \mathcal{D}(B)$.

(i) Suppose that either B is A-bounded with A-bound of B less than 1 or B is A-compact. If A is closable, so is $T = A + B$, and $\mathcal{D}(\overline{T}) = \mathcal{D}(\overline{A})$. In particular, if A is closed, then so is $A + B$.

(ii) If furthermore, A is the generator of a C_0-semigroup in X, then B is A-bounded if and only if $BR(z, A) \in \mathcal{B}(X)$, for some $z \in \rho(A)$ (and hence for all $z \in \rho(A)$). Also, in such a case, B is A-compact if and only if $BR(z, A)$ is compact for some $z \in \rho(A)$ (and hence for all $z \in \rho(A)$).

Proof. (i) Note that $\mathcal{D}(T) = \mathcal{D}(A)$ and let $\{x_n\} \subset \mathcal{D}(A)$ be a convergent sequence such that $\{Tx_n\}$ is convergent. Then the inequality (4.5) yields that

$$\begin{aligned} \|A(x_n - x_m)\| &\leq \|T(x_n - x_m)\| + \|B(x_n - x_m)\| \\ &\leq \|T(x_n - x_m)\| + a\|A(x_n - x_m)\| + b\|x_n - x_m\| \\ \text{or, } \|A(x_n - x_m)\| &\leq \frac{b}{(1-a)}\|x_n - x_m\| + \frac{1}{(1-a)}\|T(x_n - x_m)\|, \end{aligned}$$

since by hypothesis, $a < 1$, and this implies that $\{Ax_n\}$ is convergent. Thus if $x_n \to 0$, and such that $Tx_n \to y$, then Ax_n converges. Since A is closable this means that $Ax_n \to 0$ and the inequality (4.5) with x_n replacing x tells us that $Bx_n \to 0$, so that $y = 0$. This proves the closability of T. If $\overline{T}$ and $\overline{A}$ are the closures of T and A respectively, then for any $x \in \mathcal{D}(\overline{T})$, there is a sequence $\{x_n\} \in \mathcal{D}(A)$, such that $\{x_n\}$ and $\{Tx_n\}$ are both convergent. As we have seen however, this implies that $\{Ax_n\}$ is also convergent, that is, $\mathcal{D}(\overline{T}) \subset \mathcal{D}(\overline{A})$. The other inclusion follows similarly (**Exercise** 4.2.2).

Now, let B be A-compact and let $\mathcal{D}(A) \ni x_n \to 0$ and $Tx_n \to y$. We need to show that $y = 0$. We claim that the convergence of $\{x_n\}$ and of $\{Tx_n\}$ (which implies the boundedness of both of them) implies that $\{Ax_n\}$ contains a bounded subsequence. If not, then $\|Ax_n\| \to \infty$ as $n \to \infty$. Assume this and set $x'_n = x_n/\|Ax_n\|$. Then $x'_n \to 0$ and $Tx'_n = \big(\|Ax_n\|\big)^{-1}(Tx_n) \to 0$ as $n \to \infty$ and $\|Ax'_n\| = 1$. By the A- compactness of $B, \{Bx'_n\}$ contains a convergent subsequence and replacing x'_n by a suitable subsequence we may assume that $Bx'_n \to w$. Then, $Ax'_n = Tx'_n - Bx'_n \to -w$. Since $x'_n \to 0$, and since A is closable, we must have $w = 0$. This contradicts the fact that $\|Ax'_n\| = 1$ and therefore, $\{Ax_n\}$ contains a bounded subsequence. Thus, choosing that subsequence and continuing to designate the same by $\{x_n\}$, we have that $x_n \to 0$, and $Tx_n \to y$. This implies that $\|x_n\| + \|Ax_n\| \leq M$. Hence by the A-compactness of $B, \{Bx_n\}$ has a convergent subsequence. Let $Bx_n \to w$, by

renaming the subsequence. Therefore, $Ax_n = Tx_n - Bx_n \to y - w$ and since A is closable, this leads to the conclusion that $y - w = 0$. On the other hand, A-compactness of B implies the A-boundedness of B and hence (4.5) implies that $Bx_n \to 0$, so that $y = w = 0$.

(ii) Let B be A-bounded. We get from (4.5) that for $\operatorname{Re} z > \beta$ and $x \in X$,

$$\begin{aligned}\|BR(z,A)x\| &\le a\|AR(z,A)x\| + b\|R(z,A)x\| \\ &\le a\|x\| + (b + |z|a)\|R(z,A)x\| \le \|x\| \left[a + \frac{(b + a|z|)M}{(\operatorname{Re} z - \beta)}\right],\end{aligned}$$

where we have used the estimate in Theorem 2.2.7(f). This leads to the result that $BR(z,A) \in \mathcal{B}(X)$.

The converse is simpler and is left as an exercise (**Exercise** 4.2.3).

Now let $BR(z,A)$ be compact for some $z \in \rho(A)$, and let $\{x_n\} \subset \mathcal{D}(A)$ and $\{Ax_n\}$ be bounded sequences. Since $\|(z - A)x_n\| \le |z|\|x_n\| + \|Ax_n\|$, it follows that $\{(z - A)x_n\}$ is also a bounded sequence and therefore the sequence $\{Bx_n\}_n = \{BR(z,A)(z - A)x_n)\}_n$ will have a convergent subsequence. Hence B is A-compact. The converse follows by retracing the steps of this argument. That the property of boundedness or compactness of $BR(z,A)$ is independent of the choice of $z \in \rho(A)$ is an easy consequence of the resolvent equation (A.1) and in left as an exercise (**Exercise** 4.2.4).

□

When the Banach space X is a Hilbert space, there are more specific results, as given in the next theorem. But before that, we need to introduce the concept of essential spectrum of a selfadjoint operator, recalling that the spectrum of A is defined in Appendix A.1.

Definition 4.2.5. Let A be a selfadjoint operator defined on a Hilbert space H. The **discrete spectrum** $\sigma_d(A)$ of A is defined as

$$\sigma_d(A) := \{\lambda \in \sigma(A) : \lambda \text{ is an isolated eigenvalue of finite mulitplicity}\}$$

and the **essential spectrum** of A is defined as:

$$\sigma_e(A) := \sigma(A) \setminus \sigma_d(A).$$

Thus the essential spectrum of A consists of eigenvalues of infinite multiplicity, limit points of eigenvalues of A and the continuous spectrum of A.

Theorem 4.2.6. Let A be a selfadjoint operator in a Hilbert space H and let B be a symmetric operator. Then the following hold.

(i) If B is A-bounded with A-bound less than 1, then $A+B$ is selfadjoint with $\mathcal{D}(A+B)=\mathcal{D}(A)$ and furthermore, if A is bounded below, then so is $A+B$.

(ii) If B is A-compact, then the A-bound of B is 0, and hence $A+B$ is selfadjoint. Moreover, in this case $\sigma_e(A+B)=\sigma_e(A)$, that is, the essential spectrum of A is invariant under a relatively compact, symmetric perturbation.

Proof. (i) If A is a selfadjoint operator in H, then the whole imaginary axis, except possibly 0, is in the resolvent set and hence for $\alpha\in(0,\infty), y\in H$,

$$\begin{aligned}\|BR(\pm i\alpha,A)y\| &\le a\|AR(\pm i\alpha,A)y\|+b\|R(\pm i\alpha,A)y\|\\ &\le \big(a\|AR(\pm i\alpha,A)\|+b\|R(\pm i\alpha,A)\|\big)\|y\| = (a+b\alpha^{-1})\|y\|,\end{aligned}$$

where a and b are as in the inequality (4.5). This means that

$$\|B(R(\pm i\alpha,A)\|\le (a+b\alpha^{-1}),$$

which can be made less than 1 by choosing α large enough, since $a<1$. Hence, for such a choice of α, $(I+BR(\pm i\alpha,A))^{-1}$ is in $\mathcal{B}(H)$. Therefore, for $x\in\mathcal{D}(A)$, the identity

$$\big[(A+B)\pm i\alpha\big]x=\big[I+BR(\pm i\alpha,A)\big](\pm i\alpha-A)x, \tag{4.7}$$

along with the criterion of selfadjointness in Theorem A.1.12 implies that $A+B$ is selfadjoint on $\mathcal{D}(A)$. If A is bounded below, say $A\ge\gamma, (\gamma\in\mathbb{R})$ that is, there exists $\gamma\in\mathbb{R}$ such that $\langle Ax,x\rangle\ge\gamma\langle x,x\rangle$ for all $x\in\mathcal{D}(A)$, then choose $\kappa<\gamma$, so that $A-\kappa>0$. Hence $R(\kappa,A)\in\mathcal{B}(H)$ by Theorems A.1.13 and A.1.15, and the preceding argument can be repeated to show that for any $z\in\mathbb{C}$ with $\operatorname{Re} z=\kappa<\gamma$,

$$\begin{aligned}\|BR(z,A)\| &\le a\|AR(z,A)\|+b\|R(z,A)\|\\ &\le a\max\{1,|\gamma|(\gamma-\kappa)^{-1}\}+b(\gamma-\kappa)^{-1}.\end{aligned}$$

This constant can be made less than 1 by choosing κ appropriately, showing that such $z\in\rho(A+B)$ or that $A+B$ is bounded below.

(ii) As we have observed earlier in the previous theorem, if $BR(z_0, A)$ is compact for some $z_0 \in \rho(A)$, then $BR(z, A)$ is compact for every $z \in \rho(A)$. Since here A is selfadjoint, it is easily seen from Theorem A.1.15 that

$$\|(A+i)R(-in, A)\| \leq 1 \quad \forall \quad n = 1, 2, \ldots,$$

and that the sequence $\{(A+i)R(-in, A)\}$ converges strongly to 0 as $n \to \infty$ (**Exercise** 4.2.7). Since

$$BR(in, A) = -BR(i, A)\big[(A+i)R(-in, A)\big]^*$$

it follows that the sequence $\{BR(in, A)\}$ converges to 0 in operator norm as $n \to \infty$. (This implication is a consequence of the following statement whose proof is left as (**Exercise** 4.2.8): If a sequence $\{C_n\} \subset \mathcal{B}(X)$ converges strongly to C and D is a compact operator, then $DC_n^* \to DC^*$ in operator norm as $n \to \infty$.) Moreover, for $x \in \mathcal{D}(A)$,

$$\|Bx\| \leq \|BR(in, A)\|\|Ax\| + n\|BR(in, A)\|\|x\|. \tag{4.8}$$

Since $\lim_{n\to\infty} \|BR(in, A\| = 0$, the coefficient of $\|Ax\|$ in (4.8) can be made an arbitrarily small positive number. From the definition of A-bound of B, this implies that the A-bound is 0. Thus the selfadjointness of $A + B$ follows from part (i) of this theorem. The proof of the invariance of the essential spectrum is omitted and can be found in [1]. □

4.3 Convergence of semigroups

First a kind of master theorem for the convergence of a family of semigroups is proven, from which further results on the convergence of C_0-semigroups in a Banach space follow.

Theorem 4.3.1. Let $\{T^{(\lambda)}(t)\}_{t\geq 0}$, $\lambda \in (0, 1]$ be a family of C_0-semigroups in a Banach space X satisfying $\|T^{(\lambda)}(t)\| \leq Me^{\beta t}$ for some $M > 0$ and $\beta \in \mathbb{R}$, independent of λ, and let $\{T(t)\}_{t\geq 0}$ be another C_0-semigroup, acting on X_0, a closed subspace of X, with $\|T(t)\| \leq Me^{\beta t}$. Let these semigroups have generators $A^{(\lambda)}$ and A respectively. Then the following are equivalent.

(i) For some $t_0 \in (0, \infty)$ and each $x \in X_0$,

$$\lim_{\lambda\to 0^+} \sup_{t\in[0,t_0]} \big\{\|T^{(\lambda)}(t)x - T(t)x\|\big\} = 0.$$

(ii) The result (i) holds for every $t_0 \in (0, \infty)$.

(iii) The set $S = \{z \in \mathbb{C} : \operatorname{Re} z > \beta\} \subseteq \rho(A^{(\lambda)})\ \forall \lambda \in (0, 1]$ and $S \subseteq \rho(A)$, and for some $z \in S$ and each $x \in X_0$,

$$\lim_{\lambda \longrightarrow 0^+} \left(z - A^{(\lambda)}\right)^{-1} x = (z - A)^{-1} x.$$

(iv) The result (iii) holds for every $z \in S$.

(v) There exists a core $\mathcal{D}$ for A in X_0 (see Appendix A.1), such that for every $x \in \mathcal{D}$, there exists an $x^{(\lambda)} \in \mathcal{D}(A^{(\lambda)}) \subseteq X$ for each $\lambda \in (0, 1]$ satisfying $x^{(\lambda)} \longrightarrow x$ and $A^{(\lambda)} x^{(\lambda)} \longrightarrow x$ strongly as $\lambda \longrightarrow 0^+$.

Proof. That (ii) and (iv) imply (i) and (iii) respectively, is trivial. For the implication (i) $\Rightarrow$ (ii), note that for $t_0 \in (0, \infty)$, $t \in [0, t_0]$ and $x \in X_0$,

$$\begin{aligned}
&\left\|T^{(\lambda)}(t + t_0)x - T(t + t_0)x\right\| \\
&\leq \left\|T^{(\lambda)}(t)(T^{(\lambda)}(t_0)x - T(t_0)x)\right\| + \left\|(T^{(\lambda)}(t) - T(t))T(t_0)x\right\| \\
&\leq M e^{\beta t}\left(\left\|T^{(\lambda)}(t_0)x - T(t_0)x\right\|\right) + \left\|(T^{(\lambda)}(t) - T(t))T(t_0)x\right\|.
\end{aligned}$$

Therefore, $\lim_{\lambda \longrightarrow 0^+} \sup_{0 \leq t \leq 2t_0} \left\|T^{(\lambda)}(t)x - T(t)x\right\| = 0$. Proceeding this way in N steps one has that

$$\lim_{\lambda \longrightarrow 0^+} \sup_{0 \leq t \leq N t_0} \|T^{(\lambda)}(t)x - T(t)x\| = 0 \quad \forall x \in X_0,$$

and therefore (ii) holds.

(iii) $\Rightarrow$ (v): Let $\mathcal{D} \subseteq X_0$ be a core for A, and let $x \in \mathcal{D}$. Set $y = (z - A)x \in X_0$ and, for $\lambda \in (0, 1]$, $x^{(\lambda)} = (z - A^{(\lambda)})^{-1} y$. It is clear that $x^{(\lambda)} \in \mathcal{D}(A^{(\lambda)}) \subseteq X$ for each λ and that (iii) implies that $x^{(\lambda)}$ converges as $\lambda \longrightarrow 0^+$, to $(z - A)^{-1} y$, which is equal to x by the definition of y. Furthermore,

$$A^{(\lambda)} x^{(\lambda)} = A^{(\lambda)}(z - A^{(\lambda)})^{-1} y = -y + z x^{(\lambda)},$$

which converges to $-y + zx = Ax$ as $\lambda \longrightarrow 0^+$, proving (v).

(v) $\Rightarrow$ (iv): Note that, since $\mathcal{D}$ is dense in X_0 and is a core for A in X_0, $\overline{(z - A)\mathcal{D}} = X_0$ for every $z \in S$. By (v), for every $x \in \mathcal{D} \subseteq X_0$, there exists a family $\{x^{(\lambda)}\}$ with each $x^{(\lambda)} \in \mathcal{D}(A^{(\lambda)}) \subseteq X$ such that $x^{(\lambda)} \longrightarrow x$ and $A^{(\lambda)} x^{(\lambda)} \longrightarrow Ax$ as $\lambda \longrightarrow 0^+$. This implies that $(z - A^{(\lambda)}) x^{(\lambda)} \longrightarrow (z - A)x$.

Therefore, for all $x \in \mathcal{D}$,

$$\begin{aligned}&(z - A^{(\lambda)})^{-1}(z - A)x - x\\ &= (z - A^{(\lambda)})^{-1}\big[(z - A)x - (z - A^{(\lambda)})x^{(\lambda)}\big] + x^{(\lambda)} - x \longrightarrow 0 \text{ as } \lambda \longrightarrow 0^+,\end{aligned}$$

since $\big\|(z - A^{(\lambda)})^{-1}\big\| \leq (\operatorname{Re} z - \beta)^{-1}$ for every $z \in S$ and every $\lambda \in (0, 1]$. Now set $y = (z - A)x$ with $x \in \mathcal{D}$. Then the above convergence, along with the earlier observation that $\overline{(z - A)\mathcal{D}} = X_0$, yields the conclusion (iv).

Thus the implications (i) $\Leftrightarrow$ (ii) and (iii) $\Rightarrow$ (v) $\Rightarrow$ (iv) $\Rightarrow$ (iii) have been proven, and to complete the equivalence of all five statements it is sufficient to prove the equivalence (ii) $\Leftrightarrow$ (iv). For (ii) $\Rightarrow$ (iv), fix $z \in S$ and note that, by Theorem 2.2.7(c),

$$\big[(z - A^{\lambda})^{-1}x - (z - A)^{-1}x\big] = \int_0^\infty e^{-zt}\big(T^{(\lambda)}(t)x - T(t)x\big)\,dt \tag{4.9}$$

for $x \in X_0 \subseteq X$, and

$$\begin{aligned}\Big\|\int_0^\infty (T^{(\lambda)}(t)x - T(t)x)e^{-zt}\,dt\Big\| &\leq \sup_{0\leq t\leq t_0}\big\|T^{(\lambda)}(t)x - T(t)x\big\| \int_0^{t_0} e^{-(\operatorname{Re} z)t}\,dt\\ &\quad + 2M\|x\| \int_{t_0}^\infty e^{(\beta - \operatorname{Re} z)t}dt,\end{aligned} \tag{4.10}$$

where we have used the bounds for the C_0-semigroups $\big\{T^{(\lambda)}(t)\big\}$ and $\{T(t)\}$. Given $\varepsilon > 0$, choose $t_0 > 0$ sufficiently large such that the second term in the right hand side of (4.10) is $< \varepsilon/2$ and then for that t_0, choose $\lambda > 0$ sufficiently small to make the first term on the right hand side of (4.10) $< \varepsilon/2$ by using (ii). This observation and (4.9) proves (iv).

(iv) $\Rightarrow$ (ii): For $x \in X_0$, we compute the strong derivative and use the resolvent equation (A.1) to get

$$\begin{aligned}&\frac{d}{ds}\big\{T^{(\lambda)}(t - s)(z - A^{(\lambda)})^{-1}T(s)(z - A)^{-1}x\big\}\\ &= \big(T^{(\lambda)}(t - s)(-A^{(\lambda)})(z - A^{(\lambda)})^{-1}T(s)(z - A)^{-1}x\big)\\ &\qquad + \big(T^{(\lambda)}(t - s)(z - A^{(\lambda)})^{-1}T(s)A(z - A)^{-1}x\big)\\ &= T^{(\lambda)}(t - s)\big[(z - A)^{-1} - (z - A^{(\lambda)})^{-1}\big]T(s)x.\end{aligned}$$

Thus we have, by integrating the above between 0 and t, that for $x \in X_0$,

$$\begin{aligned}&(z - A^{(\lambda)})^{-1}(T^{(\lambda)}(t) - T(t))(z - A)^{-1}\\ &\qquad = \int_0^t T^{(\lambda)}(t - s)[(z - A^{(\lambda)})^{-1} - (z - A)^{-1}]T(s)x\,ds.\end{aligned} \tag{4.11}$$

Using the bound for the semigroup $T^{(\lambda)}$, one has that

$$\sup_{t\in[0,t_0]} \left\|\left(z-A^{(\lambda)}\right)^{-1}\left(T^{(\lambda)}(t)-T(t)\right)(z-A)^{-1}x\right\| \\ \leq Me^{|\beta|t_0}\int_0^{t_0}\left\|\left[(z-A)^{-1}-(z-A^{(\lambda)})^{-1}\right]T(s)x\right\|ds. \quad (4.12)$$

By (iv) and by an application of the Dominated Convergence Theorem, the right hand side of (4.12) converges to zero as $\lambda \longrightarrow 0^+$. By virtue of the facts that $Ran(z-A)^{-1} = \mathcal{D}(A)$, which is dense in X_0, and that the rest of the factors in the left hand side of (4.12) are uniformly bounded with respect to $t \in [0,t_0]$ and $\lambda \in (0,1]$, we arrive at

$$\lim_{\lambda\longrightarrow 0^+}\sup_{t\in[0,t_0]}\left\|(z-A^{(\lambda)})^{-1}(T^{(\lambda)}(t)-T(t))y\right\| = 0,\ \forall y\in X_0. \quad (4.13)$$

Next, note that for $x \in X_0$,

$$\left(T^{(\lambda)}(t)-T(t)\right)(z-A)^{-1}x \\ = T^{(\lambda)}(t)\left[(z-A)^{-1}-(z-A^{(\lambda)})^{-1}\right]x + \left(z-A^{(\lambda)}\right)^{-1}\left[T^{(\lambda)}(t)x-T(t)x\right] \\ + \left[\left(z-A^{(\lambda)}\right)^{-1}-(z-A)^{-1}\right]T(t)x. \quad (4.14)$$

Applying (iv) to the first term in (4.14) and (4.13) to the second term of (4.14) leads to the conclusion that they converge strongly to zero as $\lambda \longrightarrow 0^+$, uniformly for $t \in [0,t_0]$.

It is clear that by (iv) the third term in (4.14) converges to zero for every t as $\lambda \to 0^+$. To show that this convergence is uniform with respect to $t \in [0,t_0]$ we proceed as follows. For $x \in X_0$ and any $\varepsilon > 0$, choose $y \in \mathcal{D}(A) \subseteq X_0$ such that $\|x-y\| < \varepsilon e^{-|\beta|t_0}(\operatorname{Re} z-\beta)/8M^2$ and then note that, using (2.7),

$$\left\|\left[\left(z-A^{(\lambda)}\right)^{-1}-(z-A)^{-1}\right]T(t)x\right\| \\ \leq 2M^2e^{|\beta|t_0}(\operatorname{Re} z-\beta)^{-1}\|x-y\| \\ + \int_0^{t_0}\left\|\left[\left(z-A^{(\lambda)}\right)^{-1}-(z-A)^{-1}\right]T(s)Ay\right\|ds \\ + \left\|\left[\left(z-A^{(\lambda)}\right)^{-1}-(z-A)^{-1}\right]y\right\|$$

which can be made less than ε by choosing $\lambda > 0$ sufficiently small, uniformly in $t \in [0,t_0]$. Thus combining the above considerations along with (4.14), we get that $\lim_{\lambda\longrightarrow 0^+}\sup_{t\in[0,t_0]}\left\|(T^{(\lambda)}(t)-T(t))(z-A)^{-1}x\right\| = 0$, and the property (ii) follows by the density of $\mathcal{D}(A)$ in X_0. □

Corollary 4.3.2. Let $\{T^{(\lambda)}(t)\}_{t\geq 0}$ and $\{T(t)\}_{t\geq 0}$ be two C_0-semigroups in a Banach space X satisfying the bounds for all $\lambda \in (0,1]$ as in the previous theorem. Suppose furthermore that $\mathcal{D}$ is a core of A, such that $A^{(\lambda)}x \longrightarrow Ax$ as $\lambda \to 0^+ \quad \forall x \in \mathcal{D}$. Then $T^{(\lambda)}(t)$ converges strongly to $T(t)$ as $\lambda \longrightarrow 0^+$, uniformly for t in any compact subset of $[0,\infty)$.

Proof. In the statement (v) of the previous theorem, choose $x^{(\lambda)} = x \in X = X_0 \quad \forall \lambda \in (0,1]$. The equivalent statement (ii) gives the result. □

The next theorem is a variation on the above corollary in that we just need the strong convergence of $(z_0 - A^{(\lambda)})^{-1}$ to an operator which is injective with dense range for some z_0 with $\operatorname{Re} z_0 > \beta$.

Theorem 4.3.3. Let $\{T^{(\lambda)}(t)\}_{t\geq 0}$ for $\lambda \in (0,1]$ be a family of C_0-semigroups with generators $A^{(\lambda)}$, satisfying $\|T^{(\lambda)}(t)\| \leq Me^{\beta t}$ for some real number β and $M > 0$, uniformly in λ. Assume furthermore that for some $z_0 \in \mathbb{C}$ with $\operatorname{Re} z_0 > \beta$, the family $(z_0 - A^{(\lambda)})^{-1}$ converges strongly as $\lambda \longrightarrow 0^+$ to a bounded operator K such that K is injective and the range of K is dense in X. Then there exists a unique C_0-semigroup $\{T(t)\}_{t\geq 0}$ with generator A such that $K = (z_0 - A)^{-1}$ and $T^{(\lambda)}(t)$ converges strongly to $T(t)$ as $\lambda \longrightarrow 0^+$, uniformly for t in compact subsets of $[0,\infty)$.

Proof. Without loss of generality, set $\beta = 0$. First, we need to extend the strong convergence to an open set in $\mathbb{C}$. For this let

$$S = \big\{ z \in \mathbb{C} : |z - z_0| < M^{-1} \operatorname{Re} z_0 \big\}.$$

Then by the assumption that $\|T^{(\lambda)}(t)\| \leq M \ \forall t$, one has that

$$\big\| \big(z_0 - A^{(\lambda)}\big)^{-1} \big\| \leq M (\operatorname{Re} z_0)^{-1}.$$

This implies, therefore, that the series $\sum_{m=0}^{\infty} (z_0 - z)^m \big(z_0 - A^{(\lambda)}\big)^{-m}$ converges in operator norm and defines the operator $\big[I + (z - z_0)\big(z_0 - A^{(\lambda)}\big)^{-1}\big]^{-1}$ in $\mathcal{B}(X)$ for all $z \in S$. It follows from the relation

$$z - A^{(\lambda)} = (z_0 - A^{(\lambda)})\big[I + (z - z_0)(z_0 - A^{(\lambda)})^{-1}\big]$$

that the operator $z - A^{(\lambda)}$ is injective and that its range equals X, since the same is true for $z_0 - A^{(\lambda)}$. Hence $S \subseteq \rho(A^{(\lambda)})$ and

$$\begin{aligned}\big(z - A^{(\lambda)}\big)^{-1} &= \big(z_0 - A^{(\lambda)}\big)^{-1}\Big[I + (z - z_0)\big(z_0 - A^{(\lambda)}\big)^{-1}\Big]^{-1} \\ &= \sum_{m=0}^{\infty}(z_0 - z)^m\big(z_0 - A^{(\lambda)}\big)^{-(m+1)}. \qquad (4.15)\end{aligned}$$

Since

$$\big\|Kx\big\| = \lim_{\lambda\to 0^+}\big\|(z_0 - A^{(\lambda)})^{-1}x\big\| \le M(\operatorname{Re} z_0)^{-1}\big\|x\big\|,$$

it follows that $\big(z_0 - A^{(\lambda)}\big)^{-(m+1)}$ converges strongly to K^{m+1} as $\lambda \longrightarrow 0^+$ and

$$\big\|\big(z_0 - A^{(\lambda)}\big)^{-(m+1)} - K^{m+1}\big\| \le 2[M(\operatorname{Re} z_0)^{-1}]^{(m+1)}.$$

An application of the Dominated Convergence Theorem to (4.15) yields that for $z \in S$, $\big(z - A^{(\lambda)}\big)^{-1}$ converges strongly as $\lambda \longrightarrow 0^+$ to

$$\sum_{m=0}^{\infty}(z_0 - z)^m K^{m+1} := R(z),$$

the series converging in operator norm. Next note that for any $z, z' \in S$,

$$R(z) - R(z') = (z' - z)R(z)R(z') = (z' - z)R(z')R(z),$$

that is, $R(z)$ satisfies the resolvent equation (A.1) since for every λ, $\big(z - A^{(\lambda)}\big)^{-1}$ does the same. From this, it follows that

$$R(z) = R(z')[I + (z' - z)R(z)]$$

showing that $Ran\, R(z) \subseteq Ran\, R(z')$ and by symmetry between z and z' one has that $Ran\, R(z) = Ran\, R(z')$, that is, the range of $R(z)$ is independent of $z \in S$. Therefore $Ran\, R(z) = Ran\, K$, which is dense in X by hypothesis. Now we define a linear operator A in X as follows. Set its domain $\mathcal{D}(A) = Ran\, K$ and

$$Ax = z_0 x - K^{-1}x \;\forall\; x \in \mathcal{D}(A). \qquad (4.16)$$

Since K is injective and has dense range by hypothesis, A is a densely defined linear operator. Furthermore, from the definition of A it follows that

$$(z_0 - A)Kx = x, \text{ for all } x \in X \text{ and } K(z_0 - A)x = x \text{ for all } x \in \mathcal{D}(A),$$

and therefore $z_0 \in \rho(A)$ and $K = (z_0 - A)^{-1}$. This and the definition of $R(z)$ gives that

$$S \subseteq \rho(A) \text{ and } R(z) = (z - A)^{-1} \text{ for all } z \in S.$$

Since $M \geq 1$, there is z in S such that $\operatorname{Re} z > (1 - M^{-1}) \operatorname{Re} z_0 \geq 0$ for which one gets from (4.15) that $(z - A^{(\lambda)})^{-1}$ converges strongly to $(z - A)^{-1}$ as $\lambda \to 0^+$. Thus we conclude that $\{z : Re\, z > 0\} \subseteq \rho(A)$. Also, since $\|(z - A^{(\lambda)})^{-1}\|$ is bounded uniformly in λ for every z with $Re\, z > 0$, it follows that $(z - A^{(\lambda)})^{-m}$ converges strongly as $\lambda \to 0^+$ to $(z - A)^{-m}$ for every $m \geq 1$ and thus

$$\left\|(z - A)^{-m}\right\| \leq \liminf_{\lambda} \left\|\left(z - A^{(\lambda)}\right)^{-m}\right\| \leq M(\operatorname{Re} z)^{-m}.$$

Therefore, in order to apply Theorem 2.3.1 to show that A is the generator of a C_0-semigroup $\{T(t)\}_{t \geq 0}$ in X, it is left only to show that A is closed. For that let $y_n \in \mathcal{D}(A) \longrightarrow y$ such that $Ay_n \longrightarrow u$. By the definition of $\mathcal{D}(A)$ and (4.16), there exists a sequence $\{x_n\} \subset X$ such that $y_n = Kx_n$ for every n and in such a case $AKx_n = z_0 Kx_n - x_n$ converges to u as $n \longrightarrow \infty$. This implies that $x_n = z_0 y_n - Ay_n \longrightarrow z_0 y - u$ by hypothesis and therefore $Kx_n \longrightarrow K(z_0 y - u)$ because $K \in \mathcal{B}(X)$. Since $y_n \longrightarrow y$, one gets that $y = z_0 Ky - Ku$. This shows that

$$y \in Ran\, K = \mathcal{D}(A) \text{ and } u = z_0 y - K^{-1} y = Ay.$$

The conclusion of the theorem follows now by applying the Theorem 4.3.1. □

Further applications of many of the results in this chapter can be seen in chapters 5 through 7.

Chapter 5

Chernoff's Theorem and its applications

In this chapter, a very interesting theorem, due to Chernoff [4], is proven and some of its applications, viz. the Trotter-Kato Product Formula, the Feynman-Kac Formula and the Central Limit Theorem are given. A proof of the Mean Ergodic Theorem is also given at the end as a further application of the concept of C_0-group and Stone's Theorem.

5.1 Chernoff's Theorem

Theorem 5.1.1. Let X be a Banach space and let $F : [0, \infty) \longrightarrow \mathcal{B}(X)$ be a map satisfying $\|F(t)\| \leq 1$ for all t and $F(0) = I$. Suppose furthermore that $F'(0)$, the strong derivative of F at 0 exists on a dense set $\mathcal{D} \subseteq X$ and that $A \equiv \overline{F'(0)|_{\mathcal{D}}}$ is the generator of a contraction C_0-semigroup $\{T(t)\}_{t\geq 0}$. Then the sequence $\left\{\left[F\left(\frac{t}{n}\right)\right]^n\right\}_n$ converges strongly to $T(t)$, uniformly for t in compact subsets of $[0, \infty)$.

We start with a lemma.

Lemma 5.1.2. Let L be a contraction in X. Then for every $n \in \mathbb{N}$, and every $x \in X$,

$$\left\|\left[e^{n(L-I)} - L^n\right]x\right\| \leq \sqrt{n}\left\|(L - I)x\right\|.$$

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7_5

Proof. Note that for $x \in X$, $\big(e^{n(L-I)} - L^n\big)x = e^{-n} \sum\limits_{r=0}^{\infty} \frac{n^r}{r!}\big(L^r - L^n\big)x$ and thus

$$\begin{aligned}
\big\|\big(e^{n(L-I)} - L^n\big)x\big\| &\leq e^{-n} \sum_{r=0}^{\infty} \frac{n^r}{r!}\big\|L^{|r-n|}x - x\big\| \\
&\leq e^{-n} \sum_{r=0}^{\infty} \frac{n^r}{r!}\Big\| \sum_{j=0}^{|r-n|-1} \big(L^{j+1}x - L^j x\big)\Big\| \\
&\leq \Big(e^{-n} \sum_{r=0}^{\infty} \frac{n^r}{r!}|r-n|\Big)\big\|Lx - x\big\| \\
&\leq \sqrt{n}\big\|(L-I)x\big\|.
\end{aligned}$$

The last inequality above is obtained by using the result that

$$\Big(e^{-n} \sum_{r=0}^{\infty} \frac{n^r}{r!}|r-n|\Big)^2 \leq e^{-n} \sum_{r=0}^{\infty} \frac{n^r}{r!}(r-n)^2 = n,$$

which follows by an application of Cauchy's inequality with respect to the Poisson probability distribution $\big\{\frac{e^{-n}n^r}{r!}\big\}_{r=0,1,2,\ldots}$. □

Proof of Theorem 5.1.1. Note that for $t = 0, F(\frac{t}{n}) = F(0) = I$ and $T(0) = I$ as well and therefore, we can assume without loss of generality that $t > 0$. Set $A_n = n\big[F(\frac{t}{n}) - I\big]$ for a fixed $t > 0$ and for $n = 1, 2, \ldots$. Then $\big\|A_n\big\| \leq 2n$ and if we set for $s \in [0, \infty)$, $T^{(n)}(s) = \exp(sA_n)$, it is clear that $\{T^{(n)}(s)\}_{s\geq 0}$ is a C_0-semigroup, continuous in operator norm. From the contractivity property of $F(t)$, it follows that $\big\|T^{(n)}(s)\big\| \leq 1$ for all s and n. Since by hypothesis, $A_n x$ converges strongly to tAx as $n \longrightarrow \infty$ for any $x \in \mathcal{D}$, and $\mathcal{D}$ is a core (see Appendix A.1) for the generator tA of a C_0 contraction semigroup $\{T(st)\}_{s\geq 0}$ for fixed $t > 0$, it follows by Corollary 4.3.2 that $T^{(n)}(s)$ converges strongly to $T(st)$ as $n \longrightarrow \infty$ for the fixed $t > 0$, uniformly for s in compact subsets of $[0, \infty)$. On the other hand, by setting $L = F(\frac{t}{n})$ in Lemma 5.1.2, we get that, $n\big(L - I\big) = A_n$ and that for $x \in \mathcal{D}$

$$\begin{aligned}
\big\|\big[T^{(n)}(1) - F\big(\tfrac{t}{n}\big)^n\big]x\big\| = \big\|\big(e^{n(L-1)} - L^n\big)x\big\| &\leq \sqrt{n}\big\|\big(F\big(\tfrac{t}{n}\big) - I\big)x\big\| \\
&\leq tn^{-\frac{1}{2}}\Big\|\tfrac{n}{t}\big(F\big(\tfrac{t}{n}\big) - I\big)x\Big\| \\
&\leq t_0 n^{-\frac{1}{2}}\Big\|\tfrac{n}{t}\big(F\big(\tfrac{t}{n}\big) - I\big)x\big\|, \text{ for } 0 < t \leq t_0.
\end{aligned}$$

Since for $x \in \mathcal{D}$, $F'(0)x$ exists as a strong derivative, it follows that

$$\frac{n}{t}\big(F\big(\tfrac{t}{n}\big) - I\big)x \text{ converges to } Ax \text{ as } n \longrightarrow \infty.$$

Hence, for all $\epsilon > 0$, there exists $\delta > 0$ such that

$$\left\| \tfrac{n}{t}\big(F(\tfrac{t}{n}) - I\big)x - Ax \right\| \leq \epsilon \quad \text{if } 0 < t/n < \delta$$

whereas

$$\left\| \tfrac{n}{t}\big(F(\tfrac{t}{n}) - I\big)x \right\| \leq 2\tfrac{n}{t}\|x\| \leq 2\delta^{-1}\|x\| \quad \text{if } t/n \geq \delta.$$

Thus,

$$\sup_n \sup_{t\in[\delta,t_0]} \left\| \frac{n}{t}\big(F(\frac{t}{n}) - I\big)x \right\| \leq \max\{\epsilon + \|Ax\|, 2\delta^{-1}\|x\|\}.$$

This, combined with the earlier observation that $T^{(n)}(1)$ converges to $T(t)$, implies that for any $x \in \mathcal{D}$, $[F(\frac{t}{n})]^n x$ converges strongly to $T(t)x$ as $n \longrightarrow \infty$, uniformly for $t \in [0, t_0]$. Finally, the density of $\mathcal{D}$ in X and the facts that both $F(t)$ and $T(t)$ are contractions allow the extension of this convergence, uniformly for $t \in [0, t_0]$, to the whole space X. □

The proof of the following result uses Chernoff's Theorem 5.1.1.

Theorem 5.1.3 (**Trotter-Kato Product**). Let $\{T(t)\}_{t\geq 0}$ and $\{S(t)\}_{t\geq 0}$ be two contractive C_0-semigroups with generators A and B respectively. Assume furthermore that $\mathcal{D}(A) \cap \mathcal{D}(B) \equiv \mathcal{D}$ is dense in X and that $\overline{(A+B)|_{\mathcal{D}}}$ is the generator of a C_0-semigroup $\{Z(t)\}_{t\geq 0}$. Then the sequence $(T(t/n)S(t/n))^n$ converges strongly to $Z(t)$ for each $t \geq 0$ and the convergence is uniform for t in any compact subset of $[0, \infty)$.

Proof. Define $F : [0, \infty) \to \{$ contractions in $X\}$ by $F(t) = T(t)S(t)$ for all $t \geq 0$. Then $F(0) = I$ and for any $x \in \mathcal{D}$,

$$\begin{aligned}
&\left\| \Big(\frac{F(t) - I}{t}\Big)x - (A+B)x \right\| \\
&\leq \left\| T(t)\Big(\frac{S(t) - I}{t} - B\Big)x \right\| + \|(T(t) - I)Bx\| + \left\| \Big(\frac{T(t) - I}{t} - A\Big)x \right\| \\
&\longrightarrow 0 \text{ as } t \longrightarrow 0,
\end{aligned}$$

that is, $F'(0)x$ exists for every $x \in \mathcal{D}$ and equals $(A + B)x$. By hypothesis, $\overline{(A+B)|\mathcal{D}}$ is the generator of a C_0-semigroup $\{Z(t)\}_{t\geq 0}$ and therefore by Chernoff's Theorem,

$$s - \lim_{n\longrightarrow\infty} (T(t/n)S(t/n))^n = s - \lim_{n\longrightarrow\infty} F(t/n)^n = Z(t),$$

uniformly for t in compact subsets of $[0, \infty)$. □

Remark 5.1.4. 1. Theorem 5.1.3 subsumes the classical Lie Product Formula which motivated the above generalisation. The Lie Product theorem is as follows: let A and B be two complex $n \times n$ matrices. Then $(e^{tA/n}e^{tA/n})^n$ converges to $e^{t(A+B)}$ as $n \to \infty$, uniformly for t in compact subsets of $[0, \infty)$.

Note that even if the semigroups concerned are not necessarily contractive, but their generators are bounded operators in X, we can work with their contractive modifications, viz. $e^{t(A-\|A\|)}$ and $e^{t(B-\|B\|)}$ respectively, to get the limiting semigroup to be $e^{t(A+B-(\|A\|+\|B\|))}$, and thereby to the same result after an application of Theorem 5.1.3.

2. There are several further generalisations of Theorem 5.1.3. We state here only one of them without proof and refer the reader to the original source [14]. For definitions and explanations of the terms used in Theorem 5.1.5 and a sketch of the proof, the reader is referred to [21] and the article of [14].

Theorem 5.1.5. Let $\{T(t)\}_{t\geq 0}$ and $\{S(t)\}_{t\geq 0}$ be two holomorphic semigroups with generators A and B respectively, in a Hilbert space $\mathcal{H}$, so that $-A$ and $-B$ are m-sectorial, with associated forms a and b. Assume furthermore that $\mathcal{D}(a) \cap \mathcal{D}(b)$ is dense in $\mathcal{H}$ and such that the associated sectorial form $a + b$ is closed. Then the Trotter-Kato Product Formula is valid, that is, $s - \lim_{n\longrightarrow\infty}(T(t/n)S(t/n))^n = Z(t)$, uniformly for t in compact subset of $[0, \infty)$, where $\{Z(t)\}_{t\geq 0}$ is the holomorphic semigroup generated by C, with $-C$ the m-sectorial operator associated with the form $a + b$.

5.2 Applications of the Trotter-Kato Product Formula and of Chernoff's Theorem

Example 5.2.1. let $\mathcal{H} = L^2(\mathbb{R})$ and let V be a real-valued measurable function with $V = V_1 + V_2$ where $V_1 \in L^2(\mathbb{R})$ and V_2 is bounded. Then M_V, the operator of multiplication by the function V, is selfadjoint (see Theorem A.1.13) and we set $(T(t)f)(x) = e^{itV(x)}f(x)$ for $f \in \mathcal{H}$ to define $\{T(t)\}_{t\in\mathbb{R}}$ as a unitary C_0-group with generator $A = iM_V$. Let $\{S(t)\}_{t\in\mathbb{R}}$ be the translation group in $\mathcal{H}$ given by $(S(t)f)(x) = f(x + t)$ for all $f \in \mathcal{H}$ and $x, t \in \mathbb{R}$. The generator B of

the group $\{S(t)\}_{t\in\mathbb{R}}$ is easily seen to be $B = iP$, where $P = \overline{\big(-i\frac{d}{dx}\big)|_{\mathcal{S}(\mathbb{R})}}$ and $\mathcal{S}(\mathbb{R})$ is the class of smooth functions of rapid decrease (see Appendix A.2). Next we note that $A+B = i(P+M_V)$ and we claim that $P+M_V$ is selfadjoint on $\mathcal{D}(P)$.

This is because if $f \in \mathcal{D}(P) \subseteq L^2(\mathbb{R})$, then f is bounded, uniformly continuous and converging to 0 as $|x| \longrightarrow \infty$ (see Remark A.2.6(d)) and it follows that $\mathcal{D}(P) \subset \mathcal{D}(M_V)$. Furthermore, for $x \in \mathcal{D}(P)$,

$$\|M_V x\| \leq \|M_{V_1}(P+in)^{-1}\|\|Px\| + (\|V_2\|_\infty + n\|M_{V_1}(P+in)^{-1}\|)\|x\|$$

and since

$$\begin{aligned}\|M_{V_1}(P+in)^{-1}\| &\leq (2\pi)^{-1/2}\|V_1\|_2\big(\int_{\mathbb{R}} |k+in|^{-2}\,dk\big)^{1/2}\\ &\leq (2n)^{-1/2}\|V_1)\|_2 \to 0 \text{ as } n\to\infty,\end{aligned}$$

it follows that M_V has P-bound 0 (see Definition 4.1.2). Thus, by Theorem 4.2.6, $P + M_V$ is selfadjoint and we apply the Trotter-Kato Product Formula to get that

$$Z(t) \equiv s - \lim_{n\longrightarrow\infty}(T(t/n)\ S(t/n))^n,$$

where $\{Z(t)\}_{t\in\mathbb{R}}$ is the unitary group generated by $i(P + M_V)$. A simple computation yields that for any $f \in L^2(\mathbb{R})$ and $t \in \mathbb{R}$,

$$\big(\big(T_{t/n}S_{t/n}\big)^n f\big)(x) = \exp\left(i\sum_{j=0}^{n-1}\Big(\frac{t}{n}\Big)V\Big(x+\frac{jt}{n}\Big)\right)f(x+t), \tag{5.1}$$

for almost all $x \in \mathbb{R}$. We claim that for our choice of the function V, the right hand side of (5.1) converges as $n \to \infty$ to $\exp\Big(i\int_0^t V(x+s)\,ds\Big)f(x+t)$. For this, we note that for V in the class mentioned above one can find a sequence $\{V_n\}$ of real finitely valued functions approximating V in the $L^1_{\text{loc}}(\mathbb{R})$ topology and therefore it suffices to show that for a bounded interval $\triangle \subseteq \mathbb{R}$,

$$\int dx\Big|\sum_{j=0}^{n-1}\int_{tj/n}^{t(j+1)/n}\big[\chi_\triangle\big(x+s\big) - \chi_\triangle\big(x+tj/n\big)\big]ds\Big| \longrightarrow 0 \quad \text{as } n\longrightarrow\infty.$$

Indeed this follows from the fact that for $s \in \big(tj/n, t(j+1)/n\big)$,

$$\int dx\big|\chi_\triangle\big(x+s\big) - \chi_\triangle\big(x+tj/n\big)\big| = 2\big(s-tj/n\big)$$

and this in its turn implies that $\exp\Big(i\sum_{j=1}^{n}(t/n)V\big(x+jt/n\big)\Big)$ converges to $\exp\Big(i\int_0^t V\big(x+s\big)ds\Big)$ pointwise almost everywhere in x, by choosing a subsequence if necessary. Finally, an application of the Dominated Convergence Theorem to (5.1) shows that $\big(T(t/n)S(t/n)\big)^n$ converges strongly to the map $Z(t)$ as $n\longrightarrow\infty$, where $Z(t)f$ in $L^2(\mathbb{R})$ is given as

$$\big(Z(t)f\big)(x)=\exp\Big(i\int_0^t V\big(x+s\big)ds\Big)f(x+t). \tag{5.2}$$

However, this could also have been obtained simply by observing that if we set

$$\big(Wf\big)(x)=\exp\Big(i\int_0^x V\big(y\big)dy\Big)f(x),$$

then (i) W defines an unitary operator in $L^2(\mathbb{R})$ which leaves $\mathcal{D}(P)$ invariant and (ii) it intertwines between the selfadjoint operators P and $P+M_V$, that is, $PWf=W(P+M_V)f$ for all $f\in\mathcal{D}(P)$. Then $Z(t)=W^*S(t)W$ and a simple computation of the right hand side of this equality on a vector $f\in L^2(\mathbb{R})$ establishes (5.2).

Example 5.2.2. (**Feynman-Kac Formula**) Let $\{T(t)\}_{t\geq 0}$ be the heat semigroup in $L^2(\mathbb{R}^d)$, as described in Example 2.5.5 (see Theorem 2.5.9 in particular). Its generator is $H_0\equiv\frac{1}{2}\Delta$, the Laplacian in $\mathbb{R}^d$, with its domain of selfadjointness

$$\mathcal{D}(H_0)=\Big\{f\in L^2(\mathbb{R}^d):\ \int|k|^4|\hat{f}(k)|^2dk<\infty\Big\},$$

where $\hat{f}$ is the L^2-Fourier transform of f (see Appendix A.2). Further, let $\{S(t)\}_{t\geq 0}$ be the C_0-semigroup given by $\big(S(t)f\big)(x)=e^{-tV(x)}f(x)$ for all $f\in L^2(\mathbb{R}^d), t\in\mathbb{R}$ and $x\in\mathbb{R}^d$, where $V\in L^\infty(\mathbb{R}^d)$ is fixed (see Example 2.5.4). We note that though $\{T(t)\}_{t\geq 0}$ is a contraction semigroup, $\{S(t)\}_{t\geq 0}$ is not. However, that can be easily taken care of by setting

$$(\tilde{S}(t)f)(x)=e^{-t\big(V(x)+\|V\|_\infty\big)}f(x)=e^{-t\|V\|_\infty}(S(t)f)(x),$$

so that the new operator $\tilde{S}(t)$ differs from $S(t)$ by multiplication by a scalar and $\{\tilde{S}(t)\}_{t\geq 0}$ is now a contraction C_0-semigroup. This does not make material change in what is going to follow. In fact, now we can apply Theorem 5.1.3 to get that $Z(t)=s-\lim_{n\longrightarrow\infty}\big(T(t/n)S(t/n)\big)^n$, where $Z(t)$ is a C_0-semigroup with

generator $(H_0 + V)$, which is a selfadjoint operator with domain $\mathcal{D}(H_0)$. We know the kernel of $T(t)$ (see Example 2.5.5): if $f \in L^2(\mathbb{R}^d)$, then

$$\big(T(t)f\big)(x) = \int_{\mathbb{R}^d} K(t; x-y) f(y)\, dy, \text{ with}$$
$$K(t;x) = (2\pi t)^{-d/2} \exp\Big(-\frac{|x|^2}{2t}\Big) \quad \text{for } t > 0. \tag{5.3}$$

One can explicitly compute to get

$$\begin{aligned}
&\big(\big(T_{t/n} S_{t/n}\big)^n f\big)(x) \\
&= \int \dots \int K(t/n; x_1 - x_0) e^{-t/nV(x_1)} \dots K(t/n; x_n - x_{n-1}) \times \\
&\qquad\qquad e^{-t/nV(x_n)} f(x_n)\, dx_1 \dots\, dx_n \\
&= (2\pi t/n)^{-nd/2} \int \dots \int \exp\Big(-\frac{1}{2}\sum_{j=1}^{n} (t/n)\big(\frac{x_j - x_{j-1}}{t/n}\big)^2\Big) \times \\
&\qquad\qquad \exp\Big(-\sum_{j=1}^{n} (t/n) V(x_j)\Big) f(x_n)\, dx_1 \dots\, dx_n,
\end{aligned} \tag{5.4}$$

where we have set $x_0 = x$. Since $f \in L^2(\mathbb{R}^d)$, strong convergence implies convergence pointwise almost everywhere for a subsequence, one expects that for almost all $x \in \mathbb{R}^d$, $\big(e^{-t(H_0+V)} f\big)(x)$ will equal the limit of right hand side of (5.4), as $n \to \infty$ by choosing to enumerate the subsequence appropriately. Here we face a few difficulties in taking the limit:

- $\prod_{j=1}^{n} dx_j$, the product of n Lebesgue measures in $\mathbb{R}^d$ has no reasonable limit as a measure as $n \longrightarrow \infty$;
- the factor $(2\pi t/n)^{-nd/2}$ also does not have any limit;
- if we think of the points x_{j-1}, x_j $(j = 1, 2, \dots)$ as the initial and final points respectively of a piecewise linear path $w : [0, t] \longrightarrow \mathbb{R}^d$, then the expression in the second exponential in (5.4) formally converges to $-\int_0^t V(w(s)) ds$, while that in the first exponential formally converges to $-\frac{1}{2}\int_0^t \Big[\frac{dw}{ds}(s)\Big]^2 ds$.

Next, we note that for most (continuous) paths w, $\dot{w}$ will be quite singular leading to the formal *vanishing* of the first exponential factor in (5.4) while the

$\prod_{j=1}^{n} dx_j$ diverges as $n \longrightarrow \infty$, leaving us with the *tantalising* possibility that the product somehow will be meaningful. This we address next by constructing the Wiener measure $\mathbb{P}$ on the linear space of continuous paths w, making sense of the limit as $n \to \infty$ of the right hand side of (5.4) as

$$\int e^{-\int_0^t V(x+w(s))\,ds} f(x+w(t))\mathbb{P}(dw),$$

where the integration is over the (infinite dimensional) linear vector space $C(\mathbb{R}_+, \mathbb{R}^d)$ of all $\mathbb{R}^d$-valued continuous paths w.

There are a few different ways of constructing the Wiener measure. We shall present here in brief one such method, more probabilistic in spirit, using Kolmogorov's Extension or Consistency Theorem [17, pages 143–144 and 212–224]. For an alternative approach to the construction of the Wiener measure, the reader is referred to [21, pages 277–279].

The expressions (2.29) and the associated heat kernel K in (5.3) and the right hand side of (5.4) admits probabilistic interpretations as follows. For every Borel set B in $\underbrace{\mathbb{R}^d \times \ldots \times \mathbb{R}^d}_{n\text{-fold}}$ define the family of distributions $F_{t_1,t_2,t_3,\ldots,t_n}(B)$ for $0 < t_1 < t_2 < \cdots < t_n < t < \tau < \infty$ as

$$\begin{aligned} &F_{t_1,t_2,t_3,\ldots,t_n}(B) \\ &= \int_B p(0,x;t_1,x_1)p(t_1,x_1;t_2,x_2)\cdots p(t_n,x_n;t,y)\,dx_1\,dx_2\cdots dx_n, \end{aligned} \tag{5.5}$$

where $p(s,x;t,y) = K(t-s;y-x)$. The quantity $F_{t_1,t_2,t_3,\ldots,t_n}(B)$ is the n-dimensional probability distribution that the stochastic process, starting at time 0 at $x \in \mathbb{R}^d$ will arrive at time t at $y \in \mathbb{R}^d$, after having passed through the region B in $\mathbb{R}^{dn}$ at time-sequence $t_1 < t_2 < \ldots < t_n$. The collection

$$\left\{F_{t_1,t_2,\ldots,t_n}(B) : 0 < t_1 < t_2 < \ldots < t_n < t < \tau; B \text{ a Borel set in } (\mathbb{R}^d)^n; n \in \mathbb{N}\right\}$$

is called the set of all finite-dimensional distributions of the stochastic process. It is clear that

(i) $F_{t_1,t_2,\ldots,t_n}(B)$ is increasing if B is increasing;

(ii) $F_{t_1,t_2,,\ldots,t_n}(\cdot)$ is countably additive;

(iii) $F_{t_1,t_2,\ldots,t_n}(\mathbb{R}^{dn}) = p(0,x;t,y)$, which follows from the properties of a convolution semigroup (see Exercise 2.5.8 and Theorem 2.5.7), and

(iv) The distributions $F_{t_1,t_2,\ldots,t_m}$ are consistent. This means that for all Borel sets $E_1, E_2, \ldots E_m$ in $\mathbb{R}^d$,

$$\begin{aligned} F_{t_1,t_2,\ldots,t_m}(E_1 \times \ldots \times E_m) \\ = F_{s_1,s_2,\ldots,s_n}\big(\underbrace{\mathbb{R}^d \times \ldots \times \mathbb{R}^d}_{n_0} \times E_1 \times \underbrace{\mathbb{R}^d \times \ldots \times \mathbb{R}^d}_{n_1} \times E_2 \\ \times \ldots \times E_m \times \underbrace{\mathbb{R}^d \times \ldots \times \mathbb{R}^d}_{n_m}\big), \end{aligned}$$

where $\{t_1, t_2, \ldots, t_m\} \subset \{s_1, s_2, \ldots, s_n\} \subset [0, \tau], m < n$ such that

$$s_1^{(0)} < \ldots < s_{n_0}^{(0)} < t_1 < s_1^{(1)} < \ldots < s_{n_1}^{(1)} < t_2 < \ldots < t_m < s_1^{(m)} < \ldots < s_{n_m}^{(m)}.$$

The notation above means that there are n_0 s_j's before t_1, n_1 s_j's between t_1 and t_2 and so on. As was the case for (iii), the consistency property (iv) of the distributions F can be easily verified for the distribution generated by the heat kernel K. (**Exercise** 5.2.3).

Then by Kolmogorov's Extension Theorem [17, Chap 5, Theorem 5.1] there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P}_x)$ with a unique measure $\mathbb{P}_x$ and a stochastic process
$X\colon \mathbb{R}_+ \times \Omega \to \mathbb{R}^d$ which is $\mathcal{F}$-measurable and such that

$$\mathbb{P}(w \in \Omega : (X(0,w) = x, X(t_1,w), \ldots, X(t_n,w)) \in B) = F_{t_1,t_2,\ldots,t_{n+1}}(B)$$

for all $B \in$ Borel subsets of $(\mathbb{R}^d)^n$. This stochastic process $\{X(t)\}_{t\geq 0}$ is called the **Brownian motion**, starting at $x \in \mathbb{R}^d$, and the associated measure $\mathbb{P}_x$ the **Wiener measure** . If we set $x = 0$ in (5.5), that is, if the process starts at the origin, then $X(0) = 0$ and in such a case $\{X(t)\}_{t\geq 0}$ is the **standard Brownian motion** (S.B.M.)

Further properties of S.B.M. are as follows.

1. $X(0) = 0$,
2. $X_j(t) - X_j(s)$ is distributed as a standard normal distribution $N(0, t-s)$ for $1 \leq j \leq d$ and $0 < s < t$. This can be seen by computing the variance

$$\begin{aligned} \int_{\mathbb{R}} (x_j - y_j)^2 p(s, x_j; t, y_j)\, dy_j &= [2\pi(t-s)]^{-1/2}\Big(\int_{\mathbb{R}} x_j^2 e^{-\frac{x_j^2}{2(t-s)}}\, dx_j\Big) \\ &= t - s. \end{aligned}$$

3. $\{X(t)\}$ is an independent increment process, that is,

$$X(t_1), X(t_2) - X(t_1), X(t_3) - X(t_2), \ldots, X(t_n) - X(t_{n-1})$$

are independent $\mathbb{R}^d$-valued random variables for all $0 \leq t_1 < t_2 < t_3 \ldots < t_{n-1} < t_n$.

4. The Kolmogorov Continuity Criterion [17, Chap 7, Theorem 3.1] states that if there are constants $\alpha, \delta, \kappa > 0$ such that

$$\begin{aligned}\mathbb{E}\big(|X(t) - X(s)|^\alpha\big) &\equiv \int |u - v|^\alpha \, dF_{t,s}(u, v) \\ &\leq K(t - s)^{(1+\delta)} \quad \text{for all } 0 < t < s < \tau,\end{aligned}$$

then there exists a unique measure μ on $C[0, \tau]$ such that all the finite-dimensional distributions associated with μ equal $F_{t_1,t_2,\ldots,t_n}(\cdot)$ from which the measure $\mathbb{P}_x$ was constructed. In the above expression $\mathbb{E}$ is the expectation. It is easy to see (**Exercise** 5.2.4) that if $X(\cdot)$ are the $\mathbb{R}^d$-valued S.B.M's, then for $p \geq 2, 0 \leq s < t$, we have

$$\mathbb{E}\Big(\sum_{j=1}^{d} |X_j(t) - X_j(s)|^2\Big)^{p/2} = (t - s)^{p/2}\big(\mathbb{E}|\xi|^p\big),$$

where ξ is an $\mathbb{R}^d$-$N(0, 1)$ random variable. This implies that the Wiener measure $\mathbb{P}$, constructed to support the stochastic process which is the S.B.M. is actually equivalent to a probability measure on $C[0, \tau]$, and we shall identify the two and call it also the Wiener measure.

In some sense the original probability space is not relevant any more and we can assume that the S.B.M. (or the $\mathbb{R}^d$-valued Wiener process) is the process $\mathbb{R}_+ \times C(\mathbb{R}_+) \mapsto \mathbb{R}^d$ given by $(t, w) \mapsto w(t)$, that is, work with the continuous paths on $\mathbb{R}_+$ and the Wiener measure on them.

Now we are in a position to re-interpret (or rewrite) the expression (5.4) for $\big(\big(T(t/n)S(t/n)\big)^n f\big)(x)$ as

$$\begin{aligned}&= \int_\Omega \exp\Big\{-\sum_{j=1}^{n}(t/n)V(w(tj/n))\Big\} f(w(t))\mathbb{P}_x(dw) \\ &= \int_\Omega \exp\Big\{-\sum_{j=1}^{n}(t/n)V(x + w(tj/n))\Big\} f(x + w(t))\mathbb{P}(dw), \qquad (5.6)\end{aligned}$$

where we have written Ω for $C(\mathbb{R}_+)$. As in the case of Example 5.2.1, one can approximate V by a sequence of uniformly bounded simple functions, pointwise and since for a bounded Borel set $\triangle \subseteq \mathbb{R}^d$,

$$\sum_{j=1}^{n} \int \mathbb{P}(dw) \int_{\mathbb{R}^d} dx \int_{t(j-1)/n}^{tj/n} ds \left| (\chi_{\triangle}(x + w(s)) - \chi_{\triangle}(x + w(tj/n)) \right|$$
$$\leq \sum_{j=1}^{n} \int_{t(j-1)/n}^{tj/n} ds \int \mathbb{P}(dw) \prod_{k=1}^{d} \left| 2\big[w_k(s) - w_k(tj/n)\big] \right|. \tag{5.7}$$

By the Cauchy-Schwartz inequality, the expression on the right hand side of (5.7) is

$$\leq 2^d \sum_{j=1}^{n} \int_{t(j-1)/n}^{tj/n} ds \Big[\int \mathbb{P}(dw) \prod_{k=1}^{d} \left| w_k(s) - w_k(tj/n) \right|^2 \Big]^{1/2}$$
$$= 2^d \sum_{j=1}^{n} \int_{t(j-1)/n}^{tj/n} |s - tj/n|^{d/2} \, ds \longrightarrow 0 \text{ as } n \longrightarrow \infty. \tag{5.8}$$

To arrive at (5.8), we have used the property that for $0 \leq s < t$,

$$\int \mathbb{P}(dw) \prod_{k=1}^{d} \left| w_k(s) - w_k(t) \right|^2 = \Big[\int K(t-s; x) x^2 \, dx \Big]^d = (t-s)^d.$$

Thus it follows that for $V \in L^{\infty}(\mathbb{R}^d)$,

$$\int_{\triangle} \Big| \exp\Big\{ -\sum_{j=1}^{n} (t/n) V(x + w(tj/n)) \Big\} - \exp\Big\{ -\int_0^t V(x + w(s)) ds \Big\} \Big| \, dx$$

converges to 0 as $n \longrightarrow \infty$ for every bounded set $\triangle$ and hence the function inside $|\cdot|$ converges to 0 pointwise for almost all $x \in \mathbb{R}^d$ and for almost all w (if necessary, by choosing a subsequence). Finally, since

$$\Big| \exp\Big\{ -\sum_{j=1}^{n} (t/n) V(x + w(tj/n)) \Big\} \Big| \leq e^{t\|V\|_{\infty}} \text{ for all } x \text{ and } w,$$

an application of the Dominated Convergence Theorem to the right hand side of (5.6) leads to the conclusion that

$$((T(t/n)S(t/n))^n f)(x) \longrightarrow \int_{\Omega} \exp\big\{ -\int_0^t V(x + w(s)) ds \big\} f(x + w(t)) \mathbb{P}(dw)$$
$$= \int_{\Omega} \exp\big\{ -\int_0^t V(w(s)) ds \big\} f(w(t)) \mathbb{P}_x(dw), \tag{5.9}$$

for almost all $x \in \mathbb{R}^d$ as $n \longrightarrow \infty$. Thus,

$$(Z(t)f)(x) = \big(e^{-t(H_0+V)}f\big)(x) = \int_\Omega \exp\Big\{ -\int_0^t V(w(s))ds\Big\} f(w(t))\mathbb{P}_x(dw) \tag{5.10}$$

which is the Feynman-Kac formula.

Remark 5.2.5. Feynman's original idea ([10]) was to look for a description of the solution of the Schrödinger equation (which in quantum mechanics governs the evolution of the wave function, describing the state of the associated quantum mechanical physical system) in terms of all possible continuous paths connecting the initial space time point with the final one (see also Section 7.3). Unlike in the above discussion on the Feynman-Kac Formula, the relevant vehicle there is the unitary group given by $U(t) = \exp\big(-i(H_0+V)t\big)$, and if one tries to write down an expression similar to that in (5.4) it would be

$$(U(t)f)(x) = \lim_{n\longrightarrow\infty}\Big(\frac{2\pi it}{n}\Big)^{-nd/2}\int\ldots\int \exp\Big\{-\frac{i}{2}\sum_{j=1}^{n}(t/n)\Big(\frac{x_j - x_{j-1}}{t/n}\Big)^2\Big\}\times$$
$$\exp\Big\{-i\sum_{j=1}^{n} t/nV(x_j)\Big\}f(x_n)dx_1\ldots dx_n, \tag{5.11}$$

that is, *it* replaces t in (5.4). However, unlike in the earlier case, the *formal limit* of the expression in the right hand side of (5.11) as $n \to \infty$ cannot be made sense of. That is, there does not exist any measure ν_x on $C_x(\mathbb{R}_+)$, the space of continuous functions f on $\mathbb{R}_+$ with $f(0) = x$, such that (5.11) may be reinterpreted as

$$\int_{C_x(\mathbb{R}_+)} \exp\Big\{-i\int_0^t V(w(s))\,ds\Big\} f(w(t))\nu_x(dw),$$

as was possible in the earlier case of the construction of the Wiener measure.

Example 5.2.6 (Central Limit Theorem)**.** Let $\mathcal{X}$ and $\mathcal{Y}$ be two independent real-valued random variables with distributions G and H respectively; it is known (see [18]) that $\mathcal{X}+\mathcal{Y}$ is a real-valued random variable with distribution $G*H$. Explicitly, let

$$Probability\ \{\mathcal{X} \le x\} \equiv Pr\{\mathcal{X} \le x\} = G(x) \text{ and } Pr\{\mathcal{Y} \le y\} = H(y),$$

and then

$$Pr\{\mathcal{X}+\mathcal{Y}\le\alpha\} = Pr\{\mathcal{X}\le x \text{ and } \mathcal{Y}\le y \mid x+y\le\alpha\}$$
$$= \int_{\mathbb{R}} G(\alpha-y)H(dy) \equiv (G*H)(\alpha) = (H*G)(\alpha).$$

Next, let $\{\xi_i\}_{i=1}^n$ be n independent real-valued random variables, each with distribution G, independent of i and with mean 0 and variance 1. Then by a simple extension of the above line of reasoning, one has that for $\alpha\in\mathbb{R}$,

$$Pr\Big\{\frac{1}{\sqrt{n}}\sum_{i=1}^n \xi_i \le \alpha\Big\} = \underbrace{(G_n * G_n * ... * G_n)}_{n\text{-fold}}(\alpha),$$

where $G_n(x) = G(\sqrt{n}x)$. It is natural to set this problem up in some Banach space X, say in $X\equiv C_0(\mathbb{R})$, the Banach space of continuous complex-valued functions vanishing at infinity, equipped with the supremum norm, and to associate with the action of convolution a linear operator on X. For this, we define for $f\in X$ and any fixed distribution function G on $\mathbb{R}$, the linear operator A_G by

$$(A_G f)(x) = \int f(x-y)G(dy). \tag{5.12}$$

Then, one can verify that

(a) A_G is a bounded linear operator in X, in fact $\big\|A_G\big\|\le 1$;

(b) $A_{G_1*G_2} = A_{G_1}A_{G_2} = A_{G_2}A_{G_1} = A_{G_2*G_1}$.

For $t>0$ and $n\in\mathbb{N}$, define $F(t/n) = A_{G_{(n/t)}}$, where $G_{(n/t)}(x) = G(\sqrt{n/t}\,x) = Pr\big\{\xi_1\le\sqrt{n/t}\,x\big\}$. Then note that by the observation (b) above,

$$F(t/n)^n = \big(A_{G_{(n/t)}}\big)^n = A_{G_{(n/t)}*G_{(n/t)}*...*G_{(n/t)}} = A_{Pr\left\{\sqrt{t/n}\sum_{i=1}^n \xi_i\le\cdot\right\}},$$

thereby setting the stage for an application of Theorem 5.1.1. For that we need first to compute $F'(0)$, which is done in the next lemma.

Lemma 5.2.7. Let $F(t) = A_{G_{(1/t)}}$ for all $t>0$, and define an operator $\tilde{\Delta}$ in X by setting

$$\mathcal{D}(\tilde{\Delta}) = \big\{f\in X : f' \text{ and } f''\in X\big\} \text{ and} \tag{5.13}$$
$$(\tilde{\Delta}f)(x) = \frac{1}{2}f''(x) \text{ for any } f\in\mathcal{D}(\tilde{\Delta}) \text{ and } x\in\mathbb{R}. \tag{5.14}$$

Then $F'(0)f = \tilde{\Delta}f$ for $f \in \mathcal{D}(\tilde{\Delta})$, that is,

$$\Big\{\frac{F(t)-I}{t}f - \tilde{\Delta}f\Big\} \longrightarrow 0 \quad \text{as } t \longrightarrow 0^+.$$

Proof. For $t > 0$ and $f \in \mathcal{D}(\tilde{\Delta})$,

$$\begin{aligned} & t^{-1}\big[F(t)f - f - t\tilde{\Delta}f\big](x) \qquad (5.15)\\ & = t^{-1}[A_{G_{(1/t)}}f - f - t\tilde{\Delta}f](x) \\ & = t^{-1}\int_{\mathbb{R}} \Big[f(x-y) - f(x) + yf'(x) - \frac{y^2}{2}f''(x)\Big]G_{(1/t)}\,(dy) \\ & = t^{-1}\int_{\mathbb{R}} \big[f(x-\sqrt{t}u) - f(x) - \sqrt{t}uf'(x) - tu^2/2f''(x)\big]G(dy) \qquad (5.16) \end{aligned}$$

where we have changed variable $y = \sqrt{t}u$ and used the properties that

$$\int yG_{(1/t)}(dy) = \sqrt{t}\int yG(dy) = \sqrt{t}\mathbb{E}(\xi_1) = 0$$

and

$$\int y^2 G_{(1/t)}(dy) = t\int u^2 G(du) = t\mathbb{E}(\xi_1^2) = t.$$

Using Taylor's Theorem, there exists $\theta \in (0,1)$ such that the right hand side of (5.16)

$$\begin{aligned} & = \frac{1}{2}\int_{\mathbb{R}} \big[f''(x-\theta\sqrt{t}u) - f''(x)\big]u^2G(du) \\ & = \frac{1}{2}\int_{|u\sqrt{t}|<\delta} [f''(x-\theta\sqrt{t}u) - f''(x)]u^2G(du) \\ & \quad + \frac{1}{2}\int_{|u\sqrt{t}|\geq\delta} [f''(x-\theta\sqrt{t}u) - f''(x)]u^2G(du) \\ & = I_1(t) + I_2(t). \end{aligned}$$

Since $f'' \in X$, it follows that f'' is uniformly continuous on $\mathbb{R}$. Therefore given $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that $\sup_{x\in\mathbb{R}} \big|f''(x-\theta\sqrt{t}u) - f''(x)\big| < \varepsilon$ if $\big|\theta\sqrt{t}u\big| < |\sqrt{t}u| < \delta$ and therefore

$$\big|I_1(t)\big| < \varepsilon/2\int_{\mathbb{R}} u^2G(du) = \varepsilon/2.$$

Having chosen this δ, now one chooses t_0 with $0 < t_0 < 1$ such that

$$\big|I_2(t)\big| \leq \big\|f''\big\|_\infty \int_{|u|\geq\delta t^{-1/2}} u^2G(du) < \varepsilon/2 \quad \text{for } t < t_0.$$

This completes the proof.

□

Theorem 5.2.8 (**Central Limit Theorem**)**.** Let $\{\xi_i\}_{i\geq 1}$ be a sequence of independent and identically distributed real-valued random variables with mean 0 and variance 1. Then as $n \longrightarrow \infty$, $Pr\{\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\xi_i \leq \alpha\}$ converges to $\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\alpha} e^{-x^2/2}dx$, that is, the random variable $\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\xi_i$ converges to a $N(0,1)$-random variable in distribution.

Proof. As we have already seen, if $F(t/n) = A_{G_{(n/t)}}$ where G is the distribution of each ξ_i, then $F'(0)f = \tilde{\Delta}f$ for $f \in \mathcal{D}(\tilde{\Delta})$, a dense subset of X. Furthermore, we have seen in Chapter 2, Example 2.5.5 that $\tilde{\Delta}$ is the generator of the heat semigroup $T(t)$, a contraction C_0-semigroup. Therefore one can apply Chernoff's Theorem 5.1.1 to get that

$$F(t/n)^n = A_{G_{(n/t)}*\cdots*G_{(n/t)}} = A_{Pr\{\sqrt{t/n}\sum_{i=1}^{n}\xi_i \leq \cdot\}}$$

converges to the heat semigroup $\{T(t)\}_{t\geq 0}$, or equivalently, for any $f \in X$,

$$\big(F(t/n)^n f\big)(x) = \mathbb{E}f\Big(x - \sqrt{t/n}\sum_{i=1}^{n}\xi_i\Big) \longrightarrow \frac{1}{\sqrt{2\pi t}}\int f(x-y)e^{-y^2/2t}\,dy$$

as $n \to \infty$. □

Example 5.2.9 (Mean Ergodic Theorem)**.** Let $(\Omega, \mathcal{F}, \mu)$ be a finite measure space and let ξ be a measure preserving transformation on Ω, that is, $\mu(\xi^{-1}(\triangle)) = \mu(\triangle)$, $\forall \triangle \in \mathcal{F}$. The transformation ξ is said to be **ergodic** if the only measurable sets (up to a set of measure 0) that are invariant under ξ are the whole set Ω and the null set $\emptyset$. This action can be lifted as a map on the linear spaces of complex-valued measurable functions on $(\Omega, \mathcal{F}, \mu)$ by setting $(T_\xi f)(x) = f(\xi(x))$. In particular, if we start with $f \in L^2(\Omega, \mathcal{F}, \mu) \equiv X$, then we observe, that

$$\big\|T_\xi f\big\|^2 = \int_\Omega |f(\xi(x))|^2\mu(dx) = \int_{\xi(\Omega)} |f(x)|^2\mu\cdot\xi^{-1}(dx) \leq \big\|f\big\|^2,$$

that is, T_ξ is a contraction on X. If furthermore the map ξ is surjective, that is, $\xi(\Omega) = \Omega$, then the map T_ξ is an isometry in X. Finally, if Ω is a locally compact topological space and $\mathcal{F}$ is the Borel σ-algebra of Ω, and if we have a one-parameter continuous group $\{\xi_t\}_{t\in\mathbb{R}}$ of measure-preserving transformations on $(\Omega, \mathcal{F}, \mu)$ (that is, $\xi_t^{-1} = \xi_{-t}$, $\xi_0 = id$, $\xi_t\xi_s = \xi_{t+s}$ for all

$t, s \in \mathbb{R}$ and $\xi_t(x) \longrightarrow x \in \Omega$ as $t \longrightarrow 0$), then the family $\{U(t)\}_{t\in\mathbb{R}}$ of operators in X given by $(U(t)f)(x) = f(\xi_t(x))\ \forall\ f \in X$ and $x \in \Omega$, is a one-parameter strongly continuous group of unitaries in X. The strong continuity at $t = 0$ follows since

$$\int |((U(t) - I)f)(x)|^2 \mu(dx) = \int |f(\xi_t(x)) - f(x)|^2 \mu(dx) \longrightarrow 0 \quad \text{as } t \longrightarrow 0$$

by an application of the Dominated Convergence Theorem with simple functions f. Since the subspace of simple functions is dense in $L^2(\Omega, \mu)$, the above convergence may be extended to all of $L^2(\Omega, \mu)$. By Stone's theorem (Theorem 3.2), one has iA as the generator of $U(t)$, for some selfadjoint operator A in X.

Next, the Mean Ergodic Theorem is stated and proven.

Theorem 5.2.10 (**Mean Ergodic Theorem**)**.** Let $\{\xi_t\}_{t\in\mathbb{R}}$ be a one-parameter continuous group of measure-preserving, ergodic transformations on $(\Omega, \mathcal{F}, \mu)$ and let $X = L^2(\Omega, \mu)$. Also, let $\{U(t)\}_{t\in\mathbb{R}}$ be the associated strongly continuous unitary group in X with iA as its generator. Then for every $f \in X$, *the time average of* f under the evolution $\{U(t)\}_{t\in\mathbb{R}}$, the strong limit

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} U(s)f\, ds$$

exists and equals Pf, where P is the projection onto $\mathcal{N}(A)$, the null space of A. Furthermore, the subspace $P(X)$ consists of functions which are constant μ -almost everywhere and $(Pf)(x) = [\mu(\Omega)]^{-1}\Big[\int f(y)\mu(dy)\Big]$ (*the space average*) $\forall\ f \in X$.

Proof. First of all, let us make the simple observation that

$$f \in \mathcal{N}(A) \equiv \big\{f \in D(A) \subseteq X : Af = 0\big\} \text{ if and only if } U(t)f = f \text{ for all } t \in \mathbb{R}.$$

If $f \in \mathcal{N}(A)$ or equivalently if $U(t)f = f$, then the group property implies that given any $x \in \Omega$ there exists $y \in \Omega$ such that $\xi_t y = x$ and hence $f(y) = (U(t)f)(y) = f(\xi_t(y)) = f(x)$. This means that such an f is constant μ-almost everywhere and also that for any $g \in X$,

$$\int (Pg)(x)\mu(dx) = (Pg)(x)\mu(\Omega) = \langle Pg, 1\rangle = \langle g, P1\rangle = \langle g, 1\rangle = \int g(x)\mu(dx),$$

since $(U(t)1)(x) = 1($ that is $1(\xi_t(x)) = 1\ \forall t$, for almost all $x \in \Omega)$. It is easy to see that if $g \equiv \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} U(s)f ds$ exists, then $U(t)g = g$. In fact,

$$\begin{aligned} U(t)g &= \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} U(t+s)f ds = \lim_{T\to\infty} \frac{1}{2T}\int_{-T+t}^{T+t} U(s)f\, ds \\ &= \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} U(s)f\, ds + \lim_{T\to\infty} \frac{1}{2T}\int_{T}^{T+t} U(s)f\, ds \\ &\quad + \lim_{T\to\infty} \frac{1}{2T}\int_{-T+t}^{-T} U(s)f ds = g, \end{aligned}$$

since the second and the third limits vanish. This leads to the conclusion that $g \in \mathcal{N}(A)$, and furthermore,

$$g = Pg = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} PU(s)f\, ds = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} Pf\, ds = Pf.$$

For the penultimate equality, note that $U(t)P = P$ for all $t \in \mathbb{R}$ so that $PU(t) = P$ for all $t \in \mathbb{R}$ on taking adjoints. Hence, for every $f \in X$, if $\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} U(s)f\, ds \equiv g$ exists, then $g = Pf$, and by the earlier discussions,

$$g(x) = (Pf)(x) = [\mu(\Omega)]^{-1}\Big[\int f(y)\mu(dy)\Big].$$

Thus to complete the chain of reasoning, we have to prove the existence of the above-mentioned limit g. For this, note first that $X = \mathcal{N}(A) \oplus \overline{Ran(A)}$ since A is selfadjoint. For arbitrary$f \in X$, set $f = Pf + \tilde{f}$, with $\tilde{f} \in \overline{Ran(A)}$, and note that $\frac{1}{2T}\int_{-T}^{T} U(s)Pf\, ds = Pf$. On the other hand, if $h \in Ran(A)$, that is, $h = Au$ for some $u \in \mathcal{D}(A)$, then

$$\begin{aligned} \frac{1}{2T}\int_{-T}^{T} U(s)h\, ds &= \frac{1}{2T}\int_{-T}^{T} U(s)Au\, ds = -\frac{i}{2T}\int_{-T}^{T} \frac{d}{ds}U(s)u\, ds \\ &= -i\Big[\frac{U(T)u - U(-T)u}{2T}\Big] \longrightarrow 0 \quad \text{as } T \to \infty. \end{aligned}$$

For $\tilde{f} \in \overline{Ran(A)}$ and any $\epsilon > 0$, one can find $h \in Ran(A)$, such that $\|\tilde{f} - h\| < \epsilon$, and one has that

$$\begin{aligned} \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} U(s)\tilde{f}\, ds &= \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} U(s)(\tilde{f} - h)\, ds \\ &\quad + \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} U(s)h\, ds \\ &= \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} U(s)(\tilde{f} - h)\, ds, \end{aligned}$$

which implies that

$$\Big\| \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} U(s)\tilde{f} \Big\| \leq \Big\| \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} U(s)(\tilde{f} - h)\, ds \Big\| < \epsilon.$$

Since $\epsilon > 0$ is arbitrary, we conclude that

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} U(s)\tilde{f} = 0.$$

□

Chapter 6

Markov semigroups

In Chapter 4, we have seen methods of constructing a semigroup by perturbing a known (or given) one, where the perturbation is *small* in a certain sense. More precisely, the context were those in which the *candidate* for the generator of the new semigroup was obtained by a small additive perturbation of the generator of the known semigroup. On the other hand, in many applications, particularly in the theory of probability and stochastic processes, situations arise when the perturbation is comparable (*not small*) to the generator of the known semigroup and in such cases, the theory studied in Chapter 4 is not applicable. However, often this difficulty can be circumvented by exploiting the property of *positivity* or *positivity-preserving* of the semigroup (and of the resolvents of the generator involved.) We shall concern ourselves with such situations in this chapter, mostly restricting ourselves to a concrete model of the construction of a Markov semigroup, following Kato [13].

6.1 Probability and Markov semigroups

Any theory of classical probability starts with a state space $(\Omega, \mathcal{F}, \mu)$ where Ω is a measure space, $\mathcal{F}$ the σ-algebra of measurable subsets (events) of Ω and μ is a probability measure on $\mathcal{F}$, that is, $\mu\colon \mathcal{F} \to [0,1]$ is a countably additive map such that

$$0 \leq \mu(F) \leq 1 \quad \forall F \in \mathcal{F}, \mu(\emptyset) = 0 \text{ and } \mu(\Omega) = 1.$$

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7_6

The random variables are measurable maps or functions (real, complex or more generally vector-space valued) $f\colon \Omega \to X$, and their X- valued expectation is given by

$$\mathbb{E}(f) = \int f(w)\mu(d\,w).$$

In fact, the linear space of such functions (called random variables) and the associated probability measure are in dual relationship. More precisely, for example, if $X = \mathbb{C}$ and $\Omega = [0,1]$, equipped with the standard Borel structure, and the space of random variables on Ω is $C[0,1]$, then the Riesz-Markov Theorem [24, Theorem 6.19], tells us that every normalised positive linear functional ϕ on $C[0,1]$ is given by a regular probability measure μ_ϕ such that

$$\phi(f) = \int_{[0,1]} f(x)\mu_\phi(d\,x)$$

and conversely, given a probability measure μ on any probability space Ω, the above integral of the $\mathbb{C}$-valued random variables on Ω is a positive linear functional on the space. This kind of *duality* between functions (or random variables) and the functionals on them (or measures or states) will be exploited considerably in the sequel.

Given a probability space $\big(\Omega, \digamma, \mu\big)$, or the linear space X of $\mathbb{R}$-valued random variables on it, let us now consider a map $T\colon X \to X$ with the properties that

- $Tf \in X_+$ (**the cone of positive functions** in X) if $f \in X_+$, and
- $T(1) = 1$ (Markov map) or $T(1) \leq 1$ (semi-Markov map).

For further mathematical analysis we shall have to put some topology on X, for example, the weak-$*$ topology on $L^\infty_{\mathbb{R}}(\Omega, \digamma, \mu)$, the real $L^\infty(\Omega, \digamma, \mu)$ space, induced by its pre-dual $X_* =$ real $L^1(\Omega, \digamma, \mu) := L^1_{\mathbb{R}}(\Omega, \digamma, \mu)$, or in a simplified model where Ω is discrete, enumerated by $\mathbb{N}$, say, $X = l^\infty_{\mathbb{R}}$ and $X_* = l^1_{\mathbb{R}}$, the Banach spaces of real bounded and real summable sequences respectively. This is the set up where we shall do the analysis and we shall be concerned with a family or a semigroup of *Markov maps*:

$$T(t)\colon X \to X, \quad \forall\ t \geq 0 \text{ such that}$$

the following conditions hold.

1. $\{T(t)\}_{t\geq 0}$ is a C_0-semigroup on X with respect to the locally convex weak-$*$ topology on X induced by X_*, that is,
   (a) $T(0)f = f$ and $T(t)T(s)f = T(t+s)f$, $\forall\ t, s \geq 0$ and $f \in X$,
   (b) $\langle g, T(t)f\rangle = \int g(\omega)\overline{(T(t)f)(\omega)}\,\mu(d\omega) \to \int g(\omega)\overline{f(\omega)}\mu(d\omega)$ as $t \to 0^+$ for all $f \in X$, and $g \in X_*$.
   (c) However, note that weak-$*$ continuity of the semigroup $\{T(t)\}_{t\geq 0}$ on X is equivalent to the weak continuity of its pre-dual semigroup $T_*(t)$ on X_*. This, by Corollary 1.5.6 implies the strong continuity of $T_*(t)$ in X_*, making it a C_0-semigroup (verify (**Exercise** 6.1.1)) in the sense of Definition 2.2.5 and we shall treat it as such.
2. $T(t)f \in X_+$ whenever $f \in X_+$ $\forall\ t \geq 0$.
3. $T(t)I = I$ $\forall\ t \geq 0$.

Such a semigroup is called a **Markov semigroup**. At this point it is instructive to point out that the convolution semigroups on probability measures, as defined in Definition 2.5.6, give rise to a class of Markov semigroups as has been proven in the Theorem 2.5.7. In view of condition (1c) above, it is convenient analytically to deal with the pre-dual space (for example L^1 or l^1) and the pre-dual semigroup on it, rather than the semigroup on L^∞ or l^∞. Therefore from this point onwards we shall set $X = L^1_{\mathbb{R}}$ or $l^1_{\mathbb{R}}$ with the C_0 contraction semigroup $\{T(t)\}_{t\geq 0}$ on it and designate the dual semigroup as $\{T^*(t)\}_{t\geq 0}$ acting on $X^* = L^\infty_R$ or $l^\infty_{\mathbb{R}}$ respectively. In order to avoid confusion, the suffix $\mathbb{R}$ shall henceforth be omitted while referring to these spaces. In the context of the dual pair (l^∞, l^1) we collect a few elementary results, without proof, in the next lemma.

Let $X = l^1$ and X_+ be the **positive cone** of l^1. Then a linear map $A\colon X \to X$ is positive if $A\colon \mathcal{D}(A) \cap X_+ \to X_+$ and we note that there is a normal ordering of maps on l^1 by this cone: $A_1 \geq A_2$ if $A_1 - A_2$ is a positive map. Note also that for any $x, y \in X_+, \|x+y\|_1 = \|x\|_1 + \|y\|_1$. Then we have the following.

Lemma 6.1.2. Let x, y, y_j $(j = 1, 2, \ldots)$ belong to X_+ and let $A, B \in \mathcal{B}(X)$ be positive maps. Then the following are true.

(i) $0 \leq A \leq B$ implies that $\|A\| \leq \|B\|$;

(ii) $0 \leq y_1 \leq y_2 \leq \ldots$ and $\|y_j\|_1 \leq M \ \forall\ j$ implies that there exists a unique $y \in X_+$ such that $\|y_j - y\|_1 \to 0$ as $j \to \infty$;

(iii) $0 \leq T_1 \leq T_2 \leq \ldots$ with $T_j \in B(X)$ such that $\|T_j\| \leq M \quad \forall j$ implies that there exists a unique positive $T \in B(X)$ such that T_j converges to T strongly.

The proof of the above lemma rests essentially on the monotone convergence theorem for monotone sequences of reals, as well as for sequences in l^1, and is left as an exercise (**Exercise** 6.1.3).

6.2 Construction of Markov semigroups on a discrete state space

To fix ideas, let us start with a finite state space $\Omega = \{1, 2, \ldots, N\}$ and consider a probability distribution given by a vector $p = (p_1, p_2, \ldots p_N)$ on it and a stochastic (or Markov) matrix $(t_{ij})_{i,j=1}^N$ such that $t_{ij} \geq 0$ and $\sum_{j=1}^N t_{ij} = 1$ for each i.

(i) The probability vector p is in one-to-one correspondence with ϕ_p, the positive linear functionals on the space of real-valued functions f on Ω such that $\phi_p(f) = \sum_{i=1}^N p_i f(i)$ and $\phi_p(1) = 1$. This allows one to associate a discrete evolution T given by $T(\phi_p)(\chi_j) = \sum_i p_i t_{ij}$, where χ_j is the indicator function of the singleton set $\{j\}$.

(ii) There is a dual evolution T^* acting on the real-valued functions f on Ω given by $T^*(f)(i) = \sum_j t_{ij} f(j)$ and we observe that $(T(\phi_p))(\chi_j) = \phi_p(T^*(\chi_j))$. It can also be noted that T^* maps positive functions to positive functions and maps the identity function to itself.

(iii) $\{T^n\}_{n\in\mathbb{Z}_+}$ and $\{T^{*n}\}_{n\in\mathbb{Z}_+}$ provide two discrete (dynamical) semigroups, the second being dual to the first and clearly T^{*n} satisfies property (ii) for each n.

In probabilistic terminology, the element t_{ij} of the matrix representation of the map T denotes the (Markov) transition probability from the *initial state* i to the *final state* j, thereby explaining the requirements: $t_{ij} \geq 0$ and $\sum_{j=1}^N t_{ij} = 1$.

Now let $\mathbb{N}$ be the state space of a Markov transition over (continuous) time $t \in \mathbb{R}_+$ with transition matrix $P(t) \equiv (p_{ij}(t)),\ (i, j = 1, 2, \ldots)$ such that

it satisfies *formally* the ordinary differential equation (**Chapman-Kolmogorov equation**):

$$\frac{d}{dt}p_{ik}(t) = \sum_{j=1}^{\infty} p_{ij}(t)a_{jk} \text{ or } \frac{d}{dt}P(t) = AP(t) \tag{6.1}$$

where the matrix $(p_{ij}(t))$ acts on l^1 as the right action $(P(t)x)_j = \sum_{i=1}^{\infty} x_i p_{ij}(t)$ with initial conditions

$$\lim_{t\to 0^+} P(t) \equiv P(0) = I, \text{ or equivalently } \lim_{t\to 0^+} p_{ij}(t) = \delta_{ij}. \tag{6.2}$$

The map A appearing above is an $\mathbb{N}\times\mathbb{N}$ matrix (a_{jk}) with the properties

$$\left.\begin{array}{rlll} a_{jk} \geq 0 \text{ for } j \neq k & \text{and} & \sum_{k=1}^{\infty} a_{jk} = 0 \text{ for all } j. \\ \text{We also set} \quad a_j \equiv -a_{jj} & \text{so that} & a_j = \sum_{k\neq j} a_{jk} \geq 0. \end{array}\right\} \tag{6.3}$$

Intuitively, the condition (6.3), using (6.1), implies that

$$\frac{d}{dt}\sum_{k=1}^{\infty} p_{ik}(t) = 0,$$

that is,

$$\frac{d}{dt}\|P(t)x\|_1 = \frac{d}{dt}\sum_{j=1}^{\infty}\sum_{i=1}^{\infty} x_i p_{ij}(t) = 0 \text{ for all } x \in X_+$$

or

$$\|P(t)x\|_1 = \|P(0)x\|_1 = \|x\|_1 \ \forall x \in X,$$

where $\|x\|_1 = \sum_{j=1}^{\infty} |x_j|$, the l^1-norm. This means that the total probability that the process, starting from the state i will transit to some state $k \in \mathbb{N}$ in time t, is independent of t. Equivalently, we need to construct a C_0-semigroup $\{P(t)\}_{t\geq 0}$ acting on l^1 with generator A, which satisfies (6.3).

We remark here that Feller [9] first proved the existence of a unique minimal semigroup associated with the equations (6.3) and (6.2) for infinite discrete state space, and Kato [13] constructed the same in the framework of the theory of semigroups, exploiting the special nature of l^1. Here we follow Kato's construction.

As mentioned earlier, before the Lemma 6.1.2, we shall be dealing with the Markov semigroup $\{P(t)\}_{t\geq 0}$ acting on l^1 and its dual $\{P^*(t)\}_{t\geq 0}$ on l^∞. Also note that the semigroup $\{P(t)\}_{t\geq 0}$ or the operator A acts on the elements

of l^1 by right action while its dual acts on the elements of l^∞ by left action, that is,

$$(Ax)_j = \sum_{i=1}^{\infty} x_i a_{ij} \text{ for } x \in X \text{ and}$$
$$(A^* y)_j = \sum_{k=1}^{\infty} a_{jk} y_k \text{ for } y \in X^*.$$

Lemma 6.2.1. Let A be such that (6.3) holds. Then

(i) A is bounded if and only if $\sup_j a_j < \infty$.

(ii) The natural domain of A is the maximal domain

$$\mathcal{D}(A) := \{x \in l^1 : \sum_k |\sum_j x_j a_{jk}| < \infty\}.$$

Then A is a densely defined closed linear operator in $X = l^1$.

(iii) Let D_0 be the subspace of X consisting of all sequences with finite support. Then $A_0 \equiv A|_{\mathsf{D}_0}$ is called the minimal operator associated with A and is also a closable densely defined operator. (Note that $\overline{A_0}$ need not be equal to A.)

Proof. Let $\sup_j a_j \equiv a < \infty$. Then, since

$$(Ax)_k = \sum_j x_j a_{jk} = -x_k a_k + \sum_{j \neq k} x_j a_{jk},$$

we have that

$$\|Ax\|_1 = \sum_k |(Ax)_k| \leq a \sum_k |x_k| + \sum_k \sum_{j \neq k} |x_j| a_{jk}$$
$$= a\|x\|_1 + \sum_j |x_j| \sum_{k \neq j} a_{jk} \leq 2a\|x\|_1,$$

showing that $\|A\| \leq 2a$. Conversely, let $\sup_j a_j = \infty$. Then there exists a subsequence $\{j_l\}$, such that $a_{j_l} \uparrow \infty$. Choose $x_j^{(k)} = \delta_{jk}$ so that $\|x^{(k)}\|_1 = 1 \ \forall k$ and

$$\|Ax^{(k)}\|_1 = \sum_r |\sum_j x_j^{(k)} a_{jr}| = \sum_r |a_{kr}| = |a_{kk}| + \sum_{r \neq k} a_{kr} \geq a_k.$$

Therefore, $\|Ax^{(k_l)}\|_1 \geq a_{k_l} \uparrow \infty$ proving that A is not bounded. This proves (i).

Since $\mathsf{D}_0 \subset \mathcal{D}(A)$ and D_0 is dense in X, both A and A_0 are densely defined linear operators on X. Suppose $\{x^{(n)}\} \subset \mathcal{D}(A) \to x$ and $Ax^{(n)} \to y \in X$. This means that for each $j, r \in \mathbb{N}$,

$$|x_j^{(n)} - x_j| \leq \sum_l |x_l^{(n)} - x_l| \to 0 \text{ and}$$

$$\Big|\sum_j x_j^{(n)} a_{jr} - y_r\Big| \leq \sum_k |\sum_j x_j^{(n)} a_{jk} - y_k| = \|Ax^{(n)} - y\|_1 \to 0$$

uniformly with respect to j and r respectively, as $n \to \infty$. Therefore,

$$\Big|\sum_j x_j^{(n)} a_{jk} - \sum_j x_j a_{jk}\Big| \leq \sup_j \big|x_j^{(n)} - x_j\big| \sum_j |a_{jk}| \leq 2a_k \sup_j \big|x_j^{(n)} - x_j\big| \to 0$$

as $n \to \infty$. Hence, $y_k = \sum_j x_j a_{jk}$ and by hypothesis $y = (y_k) \in X$. Thus, $x \in \mathcal{D}(A)$ and $y = Ax$. The closability of A_0 may be proven similarly. □

Remark 6.2.2. If condition (i) of Lemma 6.2.1 above holds, then $P^*(t) = e^{tA^*}$ and

$$\frac{d}{dt}P^*(t)(I) = P^*(t)A^*(I) = 0, \text{that is,} \sum \frac{d}{dt}p_{ik}(t) = \sum_j p_{ij}(t) \sum_k a_{jk} = 0,$$

where $I = (1, 1, 1, \ldots)$, implying that $P^*(t)(I) = I, \forall\ t \geq 0$, that is, P^* is a Markov semigroup.

In view of the above lemma, two questions arise naturally :

Q1. What are the densely defined closed operators $\tilde{A}$ in X such that $A_0 \subset \tilde{A} \subset A$ and such that $\tilde{A}$ is the generator of a positive contraction C_0-semigroup?

Q2. Of all these semigroups (as possibly given as answer to Q1), which ones are Markov; that is, for which ones does the following equality hold?

$$\|P(t)x\|_1 = \|x\|_1\ \forall x \in X.$$

Remark 6.2.3. (i) The **property of conservativity**: $\|P(t)x\|_1 = \|x\|_1$ for all $x \in X$ is equivalent to $\|P(t)x\|_1 = \|x\|_1$ for all $x \in X_+$, as well as to $P^*(t)(I) = I$ for all $t \geq 0$, where P^* is the adjoint semigroup acting on $l^\infty = X^*$ and $I = (1, 1, 1 \ldots)$. The first equivalence follows from the facts that P is a positive map and the property of the l^1-norm, and the second from the duality between l^1 and l^∞, that is, $\langle P^*(t)y, x\rangle = \langle y, P(t)x\rangle$ for all $x \in X$ and $y \in X^*$.

(ii) In particular, if we choose $x = e_j$, the canonical basis (the sequence $(e_j)_k = \delta_{jk}$) in $X = l^1$, then $(P(t)e_j)_k = \sum p_{lk}(t)(e_j)_l = p_{jk}(t) > 0$, and therefore, the requirement of conservativity, $\|P(t)e_j\|_1 = \|e_j\|_1 = 1$ implies that $\sum_k p_{jk}(t) = 1\ \forall j$, while the contractivity of P implies that $\sum_k p_{jk}(t) \leq 1\ \ \forall j$.

So the problem at hand is to answer Q1 and Q2 given earlier, that is, to construct a contraction semigroup P with generator A in $X = l^1$, given the conditions (6.3), and also to determine the necessary and sufficient conditions for the constructed semigroups P to be conservative or Markov.

Let a_{jk} and $a_j \equiv -a_{jj}$ be as in (6.3) and define a positive (possibly unbounded) operator H in X by

$$\mathcal{D}(H) = \{x \in X : \textstyle\sum_j |a_j x_j| < \infty\} \text{ and} \tag{6.4}$$

$$(Hx)_j = a_j x_j\ \forall\ x \in \mathcal{D}(H). \tag{6.5}$$

Then clearly, $\mathsf{D}_0 \subsetneq \mathcal{D}(H)$. We set $\mathsf{D}_+ = \mathcal{D}(H) \cap X_+$ and $(\mathsf{D}_0)_+ = \mathsf{D}_0 \cap X_+$ leading to the next lemma.

Lemma 6.2.4. (i) H defined above, is a positive, closed operator in X with D_0 a core.

(ii) For all $\lambda > 0, (\lambda + H)^{-1}$ is a positive, bounded operator in X with

$$\|(\lambda + H)^{-1}\| \leq \lambda^{-1}.$$

(iii) If we define a linear operator K in X with domain $\mathcal{D}(H)$ by setting

$$(Kx)_k = \sum_{j \neq k} x_j a_{jk},$$

then K is a positive operator, bounded relative to H, that is, $\|Kx\| \leq \|Hx\|$ for all $x \in \mathcal{D}(H)$ (see Definition 4.1.2), and $\|Kx\| = \|Hx\|\ \forall\ x \in \mathsf{D}_+$.

(iv) $Ax = -Hx + Kx$ for any $x \in \mathcal{D}(H)$.

The proof is omitted since it is straightforward, except to note that the contraction semigroup e^{-tH} leaves D_0 invariant and that fact, by Theorem A.1.9, implies that D_0 is a core for H. The details are left for the reader to verify (**Exercise** 6.2.5).

Remark 6.2.6. (i) $\mathcal{D}(A)$ is in general strictly bigger than $\mathcal{D}(H)$ and $\mathcal{D}(H) = \mathcal{D}(-H+K)$, that is, $A \supset -H+K \supset A_0 = A|_{\mathsf{D}_0}$.

(ii) Since, in general, the bound of K relative to H is not less than 1, none of the standard results in the theory of perturbation of contraction semigroups (Theorems 4.1.3 and 4.1.5, and Corollary 4.1.4) are applicable here, though the unperturbed semigroup $T(t) = \exp^{-tH}$ is a positive contraction semigroup. The strategy, (following Feller and Kato) is to capitalise on the *positivity property* rather than the *small relative bound property.* For this, we first set $G_r := -H + rK$ for $0 \leq r < 1$ and collect some of the relevant results in the next lemma.

Lemma 6.2.7. (i) $B(\lambda) := K(\lambda + H)^{-1}$, with K and H as in Lemma 6.2.4, is a positive contraction in X for each $\lambda > 0$, satisfying $0 \leq B(\mu) \leq B(\lambda)$ for $0 < \lambda < \mu$.

(ii) For each $0 \leq r < 1, G_r$ is the generator of a contraction C_0-semigroup P_r and its resolvent $R_r(\lambda) := (\lambda - G_r)^{-1}$ is a positive bounded operator with $\|R_r(\lambda)\| \leq \lambda^{-1}$ for all $\lambda > 0$.

Proof. (i) That $B(\lambda)$ is contractive follows from Lemma 6.2.4,(iii), and that it is positive follows from the positivity of K and of $(\lambda + H)^{-1}$, and the fact that composition of two positive operators is positive. Further, by the resolvent equation, for $0 < \lambda < \mu$, $B(\lambda) - B(\mu) = (\mu - \lambda)B(\lambda)(\mu + H)^{-1}$ is positive.

(ii) It is easy to see that

$$\begin{aligned} R_r(\lambda) &= (\lambda + H - rK)^{-1} = (\lambda + H)^{-1}(1 - rB(\lambda))^{-1} \\ &= \sum_{n=0}^{\infty} r^n (\lambda + H)^{-1} B(\lambda)^n \end{aligned}$$

with the Neumann series converging in operator norm since $\|B(\lambda)\| \leq 1$. This also proves that $R_r(\lambda)$ is a positive map in X. For $f \in \mathsf{D}_+$, and $\lambda > 0$,

$$\begin{aligned} \|(\lambda - G_r)f\| &= \|(\lambda + H)f - rKf\| \geq \|(\lambda + H)f\| - r\|Kf\| \\ &= \|\lambda f\| + (\|Hf\| - r\|Kf\|) \geq \lambda \|f\| \end{aligned}$$

since $\|Kf\| \leq \|Hf\|$.

Furthermore, since $R_r(\lambda) = (\lambda - G_r)^{-1}$ is a positive bounded map,

$$(\lambda - G_r)\mathsf{D}_+ = X_+.$$

Thus for $g \in X_+$,

$$\|R_r(\lambda)g\| = \|f\| \text{ where } (\lambda - G_r)f = g \text{ for some } f \in \mathsf{D}_+.$$

Hence $\|R_r(\lambda)g\| = \|f\| \leq \lambda^{-1}\|g\|$. For any $g \in X, g = g_+ - g_-$ and $\|g\| = \|g_+\| + \|g_-\|$ with $g_\pm \in X_+$, leading to

$$R_r(\lambda)g = R_r(\lambda)g_+ - R_r(\lambda)g_-$$

and

$$\|R_r(\lambda)g\| = \|R_r(\lambda)g_+\| + \|R_r(\lambda)g_-\| \leq \lambda^{-1}(\|g_+\| + \|g_-\|) = \lambda^{-1}\|g\|.$$

While Theorem 4.2.1 shows that G_r is a closed operator for $0 \leq r < 1$, an application of the Hille-Yosida Theorem 2.3.3 (i) proves that the operator $G_r = -H + rK$ is the generator of a contraction C_0-semigroup on X.

□

If we denote by $\{P_r(t)\}_{t\geq 0}$ the contraction semigroup generated by G_r for $0 \leq r < 1$, then we have the following main result.

Theorem 6.2.8. For $0 \leq r < 1$, the contraction semigroup $\{P_r(t)\}_{t\geq 0}$ satisfies the following.

1. $P_r(t)$ is positive for every $t \geq 0$.
2. $\{P_r(t)\}$ converges strongly to a positive contraction C_0-semigroup $\{P(t)\}$ as r increases to 1 and the strong convergence is uniform for t in any compact subset of $[0, \infty)$.

Proof. Fix $\lambda > 0$. Then the map $(0, 1) \ni r \mapsto R_r(\lambda)$ is positive and increasing, since for $r > s$,

$$R_r(\lambda) - R_s(\lambda) = (\lambda + H)^{-1} \sum_{n=0}^{\infty} (r^n - s^n)B(\lambda)^n$$

is positive. Now, by Remark 2.3.4

$$P_r(t) = s - \lim_{n\to\infty} \Big[\frac{n}{t} R_r\Big(\frac{n}{t}\Big)\Big]^n$$

which implies that each $P_r(t)$ is positive and also that for $r > s$,

$$\begin{aligned} &P_r(t) - P_s(t) \\ &= s - \lim_{n\to\infty} \big[\big(\frac{n}{t}\big)^n \sum_{j=0}^{n-1} R_r\big(\frac{n}{t}\big)^{n-1-j} \big[R_r\big(\frac{n}{t}\big) - R_s\big(\frac{n}{t}\big)\big] R_s\big(\frac{n}{t}\big)^j\big] \end{aligned}$$

is positive. Thus, for a fixed $t \geq 0$, the family of contractions $[0,1) \ni r \mapsto P_r(t)$ is monotonically increasing and therefore, by Lemma 6.1.2, $P_r(t)$ converges strongly, to say $P(t)$, as $r \uparrow 1$. This implies that $\{P(t)\}_{t\geq 0}$ is a contractive positive family and $P_r(t) \leq P(t)$ for all $t \geq 0$ and $0 \leq r < 1$. That $\{P(t)\}_{t\geq 0}$ is a semigroup follows from strong convergence and the identity:

$$P_r(t+s) = P_r(t)P_r(s) \text{ for all } r \in [0,1) \text{ and } t, s \geq 0.$$

Similarly we conclude that $P(0)f = \lim_r P_r(0)f = f$ for all f.

Finally we show that the strong convergence of $P_r(t)$ to $P(t)$ as $r \uparrow 1$ is uniform for t in a bounded interval I and as a consequence of this, also prove the C_0 property. If this is not true, then there exist $\eta > 0$ and sequences $\{r_n\}$ and $\{t_n\}$ such that $t_n \to t_0 \in I$ and $r_n \uparrow 1$, with the property that

$$\|P(t_n)f - P_{r_n}(t_n)f\| \geq \eta > 0$$

for some $f \in X_+$. In such a case, since both $P_{r_n}(t_n)$ and $[P(t_n) - P_{r_n}(t_n)]$ are positive maps, one has that

$$\|P(t_n)f\| = \|P_{r_n}(t_n)f\| + \|\big[P(t_n) - P_{r_n}(t_n)\big]f\| \geq \|P_{r_n}(t_n)f\| + \eta$$

which yields that for $m < n$ (with $r_m < r_n$),

$$\|P_{r_m}(t_n)f\| \leq \|P_{r_n}(t_n)f\| \leq \|P(t_n)f\| - \eta.$$

Now, letting $n \to \infty$, so that $r_n \uparrow 1$, and $t_n \to t_0$, we have that $\|P_{r_m}(t_0)f\| \leq \|P(t_0)f\| - \eta$ which implies $\|[P(t_0) - P_{r_m}(t_0)]f\| > \eta$ and this contradicts the earlier proven fact that $P_{r_m}(t_0)$ converges strongly to $P(t_0)$ as $m \to \infty$. This implies that for $f \in X$,

$$\lim_{t\downarrow 0} P(t)f = \lim_{t\downarrow 0} \lim_{r\uparrow 1} P_r(t)f = \lim_{r\uparrow 1} \lim_{t\downarrow 0} P_r(t)f = f = P(0)f,$$

showing that $\{P(t)\}_{t\geq 0}$ is a C_0 semigroup.

□

The next theorem studies the properties of the limit semigroup P in detail; in particular it describes in the precise sense in which P is the minimal of all semigroups constructed out of the *formal sum* $-H + K$.

Theorem 6.2.9. Let G be the generator of the semigroup P constructed in the previous theorem and let $R(\lambda) = (\lambda - G)^{-1}$ for $\lambda > 0$ be its resolvent. Then the following hold.

(i) $\|R(\lambda)\| \leq \lambda^{-1}$, $0 \leq R_r(\lambda) \leq R(\lambda)$ for $0 \leq r < 1$ and $R_r(\lambda)$ converges strongly to $R(\lambda)$ as $r \uparrow 1$, for each $\lambda > 0$.

(ii) Set $R^{(n)}(\lambda) = (\lambda + H)^{-1} \sum_{j=0}^{n} B(\lambda)^j$ for $\lambda > 0$. Then $R^{(n)}(\lambda) \leq R(\lambda)$ and $R^{(n)}(\lambda)$ converges strongly to $R(\lambda)$ as $n \to \infty$.

(iii) G is a closed extension of $-H + K$ on $D(H)$ and hence also of $A_0 = A|_{\mathsf{D}_0}$. Furthermore, G is a restriction of A, that is, $A \supset G \supset -H + K \supset A_0$.

(iv) (minimality) If there exists a positive C_0-semigroup $\{P'(t)\}_{t \geq 0}$ with generator G' which extends A_0, then $P'(t) \geq P(t)$ for all $t \geq 0$.

Proof. (i) We have seen that $R(\lambda) = \displaystyle\int_0^\infty e^{-\lambda t} P(t)\, dt$ $(\lambda > 0)$, by Theorem 2.2.7(c) and similarly,

$$R_r(\lambda) = \int_0^\infty e^{-\lambda t} P_r(t)\, dt \ (\lambda > 0).$$

It is immediately clear that the positivity of the semigroups implies the same property for the resolvents and the norm bounds follow from Theorem 2.2.7(f). Note that for $f \in X$,

$$R(\lambda)f - R_r(\lambda)f = \int_0^\infty e^{-\lambda t}(P(t)f - P_r(t)f)\, dt$$

and therefore

$$\|R(\lambda)f - R_r(\lambda)f\| \leq \int_0^\alpha e^{-\lambda t}\|P(t)f - P_r(t)f\|\, dt + 2\|f\| \int_\alpha^\infty e^{-\lambda t}\, dt.$$

Given an $\epsilon > 0$, choose $\alpha > 0$ such that $2\|f\| \displaystyle\int_\alpha^\infty e^{-\lambda t}\, dt < \frac{\epsilon}{2}$ and for that α, choose $\delta > 0$, with $1 - \delta < r < 1$, so that

$$\|P(t)f - P_r(t)f\| < \frac{\epsilon}{2} \ \ \forall t \in [0, \alpha],$$

by using the result on uniform convergence in Theorem 6.2.8. This will imply that for such a choice of r, $\|R(\lambda)f - R_r(\lambda)f\| < \epsilon$, completing the proof.

(ii) Set

$$R_r^{(n)}(\lambda) = (\lambda + H)^{-1} \sum_{m=0}^{n} r^m B(\lambda)^m.$$

It is clear from the definition and part (i) above that

(1) $0 \le R_r^{(n)}(\lambda) \le R_r(\lambda) \le R(\lambda)$ for all $n \in \mathbb{N}$,

(2) $0 \le R_r^{(m)}(\lambda) \le R_r^{(n)}(\lambda)$ for $m \le n$ and that

(3) $R_r^{(n)}(\lambda)$ converges strongly to

$$R^{(n)}(\lambda) = (\lambda + H)^{-1} \sum_{j=0}^{n} B(\lambda)^j,$$

a positive bounded operator as $r \uparrow 1$.

These imply that $0 \le R^{(m)}(\lambda) \le R^{(n)}(\lambda) \le R(\lambda)$ for $\lambda > 0$ and for $n \ge m$. Thus, by Lemma 6.1.2(iii), $R^{(n)}(\lambda)$ increases strongly to a bounded operator, say $\tilde{R}(\lambda)$ and $\tilde{R}(\lambda) \le R(\lambda)$. On the other hand, $R_r^{(n)}(\lambda) \le R^{(n)}(\lambda) \le \tilde{R}(\lambda)$ and taking strong limit in the left hand side of this inequality, first as $n \to \infty$, and then as $r \uparrow 1$, we get that $R(\lambda) \le \tilde{R}(\lambda)$, leading to the required result, which means that

$$R(\lambda) = (\lambda + H)^{-1} \sum_{n=0}^{\infty} B(\lambda)^n,$$

the infinite sum converging strongly.

(iii) Note that

$$\begin{aligned} R^{(n)}(\lambda) &= (\lambda + H)^{-1} + (\lambda + H)^{-1} \Big(\sum_{j=0}^{n-1} B(\lambda)^j \Big) B(\lambda) \\ &= (\lambda + H)^{-1} + R^{(n-1)}(\lambda) B(\lambda) \end{aligned}$$

and therefore, for $f \in \mathcal{D}(H) = \mathcal{D}(K)$, we have that

$$R^{(n)}(\lambda)(\lambda + H)f = f + R^{(n-1)}(\lambda) K f.$$

Now letting $n \to \infty$ we get that

$$\begin{aligned} R(\lambda)(\lambda + H)f &= f + R(\lambda) K f, \text{ or equivalently,} \\ R(\lambda)(\lambda + H - K)f &= f \text{ for all } f \in \mathcal{D}(H). \end{aligned}$$

This implies that $f \in \mathcal{D}(G)$, that is, $\mathcal{D}(H) \subset \mathcal{D}(G)$, and for such $f, (\lambda - G)f = (\lambda + H - K)f$, so that $G = -H + K$ on $\mathcal{D}(H)$ or $G \supset -H + K$. Thus G is a closed extension of $-H + K$. Since $Ran\, R^{(n)}(\lambda) \subset \mathcal{D}(H) = \mathcal{D}(K)$, we get that for $f \in X_+$,

$$(\lambda + H)^{-1} K R^{(n)}(\lambda) f = \sum_{j=1}^{n+1} (\lambda + H)^{-1} B(\lambda)^j f = R^{(n+1)}(\lambda) f - (\lambda + H)^{-1} f. \tag{6.6}$$

The right hand side in (6.6) converges to $R(\lambda)f - (\lambda + H)^{-1} f$ as $n \to \infty$. Since l^1-convergence implies pointwise convergence (choosing a subsequence if necessary), this means that the left hand side vector in (6.6), represented by the sequence

$$(\lambda + a_k)^{-1} \sum_{j \neq k} a_{jk} y_j^{(n)}, \text{ converges to } (\lambda + a_k)^{-1} \sum_{j \neq k} a_{jk} y_j,$$

where

$$y_j^{(n)} = \left(R^{(n)}(\lambda) f\right)_j, \; y_j = \left(R(\lambda) f\right)_j \text{ and } a_{jk} > 0 \; (j \neq k).$$

This observation and (6.6) implies that

$$(\lambda + a_k)^{-1} \sum_{j \neq k} a_{jk} y_j = y_k - (\lambda + a_k)^{-1} f_k$$

and thus

$$(Ky)_k = (\lambda + a_k) y_k - f_k = \left((\lambda + H) y\right)_k - f_k \text{ or}$$

$$f_k = ((\lambda + H - K) y)_k = \lambda y_k - \sum_{j=1}^{\infty} y_j a_{jk}.$$

For any $f \in X$, we write $f = f_+ - f_-$ and conclude similarly that

$$(f_+)_k = ((\lambda + H - K) y_+)_k \text{ and } (f_-)_k = ((\lambda + H - K) y_-)_k,$$

where $y_\pm = R(\lambda) f_\pm$. This gives the result that $RanR(\lambda) \subset \mathcal{D}(A)$ and $(\lambda + H - K) R(\lambda) f = f$ for all $f \in X$. Consequently,

$$A \supset G \supset -H + K \supset A_0.$$

(iv) Let G' be the generator of another C_0-semigroup P' extending $-H+K$. Then for $x \in \mathcal{D}(H)$, there exists a sequence $\{x_n\} \subset \mathsf{D}_0$ such that $x_n \to x$ and $Hx_n \to Hx$ since D_0 is a core for H by Lemma 6.2.4(i), which implies that

$$\|K(x-x_n)\| \leq \|H(x-x_n)\| \to 0 \text{ as } n \to \infty.$$

This leads to the conclusion that

$$A_0 x_n = (-H+K)x_n \to (-H+K)x \text{ as } n \to \infty.$$

This fact along with the assumption of G' being a closed extension of A_0 implies that

$$\mathcal{D}(G') \supset \mathcal{D}(H) \text{ and } G'x = (-H+K)x \text{ for all } x \in \mathcal{D}(H),$$

that is, $G' \supset -H+K$. If the C_0-semigroup P' is of type β, then

$$\{\lambda \in \mathbb{R} : \lambda > \beta\} \subset \rho(G')$$

and we set $R'(\lambda) = (\lambda - G')^{-1}$. Then we have that

$$R'(\lambda) - R_r(\lambda) = R'(\lambda)(G'-G_r)R_r(\lambda) = (1-r)R'(\lambda)KR_r(\lambda),$$

which is positive since $R'(\lambda)$, the resolvent of a positive semigroup P', is also positive for $\lambda > \lambda_0 := \max(0,\beta)$ and since $0 \leq r < 1$. This implies that, $R_r(\lambda) \leq R'(\lambda)$, leading to $R(\lambda) \leq R'(\lambda)$, for each $\lambda > \lambda_0$. Fix $t > 0$, and note that for all large enough $n \in \mathbb{N}, n/t > \lambda_0$. Therefore,

$$\frac{n}{t}\Big(\frac{n}{t} - G'\Big)^{-1} \geq \frac{n}{t}\big(\frac{n}{t} - G\big)^{-1},$$

which implies that

$$\begin{aligned}
&\Big[\frac{n}{t}\Big(\frac{n}{t} - G'\Big)^{-1}\Big]^n - \Big[\frac{n}{t}\Big(\frac{n}{t} - G\Big)^{-1}\Big]^n \\
&= \Big(\frac{n}{t}\Big)^n \sum_{j=0}^{n-1} \Big(\frac{n}{t} - G'\Big)^{-(n-j-1)} \Big[\Big(\frac{n}{t} - G'\Big)^{-1} - \Big(\frac{n}{t} - G\Big)^{-1}\Big] \Big(\frac{n}{t} - G\Big)^{-j}
\end{aligned} \tag{6.7}$$

is positive. Taking strong limit as $n \to \infty$ on the left hand side of (6.7), as in the proof of Theorem 6.2.8, we get that

$$P'(t) \geq P(t) \ \forall t > 0.$$

Note that for $t = 0, P'(0) = I = P(0)$.

□

The contractive C_0-semigroup $\{P(t)\}_{t\geq 0}$ constructed in Theorem 6.2.8 is thus naturally called the **minimal semigroup**.

As explained earlier, unlike in the case of a finite state space, or equivalently, unlike when the underlying space of analysis is finite-dimensional, in the more general case, the condition (6.2) that $\sum_{k=1}^{\infty} a_{jk} = 0$ for all j, *does not necessarily imply the conservation of probability*, that is, $\|P(t)f\|_1$ may not be equal to $\|f\|_1$ for all $f \in X$, and $t > 0$, or for the dual semigroup on l^∞, $P^*(t)I \neq I$ for all $t > 0$, where I is the sequence $(1, 1, 1, \ldots) \in l^\infty$. Next, the necessary and sufficient conditions for the conservativity or the Markov property of the minimal semigroup P are studied.

Lemma 6.2.10. (i) For all $x \in X_+$ and $\lambda > 0$, $b(\lambda; x) \equiv \lim_{n\to\infty} \|B(\lambda)^n x\|$ exists, and defines the map $b(\lambda, \cdot)$ as a map from the positive cone X_+ into the cone $\mathbb{R}_+$, such that $b(\lambda; \alpha x + y) = \alpha b(\lambda; x) + b(\lambda; y)$ for all $x, y \in X_+$ and $\alpha > 0$.

(ii) For any fixed $x \in X_+$, either $b(\lambda; x) = 0$ for all $\lambda > 0$ or $b(\lambda; x) > 0$ for all $\lambda > 0$. Also, $b(\lambda; x)$ is a non-increasing function of λ and $\lim_{\lambda\downarrow 0} b(\lambda; x)$ exists.

(iii) For any $x \in X_+$, the function $[0, \infty) \ni t \mapsto \|P(t)x\|$ is a non-increasing function and $\lim_{t\to\infty} \|P(t)x\| = \|x\| - \lim_{\lambda\downarrow 0} b(\lambda; x)$.

(iv) Furthermore, for any $\lambda > 0$, and $x \in X_+$,
$(0, \infty) \ni \lambda \mapsto \lambda^{-1}[\|x\| - b(\lambda; x)] \in (0, \infty]$ is also a non-increasing function and

$$\int_0^\infty \|P(t)x\|\, dt = \lim_{\lambda\downarrow 0} \lambda^{-1}[\|x\| - b(\lambda; x)],$$

which may be $+\infty$.

Proof. (i) Fix $\lambda > 0$, and note that

$$I + KR^{(n)}(\lambda) = \sum_{j=0}^{n+1} B(\lambda)^j = (\lambda + H)R^{(n)}(\lambda) + B(\lambda)^{n+1}, \tag{6.8}$$

and thus for $x \in X_+$,

$$\|x\| + \|KR^{(n)}(\lambda)x\| = \lambda\|R^{(n)}(\lambda)x\| + \|HR^{(n)}(\lambda)x\| + \|B(\lambda)^{n+1}x\|,$$

where we have used the fact that for $x, y \in X_+$, $\|x + y\| = \|x\| + \|y\|$.

Since $R^{(n)}(\lambda)x \in \mathsf{D}_+$, we have by Lemma 6.2.4 (iii) that

$$\|KR^{(n)}(\lambda)x\| = \|HR^{(n)}(\lambda)x\|$$

and it follows that

$$\|B(\lambda)^{n+1}x\| = \|x\| - \lambda\|R^{(n)}(\lambda)x\|.$$

Therefore, by Theorem 6.2.9(ii), $b(\lambda; x)$ exists for every $x \in X_+$ and

$$b(\lambda; x) = \|x\| - \lambda\|R(\lambda)x\| = \lambda \int_0^\infty e^{-\lambda t}\big(\|x\| - \|P(t)x\|\big)\, dt, \tag{6.9}$$

by Theorem 2.2.7(e). The final claim follows immediately from (6.9).

(ii) If $b(\lambda; x) = 0$ for some $\lambda > 0$, then it follows from (6.9) that for almost all t, $\|x\| - \|P(t)x\| = 0$, and therefore, $b(\lambda; x) = 0$ for all $\lambda > 0$. That $b(\lambda; x)$ if not identically zero, is non-increasing in λ follows from the same property of $B(\lambda)$ (see Lemma 6.2.7(i)) and their positivity. The existence of $\lim_{\lambda \downarrow 0} b(\lambda; x)$ follows from the fact that it is a non-increasing function of λ.

(iii) That $\|P(t)x\|$ is a non-increasing function of t follows from the contraction semigroup property of P; this and (6.9) leads to

$$\|x\| - b(\lambda; x) = \lambda \int_0^\infty e^{-\lambda t}\|P(t)x\|\, dt.$$

Letting $\lambda \downarrow 0$, in above gives that $\lim\limits_{\lambda \to 0} b(\lambda; x) = \|x\| - \lim\limits_{t \to \infty} \|P(t)x\|$, (the reader may verify this equality (**Exercise** 6.2.11)) showing that $\lim\limits_{t \to \infty} \|P(t)x\|$ exists for $x \in X_+$.

(iv) This follows in a manner similar to above and the details are left for the reader to work out (**Exercise** 6.2.12).

□

Remark 6.2.13. (i) For $x \in X_+, \|P(t)x\| = \|x\|$ if and only if $b(\lambda; x) = 0$ for some, and hence for all $\lambda > 0$, or equivalently if and only if $B(\lambda)^n x \to 0$ as $n \to \infty$. This follows clearly from the proof of Lemma 6.2.10, in particular from (6.9).

(ii) In the theory of perturbation of semigroups or of the generators of semigroups, it is the *smallness of the perturbation relative to the unperturbed*

generator in some sense, that is used. Here, the positive operator H is the generator of the unperturbed semigroup and K is the perturbation. But as noted in Lemma 6.2.4(iii), K is not *small* relative to H and hence the methods of Chapter 4 are not available. However, the fact that both H and K are positive maps comes to our rescue. Nevertheless, a price has to be paid in the sense that the *uniqueness* of the perturbed semigroups is lost and one needs to concentrate on the minimal one only.

(iii) If we choose, in particular, $x = e_j$, the j-th member of the canonical basis of X, then $x \in X_+$ and $b(\lambda, e_j) = 0$ if and only if $\|P(t)e_j\| = \sum_k p_{jk}(t) = 1$ for all t and each j. This follows from (6.9).

The next theorem collects all these results relevant to the conservativity of the minimal semigroup.

Theorem 6.2.14. The minimal semigroup P is conservative (or Markov) that is, $\|P(t)x\| = \|x\|\ \forall x \in X_+$, if and only if any one of the following equivalent conditions is satisfied.

(i) For some $\lambda > 0$, and hence for all $\lambda > 0$ and all $x \in X_+$, $b(\lambda; x) \equiv \lim_{n\to\infty} \|B(\lambda)^n x\| = 0$.

(ii) For some $\lambda > 0$, the dual eigenvalue equation $A_0^* x^* = \lambda x^*$ has no non-zero solution in $X^* = l^\infty$.

(iii) For some $\lambda > 0$, $Ran(\lambda - A_0)$ is dense in X where $A_0 = A|_{\mathsf{D}_0}$.

Furthermore, when any of the above conditions (and hence all) is satisfied, then the minimal semigroup P is the only semigroup, the generator of which is an extension of A_0.

Proof. In view of Remark 6.2.13(i), it is sufficient to prove the equivalence of the statements (i), (ii) and (iii). That (ii) is equivalent to (iii) follows by using the equality

$$\mathcal{N}((A_0 - \lambda)^*) = Ran(A_0 - \lambda)^\perp,$$

where $\mathcal{N}(T)$ is the null space of the operator T and $Ran(T)^\perp$ is the annihilator of the range of T, together with an application of the Hahn-Banach theorem. (**Exercise** 6.2.15)

(i) $\Leftrightarrow$ (iii). From (6.8) one has that

$$(\lambda + H - K)R^{(n)}(\lambda)x = x - B(\lambda)^{n+1}x,$$

and therefore (i) implies that $(\lambda + H - K)R^{(n)}(\lambda)x$ converges strongly to x as $n \to \infty$ for every $x \in X$. Since $\mathcal{D}(H) \supset R^{(n)}(\lambda)x$ converges to $R(\lambda)x$ as $n \to \infty$ and since $RanR(\lambda) = \mathcal{D}(H)$, this means that

$$Ran(\lambda + H - K) \text{ is dense in } X.$$

Since by Lemma 6.2.4, D_0 is a core for H, and since for $x \in \mathsf{D}_0 \subset \mathcal{D}(H)$, $\|Kx\| \leq \|Hx\|$, it follows that

$$\overline{(\lambda + H - K)\mathcal{D}(H)} = \overline{(\lambda + H - K)\mathsf{D}_0}.$$

Thus we have that $\overline{(\lambda - A_0)\mathsf{D}_0} = X$.
Conversely, let $\overline{(\lambda - A_0)\mathsf{D}_0} = X$. Then, since $(\lambda + H)\mathcal{D}(H) = X$, one has that

$$(I - B(\lambda))X = [I - B(\lambda)](\lambda + H)\mathcal{D}(H) = (\lambda + H - K)\mathcal{D}(H) \supseteq (\lambda - A_0)\mathsf{D}_0,$$

which is assumed to be dense and thus $Ran[I - B(\lambda)]$ is dense in X. Set

$$B_n(\lambda) = \frac{1}{(n+1)} \sum_{j=0}^{n} B(\lambda)^j$$

so that for every $x \in X$,

$$B_n(\lambda)[I - B(\lambda)]x = (n+1)^{-1}[x - B(\lambda)^{n+1}x],$$

which converges to 0. Since $Ran[I - B(\lambda)]$ is dense and $\|B_n(\lambda)\| \leq 1 \quad \forall\, n$, it then follows that $B_n(\lambda)x \to 0$ for as $n \longrightarrow \infty$ for any $x \in X$. Using the contractivity of $B(\lambda)$, one notes that

$$\|B(\lambda)^n x\| \leq \|B(\lambda)^j x\| \quad \text{ if } 0 \leq j \leq n.$$

Hence for $x \in X_+$,

$$\|B_n(\lambda)x\| = \frac{1}{n+1} \sum_{j=0}^{n} \|B(\lambda)^j x\| \geq \|B(\lambda)^n x\|$$

and one concludes that

$$b(\lambda; x) = \lim_{n \longrightarrow \infty} \|B(\lambda)^n x\| = 0 \quad \forall\, x \in X_+.$$

This completes the proof. □

To end the chapter, we discuss a few concrete examples with infinite state space.

Example 6.2.16. (The bounded case, i.e., $\sup|a_j| \equiv a < \infty$)
We have already seen that since this implies that H and K are both bounded positive operators, there is only one (uniformly continuous) semigroup P and that is clearly Markov. However, we discuss this now with Theorem 6.2.14 as the backdrop. Note that

$$\begin{aligned}\big\|B(\lambda)x\big\| &= \big\|K(\lambda+H)^{-1}x\big\| \leq \big\|H(\lambda+H)^{-1}x\big\| \\ &= \sum_j \big|\big(\frac{a_j}{(\lambda+a_j)}\big)x_j\big| \leq \big(\frac{a}{(\lambda+a)}\big)\|x\|\end{aligned}$$

implying that $\big\|B(\lambda)\big\| \leq a(\lambda+a)^{-1}$ and therefore $\big\|B(\lambda)^n\big\| \longrightarrow 0$ as $n \longrightarrow \infty$ for every $\lambda > 0$.

Example 6.2.17. (The death process) Here assume that the infinite matrix A is lower triangular, that is, $a_{jk} = 0 \,\forall\, k > j$, and therefore $\big(B(\lambda)x\big)_k = \sum_{j\neq k} x_j b_{jk}(\lambda)$ with the matrix element $b_{jk}(\lambda) = 0$ if $k > j$ and $= a_{jk}/\lambda + a_j$ if $k < j$. Then

$$\Big(B(\lambda)^n e_j\Big)_k = \sum \Big(\frac{a_{j_n j_{n-1}}}{\lambda + a_{j_n}}\Big)\Big(\frac{a_{j_{n-1}j_{n-2}}}{\lambda + a_{j_{n-1}}}\Big)\cdots\Big(\frac{a_{j_2 j_1}}{\lambda + a_{j_2}}\Big)\Big(\frac{a_{j_1 k}}{\lambda + a_{j_1}}\Big),$$

where the summation above is taken over the n indices $j_1, j_2, \ldots, j_n$, subject to the conditions $j = j_n \neq j_{n-1}, j_{n-1} \neq j_{n-2}, \ldots j_2 \neq j_1, j_1 \neq k$. By virtue of the assumption on the matrix elements a_{jk}, the above sum is non-zero only if $j = j_n \geq j_{n-1} + 1 \geq j_{n-2} + 2 \geq \ldots \geq j_1 + n - 1 \geq k + n$. This implies that $\big(B(\lambda)^n e_j\big)_k = 0$ if $j < k + n$ and for each fixed j,

$$\big\|B(\lambda)^n e_j\big\| = \sum_{k=1}^{\infty} \big(B(\lambda)^n e_j\big)_k = 0$$

if we choose n large enough. Therefore, for $x = \sum_{j=1}^{\infty} \alpha_j e_j$, since $B(\lambda)$ is a contraction,

$$\big\|B(\lambda)^n x\big\| \leq \big\|B(\lambda)^n \sum_{j=1}^{N} \alpha_j e_j\big\| + \big\|x - \sum_{j=1}^{N} \alpha_j e_j\big\|,$$

and we have that $B(\lambda)^n$ converges strongly to 0. Equivalently, by Theorem 6.2.14, the associated minimal semigroup is Markov.

Example 6.2.18. (The birth process) In this example, the matrix a_{jk} has the property that $a_{j,j+1} = -a_{jj} = a_j \geq 0$ and all other $a_{jk} = 0$. This means that for the matrix $\big(b_{jk}(\lambda) \equiv \big(B(\lambda)e_j\big)_k\big)$ all $b_{jk}(\lambda) = 0$ except for $b_{j,j+1}(\lambda)$ which is equal to $\frac{a_j}{\lambda + a_j}$. Therefore, $\big(B(\lambda)^n e_j\big)_k = 0$ unless $k = j + n$ and

$$\begin{aligned} \big\|B(\lambda)^n e_j\big\| = \big(B(\lambda)^n e_j\big)_{j+n} &= \prod_{m=0}^{n-1} \big(\frac{a_{j+m}}{\lambda + a_{j+m}}\big) \\ &= \prod_{m=0}^{n-1} \big(1 + \lambda a_{j+m}^{-1}\big)^{-1}. \end{aligned}$$

For fixed j, the infinite product above converges to 0 as $n \to \infty$ if and only if $\sum_{m=0}^{\infty} \log\big(1 + \lambda a_{j+m}^{-1}\big) = \infty$, which happens if and only if either $a_j = 0$ for infinitely many j or $\sum_{j=N}^{\infty} {a_j}^{-1} = \infty$ for some N such that $a_j > 0$ for all $j \geq N$. On the other hand if $\sum_{m=0}^{\infty} a_{j+m}^{-1} < \infty$ for some j, then

$$\begin{aligned} b_j(\lambda) \equiv \lim_{n \longrightarrow \infty} \big\|B(\lambda)^n e_j\big\| &= \prod_{m=0}^{\infty} \big(1 + \lambda a_{j+m}^{-1}\big)^{-1} \\ &= 1 - \lambda \sum_{m=0}^{\infty} a_{j+m}^{-1} + o(\lambda^2) \quad \text{for small } \lambda > 0. \end{aligned}$$

In such a case, by Lemma 6.2.10(iii),

$$\lim_{t \longrightarrow \infty} \big\|P(t)e_j\big\| = \big\|e_j\big\| - \lim_{\lambda \downarrow 0} b_j(\lambda) = 0, \text{ that is, } \lim_{t \longrightarrow \infty} \sum_k p_{jk}(t) = 0.$$

It also follows from Lemma 6.2.10(iv), that

$$\begin{aligned} \sum_k \int_0^{\infty} p_{jk}(t)dt = \int_0^{\infty} \big\|P(t)e_j\big\|dt &= \lim_{\lambda \downarrow 0} \big\{\lambda^{-1}\big(\big\|e_j\big\| - b_j(\lambda)\big)\big\} \\ &= \sum_{m=0}^{\infty} a_{j+m}^{-1} < \infty. \end{aligned}$$

Remark 6.2.19. Many results, similar to the ones derived in this chapter, can be obtained also in the non-commutative context. For example, one can study the Markov or semi-Markov semigroups on $\mathcal{B}(H)$, or its pre-dual semigroup on $\mathcal{B}_1(H)$ [6]. A more general extension of this theory to an arbitrary von Neumann algebra or its pre-dual Banach space, respectively, can be found in [25, Section 3.2].

Chapter 7

Applications to partial differential equations

While semigroups of operators are interesting objects of study by themselves, one of the main reasons why they are studied so extensively is due to the important role they play in the study of partial differential equations.

Consider the well-known initial value problem:

$$\begin{cases} \dfrac{\partial u}{\partial t}(x,t) &= \Delta u(x,t), \quad x \in \mathbb{R}^d, t \geq 0 \\ u(x,0) &= f(x), \qquad x \in \mathbb{R}^d, \end{cases} \tag{7.1}$$

where

$$\Delta u = \sum_{i=1}^{d} \frac{\partial^2 u}{\partial x_i^2}.$$

Finding solutions of the above using the method of separation of variables, which is an effective method only in the presence of some symmetry in the system, is part of every basic course on partial differential equations, often leaving many fundamental issues like uniqueness, completeness and so on unanswered. However, the theory of C_0-semigroups can be employed to treat an equation such as this in an elegant manner, yielding, in most cases pay-offs beyond just existence and uniqueness of solutions. To begin with, given the linear partial differential equation, one tries to write it as an abstract Cauchy problem on a Banach space X,

$$\begin{cases} u'(t) &= Au(t), \quad t \geq 0 \\ u(0) &= f \end{cases} \tag{7.2}$$

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7_7

where the space X is a suitably chosen concrete Banach space of functions in which the initial value lies and A is the differential operator (from the given partial differential equation) with an appropriate domain $\mathcal{D}(A) \subset X$. The solution of the given differential equation is then a trajectory in X. Depending on the nature and properties of the operator A, the trajectory may be described by a C_0-semigroup, which is a very convenient tool. Indeed, if the operator A is the infinitesimal generator of a C_0-semigroup on X, say $\{T(t)\}_{t\geq 0}$, then $u(t) = T(t)f$ gives mild ($f \in X$) or classical ($f \in \mathcal{D}(A)$) solutions of the Cauchy problem and hence of the original problem.

We now discuss some simple differential operators and show that they generate C_0-semigroups on appropriate function spaces.

7.1 Parabolic equations

By a multi-index α we shall mean that $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d)$ where $\alpha_1, \ldots, \alpha_d \in \mathbb{N} \cup \{0\}$. We set $|\alpha| = \sum_{\mathrm{J}=1}^{j=d} \alpha_j$, and $x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_d^{\alpha_d}$, for $x \in \mathbb{R}^d$ and $\tilde{D}^\alpha = \tilde{D}_1^{\alpha_1} \cdots \tilde{D}_d^{\alpha_d}$, where $\tilde{D}_j = -i\frac{\partial}{\partial x_j}$. If P is the polynomial defined by $P(x) = \sum_{|\alpha|\leq m} a_\alpha x^\alpha$ where $a_\alpha \in \mathbb{R}$, then we set $P(\tilde{D}) = \sum_{|\alpha|\leq m} a_\alpha \tilde{D}^\alpha$. Note that in contrast to the notation D_j and D^α introduced in Section A.2, here $\tilde{D}_j$ has a factor $-i$ in front of $\dfrac{\partial}{\partial x_j}$. This is done to avoid the appearance of the powers of i in the polynomial P in the Fourier variable (see Lemma A.2.3).

If P is a polynomial, then $P(\tilde{D})$ is a *linear differential operator with constant coefficients*. The *order* of $P(\tilde{D})$ is the degree of the polynomial. The polynomial P is called *homogeneous* if it is of the form $P(x) = \sum_{|\alpha|=m} a_\alpha x^\alpha$. We shall call a homogeneous polynomial of degree $2m$ *weakly elliptic* if $P(x) \leq 0$ and *strictly elliptic* if $P(x) < 0$ for all $x \in \mathbb{R}^d \setminus \{0\}$. Note here that a non-zero homogeneous polynomial satisfying $P(x) \leq 0$, for all $x \in \mathbb{R}^d \setminus \{0\}$ is necessarily of even order. If P is strictly elliptic of order $2m$, then P is non positive on the unit sphere of $\mathbb{R}^d$, which is compact. It follows therefore that there is a constant $\theta > 0$ such that $P(x) \leq -\theta|x|^{2m}$ for all $x \in \mathbb{R}^d$, where $|x|^2 = x_1^2 + \cdots + x_d^2$.

From the discussions in Appendix A.2 it follows that $P(\tilde{D})$ has a natural extension as an operator acting on the space $L^2(\mathbb{R}^d)$. Precisely, we have that

$$\mathcal{D}(P(\tilde{D})) = \{f \in L^2(\mathbb{R}^d) : P(\cdot)\mathcal{F}f \in L^2(\mathbb{R}^d)\},$$
$$P(\tilde{D}) = \mathcal{F}^{-1}M_P\mathcal{F},$$

where M_P is the multiplication operator (see Appendix A.1 and also Section 2.5) given by

$$(M_P f)(x) = P(x)f(x), \text{ for all } x \in \mathbb{R}^d,$$

$$\text{for all } f \in \mathcal{D}(M_P) = \Big\{f \in L^2(\mathbb{R}^d) : \int_{\mathbb{R}^d} |P(x)f(x)|^2\,dx < \infty\Big\}.$$

Thus M_P and $P(\tilde{D})$ are unitarily equivalent.

If $Q(x,y) = \sum b_\alpha(x)y^\alpha$ is a polynomial in $y \in \mathbb{R}^d$ such that the coefficients b_α are functions of $x \in \mathbb{R}^d$, then the operator $Q(\cdot,\tilde{D})$ given by

$$(Q(\cdot,\tilde{D})f)x = \sum b_\alpha(x)\tilde{D}^\alpha f(x)$$

is called a *linear partial differential operator with variable coefficients*. Note here that $Q(\cdot,\tilde{D})$ can be written as $Q(\cdot,\tilde{D}) = \sum M_{b_\alpha}\tilde{D}^\alpha$.

Theorem 7.1.1. If P is a weakly elliptic homogeneous polynomial of degree $2m$ $(m = 1, 2, \ldots)$ with real coefficients, then $P(\tilde{D})$ generates a holomorphic semigroup on $L^2(\mathbb{R}^d)$ on the sector $\operatorname{Re} z > 0$ (that is, of angle $\pi/2$) and a unitary group for $\operatorname{Re} z = 0$. Furthermore, if P is strictly elliptic and $Q(x,y) = \sum_{|\alpha|\le 2m-1} b_\alpha(x)y^\alpha$, where $b_\alpha \in L^\infty(\mathbb{R}^d)$, for all α, then $P(\tilde{D}) + Q(\cdot,\tilde{D})$ also generates a holomorphic semigroup.

Proof. Since P is weakly elliptic, it is in particular real valued and therefore, from Theorem A.1.13, it follows that M_P and hence $P(\tilde{D})$ is selfadjoint. Moreover, $P(\tilde{D}) \le 0$, since P being weakly elliptic implies

$$\langle M_P f, f\rangle_{L^2(\mathbb{R}^n)} = \int_{\mathbb{R}^n} P(x)|f(x)|^2\,dx \le 0 \ \ \forall\ f \in \mathcal{D}(M_P),$$

that is, $M_P \le 0$. Therefore, by Example 3.4.2, it follows that $P(\tilde{D})$ generates a holomorphic semigroup of angle $\pi/2$. By Stone's Theorem, $iP(\tilde{D})$ generates a unitary C_0-group on $\mathbb{R}$ or equivalently, $P(\tilde{D})$ generates a unitary C_0-group for $\operatorname{Re} z = 0$.

We shall prove that $Q(\cdot,\tilde{D}) = \sum M_{b_\alpha}\tilde{D}^\alpha$ is relatively bounded with respect to $P(\tilde{D})$, in fact that its $P(\tilde{D})$-bound is less than 1. Let α be a multi-index with $|\alpha| \le 2m-1$. Since the map $\mathbb{R}^d \ni y \mapsto y^\alpha$ is continuous where $|\cdot|$ is the Euclidean norm in $\mathbb{R}^d$, and since $\big||y^\alpha|/|y|^{2m}\big| = |y|^{|\alpha|-2m}$ converges to 0 as $|y| \to \infty$, we conclude that for every $\epsilon > 0$, we can find $c_\epsilon > 0$ such that

$$|y^\alpha|^2 \le c_\epsilon^2 + \epsilon^2|y|^{4m} \quad \forall \quad y \in \mathbb{R}^d.$$

Since P is strictly elliptic, there exists $\theta > 0$ such that $P(y) \le -\theta|y|^{2m}$ for all $y \in \mathbb{R}^d$, so $|P(y)|^2 \ge \theta^2|y|^{4m}$. Therefore, by the results in Appendix A.2, and in view of the discussion immediately preceding Theorem 7.1.1, it follows, on setting $\phi_\alpha(y) = y^\alpha$, that

$$\begin{aligned}
\|\tilde{D}^\alpha f\|^2_{L^2(\mathbb{R}^d)} &= \|\mathcal{F}^{-1}M_{\phi_\alpha}\mathcal{F}f\|^2_{L^2(\mathbb{R}^d)} = \int_{\mathbb{R}^d} |(M_{\phi_\alpha}\mathcal{F}f)(y)|^2\,dy \\
&= \int_{\mathbb{R}^d} |y^\alpha|^2|\mathcal{F}f(y)|^2\,dy \\
&\le c_\epsilon^2\|\mathcal{F}f\|^2 + \epsilon^2\int_{\mathbb{R}^n} |y|^{4m}|\mathcal{F}(y)|^2\,dy \\
&\le c_\epsilon^2\|f\|^2 + \frac{\epsilon^2}{\theta^2}\|P(\tilde{D})f\|^2 \qquad (7.3)\\
&\le \big(c_\epsilon\|f\| + \frac{\epsilon}{\theta}\|P(\tilde{D})f\|\big)^2. \qquad (7.4)
\end{aligned}$$

Since $b_\alpha \in L^\infty(\mathbb{R}^d), M_{b_\alpha}$ is a bounded operator on $L^2(\mathbb{R}^d)$, and $\|M_{b_\alpha}\| \le \|b_\alpha\|_\infty$. Therefore, using (7.4), we get, for $f \in \mathcal{D}(P(\tilde{D}))$,

$$\|M_{b_\alpha}(\tilde{D}^\alpha f)\| \le \|b_\alpha\|_\infty\|\tilde{D}^\alpha f\| \le \epsilon\|b_\alpha\|_\infty\theta^{-1}\|P(\tilde{D})f\| + \|b_\alpha\|_\infty c_\epsilon\|f\|.$$

This holds for every α with $|\alpha| \le 2m-1$. Therefore,

$$\begin{aligned}
\|Q(\cdot,\tilde{D})f\| &= \Big\|\sum_{|\alpha|\le 2m-1} M_{b_\alpha}\tilde{D}^\alpha f\Big\| \le \sum_{|\alpha|\le 2m-1}\|b_\alpha\|_\infty\|\tilde{D}^\alpha f\| \\
&\le \Big(\sum_{|\alpha|\le 2m-1}\|b_\alpha\|_\infty\Big)\big(\epsilon\theta^{-1}\|P(\tilde{D})f\| + c_\epsilon\|f\|\big).
\end{aligned}$$

It follows that $\mathcal{D}(P(\tilde{D})) \subset \mathcal{D}(Q(\cdot,\tilde{D}))$ and $Q(\cdot,\tilde{D})$ is $P(\tilde{D})$-bounded, and since $\epsilon > 0$ is arbitrary, the $P(\tilde{D})$-bound of $Q(\cdot,\tilde{D})$ is 0. Thus, by Remark 4.1.6, it follows that $P(\tilde{D}) + Q(\cdot,\tilde{D})$ generates a semigroup holomorphic in the right half plane. □

Example 7.1.2. Let $P(k) = -\sum_{j=1}^{d} k_j^2$ and $Q(x,k) = \sum_{j=1}^{d} b_j(x)k_j$, where $b_j \in L^\infty(\mathbb{R}^d)$. Then P, Q satisfy all the conditions of Theorem 7.1.1. Thus $A = P(\tilde{D}) + Q(\cdot, \tilde{D})$ generates a holomorphic semigroup of angle $\pi/2$, that is, $A = \sum_{j=1}^{d} \tilde{D}_j^2 + \sum_{j=1}^{d} b_j(x)\tilde{D}_j$ is the generator of a holomorphic semigroup of angle $\pi/2$.

7.2 The wave equation

We consider the following initial value problem for the wave equation:

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2}(x,t) &= \Delta u(x,t), \quad x \in \mathbb{R}^d,\ t > 0, \\ u(x,0) &= f(x), \qquad x \in \mathbb{R}^d, \\ \dfrac{\partial u}{\partial t}(x,0) &= g(x), \qquad x \in \mathbb{R}^d. \end{cases} \tag{7.5}$$

We rewrite (7.5) as a first order problem in two dimensions, by setting $u_1(x,t) = u(x,t)$ and $u_2(x,t) = \frac{\partial u}{\partial t}(x,t)$, to get that

$$\begin{cases} \dfrac{\partial}{\partial t}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} &= \begin{pmatrix} 0 & I \\ \Delta & 0 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad x \in \mathbb{R}^d,\ t > 0, \\ \begin{pmatrix} u_1(x,0) \\ u_2(x,0) \end{pmatrix} &= \begin{pmatrix} f(x) \\ g(x) \end{pmatrix}, \qquad x \in \mathbb{R}^d. \end{cases} \tag{7.6}$$

As mentioned earlier, we will write (7.6) as an abstract Cauchy problem on an appropriate Banach space and show that the corresponding operator A is the generator of a C_0-semigroup. The right space in this instance is the Hilbert space $\mathcal{H} = H^1(\mathbb{R}^d) \oplus L^2(\mathbb{R}^d)$ (see Appendix A.3).

Note that if we define, for $(u_1, u_2) \in Y := C_c^\infty(\mathbb{R}^d) \oplus C_c^\infty(\mathbb{R}^d)$, the norm $\|\cdot\|$ by

$$\|(u_1, u_2)\|^2 = \|u_1\|^2_{H^1(\mathbb{R}^d)} + \|u_2\|^2_{L^2(\mathbb{R}^d)},$$

then the completion of the normed space $\big(Y, \|\cdot\|\big)$ is the Hilbert Space $\mathcal{H}$.

We define the operator A associated with (7.6) as follows:

Definition 7.2.1. Let $\mathcal{D}(A) = H^2(\mathbb{R}^d) \oplus H^1(\mathbb{R}^d) \subset \mathcal{H}$ and define

$$A(u_1, u_2) = (u_2, \Delta u_1) \text{ for all } (u_1, u_2) \in \mathcal{D}(A).$$

Note that then the above problem may be written as the abstract Cauchy problem $u'(t) = Au(t), u(0) = u_0 = (f, g)$ on the Hilbert space $\mathcal{H}$.

Theorem 7.2.2. The operator A defined above is the infinitesimal generator of a C_0-semigroup $\{T(t)\}_{t\geq 0}$ on $\mathcal{H} = H^1(\mathbb{R}^d) \oplus L^2(\mathbb{R}^d)$ satisfying

$$\|T(t)\| \leq e^{2t}.$$

In fact, $\{T(t)\}_{t\geq 0}$ extends to a C_0-group on $\mathcal{H}$.

Proof. We show first that for every $f = (f_1, f_2) \in C_c^\infty(\mathbb{R}^d) \oplus C_c^\infty(\mathbb{R}^d)$ and real $\lambda \neq 0$, there is a unique element $u = (u_1, u_2) \in \mathcal{D}(A)$ such that $(I - \lambda A)u = f$ and

$$\|u\| \leq (1 - 2|\lambda|)^{-1}\|f\| \quad \text{for } 0 < |\lambda| < \frac{1}{2}.$$

Let $f = (f_1, f_2) \in C_c^\infty(\mathbb{R}^d) \oplus C_c^\infty(\mathbb{R}^d)$, and let $k > 2$. Set

$$\tilde{w}_i(\eta) = (1 + \lambda^2|\eta|^2)^{-1}\hat{f}_i(\eta).$$

Since $f_i \in C_c^\infty(\mathbb{R}^d) \subset H^2(\mathbb{R}^d), (1 + |\cdot|^2)^{k/2}\hat{f}_i \in L^2(\mathbb{R}^d)$, by Lemma A.3.1 and thus $(1 + \lambda^2|\eta|^2)^{(k+2)/2}\tilde{w}_i(\eta) \in L^2(\mathbb{R}^d)$.

If we set

$$w_i(x) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{i\langle x,\eta\rangle}\tilde{w}(\eta)\, d\eta \text{ for } \ i = 1, 2,$$

then $w_i \in H^{k+2}(\mathbb{R}^d)$, again by Lemma A.3.1, and

$$w_i - \lambda^2\Delta w_i = f_i, \quad i = 1, 2.$$

Setting $u_1 = w_1 + \lambda w_2, u_2 = w_2 + \lambda\Delta w_1$ and $u = \big(u_1, u_2\big)$ we see that $u \in H^k(\mathbb{R}^d) \oplus H^{k-2}(\mathbb{R}^d) \subset \mathcal{D}(A) = H^2(\mathbb{R}^d) \oplus H^1(\mathbb{R}^d)$ for $k \geq 3$, and

$$\begin{aligned}(I - \lambda A)u &= \big(u_1 - \lambda u_2, u_2 - \lambda\Delta u_1\big) = \big(w_1 - \lambda^2\Delta w_1, w_2 - \lambda^2\Delta w_2\big) \\ &= \big(f_1, f_2\big) = f.\end{aligned}$$

Observe that since $f_i \in C_c^\infty(\mathbb{R}^d), w_i \in S(\mathbb{R}^d)$, for $i = 1, 2$ and hence u_1, u_2 are in the domain of Δ. We also note here that $\langle u_2, \Delta u_1\rangle_{L^2(\mathbb{R}^d)} = \langle \Delta u_2, u_1\rangle_{L^2(\mathbb{R}^d)}$.

Moreover, since $f_1 = u_1 - \lambda u_2$ and $f_2 = u_2 - \lambda \Delta u_1$,

$$\begin{aligned}
\|f\|_{\mathcal{H}}^2 &= \langle f_1, f_1\rangle_{H^1(\mathbb{R}^d)} + \langle f_2, f_2\rangle_{L^2(\mathbb{R}^d)} \\
&= \langle f_1, f_1\rangle_{L^2(\mathbb{R}^d)} + \langle \nabla f_1, \nabla f_1\rangle_{L^2(\mathbb{R}^d)} + \langle f_2, f_2\rangle_{L^2(\mathbb{R}^d)} \\
&= \langle f_1, f_1\rangle_{L^2(\mathbb{R}^d)} - \langle \Delta f_1, f_1\rangle_{L^2(\mathbb{R}^d)} + \langle f_2, f_2\rangle_{L^2(\mathbb{R}^d)} \\
&= \langle u_1 - \lambda u_2 - \Delta u_1 + \lambda \Delta u_2, u_1 - \lambda u_2\rangle_{L^2(\mathbb{R}^d)} \\
&\quad + \langle u_2 - \lambda \Delta u_1, u_2 - \lambda \Delta u_1\rangle_{L^2(\mathbb{R}^d)} \\
&= \langle u_1 - \Delta u_1, u_1\rangle_{L^2(\mathbb{R}^d)} + \|u_2\|_{L^2(\mathbb{R}^d)}^2 - 2\lambda \operatorname{Re}\langle u_1, u_2\rangle_{L^2(\mathbb{R}^d)} \\
&\quad + \lambda^2 \|\Delta u_1\|_{L^2(\mathbb{R}^d)}^2 + \lambda^2 \|u_2\|_{L^2(\mathbb{R}^d)} \\
&\quad - \lambda^2 \langle \Delta u_2, u_2\rangle_{L^2(\mathbb{R}^d)} - \lambda\big[\langle u_2, \Delta u_1\rangle_{L^2(\mathbb{R}^d)} - \langle \Delta u_2, u_1\rangle_{L^2(\mathbb{R}^d)}\big].
\end{aligned}$$

Since $-\Delta$ is a positive operator we conclude that

$$\begin{aligned}
\|f\|_{\mathcal{H}}^2 &\geq \langle u_1 - \Delta u_1, u_1\rangle_{L^2(\mathbb{R}^d)} + \|u_2\|_{L^2(\mathbb{R}^d)}^2 - 2|\lambda| \operatorname{Re}\langle u_1, u_2\rangle_{L^2(\mathbb{R}^d)} \\
&\geq (1 - |\lambda|)||u||_{\mathcal{H}}^2.
\end{aligned}$$

Therefore, if $\lambda \in \mathbb{R}$ and $0 < |\lambda| < \frac{1}{2}$, then

$$\|f\|_{\mathcal{H}}^2 \geq (1 - 2|\lambda|)^2 \|u\|_{\mathcal{H}}^2. \tag{7.7}$$

Thus, whenever λ is real and $0 < |\lambda| < \frac{1}{2}$, the range of $(I - \lambda A)$ contains $C_c^\infty(\mathbb{R}^d) \oplus C_c^\infty(\mathbb{R}^d)$, which is dense in $\mathcal{H}$. But A is closed, and we have that the range of $(I - \lambda A)$ is closed and hence it equals $\mathcal{H}$. Therefore, for every $f \in \mathcal{H}$ and real λ satisfying $0 < |\lambda| < \frac{1}{2}$, there is a unique $u \in \mathcal{D}(A)$ such that

$$(I - \lambda A)u = f \text{ and } \|u\|_{\mathcal{H}} \geq (1 - 2|\lambda|)^{-1} \|f\|_{\mathcal{H}}.$$

This implies that $(\mu - A)^{-1}$ exists as a bounded operator on $\mathcal{H}$, for real μ satisfying $|\mu| > 2$, and

$$\|(\mu - A)^{-1}\| \leq (|\mu| - 2)^{-1} \text{ for such } \mu. \tag{7.8}$$

Thus, $(-\infty, -2) \cup (2, \infty) \subset \rho(A)$ and $\|R(\mu, A)\| \leq (|\mu| - 2)^{-1}$ if $|\mu| > 2$.
Noting that $\mathcal{D}(A) = H^2(\mathbb{R}^d) \oplus H^1(\mathbb{R}^d)$ is dense in H, and A is closed, it follows from Theorem 2.3.3 that A generates a C_0-semigroup $\{T(t)\}_{t\geq 0}$ on $\mathcal{H}$ satisfying $\|T(t)\| \leq e^{2t}$, for all $t \geq 0$. Furthermore, since $-A$ also generates a C_0-semigroup, $\{T(t)\}_{t\geq 0}$ extends to a C_0-group.

□

7.3 Schrödinger equation

In this section we consider the Schrödinger equation

$$\frac{1}{i}\frac{\partial u}{\partial t} = \Delta u - Vu, \tag{7.9}$$

where the function V is called the potential. As before, we write (7.9) as an abstract problem on a Hilbert space, $L^2(\mathbb{R}^d)$, and use the theory of C_0-semigroups developed in the previous chapters. Let $H = L^2(\mathbb{R}^d)$. Define the operators A and B as follows:

Definition 7.3.1. Let $\mathcal{D}(A) = H^2(\mathbb{R}^d)$ and set for $u \in \mathcal{D}(A)$,

$$Au = i\Delta u = i\sum_{j=1}^{d} \frac{\partial^2 u}{\partial x_j}.$$

Definition 7.3.2. Let V be a measurable, finite almost everywhere defined function on $\mathbb{R}^d$, with $\mathcal{D}(B) = \{u \in L^2(\mathbb{R}^d), Vu \in L^2(\mathbb{R}^d)\}$ and set

$$(Bu)(x) = V(x)u(x) \text{ for all } x \in \mathbb{R}^d.$$

Theorem 7.3.3. The operator A defined above generates a C_0-group of unitary operators.

Proof. The result follows from Stone's Theorem 3.2 if we can show that $iA = -\Delta$ is selfadjoint. By the results on the Fourier transform in $L^2(\mathbb{R}^n)$ in Lemma A.2.3 and Remark A.2.6(d), it follows that iA is unitarily equivalent to the operator of multiplication by the measurable non-negative function $\phi(k) = k^2$, that is,

$$\mathcal{F}(iAf) = M_\phi(\mathcal{F}f)(k) \text{ for all } f \in \mathcal{D}(iA), \text{ where}$$
$$\mathcal{D}(iA) = \big\{f \in L^2(\mathbb{R}^d : \int \big|\phi(k)(\mathcal{F}f)(k)\big|^2\, dk < \infty\big\}.$$

On the other hand, Theorem A.1.13 tells us that such an M_ϕ is selfadjoint (and is also non-negative). □

We remark here that the operator $-iA$ discussed in this section coincides with the operator $P(\tilde{D})$ discussed in Section 7.1, with the polynomial P as in Example 7.1.2. Hence the conclusion in Theorem 7.3.3 may alternatively be deduced from the proof of Theorem 7.1.1, where it is shown that $iP(\tilde{D})$ generates a unitary C_0-group on $\mathbb{R}$.

Theorem 7.3.4. Let $\phi \in L^p(\mathbb{R}^d)$ where $p \geq 2$ and $p > d/2$. Then $M_\phi(iA+1)^{-1}$ is compact in $L^2(\mathbb{R}^d)$.

Proof. Using results on the Fourier transform in Appendix (A.2) one concludes that $\mathcal{F}(iA+1)^{-1}\mathcal{F}^{-1} = M_{(k^2+1)^{-1}}$ in $L^2(\mathbb{R}^d)$. For any measurable function $\psi\colon \mathbb{R}^d \to \mathbb{C}$, the operator $M_\phi\psi(iA)$ is an integral operator with kernel $(2\pi)^{-d/2}\phi(x)\hat{\psi}(y-x)$ and therefore it follows easily that $M_\phi\psi(iA)$ is Hilbert-Schmidt and hence compact if both ϕ and $\psi \in L^2(\mathbb{R}^d)$. Indeed, for $\psi \in L^p(\mathbb{R}^d)$ with $p > d/2$ and $p \geq 2$, we have $\psi(iA)\hat{f} \in L^q(\mathbb{R}^d)$, by the generalised Hölder inequality, where $q^{-1} = p^{-1} + 1/2$. Thus $1 \leq q \leq 2$, and therefore, by the Hausdorff-Young inequality (A.14) in Appendix A.2, the inverse Fourier transform $\mathcal{F}^{-1}(\psi(iA)\hat{f}) \in L^r(\mathbb{R}^d)$, with $r^{-1} + q^{-1} = 1$, such that

$$\|\mathcal{F}^{-1}(\psi(iA)\hat{f})\|_r \leq C(p)\|\psi(iA)\hat{f}\|_q.$$

From the two relations among p, q and r it follows that $r^{-1}+p^{-1} = 1/2$ and another application of the generalised Hölder inequality yields that $M_\phi\psi(iA)f \in L^2(\mathbb{R}^d)$, and that

$$\begin{aligned}\|M_\phi\psi(iA)f\|_2 &\leq \|\phi\|_p\big\|\mathcal{F}^{-1}(\psi(iA)\hat{f})\big\|_r \\ &\leq C(p)\|\phi\|_p\|\psi(iA)\hat{f}\|_q \leq C(p)\|\phi\|_p\|\psi\|_p\|f\|_2.\end{aligned}$$

This leads to the conclusion that $M_\phi\psi(iA) \in \mathcal{B}(H)$ and

$$\|M_\phi\psi(iA)\| \leq C(p)\|\phi\|_p\|\psi\|_p. \tag{7.10}$$

For $\phi, \psi \in L^p(\mathbb{R}^d)$, set $\phi_m(x) = \chi_m(x)\phi(x)$, and $\psi_m(iA) = \chi_m(iA)\psi(A)$, where $m \in \mathbb{N}$, and $\chi_m(x) = \chi_{\{x:|x|\leq m\}}(x)$ for all $x \in \mathbb{R}^d$. Then ϕ_m and $\psi_m \in L^2(\mathbb{R}^d)$ for every m. By the earlier discussion, it follows that $M_{\phi_m}\psi_m(iA)$ is Hilbert Schmidt and hence compact in $L^2(\mathbb{R}^d)$. By an application of the Dominated Convergence Theorem one has that $\|\phi-\phi_m\|_p$ and $\|\psi-\psi_m\|_p \to 0$, as $m \to \infty$. Therefore, by (7.10),

$$\begin{aligned}\|[M_\phi\psi(iA)] - [M_{\phi_m}\psi_m(iA)]\| &\leq \|(M_\phi - M_{\phi_m})\psi(iA)\| \\ &\quad + \|M_{\phi_m}(\psi(iA) - \psi_m(iA))\| \\ &\leq C(p)\big(\|\phi-\phi_m\|_p\|\psi\|_p + \|\phi_m\|_p\|\psi-\psi_m\|_p\big).\end{aligned}$$

This estimate, together with the uniform boundedness of $\|\phi_m\|_p$, implies that $M_\phi\psi(iA)$ is the limit in operator norm of the sequence $\{M_{\phi_m}\psi_m(iA)\}_m$ of

Hilbert Schmidt operators and hence is compact. For the final conclusion it remains only to convince ourselves that if $\psi(k) = (k^2 + 1)^{-1}$ for all $k \in \mathbb{R}^d$, then $\psi \in L^p(\mathbb{R}^d)$ with $p > d/2$ and $p \geq 2$. □

Theorem 7.3.5. Suppose $V(x)$ is real for all $x \in \mathbb{R}^d$ and $V \in L^p(\mathbb{R}^d)$, where $p > d/2$ and $p \geq 2$, and let $B = M_V$, the operator of multiplication by V. Then $A - iB$ is the infinitesimal generator of a group of unitary operators $L^2(\mathbb{R}^2)$ and $\sigma_e(iA + B) = \sigma_e(iA)$.

Proof. By Theorem 7.3.4, $M_V(iA+1)^{-1} = B(iA+1)^{-1}$ is compact and therefore, by Theorem 4.2.1, B is iA compact. Therefore, by Theorem 4.2.6, $iA + B$ is selfadjoint and since $iA = -\Delta \geq 0, iA + B$ is also bounded below, and $\sigma_e(iA + B) = \sigma_e(iA)$. The selfadjointness of $iA + B$ together with Stone's Theorem 3.2 implies that $A - iB = -i(iA + B)$ generates a group of unitary operators. □

Appendix

We collect in this appendix concepts and results that are essential for the development of the main theme of the book, though, in most cases, they constitute basic materials. Therefore, very few proofs are included but relevant references are given where required.

A.1 Unbounded operators

A linear operator A is bounded if it maps bounded sets into bounded sets and this property is equivalent to saying that A is a continuous linear map, that is, if $x_n \to x \in X$, then $Ax_n \to Ax$. Also, this property allows A to be defined, unambiguously, on whole of X, preserving the norm and hence, one can assume without loss of generality that all such operators are defined everywhere, constituting the Banach space $\mathcal{B}(X)$. However, a linear operator, when not bounded, cannot be continuous and the property *nearest to continuity* is that of **closedness**:

Definition A.1.1. Let X be a Banach space. A linear operator $A\colon \mathcal{D}(A) \subset X \to X$ is said to be

1. **closed** if for every Cauchy sequence $\{x_n\} \subset \mathcal{D}(A)$ for which $\{Ax_n\}$ is also a Cauchy sequence, one has $x \equiv \lim_{n\to\infty} x_n \in \mathcal{D}(A)$ and $\lim_n Ax_n = Ax$;
2. **closable** if $\{x_n\} \subset \mathcal{D}(A)$ is such that $x_n \to 0$ and $Ax_n \to y$, then $y = 0$.

Just as a bounded operator, a priori not defined everywhere, can be extended to an element of $\mathcal{B}(X)$, a linear operator A, if closable, can be extended to a closed operator but there can be many closed extensions; the smallest of these extensions (that is, the one with the smallest domain), called the **closure**

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7

of A and denoted $\overline{A}$, is obtained as follows:

$$\begin{aligned}\mathcal{D}(\overline{A}) := \{x \in X : \exists\, \{x_n\} \subset \mathcal{D}(A) \quad &\text{and } y \in X \\ &\text{such that } \{x_n\} \to x \text{ and } Ax_n \to y\},\\ \overline{A}x = y \text{ for all } x \in \mathcal{D}(\overline{A}).&\end{aligned}$$

One can check that the closability implies that $\overline{A}$ is a well defined operator with $\mathcal{D}(A) \subset \mathcal{D}(\overline{A})$, and also that A is closed if and only if $A = \overline{A}$ (**Exercise** A.1.2).

As an example, consider the (classical) differential operator in $L^2(\mathbb{R})$, given by

$$A = -i\frac{d}{dx} \text{ with } \mathcal{D}(A) = C_c^\infty(\mathbb{R}),$$

the set of smooth functions on $\mathbb{R}$ with compact support, which is dense in $L^2(\mathbb{R})$. It can be shown (**Exercise** A.1.3) that A is closable but not closed. For this, one may note that the function $f_0(x) = \exp(-|x|)$ is differentiable everywhere except at $x = 0$ and $f_0' \in L^2(\mathbb{R})$ but f_0 does not belong in $C_c^\infty(\mathbb{R})$. We shall discuss this operator again in the next section on Fourier transforms.

Definition A.1.4. For a closed operator A on a Banach space X, the resolvent set $\rho(A)$ and the resolvent operator $R(\lambda, A)$ of A are respectively defined as:

$$\rho(A) = \{\lambda \in \mathbb{C} : (\lambda - A)^{-1} \in \mathcal{B}(X)\} \text{ and } R(\lambda, A) = (\lambda - A)^{-1}.$$

Further, the spectrum of A shall be denoted by $\sigma(A) := \mathbb{C} \setminus \rho(A)$.

Then the following facts can be easily verified (**Exercise** A.1.5).

(1) For each $\lambda, \mu \in \rho(A)$, the following **resolvent equation** holds.

$$R(\lambda, A) - R(\mu, A) = (\mu - \lambda)R(\lambda, A)R(\mu, A). \tag{A.1}$$

(2) $\rho(A)$ is open in $\mathbb{C}$ and the map $\rho(A) \ni \lambda \mapsto R(\lambda, A) \in \mathcal{B}(X)$ is norm-holomorphic in each connected component of $\rho(A)$. In particular, $R(\lambda, A)$ admits a power series expansion

$$R(\lambda, A) = \sum_{n=0}^{\infty} R(\lambda_0, A)^{n+1}(\lambda_0 - \lambda)^n, \text{ for } |\lambda - \lambda_0| \leq \|R(\lambda_0, A)\|^{-1}. \tag{A.2}$$

(3) $\sigma(A)$ is a closed subset of $\mathbb{C}$.

If the underlying space X is a Hilbert space, then the **adjoint** A^* of a densely defined linear operator A is defined as follows:

$$\mathcal{D}(A^*) = \{g \in H : \exists\, g^* \in H \text{ such that } \langle g, Af\rangle = \langle g^*, f\rangle \ \forall f \in \mathcal{D}(A)\},$$
$$A^* g = g^* \ \forall g \in \mathcal{D}(A^*).$$

For such an operator, one has the result that A is closable if and only if its adjoint A^* is densely defined and in such a case $\overline{A} = A^{**} = (A^*)^*$ (**Exercise** A.1.6).

In a similar spirit, the adjoint A^* for a linear, densely defined operator A can be defined on a Banach space X :

$$\mathcal{D}(A^*) = \{g^* \in X^* : \exists h^* \in X^* \text{ such that } \langle g^*, Af\rangle = \langle h^*, f\rangle \ \forall f \in \mathcal{D}(A) \subset X\},$$
$$A^* g^* = h^* \quad \forall g^* \in \mathcal{D}(A^*) \subset X^*.$$

An example of a linear (unbounded) operator which is not closable is the following.

Example A.1.7. Let $\{e_j\}_1^\infty$ be an orthonormal basis in an infinite-dimensional Hilbert space, let $\mathcal{D}(A)$ be the linear manifold of all finite linear combinations of $\{e_j\}$, and set $Ae_j = je_1$ and extend linearly. It can be verified easily that $\mathcal{D}(A^*)$ is not dense and A is not closable (**Exercise** A.1.8).

A dense subset $S \subset \mathcal{D}(A)$ is said to be a **core** for A if the closure $\overline{A|_S}$ of A restricted to S is exactly A. In the example of the differential operator A in the paragraph preceding Definition A.1.4, $C_c^\infty(\mathbb{R})$ is a core for the closure of A. A nice sufficient condition for a dense subset $S \subset \mathcal{D}(A)$ to be a core for A is given in the following theorem, in the case when A is the generator of a contraction semigroup $\{T(t)\}_{t\geq 0}$.

Theorem A.1.9. Let $\{T(t)\}_{t\geq 0}$ be a contraction C_0-semigroup with generator A in a Banach space X, and let $S \subset \mathcal{D}(A)$ be a dense subset in X such that $T(t)x \in S$ for every $x \in S$, for each $t > 0$. Then S is a core for A.

Proof. First we observe the following.

(i) For every $y \in X$, and $t > 0$, the strong integral $\displaystyle\int_0^t T(s)y\, ds$ is defined as the limit of the Riemann sum $\sum_P \left[T(s_j)y(s_{j+1} - s_j)\right] \equiv \Psi(y, P; t)$ in X as the width of the partition $P \equiv \{0 \leq s_1 < s_2 < \ldots < s_n = t\}, |P| = \max(s_{j+1} - s_j) \to 0$, and this limit exists in X since $s \mapsto T(s)y \in X$ is continuous (see Lemma 1.2.5).

(ii) $t^{-1}\int_0^t T(s)y\,ds$ converges to y as $t \to 0^+$, by Lemma 2.1.2.

(iii) By Lemma 1.2.5 and Theorem 2.2.7(c),

$$A\int_0^t T(s)y\,ds = \int_0^t AT(s)y\,ds = (T(t)-I)y,$$

for every $y \in \mathcal{D}(A)$.

Let $x \in \mathcal{D}(A)$, and let $\{x_n\}$ be a sequence of vectors in S such that $x_n \to x$ and $Ax_n \to u$ as $n \to \infty$. Then by the hypothesis of the theorem, $\Psi(x_n, P; t) \in S$ for each n, and

$$\lim_{n\to\infty}\lim_{|P|\to 0} t^{-1}\Psi(x_n, P; t) = t^{-1}\int_0^t T(s)x\,ds, \tag{A.3}$$

while

$$\begin{aligned}\lim_{n\to\infty}\lim_{|P|\to 0} t^{-1}A\Psi(x_n, P; t) &= \lim_{n\to\infty}\lim_{|P|\to 0} t^{-1}\Psi(Ax_n, P; t) \\ &= t^{-1}(T(t)-I)x,\end{aligned} \tag{A.4}$$

for every $t > 0$. On the other hand, since $Ax_n \to u$ as $n \to \infty$, it follows that

$$\lim_{n\to\infty}\lim_{|P|\to 0} t^{-1}\Psi(Ax_n, P; t) = t^{-1}\int_0^t T(s)u\,ds$$

for every $t > 0$. Therefore,

$$t^{-1}\int_0^t T(s)u\,ds = t^{-1}(T(t)-I)x = t^{-1}\int_0^t T(s)Ax\,ds.$$

Taking the limit $t \to 0^+$ in the above equality we have using (ii) that $u = Ax$. This proves that S is a core for A. □

Let H be a Hilbert space. A densely defined operator A is said to be a **symmetric operator** if $A \subset A^*$, that is, $\mathcal{D}(A) \subset \mathcal{D}(A^*)$ and $Af = A^*f$ for all $f \in \mathcal{D}(A)$, and it is said to be **selfadjoint** if furthermore $\mathcal{D}(A) = \mathcal{D}(A^*)$. While every bounded symmetric operator with $\mathcal{D}(A) = H$ is selfadjoint, the condition $A = A^*$ puts severe restrictions in the unbounded case. A symmetric operator may have many closed symmetric extensions, including selfadjoint ones, as the following example shows.

Example A.1.10. Let $H = L^2[0,1]$, and set

$$\mathcal{D}(A_0) = \{f \in H : f \text{ absolutely continuous}, f' \in L^2, f(0) = f(1) = 0\},$$
$$(A_0 f)x = -if'(x).$$

One can compute the adjoint A_0^* to be

$$\mathcal{D}(A_0^*) = \{f \in H : f \text{ absolutely continuous}, f' \in L^2\}$$
$$(A_0^* f)x = -if'(x),$$

showing that A_0 is symmetric but not selfadjoint, that is, $A_0 \subsetneq A_0^*$. One can compute all the selfadjoint extensions of A_0 (**Exercise** A.1.11) to find that there are uncountably many selfadjoint extensions, denoted by A_α, for $\alpha \in \mathbb{C}, |\alpha| = 1$, with

$$\mathcal{D}(A_\alpha) = \{f \in H : f \text{ absolutely continuous}, \ f' \in L^2, f(1) = \alpha f(0)\}.$$

A criterion for selfadjointness is included in the next result:

Theorem A.1.12. (i) For a symmetric operator $A, x \in \mathcal{D}(A)$ and $z \in \mathbb{C}$, with $\operatorname{Im} z \neq 0$,

$$\|(A + z)x\|^2 = \|(A + \operatorname{Re}\ z)x\|^2 + |\operatorname{Im} z|^2 \|x\|^2.$$

(ii) A closed symmetric operator is selfadjoint if and only if the ranges of the operators $A \pm i$ are both H.

From Definition A.1.4 and Theorem A.1.12, it can be shown that (see for example [1, pages 56 and 200]) the open upper and lower half planes in $\mathbb{C}$ are contained in the resolvent set of a selfadjoint operator in a Hilbert space and hence its spectrum is a closed subset of $\mathbb{R}$.

One of the simplest, in fact, almost canonical examples of a selfadjoint operator in a Hilbert space $L^2(\mu)$ is that of an operator of multiplication by a real-valued, measurable function $\phi\colon \Omega \to \mathbb{C}$ with

$$\mathcal{D}(M_\phi) = \{f \in L^2(\mu) : \phi f \in L^2(\mu)\}$$
$$(M_\phi f)(x) = \phi(x) f(x) \quad \forall f \in \mathcal{D}(M_\phi), \text{ and } x \in \Omega.$$

In fact, the above definition extends to $L^p(\mu), 1 \leq p < \infty$, with

$$\mathcal{D}(M_\phi) = \{f \in L^p(\mu) : \phi f \in L^p(\mu)\}.$$

Theorem A.1.13. Let $1 \leq p < \infty$ and ϕ be as above. Then

(i) M_ϕ in $L^p(\mu)$ is bounded if and only if $\phi \in L^\infty$ and $\|M_\phi\| = \|\phi\|_\infty$;

(ii) M_ϕ in $L^p(\mu)$ is a closed, densely defined operator and

$$\begin{aligned}\sigma(M_\phi) = &\text{ essential range of } \phi \\ \equiv &\bigcap_{\epsilon>0} \big\{\lambda \in \mathbb{C} : \mu\{x \in \Omega : |\phi(x) - \lambda| < \epsilon\} > 0\big\};\end{aligned}$$

(iii) if $p = 2$ and if ϕ is real valued, then M_ϕ is selfadjoint in $L^2(\mu)$.

Proof. (i) It is clear that M_ϕ is bounded if $\phi \in L^\infty(\mu)$, and in such a case, $\|M_\phi\| \leq \|\phi\|_\infty$. The equality follows from the definition of L^∞ norm as essential supremum (**Exercise** A.1.14). Conversely, if ϕ is not essentially bounded, then for every $m \in \mathbb{N}$, if we set $S_m = \{x \in \Omega : |\phi(x)| \geq m\}$, S_m will have a non-zero positive measure and therefore $\|M_\phi f\|_p \geq m\|f\|_p$, if $f = \chi_{S_m} \neq 0$.

(ii) Let $f \in L^p(\mu)$. The sequence $\{f_m\}$, given by

$$f_m(x) = \Big(1 + \frac{|\phi(x)|}{m}\Big)^{-1} f(x), \quad m = 1, 2, \ldots,$$

is well defined, measurable and satisfies $|f_m(x)| \leq |f(x)| \quad \forall x \in \Omega$ and $f_m(x) \to f(x)$ pointwise as $m \to \infty$. An application of the Dominated Convergence Theorem proves the convergence of f_m to f in $L^p(\mu)$. It is clear that for each m, $f_m \in \mathcal{D}(M_\phi)$, proving that M_ϕ is densely defined. That M_ϕ is closed is a simple consequence of the fact that if $g_m \to g$, in $L^p(\mu)$ norm, as $m \to \infty$, then $g_{m_k}(x) \to g(x)$ pointwise μ-almost everywhere, as $k \to \infty$, for a subsequence $\{m_k\} \subset \{m\}$.

Now suppose $\lambda \notin$ ess ran ϕ. Then there exists $\delta > 0$ such that

$$|\phi(x) - \lambda| > \delta \text{ for } \mu - \text{ almost all } x.$$

Thus $\psi(x) = (\phi(x) - \lambda)^{-1}$ is in $L^\infty(\mu)$ and by (i) above, M_ψ is a bounded inverse of $M_\phi - \lambda$. Therefore, $\lambda \in \rho(M_\phi)$. Conversely, if $\lambda \in \rho(M_\phi)$, then $M_{(\phi-\lambda)^{-1}}$ is a bounded operator, which on using (i) shows that $\lambda \notin$ ess ran ϕ.

(iii) If ϕ is real valued, then it follows from the definition that M_ϕ is symmetric in the Hilbert space $L^2(\mu)$ and is closed. Furthermore, $\phi(x) \pm i \neq 0$, and $|(\phi(x) \pm i)^{-1}| \leq 1 \ \forall x \in \Omega$. Therefore every $f \in L^2(\mu)$ may be written as $f(x) = \big((M_\phi \pm i)g\big)(x)$ for $g(x) = (\phi(x) \pm i)^{-1} f(x) \in L^2(\mu)$, and this shows that $Ran(M_\phi \pm i) = L^2(\mu)$. Hence self-adjointness follows from Theorem A.1.12(ii).

□

We shall often need the following form of the Spectral Theorem.

Theorem A.1.15. Each selfadjoint operator is unitarily equivalent to a real multiplication operator. More precisely, if A is a selfadjoint operator on a Hilbert space H, then there exist a σ-finite measure space $(\Omega, \mathcal{F}, \mu)$ and a measurable function $\phi\colon \Omega \to \mathbb{R}$, such that A is unitarily equivalent to the multiplication operator M_ϕ on $L^2(\mu)$.

We refer to [1], [20] and [21] for proofs and discussions on the material presented in Appendix A.1 as well as in Appendix A.2.

A.2 Fourier transforms

Here we shall define and discuss the Fourier transform, which has very wide applications in various areas of analysis.

By a *multi-index* α we shall mean that $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d)$ where $\alpha_1, \ldots, \alpha_d \in \mathbb{N} \bigcup \{0\}$. Define $|\alpha| := \sum_{j=1}^{j=d} \alpha_j$, and $x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_d^{\alpha_d}$, for all $x \in \mathbb{R}^d$ and $D^\alpha = D_1^{\alpha_1} \cdots D_d^{\alpha_d}$, where $D_j = \frac{\partial}{\partial x_j}$. Thus,

$$D^\alpha = D_1^{\alpha_1} \ldots D_d^{\alpha_d} = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \frac{\partial^{\alpha_2}}{\partial x_2^{\alpha_2}} \cdots \frac{\partial^{\alpha_d}}{\partial x_d^{\alpha_d}}.$$

First we define the class $\mathcal{S}(\mathbb{R}^d)$, called the Schwartz class of functions on $\mathbb{R}^d$, the linear space of arbitrarily often differentiable complex-valued functions of rapid decrease at infinity as:

$$\mathcal{S}(\mathbb{R}^d) \equiv \big\{ f \in C^\infty(\mathbb{R}^d) : c_{\alpha,l}(f) < \infty \text{ for every } l \in \mathbb{N} \cup \{0\} \text{ and multi-index } \alpha \big\}$$

where

$$c_{\alpha,l}(f) := \sup_{x \in \mathbb{R}^d} \big| \big(1 + |x|^2\big)^{l/2} (D^\alpha f)(x) \big|. \tag{A.5}$$

A typical member of $\mathcal{S}(\mathbb{R}^d)$ is the Gaussian function $h_0(x) = \exp\left(-\frac{1}{2}|x|^2\right)$ and also note that $C_c^\infty(\mathbb{R}^d)$ is a strict subset of $\mathcal{S}(\mathbb{R}^d)$ and therefore $\mathcal{S}(\mathbb{R}^d)$ is dense in $L^2(\mathbb{R}^d)$, in fact in every $L^p(\mathbb{R}^d)$ for $1 \leqslant p < \infty$. For $f \in L^1(\mathbb{R}^d)$, define the Fourier transform $\hat{f}$ of f as:

$$\hat{f}(k) = (2\pi)^{-d/2} \int f(x) e^{-i\langle k,x\rangle}\, dx, \tag{A.6}$$

where $k \in \mathbb{R}^d$ and $\langle .,. \rangle$ is the Euclidean inner product on $\mathbb{R}^d$. It is clear that $\hat{f}$ is well defined everywhere in $\mathbb{R}^d$ as a bounded continuous function which converges to 0 at ∞ and hence is uniformly continuous in $\mathbb{R}^d$. (**Exercise** A.2.1). It is useful to compute the Fourier transform of h_0 to find that

$$\hat{h}_0(k) = \exp\Big(-\frac{1}{2}|k|^2\Big) = h_0(k). \tag{A.7}$$

The details of this computation are left as (**Exercise** A.2.2).

Lemma A.2.3. Let $f \in \mathcal{S}(\mathbb{R}^d)$. Then
(i) $\widehat{\big(\frac{\partial^l f}{\partial x_j{}^l}\big)}\big(k\big) = \big(ik_j\big)^l \hat{f}\big(k\big) \quad \forall\ k \in \mathbb{R}^d,\ j \in \{1, 2, \ldots, d\},\ l \in \mathbb{N};$

(ii) $\widehat{\big(x_j{}^l f\big)}\big(k\big) = i^l \big(\frac{\partial^l f}{\partial k_j{}^l}\big)\big(k\big) \quad \forall\ k \in \mathbb{R}^d,\ j \in \{1, 2, \ldots, n\},\ l \in \mathbb{N}.$

The proof consists of treating the defining integral (A.6) for $\widehat{\big(\frac{\partial^l f}{\partial x_j{}^l}\big)}\big(k\big)$ and $\widehat{\big(x_j{}^l f\big)}\big(k\big)$ respectively as an iterated integral, performing an integration by parts in the jth integral and using the property (A.5) of rapid decrease at infinity of f and all its partial derivatives. The details are left as (**Exercise** A.2.4).

The next theorem is an immediate consequence of this lemma and sums up the essential properties of the Fourier transform map in the Hilbert space $L^2(\mathbb{R}^d)$.

Theorem A.2.5. (i) $\mathcal{S}(\mathbb{R}^d)$ is invariant under the Fourier transformation and the map defined as $\mathcal{F}f(k) = \hat{f}(k)\ \forall\ k \in \mathbb{R}^d$ and $f \in \mathcal{S}(\mathbb{R}^d)$ maps $\mathcal{S}(\mathbb{R}^d)$ bijectively to $\mathcal{S}(\mathbb{R}^d)$.

(ii) (Plancherel) Furthermore, $\mathcal{F}$ is an isometry in $L^2(\mathbb{R}^d)$-norm on $\mathcal{S}(\mathbb{R}^d)$ and thus extends to a unitary operator on $L^2(\mathbb{R}^d)$. (We denote the unitary extension by the same symbol $\mathcal{F}$).

Proof. (i) By virtue of Lemma A.2.3 and the definition of the Schwartz class (A.5), it follows easily that $\hat{f} \in C^\infty(\mathbb{R}^d)$ for every $f \in \mathcal{S}(\mathbb{R}^d)$. Combining both

the results of Lemma A.2.3, we get that, for any $l \in 2\mathbb{N}$,

$$\big(1+|k|^2\big)^{l/2}\Big(\frac{\partial^{|m|}\hat{f}}{\partial k_1^{m_1}...\,\partial k_d^{m_d}}\Big)(k) = \mathcal{F}\big\{(1-\Delta)^{l/2}\big[(-ix_1)^{m_1}...(-ix_d)^{m_d}f\big]\big\}(k),$$

proving that $\hat{f} \in \mathcal{S}(\mathbb{R}^d)$.

For the second part of (i), we need to construct the inverse Fourier transform. Let f, $g \in \mathcal{S}(\mathbb{R}^d)$, $\alpha > 0$ and $y \in \mathbb{R}^d$. Then

$$\begin{aligned}\int_{\mathbb{R}^d} g(\alpha k)\hat{f}(k)e^{i\langle k,y\rangle}dk &= (2\pi)^{-d/2}\int g(\alpha k)\int f(x)e^{i\langle k,y-x\rangle}\,dx\,dk \\ &= \alpha^{-d}\int f(x)\big\{(2\pi)^{-d/2}\int g(k)e^{i\langle\frac{y-x}{\alpha},k\rangle}\,dk\big\}\,dx \quad \text{(A.8)}\end{aligned}$$

where we have used Fubini's theorem to change the order of integration and made a change of variable $k \longrightarrow \alpha k \in \mathbb{R}^d$. Continuing with the above computation, we get that the right hand side of (A.8) equals

$$\alpha^{-d}\int f(x)\hat{g}\Big(\frac{x-y}{\alpha}\Big)\,dx = \int f(\alpha u+y)\hat{g}(u)\,du. \tag{A.9}$$

Letting $\alpha \longrightarrow 0^+$, by the Dominated Convergence Theorem and the properties of the function f and g, the right hand side of (A.9) converges to $f(y)\int \hat{g}(u)\,du$. Thus, letting $\alpha \to 0^+$ in (A.8) gives

$$g(0)\int \hat{f}(k)e^{i\langle k,y\rangle}dk = f(y)\int \hat{g}(u)\,du. \tag{A.10}$$

Next we pick g to be the special function h_0 and note from (A.7) that

$$\int \hat{h}_0(u)du = \int h_0(u)du = \int e^{-|u|^2/2}\,du = (2\pi)^{d/2}$$

to lead to the inverse Fourier transform:

$$f(y) = (2\pi)^{-d/2}\int \hat{f}(k)e^{i\langle k,y\rangle}\,dk. \tag{A.11}$$

This completes the proof of (i).

(ii) Using the symbol $\mathcal{F}$ for the map of the Fourier transform in equation (A.8) and setting $\alpha = 1$ and $y = 0 \in \mathbb{R}^d$, we get

$$\int g(k)(\mathcal{F}f)(k)\,dk = \int f(u)(\mathcal{F}g)(u)\,du. \tag{A.12}$$

Now set $\bar{h} = \mathcal{F}g$ (which belongs to $\mathcal{S}(\mathbb{R}^d)$ by part (i)) and observe that

$$g(k) = (\mathcal{F}^{-1}\bar{h})(k) = (2\pi)^{-d/2} \int \bar{h}(u) e^{i\langle k,u\rangle}\, du$$
$$= \overline{\left\{(2\pi)^{-d/2} \int h(u) e^{-i\langle k,u\rangle} du\right\}} = \bar{\hat{h}}(k).$$

Substituting the above in (A.12) leads to the **Plancherel relation**, viz.

$$\big\langle \mathcal{F}f, \mathcal{F}h \big\rangle_{L^2(\mathbb{R}^d)} = \int \hat{f}(k)\overline{\hat{h}(k)}\, dk = \big\langle f, h \big\rangle_{L^2(\mathbb{R}^d)}. \tag{A.13}$$

Thus, $\mathcal{F}$ defined on $\mathcal{S}(\mathbb{R}^d)$ is an isometric map with range equal to $\mathcal{S}(\mathbb{R}^d)$, which is dense in $L^2(\mathbb{R}^d)$, and therefore, by standard arguments, extends to a unitary operator in $L^2(\mathbb{R}^d)$. □

Remark A.2.6. (a) Note that, though the expression for $f \in \mathcal{S}(\mathbb{R}^d)$,

$$\hat{f}(k) \equiv \mathcal{F}f(k) = (2\pi)^{-d/2} \int f(x) e^{-i\langle k,x\rangle}\, dx,$$

makes sense for every $k \in \mathbb{R}^d$, the extension, viz. $\mathcal{F}f(k)$ for $f \in L^2(\mathbb{R}^d)$ makes sense for almost all (Lebesgue) $k \in \mathbb{R}^d$ and the right hand side of the above identity has to be interpreted as such. However, the sequence of characteristic functions $\{\chi_n\}$ of balls of radius N about the origin in $L^2(\mathbb{R}^d)$ converges strongly to the identity operator in $L^2(\mathbb{R}^d)$ and we have that

$$\big\|\mathcal{F}f - \mathcal{F}(\chi_N f)\big\|_2^2 = \big\|f - \chi_N f\big\|_2^2 \longrightarrow 0$$

as $N \longrightarrow \infty$. Hence as $N \longrightarrow \infty$, $(2\pi)^{-d/2} \int \chi_N f(x) e^{-i\langle k,x\rangle} dx$ converges to $\mathcal{F}f(k)$ in the L^2-sense. This in turn implies that for a suitable subsequence $\{N_l\} \subset \{N\}$, $(2\pi)^{-d/2} \int_{|x|\le N_l} f(x) e^{-i\langle k,x\rangle}\, dx$ converges pointwise almost everywhere to $(\mathcal{F}f)(k)$.

(b) Consider $\mathbb{R}^d$ as the locally compact abelian group under addition (called the translation group in $\mathbb{R}^d$) with the Lebesgue measure as the Haar (invariant) measure on that group. Denote the associated transformation T_y (for $y \in \mathbb{R}^d$) as

$$\big(T_y f\big)(x) := f(x - y).$$

Then it is clear that for $f, g \in L^2(\mathbb{R}^d)$, $T_y f \in L^2(\mathbb{R}^d)$, $\langle T_y f, T_y g\rangle = \langle f, g\rangle$ and

$$\begin{aligned}\widehat{(T_y f)}(k) &= (2\pi)^{-d/2}\int T_y f(x) e^{-i\langle k,x\rangle}dx \\ &= (2\pi)^{-d/2}\int f(x-y)e^{-i\langle k,x-y+y\rangle}dx \\ &= e^{-i\langle k,y\rangle}\hat{f}(k),\end{aligned}$$

that is, the factor $\exp\big(-i\langle k,.\rangle\big)$ is the (one-dimensional irreducible representation) character of the translation group $\mathbb{R}^d$ in its regular representation in $L^2(\mathbb{R}^d)$. This result can easily be extended to any $L^p(\mathbb{R}^d)$ (except for the Plancherel relation) whenever translation can be defined.

(c) So far we have only talked about the Fourier transform on $L^2(\mathbb{R}^d)$ and the question arises about how much of these results can be extended to L^p with arbitrary $p \in [1, \infty)$. By what we have said so far, it is clear that the Fourier transform map $\mathcal{F}$ has the following properties.

(i) $\mathcal{F}\colon L^1(\mathbb{R}^d) \longrightarrow L^\infty(\mathbb{R}^d)$ and $\|\mathcal{F}f\|_\infty \leqslant (2\pi)^{-d/2}\|f\|_1$;

(ii) $\mathcal{F}\colon L^2(\mathbb{R}^d) \longrightarrow L^2(\mathbb{R}^d)$ and $\|\mathcal{F}f\|_2 = \|f\|_2$.

Therefore, interpolating (see pages 11–34 of [21]) between $p = 1$ and $p = 2$, one gets that

$$\mathcal{F}\colon L^p(\mathbb{R}^d) \longrightarrow L^q(\mathbb{R}^d) \text{ and } \|\mathcal{F}f\|_q \leqslant (2\pi)^{d\left(\frac{1}{2}-p^{-1}\right)}\|f\|_p, \qquad \text{(A.14)}$$

where $1 \leqslant p \leqslant 2$ and $q^{-1} + p^{-1} = 1$ (called the **Hausdorff-Young inequality**). For $2 < p < \infty$, in general, the Fourier transform of $f \in L^p(\mathbb{R}^d)$ need not be a function, defined almost everywhere. However, a wider meaning can be associated to the Fourier transform, making it meaningful, in the sense of distributions (see, for example, [22]).

(d) The results of Lemma A.2.3 can be used to give meaning to *the operation of differentiation*, well beyond the usual *classical* meaning of it. In other words, if we rewrite the result

$$\left(\mathcal{F}\Big(\frac{\partial}{\partial x_j}f\Big)\right)(k) = ik_j\big(\mathcal{F}f\big)(k) \text{ as } \mathcal{F}\frac{\partial}{\partial x_j} = \big(ik_j\big)\mathcal{F}, \text{ for each } j = 1, 2\ldots d,$$

then this gives rise to the following.

$$\mathcal{D}\Big(\frac{\partial}{\partial x_j}\Big) = \big\{f \in L^2(\mathbb{R}^d) : \int \big|k_j\hat{f}(k)\big|^2 dk < \infty\big\}$$

and the action is given as above. From this one can show, for example, that if $f \in \mathcal{D}\big(\frac{\partial}{\partial x_j}\big)$ $(j = 1, 2, ..., d)$, then f is bounded and continuous, but in general need not be *classically* differentiable (**Exercise** A.2.7).

A.3 Sobolev spaces

In this section we define some Sobolev spaces and some results concerning them that we need. Although the ideas are implicit in our treatment, we avoid using explicitly the language of distributions. For details and proofs of the results quoted here we refer the reader to [3] and [22].

Let $\Omega = \mathbb{R}^d$ for $d > 1$, or let Ω be an open interval of $\mathbb{R}$, finite or infinite. Recall that $C_c^\infty(\Omega)$ is the space of infinitely differentiable functions with compact support in Ω (also called the space of **test functions** on Ω) while $L^p(\Omega)$ denotes, as usual, the Lebesgue measurable functions on Ω whose pth power is Lebesgue integrable. For $k \in \mathbb{N}$ and $1 \le p \le \infty$, the **Sobolev space** $W^{k,p}(\Omega)$ is defined as

$$W^{k,p}(\Omega) = \Big\{ f \in L^p(\Omega) : \text{ for each multi-index } \alpha \text{ with } |\alpha| \le k, \text{ there exists } f_\alpha \in L^p(\Omega) \text{ such that } \int_\Omega f_\alpha \overline{\phi}\, dx = (-1)^{|\alpha|} \int_\Omega f\big(\overline{D^\alpha \phi}\big)\, dx \ \forall \phi \in C_c^\infty(\Omega) \Big\},$$

where D^α has been defined earlier in Section A.2. The element f_α in the above definition is called the **weak derivative** of f of order α and we write $D^\alpha f := f_\alpha$.

Thus, $W^{k,p}(\Omega)$ consists of those functions f in $L^p(\Omega)$ whose weak derivative $D^\alpha f$ lies in $L^p(\Omega)$ for every multi-index α with $|\alpha| \le k$; clearly, $W^{0,p}(\Omega) = L^p(\Omega)$.

$W^{k,p}(\Omega)$ is a Banach space with respect to the norm

$$\|f\|_{k,p} \equiv \|f\|_{W^{k,p}(\Omega)} := \sum_{|\alpha| \le k} \|D^\alpha f\|_p$$

Further, we denote by $W_0^{k,p}(\Omega)$ the closure of $C_c^\infty(\Omega)$ in $W^{k,p}(\Omega)$.

For $p = 2$, the following defines an equivalent norm on $W^{2,p}(\Omega)$:

$$\|f\|_{k,2} := \Big(\sum_{|\alpha| \le k} \|D^\alpha f\|_2^2 \Big)^{1/2}.$$

This makes $W^{2,p}(\Omega)$ a Hilbert space with the inner product

$$\langle f, g\rangle = \sum_{|\alpha|\le k} \int_\Omega D^\alpha f \overline{D^\alpha g}\, dx.$$

We write

$$H^k(\Omega) := W^{k,2}(\Omega) \quad \text{and} \quad H_0^k := W_0^{k,2}(\Omega),$$

as has been noted earlier in Chapter 7.

The following characterisation in terms of Fourier transforms holds for the spaces $H^k(\mathbb{R}^d)$, as has been noted earlier in Remark A.2.6(d).

Lemma A.3.1. Let $k \in \mathbb{N} \cup \{0\}$ and $f \in L^2(\mathbb{R}^d)$. Then $f \in H^k(\mathbb{R}^d)$ if and only if the function $x \mapsto (1 + |x|^2)^{k/2} \mathcal{F}f(x) \in L^2(\mathbb{R}^d)$.

When $d = 1$ and Ω is an interval in $\mathbb{R}$ (bounded or unbounded) we have the following useful characterisation of $W^{k,p}(\Omega)$.

Lemma A.3.2. Let $1 \le p < \infty$ and Ω be an interval in $\mathbb{R}$. Then $f \in W^{k,p}(\Omega)$ if and only if f has continuous derivatives up to order $(k-1)$, $f^{(k-1)}$ is absolutely continuous and $f^{(j)} \in L^p(\Omega)$ for $j \in \{0, 1, \ldots, k\}$.

Proof. We include a brief sketch of a proof. From the definition of $W^{k,p}(\Omega)$, it follows that for $k \ge 2$,

$$W^{k,p}(\Omega) = \{f \in W^{(k-1),p}(\Omega) : Df \in W^{(k-1),p}(\Omega)\}.$$

Thus, it can be seen that if the statement of the above theorem holds for $k \in \mathbb{N}$, then it holds for $k + 1$ as well. So it is enough to prove the result for $k = 1$. Now if f is absolutely continuous, and $f' \in L^p(\Omega)$, then it is clear that $f \in W^{1,p}(\Omega)$. Conversely, $f \in W^{1,p}(\Omega)$ implies that $f' \in L^p(\Omega)$. Therefore, by Hölder's inequality, it follows that f' is in $L^1_{\mathrm{loc}}(\Omega)$. The fundamental theorem of integral calculus now implies that f is absolutely continuous. □

Finally, for a function $f \in W^{2,p}(\Omega)$ where $1 \le p < \infty$, we define the **Laplacian** Δf of f as

$$\Delta f = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial {x_j}^2}. \tag{A.15}$$

For $f, g \in H^2(\Omega)$, using Lemma A.2.3 and Plancherel's relation (Theorem A.2.5(ii)) we have:

$$\langle \Delta f, g\rangle_{L^2(\Omega)} = \langle \mathcal{F}(\Delta f), \mathcal{F}g\rangle_{L^2} = -\langle \mathcal{F}\nabla f, \mathcal{F}\nabla g\rangle_{L^2} = -\langle \nabla f, \nabla g\rangle_{L^2(\Omega)}, \tag{A.16}$$

where

$$\nabla f = \text{gradient } f = \Big(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_d}\Big).$$

References

[1] W.O. Amerin, J.M. Jauch, and K.B. Sinha. *Scattering Theory in Quantum Mechanics.* Reading, M.A; W.A. Benjamin, 1977.

[2] W. Arendt, C.J.K. Batty, M. Hieber and F. Nuebrander. *Vector-valued Laplace Transforms and Cauchy Problems.* 2nd Edition, Monographs in Mathematics, Birkhauser, Basel 2011.

[3] H. Brezis. *Functional Analysis, Sobolev Spaces and Partial Differential Equations.* Springer, 2011.

[4] P.R. Chernoff. *Note on product formulas for operator semigroups.* Journal of Functional Analysis, 2 (1968), 238–242.

[5] J.B. Conway. *A Course in Functional Analysis.* Springer (India), 2008.

[6] E.B. Davies. *Quantum Dynamical Semigroups and Neutron diffusion Equations.* Rep. Math. Physics, 11 (1977) 169–89.

[7] E.B. Davies. *One-Parameter Semigroups.* Academic Press Inc. 1980.

[8] K.J. Engel and R. Nagel. *One-Parameter Semigroups for Linear Evolution Equations.* Springer-Verlag, Berlin, 2000.

[9] W. Feller. *An Introduction to Probability Theory and its Applications.* Vol 2, John Wiley, New York, 1966.

[10] R.P. Feynman and A.R. Hibbs. *Quantum Mechanics and Path Integrals.* Dover Books on Physics, 2010.

[11] J.A. Goldstein. *Semigroups of Linear Operators and Applications.* Oxford Univ. Press, Oxford, 1985.

[12] E. Hille and R.S. Phillips. *Functional Analysis and Semi-groups.* Amer. Math. Soc., Providence, 1957.

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7

[13] T. Kato. *On the semigroups generated by Kolmogorov's differential equations.* J. Math. Soc. Japan, 11, 1954, 169–189.

[14] T. Kato. *Trotter Product Formula for an arbitrary pair of selfadjoint contractions.* Topics in functional analysis (essays dedicated to M. G. Kreĭn on the occasion of his 70th birthday), Adv. in Math. Suppl. Stud., 3, Academic Press, New York-London, 1978, 185–195.

[15] T. Kato. *Perturbation Theory of Linear Operators.* Springer Verlag, New York, 2nd Edition, 1980.

[16] G. Lumer and R.S. Phillips. *Dissipative operators in a Banach space.* Pacific J. Math. No. 11, 1961, 679–698.

[17] K. R. Parthasarathy. *Probability Measures on Metric Spaces.* AMS Chelsea Publishing, 2005.

[18] K. R. Parthasarathy. *Introduction to Probability and Measure.* Texts and Readings in Mathematics, Hindustan Book Agency, 2005.

[19] A. Pazy. *Semigroups of Linear Operators and Applications to Partial Differential Equations.* Springer-Verlag, Berlin, 1983.

[20] M. Reed and B. Simon. *Methods of Modern Mathematical Physics.* Vol I. Academic Press, New York, 1972.

[21] M. Reed and B. Simon. *Methods of Modern Mathematical Physics.* Vol II. Academic Press, New York, 1975.

[22] M. Renardy and R.C. Rogers. *An Introduction to Partial Differential Equations.* Texts in Applied Mathematics, Vol. 13, 2nd Edition, Springer-Verlag New York, 2004.

[23] H. Royden and P. Fitzpatrick. *Real Analysis.* Pearson, 4th Edition, 2010.

[24] W. Rudin. *Real and Complex Analysis.* Higher Mathematics series, McGraw-Hill Education, 3rd Edition, 1987.

[25] K.B. Sinha and D. Goswami. *Quantum Stochastic Processes and Non-Commutative Geometry.* Cambridge Tracts in Maths 169, Cambridge University Press, 2007.

[26] K. Yosida. *On the differentiability and the representation of one-parameter semigroups of linear operators.* J. Math. Soc. Japan, No. 1, (1948), 15–21.

[27] K. Yosida. *Functional Analysis.* Springer Verlag, New York, 1974.

Index

A^*, 36, 149
$\overline{A}$, 148
$BUC(\mathbb{R}_+)$, 19
$BUC^1(\mathbb{R}_+)$, 38
$\mathcal{B}(X)$, 1
$C^1[0,1]$, 60
$C[0,1]$, 60
$C_c^\infty(\mathbb{R}_+)$, 39
$C_0(\mathbb{R}_+)$, 16
C_0, 19
$\mathcal{D}(\cdot)$, 22
Δ, 47, 159
$\tilde{D}$, 138
D^α, 153
$\mathbb{E}$, 106, 116
$\mathcal{F}f$, 64, 154
$H^1(\mathbb{R}^d)$, 141
$H^2(\mathbb{R}^d)$, 63
$H^k(\Omega)$, 159
$l^1_{\mathbb{R}}$, 116
$l^\infty_{\mathbb{R}}$, 116
$L^p[a,b]$, 42
$L^p(\mathbb{R}_+)$, 39
$L^p(\mathbb{R})$, 41
$L^1_{loc}(\mathbb{R})$, 48
$L^1_{\mathbb{R}}(\Omega, \digamma, \mu)$, 116
$L^\infty_{\mathbb{R}}(\Omega, \digamma, \mu)$, 116
M_ϕ, 42, 77, 151
$M_n(\mathbb{C})$, 21
$R(\lambda, A)$, 148
$\mathbb{R}_+$, 5
$\rho(A)$, 26, 148
S_θ, 67
$\mathcal{S}(\mathbb{R}^d)$, 153
$\sigma(A)$, 26, 148
$\sigma_d(A)$, 88
$\sigma_e(A)$, 88
T, 13
T^*, 36
T_*, 117
$W^{1,p}(\mathbb{R}_+)$, 39
$W^{k,p}(\Omega)$, 158
X^*, 2
X_*, 116
X, 1
$\chi_\triangle$, 1, 2, 5

almost separably valued, 2

Banach algebra, 10
birth process, 135
Bochner integrable, 5
Bochner integral, 6

© Springer Nature Singapore Pte Ltd. 2017 and Hindustan Book Agency 2017

K.B. Sinha and S. Srivastava, *Theory of Semigroups and Applications*,
Texts and Readings in Mathematics, DOI 10.1007/978-981-10-4864-7

Brownian motion, 44, 105
    standard, 105

Cauchy measure, 47
Central Limit Theorem, 108, 111
Chapman-Kolmogorov equation, 119
Chernoff's Theorem, 97
convergence of semigroups, 90

death process, 134
determining set, 3
differential operator, 138
discrete state space, 118
dual injection, 59

ergodic, 111
essential range, 42
expectation, 106, 116

Feynman-Kac Formula, 102
finitely valued, 2
Fourier transform, 154

Gaussian function, 154
generator
    heat semigroup, 47
    holomorphic semigroup, 67
    multiplication semigroup, 43
    perturbation of, 81
    relatively bounded perturbation of, 84
group
    C_0, 31
    generator, 31
    isometry, 41
    Schrödinger free evolution, 79
    unitary, 64

Hausdorff-Young inequality, 157
Hille Yosida Theorem, 32, 35
holomorphic function, 66
    weakly, 66
    strongly, 66

Jensen's inequality, 45, 49

Laplacian, 47, 159
Lie Product Formula, 100
Lumer Phillips Theorem, 61

Markov maps, 116
Markov semigroup, 117
Mean Ergodic Theorem, 111, 112
multi-index, 153

normal probability measure, 47

operator
    adjoint of, 149
    accretive, 53
    closable, 147
    closed, 147
    closure of, 148
    core of, 98, 149
    dissipative, 53, 59
    maximal dissipative, 53
    multiplication, 152
    relatively bounded, 84, 86
    relatively compact, 86
    resolvent of, 148
    sectorial, 69
    selfadjoint, 78, 150
    symmetric, 150

unbounded, 147
operator valued function, 12
strongly continuous, 13
strongly measurable, 13
uniformly continuous, 13
uniformly measurable, 12
weakly continuous, 13
weakly measurable, 13

Plancherel Relation, 156
positive cone, 116, 117

relative bound, 84
resolvent, 26
resolvent equation, 148
resolvent set, 26

Schrödinger equation, 144
Schwartz class, 153
semigroup, 13
C_0, 19, 25
adjoint, 36
bounded holomorphic, 67
contraction, 19
convolution, 44
diffusion, 44
exponential growth bound, 17
Gaussian, 44
generator, 26
heat, 44, 47
holomorphic, 66, 67
left shift, 41
minimal, 130
multiplication, 42, 43, 77
nilpotent, 42
right shift, 41
strongly continuous, 19
translation, 41
type, 17
separably valued, 2
simple function, 2
Sobolev space, 158
Spectral Theorem, 78, 153
spectrum, 26
discrete, 88
essential, 88
of selfadjoint operator, 151
Stone's Theorem, 64
strictly elliptic, 138
strongly measurable, 2
subadditive function, 10

test functions, 158
Trotter-Kato Product, 99

wave equation, 141
weak derivative , 158
weakly elliptic, 138
weakly measurable, 2
Wiener measure, 103, 105

Texts and Readings in Mathematics

1. R. B. Bapat: Linear Algebra and Linear Models (Third Edition)
2. Rajendra Bhatia: Fourier Series (Second Edition)
3. C. Musili: Representations of Finite Groups
4. H. Helson: Linear Algebra (Second Edition)
5. D. Sarason: Complex Function Theory (Second Edition)
6. M. G. Nadkarni: Basic Ergodic Theory (Third Edition)
7. H. Helson: Harmonic Analysis (Second Edition)
8. K. Chandrasekharan: A Course on Integration Theory
9. K. Chandrasekharan: A Course on Topological Groups
10. R. Bhatia (ed.): Analysis, Geometry and Probability
11. K. R. Davidson: C*-Algebras by Example (Reprint)
12. M. Bhattacharjee et al.: Notes on Infinite Permutation Groups
13. V. S. Sunder: Functional Analysis – Spectral Theory
14. V. S. Varadarajan: Algebra in Ancient and Modern Times
15. M. G. Nadkarni: Spectral Theory of Dynamical Systems
16. A. Borel: Semisimple Groups and Riemannian Symmetric Spaces
17. M. Marcolli: Seiberg-Witten Gauge Theory
18. A. Bottcher and S. M. Grudsky: Toeplitz Matrices, Asymptotic Linear Algebra and Functional Analysis
19. A. R. Rao and P. Bhimasankaram: Linear Algebra (Second Edition)
20. C. Musili: Algebraic Geometry for Beginners
21. A. R. Rajwade: Convex Polyhedra with Regularity Conditions and Hilbert's Third Problem
22. S. Kumaresan: A Course in Differential Geometry and Lie Groups
23. Stef Tijs: Introduction to Game Theory
24. B. Sury: The Congruence Subgroup Problem
25. R. Bhatia (ed.): Connected at Infinity
26. K. Mukherjea: Differential Calculus in Normed Linear Spaces (Second Edition)
27. Satya Deo: Algebraic Topology: A Primer (Corrected Reprint)
28. S. Kesavan: Nonlinear Functional Analysis: A First Course
29. S. Szabó: Topics in Factorization of Abelian Groups
30. S. Kumaresan and G. Santhanam: An Expedition to Geometry
31. D. Mumford: Lectures on Curves on an Algebraic Surface (Reprint)
32. J. W. Milnor and J. D. Stasheff: Characteristic Classes (Reprint)
33. K. R. Parthasarathy: Introduction to Probability and Measure (Corrected Reprint)
34. Amiya Mukherjee: Topics in Differential Topology
35. K. R. Parthasarathy: Mathematical Foundations of Quantum Mechanics
36. K. B. Athreya and S. N. Lahiri: Measure Theory
37. Terence Tao: Analysis I (Third Edition)
38. Terence Tao: Analysis II (Third Edition)
39. W. Decker and C. Lossen: Computing in Algebraic Geometry
40. A. Goswami and B. V. Rao: A Course in Applied Stochastic Processes

41. K. B. Athreya and S. N. Lahiri: Probability Theory
42. A. R. Rajwade and A. K. Bhandari: Surprises and Counterexamples in Real Function Theory
43. G. H. Golub and C. F. Van Loan: Matrix Computations (Reprint of the Fourth Edition)
44. Rajendra Bhatia: Positive Definite Matrices
45. K. R. Parthasarathy: Coding Theorems of Classical and Quantum Information Theory (Second Edition)
46. C. S. Seshadri: Introduction to the Theory of Standard Monomials (Second Edition)
47. Alain Connes and Matilde Marcolli: Noncommutative Geometry, Quantum Fields and Motives
48. Vivek S. Borkar: Stochastic Approximation: A Dynamical Systems Viewpoint
49. B. J. Venkatachala: Inequalities: An Approach Through Problems
50. Rajendra Bhatia: Notes on Functional Analysis
51. A. Clebsch (ed.): Jacobi's Lectures on Dynamics (Second Revised Edition)
52. S. Kesavan: Functional Analysis
53. V. Lakshmibai and Justin Brown: Flag Varieties: An Interplay of Geometry, Combinatorics, and Representation Theory
54. S. Ramasubramanian: Lectures on Insurance Models
55. Sebastian M. Cioaba and M. Ram Murty: A First Course in Graph Theory and Combinatorics
56. Bamdad R. Yahaghi: Iranian Mathematics Competitions, 1973-2007
57. Aloke Dey: Incomplete Block Designs
58. R. B. Bapat: Graphs and Matrices (Second Edition)
59. Hermann Weyl: Algebraic Theory of Numbers (Reprint)
60. Carl Ludwig Siegel: Transcendental Numbers (Reprint)
61. Steven J. Miller and Ramin Takloo-Bighash: An Invitation to Number Theory (Reprint)
62. John Milnor: Dynamics in One Complex Variable (Reprint)
63. R. P. Pakshirajan: Probability Theory: A Foundational Course
64. Sharad S. Sane: Combinatorial Techniques
65. Hermann Weyl: The Classical Groups: Their Invariants and Representations (Reprint)
66. John Milnor: Morse Theory (Reprint)
67. R. Bhatia (ed.): Connected at Infinity II
68. Donald Passman: A Course in Ring Theory (Reprint)
69. Amiya Mukherjee: Atiyah-Singer Index Theorem: An Introduction
70. Fumio Hiai and Dénes Petz: Introduction to Matrix Analysis and Applications
71. V. S. Sunder: Operators on Hilbert Spaces
72. Amiya Mukherjee: Differential Topology
73. David Mumford and Tadao Oda: Algebraic Geometry II